Proofs, Counterexamples, and Structural Reductions for Written on the Wall II

An integrated account of Rounds 1–22 with structural extensions

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Revised integrated edition

Abstract

We consolidate the mathematical results in twenty-two rounds of work on the graph-conjecture catalogue *Written on the Wall II*. Complete arguments establish the archived statements 434d, 434e, and 298, and explicit counterexample families refute 394, 392e, 392f, 395a, 382c, 269, 302, 171, 382e, 154, and the specified historical form of 354. The latter uses coefficient one-half and a union of open neighborhoods. These fourteen source-qualified conclusions were reached by Round 14; later rounds strengthen partial results without increasing that count. This revised edition also proves triangle-free counterexample families for 394, its validity on bipartite graphs, and an exact bound using an odd-cycle transversal, together with further independent-domination corollaries. Further restricted results concern 40, 61, 165, 199, and 384b. An eleven-vertex counterexample refutes the lower-rank quartile reading of 389a, while the upper-rank quartile-one case is proved; neither result is counted as an unqualified resolution of 389a. For statement 133 we prove the original unrounded inequality whenever average local independence is at most five, with larger ranges at radii two and three. The proof combines density bounds, radius classifications, and closed finite extension certificates. For 384b we show that any failure has a connected representative of order at most 37, with between 11 and 33 vertices of degree at least three. For 199 we establish several split-graph traceability cases and derive strict Hall and capacity conditions for any residual obstruction; cycle elimination remains unresolved. Further results concern induced forests and residue, triangle-free graphs, spanning-tree leaves, claw-free graphs, even-distance traceability, and degree frequencies. The paper integrates complete analytic arguments and explicitly identified finite certificates. Established inputs and adapted constructions are credited, and the accompanying package records source integrity, verification programs, and their outputs. No claim of worldwide priority, formal proof-assistant certification of the whole paper, or unrestricted resolution of the remaining conjectures is made.

Keywords. graph conjectures; domination; annihilation number; induced paths; traceability; degree sequences; computer-assisted proof; reproducible certificates.

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1 Introduction and exact results ledger

Written on the Wall II (WoWII) is the graph-conjecture catalogue associated with E. DeLaViña and Graffiti.pc. Its formulas combine classical graph invariants with less standard quantities such as degree-class frequencies, radial-circle orders, and neighborhood statistics. The precise definitions matter: changing a set's open neighborhood to its external neighborhood, interpreting a frequency as a matching number, or rounding an average can change the statement being tested. We retain the archived formulations and give each needed definition at its point of use [2, 3].

This paper was prepared from five supplied archive bundles covering Rounds 1–22. The bundles contain 51 main LaTeX manuscripts together with proof notes, exact examples, checkers, and certificates. The extracted source files used here were matched against the archive bytes by SHA-256. The proofs are reorganized by mathematical theme: common constructions are proved once, later statements replace superseded intermediate bounds, and previously separate dependencies are connected by internal references. Auxiliary notes with proved statements are included even when they were not separate main papers. Appendix A gives the source map and reproduction scope.

1.1 Fourteen complete conclusions about original formulations

The affirmative conclusions are the following. Write i for independent domination, d_* for degeneracy, B for the periphery, C for the center, $\rho = \max_x d(x, C)$, and a for the annihilation number. The notation is defined fully in the proof sections.

Item	Established inequality and strengthening	Proof
434d	$i(G) \le d_*(\overline{G}) - c(G[N[B]]) + 2$. More generally, $i(G) + p_4(G) - 2 \le d_*(\overline{G})$ for a maximum distance-four packing p_4.	Theorem 2.1; Corollary 2.2
434e	$i(G) \le d_*(\overline{G}) - \rho(G) + 2$. The quantity ρ is maximum distance *to the center*.	Theorem 2.10
298	$\gamma_t(G) \le a(G) + \chi_{C_4}(G)$. The stronger inequality $\gamma_t \le a$ holds whenever a cycle of length divisible by four occurs.	Theorem 7.11; Corollary 7.12

Here $\chi_{C_4} = 1$ when there is no ordinary four-cycle, and zero otherwise. The proof of 298 includes the antecedent annihilation and total-domination argument with explicit attribution; the global $a + 1$ bound is not presented as a new contribution.

The eleven refutations are witnessed by the following explicit graphs. Every row is supported by a general construction and an exact invariant calculation in the indicated section. The displayed order need not be the smallest possible order of a counterexample.

Item	Order	Reversed comparison	Location and qualification
394	15	$9 > 8$	Corollary 3.3
392e	20	$11 > 10$	Corollary 3.8
392f	39	$22 > 21$	Corollary 3.8
395a	33	$24 > 23$	Proposition 3.9; a unique-mode witness also has order 34
382c	24	$16 > 15$	Corollary 3.11; an integer-valued violation avoids rounding ambiguity
269	268	$4 < 5$	Theorem 5.9; minimum only within the specified family
302	40	$8 > 7$	Corollary 6.3
171	407	$53 < 54$	Subsection 8.1; periphery distances and even-distance counts use the original conventions
382e	21	$4 > 3$	Subsection 8.2; all Maxine runs have the same output size
154	20	$13 < 14$	Subsection 8.3; the ratio of radial-circle orders, distinct from item 155
354	36	$14 < 15$	Subsection 8.4; coefficient $1/2$ and open neighborhood union, using a credited Feit construction

The count is therefore three proofs and eleven refutations of specified original formulations. It was reached in Round 14. It counts mathematical conclusions established in this source corpus, not independently certified new discoveries, journal acceptances, or catalogue updates. Previously known statements merely used or reproduced, including the separately recorded items 299, 412d, and 310, are not added to it. Rounds 15–22 add substantial theorems but no further complete original-item conclusion.

1.2 Structural extensions in the revised edition

Section 4 allows arbitrary internal graphs of minimum degree at least two in the 394 spoke construction, gives triangle-free counterexamples on 15 and 19 vertices and corresponding infinite families, proves 394 on bipartite graphs, and derives an odd-cycle-transversal bound attained by both new families. The independent-domination section also extends the packing inequality to nonempty graphs without isolated vertices and proves $\operatorname{diam}(G) \le 2(d_*(\overline{G}) + 1 - i(G)) + 3$ for connected nontrivial graphs. These are stronger structural results for already counted entries. They do not increase the fourteen-item count.

1.3 The strongest retained partial results

The following table records the terminal claims. Their conditions are part of the conclusions, rather than qualifications to be dropped when counting resolved items.

Item	Proved scope	Location
40	Graphs with a nonempty vertex cover whose every vertex has at least two pendant neighbors.	Proposition 21.4

Item	Proved scope	Location
61	Triangle-free graphs with diameter at most six; triangle-free graphs with feedback vertex number at most one; and triangle-free graphs whose ordinary matching number is $\lceil \operatorname{diam}/2 \rceil$. The last two regimes give $f - \operatorname{res} \geq \lceil \operatorname{diam}/2 \rceil$. In addition, all support-graph expansions with private leaves in Theorem 21.1, without a triangle-free assumption.	§10.3, 10.4, 10.5, 21.1
133	The original unrounded local-independence inequality for $\ell \leq 5$, and additionally $r = 2, \ell \leq 7$ and $r = 3, \ell \leq 6$. Sharp density statements, radius classifications, and four- and eleven-core theorems are proved on the way.	Theorem 15.8
155	All nontrivial connected graphs whose maximum spanning-tree leaf number is at most three.	Theorem 11.1
165	Maximum degree at most two; or independent maximum-degree set M, $3 \leq \Delta \leq 5$, with every vertex outside M of degree at most two. The latter class satisfies $L \geq 2 + \lvert N(M) \setminus M \rvert/4$.	§21.2
177, 180	All connected claw-free graphs in the original domain.	Section 9
189	Every deficient independent set forces an explicitly traceable graph of order $2d_2 + 1$; every possible counterexample has a perfect 2-matching. Infinite example families exclude several automatic stronger reductions.	Section 20
199	Split-graph cases include $\kappa \leq 3$, the stated maximum-column bound, balanced sides and their padding range, $\lvert C \rvert \leq 2\kappa$, equal sides, and Hall equality. The residual case satisfies strict Hall, an exact degree-two factor condition, strict capacity, and directed alternating reach. The revision also proves $\alpha \leq \kappa + 3$ under the original premise, using the Tan–Hung classification for the last boundary.	§17–19, 21.4
384b	The inequality holds when the number h of degree-at-least-three vertices is at most 10 or at least 34. If a counterexample exists, a connected representative has order at most 37 and $11 \leq h \leq 33$. At $h = 33$, a counterexample has exactly two cubic vertices with triangle counts two and three, and the stated attachment restrictions.	Theorem 16.1; §21.5
389a	Sufficient conditions in terms of degree frequency, the number of low-degree vertices, and a specified first-quartile convention. The upper-rank quartile-one case holds. An explicit eleven-vertex counterexample refutes the lower-rank reading; the source ambiguity is retained and no unqualified closure is counted.	§12, 21.6, 21.7

The residue material also includes exact forest equality, matching-padding and weak-majorization consequences, inheritance under split composition, and the corresponding unigraph result. These supporting theorems are included with proofs and are not separate original-item closures.

1.4 Notation and logical distinctions

All graphs are finite, simple, and undirected. A connected graph's order and size are n and m. The neighborhood $N(v)$ is open. Unless a section explicitly says otherwise, $N(A) = \bigcup_{a \in A} N(a)$ is the union of open neighborhoods and can intersect A; it does not automatically mean $N(A) \setminus A$. The closed neighborhood is $N[A]$. Distance is measured in the ambient graph. The center minimizes vertex eccentricity, and the periphery maximizes it. The historical catalogue term "boundary" is interpreted according to that periphery definition where the archived formulation requires it.

We distinguish ordinary domination γ, total domination γ_t, and two-domination γ_2. An independent dominating set is an inclusion-maximal independent set, and i minimizes its size. The ordinary matching number is denoted by ν or μ in the locally defined settings; induced and acyclic matchings are separately defined. The residue res comes from the Havel–Hakimi elimination process. The largest induced forest has order f, while τ in the traceability sections is the largest *connected* induced-tree order. The longest induced path p is counted in vertices; diameter is counted in edges.

For Conjecture 133, the local-independence sum is denoted by Λ and its unrounded mean by ℓ. This avoids confusing that sum with spanning-tree leaves or low-degree vertex sets. Notation used only in a construction, such as its parameter r or its low-degree set L, is explicitly reset there. No proof passes between these meanings without a new definition.

1.5 Proof organization and verification

The affirmative independent-domination results precede the shared two-domination modules and the total-domination arguments. The remaining counterexample families and structural leaf and residue results follow. The longer partial programs for 133, 384b, and 199 are presented as connected proof chains, and the even-distance reduction for 189 closes the mathematical development.

An analytic proof is distinguished from a finite supplement to it. For the P_8 and P_9 core classifications and the exact finite kernels and rational inequalities in 384b, the specified certificate is itself a proof dependency. The accompanying verifier checks its complete coverage and exact arithmetic. Construction programs, canonical-labeling heuristics, and exploratory bounded searches are not silently promoted to proof dependencies. In particular, the order-37 result for 384b is an existence-of-a-representative reduction; it is neither an assertion that every counterexample has that order bound nor a completed census through that order. For 199, a factor containing cycles is explicitly distinguished from a Hamiltonian path.

2 Independent domination, complement degeneracy, and the center

We begin with the two affirmative results numbered 434d and 434e. For a graph F, let $i(F)$ be the minimum cardinality of an inclusion-maximal independent set, and put $i(\varnothing) = 0$. For a nonempty graph F write

$$d_*(F) = \max_{\varnothing \neq X \subseteq V(F)} \delta(F[X]).$$

Thus d_* is degeneracy itself. In particular, the catalogue's Szekeres–Wilf quantity in these two statements is not degeneracy plus one. For connected G, write $C(G)$ and $B(G)$ for its center

and periphery, and define

$$\rho(G) = \max_{x \in V(G)} d_G(x, C(G)), \qquad d_G(x, C(G)) = \min_{c \in C(G)} d_G(x, c).$$

The set eccentricity $\rho(G)$ is different from the radius. It is zero when every vertex is central, and always satisfies $\rho(G) \le \mathrm{rad}(G)$. For a vertex set A, $N[A]$ denotes its closed neighborhood, and $\mathrm{comp}(F)$ is the number of connected components of F.

With these conventions, the two original inequalities, stated for connected graphs of order greater than three, are

$$i(G) \le d_*(\overline{G}) - \mathrm{comp}(G[N[B(G)]]) + 2, \tag{1}$$
$$i(G) \le d_*(\overline{G}) - \rho(G) + 2. \tag{2}$$

These conventions agree with the preserved catalogue and with the independent-domination paper of DeLaViña, Pepper, and Waller [2, 4]. Both inequalities will be proved for all connected graphs of order at least two.

2.1 Distance-four packings: statement 434d

Let $p_4(G)$ be the maximum cardinality of a vertex set whose distinct members have pairwise distance at least four.

Theorem 2.1 (Packing strengthening of 434d)**.** *Every connected graph G of order at least two satisfies*

$$i(G) + p_4(G) - 2 \le d_*(\overline{G}).$$

Proof. Put $H = \overline{G}$ and $k = i(G)$. Every maximal clique of H has at least k vertices. If $p_4(G) = 1$, a maximal clique immediately gives $d_*(H) \ge k - 1$, which is the required inequality.

Now take $S = \{s_1, \dots, s_c\}$ with pairwise distances at least four, where $c \ge 2$. For each j, choose a neighbor t_j of s_j in G. The closed neighborhoods $N_G[s_j]$ are disjoint, and there are no G-edges between two of them: any such edge would give a path of length at most three between the corresponding members of S. Consequently $T = \{t_1, \dots, t_c\}$ and S are cliques of H, $s_i t_j$ is an H-edge for $i \ne j$, and $s_j t_j$ is not an H-edge.

For a clique K of H, put

$$C_H(K) = \{x : K \subseteq N_H[x]\}, \qquad U = \bigcup_{j=1}^{c} C_H(T \setminus \{t_j\}).$$

In particular $S \subseteq U$. We show that every vertex of $H[U]$ has degree at least $k + c - 2$. Fix $x \in U$. There is an index j such that

$$T \setminus \{t_j\} \subseteq N_H[x], \qquad S \setminus \{s_j\} \subseteq N_H(x).$$

Indeed, if $x \in N_G[s_j]$ for some j, the disjointness and absence of edges between these closed neighborhoods give both inclusions for that index. If x lies outside their union, it is H-adjacent to all of S, and any index witnessing $x \in U$ suffices.

Extend the clique $\{x\} \cup (T \setminus \{t_j\})$ to a maximal clique Q of H. Then $|Q| \ge k$ and $Q \subseteq U$. None of the vertices s_i with $i \ne j$ belongs to Q, because $t_i \in Q$ and $s_i t_i$ is not an H-edge. Thus x has the disjoint sets $Q \setminus \{x\}$ and $S \setminus \{s_j\}$ among its neighbors in $H[U]$. This gives $d_{H[U]}(x) \ge (k-1)+(c-1)$. The definition of degeneracy now proves the result with $c = p_4(G)$. □

Corollary 2.2 (Statement 434d)**.** *For every nonempty $A \subseteq V(G)$ in a connected graph of order at least two,*

$$i(G) \le d_*(\overline{G}) - \mathrm{comp}(G[N[A]]) + 2.$$

In particular (1) *holds.*

Proof. Choose one vertex of A from each component of $G[N[A]]$. Each component contains a member of A. Representatives from different components cannot be at distance at most three: a path of length at most three between two vertices of A lies entirely in $N[A]$. They therefore form a distance-four packing. Apply Theorem 2.1 and then take $A = B(G)$. □

Example 2.3 (Sharpness). Let $c \ge 2$, and subdivide every edge of $K_{1,c}$ once. Denote its vertices by $z, t_1, s_1, \ldots, t_c, s_c$, with edges zt_j, t_js_j. Every independent dominating set either omits z and uses at least one of t_j, s_j for each j, or contains z and all s_j. The set of all t_j is independent and dominating, so $i(G) = c$. The leaves s_j are pairwise at distance four, while no other vertex can coexist in a distance-four packing with another vertex; hence $p_4(G) = c$. In the complement, deleting z leaves the complement of cK_2, a $(2c-2)$-regular graph. The complement degree of z is $c \le 2c-2$, so this also proves $d_*(\overline{G}) = 2c-2$. The periphery consists of the s_j, and its closed neighborhood induces cK_2 in G. Thus both the packing bound and statement 434d hold with equality.

2.2 Minimal vertex covers and distance to the center

For a graph F, define

$$q(F) = \max\{|Q| : Q \text{ is an inclusion-minimal vertex cover of } F\}.$$

Minimality is by inclusion, not by cardinality. Complements interchange minimal vertex covers and maximal independent sets, so

$$q(F) = |V(F)| - i(F), \qquad q(\varnothing) = 0. \tag{3}$$

Lemma 2.4. *If b is a nonisolated vertex of F, some minimal vertex cover of F contains b.*

Proof. Extend a neighbor of b to a maximal independent set. Its complement is the required cover. □

Lemma 2.5. *If F is connected and $Q \ne \varnothing$ is a vertex cover, there is a tree $T \subseteq F$ containing Q, all of whose leaves belong to Q, such that* $\mathrm{diam}(T) \le 2|Q| - 2$. *A one-vertex tree is allowed.*

Proof. Choose a tree containing Q with the fewest vertices. Deleting a leaf outside Q would contradict this choice. Vertices outside Q are independent. A longest path therefore has endpoints in Q, at most $|Q|$ vertices in Q, and at most $|Q| - 1$ vertices outside Q. □

Theorem 2.6 (A local cover bound)**.** *Let G be connected of order at least two, let $v \in V(G)$, and put $R = G - N_G[v]$. Then*

$$\rho(G) \le \max\{2, q(R) + 1\}. \tag{4}$$

If $q(R) > 0$, then for every prescribed $u \in V(G)$ there is a tree subgraph $T \subseteq G$ whose vertex set dominates G, which contains u, and which satisfies $\mathrm{diam}(T) \le 2q(R) + 2$.

Proof. Put $A = N_G(v)$ and $q_R = q(R)$. If $q_R = 0$, R is independent. Each vertex of R has a neighbor in A, by connectivity, so $\mathrm{ecc}_G(v) \le 2$ and $\rho(G) \le \mathrm{rad}(G) \le 2$.

Suppose $q_R \ge 1$. Let $F_1, \dots, F_s$ be the nontrivial components of R. Choose in each F_j a vertex b_j adjacent to some $a_j \in A$, and a minimal vertex cover Q_j containing b_j by Lemma 2.4. Their union is a minimal vertex cover of R. If $q_j = |Q_j|$ and $q = \sum_j q_j$, then $1 \le q \le q_R$. Choose a tree T_j as in Lemma 2.5. Form T_0 from the star with center v and leaves A, all the T_j, and the single attachment edges $a_j b_j$. This is a tree, even when some a_j coincide. Its vertex set dominates G: it contains v, A, and the covers Q_j, while an isolated vertex of R has a neighbor in A.

Two vertices in distinct T_j, T_ℓ have distance at most $(2q_j - 2) + 4 + (2q_\ell - 2) \le 2q$. A vertex of T_j is at distance at most $2q_j + 1$ from any vertex of the star. Hence $\mathrm{diam}(T_0) \le 2q + 1$. If $u \notin T_0$, it lies in R; attach it as a single leaf to a neighbor in $\bigcup_j Q_j$ if it is nonisolated in R, and to a neighbor in A otherwise. The resulting dominating tree T contains u and has diameter at most $2q_R + 2$.

A center z of T has tree eccentricity at most $q_R + 1$. Because T dominates G,

$$\mathrm{ecc}_G(z) \le q_R + 2, \qquad d_G(u, z) \le q_R + 1.$$

Thus $\mathrm{rad}(G) \le q_R + 2$. If the radius is at most $q_R + 1$, then the same is true of $\rho(G)$. In the remaining integer case $\mathrm{rad}(G) = q_R + 2$, the vertex z is a center of G. Since u was arbitrary, every vertex is within $q_R + 1$ of a center, proving the result. □

Corollary 2.7. *Every connected graph G of order $n \ge 2$ satisfies*

$$\rho(G) \le \max\{2, n - \Delta(G) - i(G) + 1\} = \max\{2, \delta(\overline{G}) - i(G) + 2\}.$$

Proof. Choose v of maximum degree. If $R = G - N_G[v]$, adjoining v to an independent dominating set of R gives one of G. Therefore $i(R) \ge i(G) - 1$ and $q(R) = n - \Delta(G) - 1 - i(R) \le n - \Delta(G) - i(G)$. Apply Theorem 2.6. □

2.3 The equality boundary and statement 434e

The following boundary argument was already present in the supplied baseline [5]. We include its proof because it is needed to complete the unrestricted result. The local cover theorem above is the ingredient that treats the remaining positive-slack cases.

Lemma 2.8. *Suppose every maximal clique of a $(k-1)$-degenerate graph H has order k, and $G = \overline{H}$ is connected of order at least two. Put*

$$M = \{m : d_H(m) = k - 1\}, \qquad W = V(G) \setminus N_G[M].$$

The sets $N_G(w)$, for $w \in W$, are linearly ordered by inclusion. Consequently, if $W \ne \varnothing$, its members have a common neighbor.

Proof. Every vertex of H lies in a k-clique. Thus $\delta(H) \ge k - 1$, with equality by degeneracy, and $M \ne \varnothing$. For $m \in M$, $N_H[m]$ is a k-clique. The set W is disjoint from M, is a clique of H, and every $m \in M$ is H-adjacent to all of W.

For nonempty $P \subseteq W$, let

$$L = \{x \notin P : P \subseteq N_H(x)\}, \quad Z = V(H) \setminus (P \cup L), \quad L_0 = \{x \in L : N_H(x) \cap Z = \varnothing\}.$$

We have $M \subseteq L_0$: a neighbor in Z of a simplicial $m \in M$ would be adjacent to every vertex of P, contrary to its membership in Z. Suppose each member of P has an H-neighbor in Z.

In a degeneracy ordering of H, let x be the first vertex outside L_0. If $x \in Z$, it has no earlier neighbor and, as $x \notin M$, still has degree at least k, a contradiction. Otherwise $x \in P \cup (L \setminus L_0)$ and has a neighbor $z \in Z$. Extend xz to a maximal clique K, of order k. None of K belongs to L_0, because its vertices are adjacent to z; all of K therefore survives. The remaining closed neighborhood of x has at most k vertices and contains K, so it equals K. It also contains all of P. This forces z to be adjacent to all of P, a contradiction.

Some $p \in P$ consequently has no H-neighbor in Z. Since P is independent in G, we have $Z = \bigcup_{w \in P} N_G(w)$, and hence $N_G(p) = \bigcup_{w \in P} N_G(w)$. Apply this conclusion to every pair in W to obtain pairwise comparability. Connectivity makes every such neighborhood nonempty, and a smallest one provides a common neighbor. □

Theorem 2.9 (Previously established zero-slack boundary)**.** *If G is connected of order at least two and $i(G) = d_*(\overline{G}) + 1$, then $C(G)$ is a dominating set and $\rho(G) \le 1$.*

Proof. Put $H = \overline{G}$ and $k = i(G)$. Every maximal clique of H has at least k vertices, while $(k-1)$-degeneracy forbids a larger clique. If $k = 1$, G is complete. Otherwise apply Lemma 2.8.

For each $m \in M$, the vertices nonadjacent to m in G form an independent set. Each such vertex has a neighbor, which must itself be adjacent to m. Thus $\mathrm{ecc}_G(m) \le 2$. There is no universal vertex because $i(G) \ge 2$, so $\mathrm{rad}(G) = 2$ and $M \subseteq C(G)$. If $W = \varnothing$, these centers dominate G. Otherwise let y be a common neighbor of W. For any $m \in M$ and $w \in W$, the edge wy forces $my \in E(G)$, by the independence just noted; also $y \ne m$. Hence y is adjacent to all of M, is at distance at most two from every vertex of $N_G[M]$, and is adjacent to every vertex of W. Therefore $y \in C(G)$. The centers $M \cup \{y\}$ dominate G. □

Theorem 2.10 (Statement 434e)**.** *Every connected graph G of order at least two satisfies*

$$i(G) \le d_*(\overline{G}) - \rho(G) + 2.$$

Proof. Write $d = d_*(\overline{G})$ and $h = d + 1 - i(G)$. A maximal clique of $\overline{G}$ has at least $i(G)$ vertices, so $h \ge 0$. If $h = 0$, Theorem 2.9 gives $\rho(G) \le 1 = h + 1$. If $h \ge 1$, Corollary 2.7 gives

$$\rho(G) \le \max\{2, \delta(\overline{G}) - i(G) + 2\} \le \max\{2, h+1\} = h + 1.$$

Rearranging proves the assertion. □

Example 2.11 (Sharpness and the two eccentricities). For P_7, an independent dominating set in positions $1, 4, 7$ has size three, while two closed neighborhoods cover at most six vertices. Thus $i(P_7) = 3$. Its complement has minimum degree four; deleting the middle vertex first leaves the complement of $2P_3$, of maximum degree four. Consequently $d_*(\overline{P_7}) = 4$. The unique center has eccentricity three, so $\rho(P_7) = 3$ and statement 434e is sharp with positive slack.

For the subdivided stars of Example 2.3, taking $v = z$ gives $q(G - N[v]) = 0$ but $\rho(G) = 2$. The constant two in (4) is therefore necessary. Radius cannot replace ρ in statement 434e either. The tree with vertex set $\{0, \dots, 6\}$ and edges $01, 13, 14, 26, 36, 45$ has $i = 3$: $\{1, 2, 5\}$ is an independent dominating set. A dominating pair would have to contain vertex 1, since every other closed neighborhood has size at most three. But 1 dominates neither 2 nor 5, and no second vertex can dominate both, as their distance is five. Its complement has degeneracy three, as witnessed below by its minimum degree and above by the deletion order $1, 6, 4$, followed by the four remaining vertices. Its center is $\{1, 3\}$, its radius is three, and $\rho = 2$.

2.4 Further consequences of the packing and cover arguments

The following two consequences extend the preceding arguments. They do not add catalogue identifiers to the fourteen-item ledger.

Corollary 2.12 (Disconnected graphs without isolated vertices)**.** *Let G be a nonempty finite simple graph without isolated vertices, and put the distance between vertices in different components equal to infinity. Then*

$$i(G) + p_4(G) - 2 \le d_*(\overline{G}).$$

Consequently, for every nonempty $A \subseteq V(G)$,

$$i(G) \le d_*(\overline{G}) - c(G[N[A]]) + 2.$$

Proof. The complement-core proof of Theorem 2.1 uses connectivity only to choose a neighbor of each packing vertex. The absence of isolated vertices supplies exactly that condition. Distinct closed neighborhoods of distance-four-separated vertices are still disjoint and anticomplete, including vertices in different components. All clique extensions and degree counts therefore apply unchanged. Selecting one member of A from each component of $G[N[A]]$ proves the second assertion. □

Corollary 2.13 (Diameter controlled by the degeneracy slack)**.** *For every connected graph G of order at least two, put $h = d_*(\overline{G}) + 1 - i(G)$. Then*

$$\operatorname{diam}(G) \le 2h + 3.$$

Proof. The packing bound gives $p_4(G) \le h + 1$. If $h = 0$, no two vertices are at distance at least four, giving the result. Suppose $h \ge 1$. Choose a maximum-degree vertex v and let $R = G - N[v]$. Corollary 2.7 gives $q(R) \le n - \Delta(G) - i(G) \le h$. If $q(R) = 0$, then $\operatorname{ecc}_G(v) \le 2$, so the diameter is at most four. Otherwise, the base dominating tree T_0 in the proof of Theorem 2.6 has diameter at most $2q(R) + 1$. Each graph vertex belongs to T_0 or has a neighbor in T_0, and consequently $\operatorname{diam}(G) \le \operatorname{diam}(T_0) + 2 \le 2q(R) + 3 \le 2h + 3$. □

3 Rooted constructions for two-domination counterexamples

A two-dominating set D requires every vertex outside D to have at least two neighbors in D; there is no condition on vertices of D. Its minimum size is $\gamma_2(G)$. We use $\nu(G)$ for the matching number, corresponding to μ in the original rows. Put

$$A_\delta(G) = \{v : d_G(v) = \delta(G)\}, \qquad A_2(G) = \{v : d_G(v) = 2\}, \qquad \lambda_G(v) = \alpha(G[N_G(v)]).$$

In particular, A_2 here means degree exactly two. The constructions below refute statements 394, 392e, 392f, 395a, and 382c under their original definitions [2, 6, 7]. All five refutations have unbounded additive deficit. The stated small witnesses are explicit examples, not assertions of global minimal order.

3.1 A clique module and statement 394

Let $M_{\min\lambda}(G) = \{v : \lambda_G(v) = \min_x \lambda_G(x)\}$. The original statement 394 is

$$\gamma_2(G) \le \nu(G) + |E(G[A_2(G)])| + |M_{\min\lambda}(G)|. \tag{5}$$

The final set is denoted by a capital M in the original row; reading it as a second matching-number symbol changes the proposed inequality.

Definition 3.1. Fix $r \geq 2$, $k \geq 3$, and a connected graph H on $\{c_1, \dots, c_r\}$. For each i, introduce a k-clique $Q_i = \{q_{i1}, \dots, q_{ik}\}$ and an independent set $T_i = \{t_{i1}, \dots, t_{ik}\}$. Add the edges $c_i t_{ij}$ and $t_{ij} q_{ij}$ for all i, j. Finally attach a leaf p to c_1. There are no other edges except the edges of H and the cliques Q_i. Write $X(H, k)$ for this graph and H^+ for H with the leaf p attached.

Proposition 3.2. *The graph $X = X(H, k)$ is connected and satisfies*

$$|V(X)| = r(2k+1)+1, \qquad |E(X)| = |E(H)| + r\big(2k + \tbinom{k}{2}\big) + 1,$$
$$\gamma_2(X) = r(k+1)+1, \qquad \nu(X) = rk + \nu(H^+),$$
$$E(X[A_2(X)]) = \varnothing, \qquad M_{\min\lambda}(X) = \{p\}.$$

Proof. The order, size, and connectivity follow directly from the construction. Every two-dominating set contains p. Consider the vertices $\{c_i\} \cup T_i \cup Q_i$. If c_i is selected, each pair $\{t_{ij}, q_{ij}\}$ contributes at least one selected vertex, since an unselected t_{ij} needs both its neighbors selected. This gives at least $k+1$ vertices in the module. If c_i is unselected, all of T_i must be selected. At least one member of Q_i must also be selected, since otherwise each vertex of Q_i has only its one selected neighbor in T_i. This again gives $k+1$. Conversely, p, all the centers c_i, and all the vertices in all the Q_i form a two-dominating set, proving the formula for γ_2.

The rk edges $t_{ij} q_{ij}$ together with a maximum matching of H^+ give a matching of size $rk + \nu(H^+)$. In any matching, the edges outside H^+ use the $2k+1$ vertices of one module and therefore contribute at most k per module. The remaining matching edges belong to H^+, proving the reverse bound.

The only degree-two vertices are those in the T_i, and they form an independent set. The local independence numbers are

$$\lambda_X(p) = 1, \quad \lambda_X(t_{ij}) = 2, \quad \lambda_X(q_{ij}) = 2, \quad \lambda_X(c_i) \geq k.$$

For q_{ij} its neighborhood is a clique of order $k-1$ together with the isolated vertex t_{ij}; for c_i the set T_i is independent. These observations prove the last assertion. □

Corollary 3.3 (Refutation of 394)**.** *Every graph in Definition 3.1 violates (5), with deficit $r - \nu(H^+) > 0$. For $r = 2$, $k = 3$, and $H = K_2$, it has order 15, size 20, $\gamma_2 = 9$, and proposed right side $7 + 0 + 1 = 8$.*

Proof. Subtract the right side from the left using Proposition 3.2. Since H^+ has $r+1$ vertices, $\nu(H^+) \leq \lfloor (r+1)/2 \rfloor$, and the deficit is at least $\lfloor r/2 \rfloor$. Alternatively choose H to be the star centered at c_1 and fix $k = 3$. Then H^+ is also a star, so $\nu(H^+) = 1$. This yields graphs of order $7r+1$ with $\gamma_2 = 4r+1$ and proposed right side $3r+2$, an unbounded deficit $r-1$. □

3.2 One nine-vertex module for 392e, 392f, and 395a

The next three counterexamples use the same module; consolidating it avoids repeating its local two-domination and matching arguments.

Definition 3.4. Let F have vertices $c, t_1, t_2, q_1, q_2, w_1, w_2, w_3, w_4$ and edges

$$ct_1, ct_2, t_1q_1, t_2q_2, w_1w_2, w_2w_3, w_3w_4, w_4w_1, q_1w_1, q_1w_3, q_2w_2, q_2w_4.$$

The vertex c is its root. When copies of F are used, only roots receive neighbors outside their own copies.

Lemma 3.5 (Local cost of the module)**.** *In any graph containing such a copy of F, every two-dominating set contains at least five of its nine vertices. The set $\{c, q_1, q_2, w_1, w_3\}$ two-dominates the module. The module also has a matching of size four avoiding its root.*

Proof. Put $U = \{w_1, w_3\}$ and $W = \{w_2, w_4\}$. At least two of the four w-vertices must be selected. With none selected each has at most its one q-neighbor selected; with just one selected, the other vertex in the same part U or W has the same obstruction.

If c is selected, each pair $\{t_j, q_j\}$ contributes at least one selected vertex, so there are at least five in all. If c is not selected, both t_j are forced. A selection of at most four module vertices would therefore consist of the two t_j and exactly two w-vertices, with no q_j. The two unselected q_j then require at least one selected vertex in each of U, W. An unselected w-vertex would consequently see just one selected vertex in the opposite part and no selected q-neighbor, a contradiction.

The displayed five-set works by direct inspection of the unselected vertices. Finally $t_1q_1, t_2q_2, w_1w_2, w_3w_4$ is the asserted matching. □

The original statements 392e and 392f are, respectively,

$$\gamma_2(G) \le \nu(G) + \nu(G[N(A_2(G))]) + |A_\delta(G)|, \tag{6}$$

$$\gamma_2(G) \le \nu(G) + \nu(G[N(A_2(G))]) + \big|\{v : |N(v) \cap N(A_\delta(G))| = 1\}\big|. \tag{7}$$

Neighborhoods in these formulas are open neighborhood unions.

Definition 3.6. Take $r \ge 2$ disjoint rooted copies F_i of F, and join their roots c_i to a new vertex x. For G_r, add the single edge xp to a new leaf p. For J_r, instead add the path xsp on two new vertices s, p.

Proposition 3.7. *The two families have the following invariants, where* $B_1 = \{v : |N(v) \cap N(A_\delta)| = 1\}$*:*

	G_r	J_r
$\lvert V\rvert$	$9r+2$	$9r+3$
$\lvert E\rvert$	$13r+1$	$13r+2$
γ_2	$5r+1$	$5r+2$
ν	$4r+1$	$4r+2$
$\lvert A_\delta\rvert$	1	1
$\nu(G[N(A_2)])$	0	1
$\lvert B_1\rvert$	$r+1$	2

Proof. The leaf p is forced in a two-dominating set. Each module contributes at least five by Lemma 3.5. For G_r, use p and the indicated five-set in every module; the unselected hub sees all selected roots, and the lower bound is attained. In J_r, the path requires at least one of x, s in addition to p: otherwise s has only one selected neighbor. Choosing p, x and the five-sets attains $5r + 2$.

Each module contributes at most four matching edges. The edges outside the modules all meet x in G_r, and all meet $\{x, s\}$ in J_r. Thus the matching upper bounds are $4r + 1$ and $4r + 2$. They are attained by the root-avoiding four-matchings, together with xp in G_r, or xc_1, sp in J_r.

The unique minimum-degree vertex is p. The degree-two vertices of G_r are exactly the t-vertices. Their open neighborhood consists of roots and q-vertices, an independent set. In J_r, the vertex s also has degree two. Its neighbors add x, p to this set, whose induced graph is a star with center x and leaves c_i, together with isolated vertices. Its matching number is one. Finally $N(A_\delta) = \{x\}$ for G_r, giving $B_1 = N(x)$ of size $r + 1$; for J_r it is $\{s\}$, giving $B_1 = \{x, p\}$. □

Corollary 3.8 (Refutations of 392e and 392f)**.** *Statement 392e fails on G_r for every $r \ge 2$, with deficit $r - 1$, and on J_r for every $r \ge 3$, with deficit $r - 2$. Statement 392f fails on J_r for every $r \ge 4$, with deficit $r - 3$. The displayed deficit formulas remain valid at the smaller defined values of r, but a zero or negative deficit is not a counterexample. In particular, G_2 has order 20, size 27, and $\gamma_2 = 11 > 9 + 0 + 1 = 10$. The graph J_4 has order 39, size 54, and $\gamma_2 = 22 > 18 + 1 + 2 = 21$ for 392f.*

Proof. Substitute the invariant values in Proposition 3.7. The deficits are unbounded as r increases. □

For statement 395a, let the *mode-min degree* be the smallest degree whose frequency is maximum, and let $\mathcal{M}(G)$ be its degree class. The proposed inequality is

$$\gamma_2(G) \le |\mathcal{M}(G)| + \nu(G - A_\delta(G)) + \nu(G[N(A_2(G))]). \tag{8}$$

The cardinality of the mode class is independent of the choice among tied modal degrees; the following construction also supplies witnesses with a unique modal degree.

Proposition 3.9 (Refutation of 395a)**.** *Take $r \ge 2$ copies of F, join their roots to a hub x, and attach $L \ge 7r$ leaves to x. Denote the resulting graph by $K_{r,L}$. Then*

$$|V(K_{r,L})| = 9r + 1 + L, \qquad |E(K_{r,L})| = 13r + L,$$
$$\gamma_2(K_{r,L}) = 5r + L, \qquad |\mathcal{M}(K_{r,L})| = L,$$
$$\nu(K_{r,L} - A_\delta) = 4r + 1, \qquad \nu(K_{r,L}[N(A_2)]) = 0.$$

Consequently the deficit in (8) is $r - 1$.

Proof. All L leaves are forced, and each module costs five. Conversely the leaves and the five-sets from Lemma 3.5 work, including at the unselected hub. The degree frequencies are

degree	1	2	3	$L + r$
frequency	L	$2r$	$7r$	1.

Here A_δ is the leaf set and the mode-min class is that same set. The mode is unique if $L > 7r$. Deleting the leaves leaves r modules and the star edges at x. At most one star edge belongs to a matching, so its matching number is at most $4r + 1$, attained by xc_1 and the four root-avoiding edges in each module. As before, the open neighborhood of the degree-two vertices consists of independent roots and q-vertices, giving the final zero. Subtraction proves the deficit. □

For $(r, L) = (2, 14)$ this gives order 33, size 40, and $\gamma_2 = 24 > 14 + 9 = 23$. The choice $(2, 15)$ gives the unique-mode witness of order 34, size 41, and $\gamma_2 = 25 > 15 + 9 = 24$. Taking $L = 7r$ gives order $16r + 1$ and arbitrarily large deficit.

3.3 Triangle coronas and statement 382c

If $A \neq \varnothing$, define its average vertex eccentricity by

$$\overline{\mathrm{ecc}}_G(A) = \frac{1}{|A|} \sum_{v \in A} \mathrm{ecc}_G(v).$$

This averages eccentricities of individual vertices; it is neither set eccentricity nor average pairwise distance. Statement 382c is

$$\gamma_2(G) \le \frac{\alpha(G)\, \overline{\mathrm{ecc}}_G(A_\delta(G)) + |B(G)|}{2}. \tag{9}$$

The outer brackets in the catalogue serve as grouping symbols. We give an integer-valued violation as well, so the refutation does not depend on a possible rounding interpretation.

Proposition 3.10 (Exact reduction for a triangle corona)**.** *Let H be connected on $m \geq 2$ vertices. Form $G = H \circ K_2$ by attaching to each vertex z a private pair p_z, q_z, adjacent to one another and to z. Then*

$$\gamma_2(G) = 2m, \qquad \alpha(G) = m,$$
$$A_\delta(G) = \{p_z, q_z : z \in V(H)\}, \qquad B(G) = \{p_z, q_z : z \in B(H)\},$$
$$\mathrm{ecc}_G(z) = \mathrm{ecc}_H(z) + 1, \qquad \mathrm{ecc}_G(p_z) = \mathrm{ecc}_G(q_z) = \mathrm{ecc}_H(z) + 2.$$

For these graphs, (9) is equivalent to

$$2m \leq \sum_{z \in V(H)} \mathrm{ecc}_H(z) + 2|B(H)|. \tag{10}$$

Proof. In each private triangle, a two-dominating set must select at least two vertices: if at most one were selected, an unselected private vertex would have fewer than two selected neighbors. Selecting all private vertices attains $2m$. An independent set has at most one vertex in each private triangle, and selecting one private vertex for each z attains m. Private vertices have degree two, while core vertices have degree at least three. Distances between different attached triangles pass through their core vertices. The eccentricity formulas follow, as does the description of the periphery. In particular

$$\overline{\mathrm{ecc}}_G(A_\delta) = 2 + \frac{1}{m} \sum_z \mathrm{ecc}_H(z), \qquad |B(G)| = 2|B(H)|.$$

Substituting these values and $\gamma_2 = 2m$, $\alpha = m$ gives (10). □

Corollary 3.11 (Refutation of 382c)**.** *Let $H_r = K_r \vee 2K_1$ for $r \geq 1$, where the two vertices outside the clique are nonadjacent, and put $L_r = H_r \circ K_2$. Then*

$$|V(L_r)| = 3r + 6, \qquad |E(L_r)| = \tbinom{r}{2} + 5r + 6,$$
$$\gamma_2(L_r) = 2r + 4, \qquad \alpha(L_r) = r + 2,$$
$$|A_\delta(L_r)| = 2r + 4, \qquad |B(L_r)| = 4,$$
$$\overline{\mathrm{ecc}}_{L_r}(A_\delta) = \frac{3r + 8}{r + 2}.$$

The proposed right side is $(3r + 12)/2$, so its deficit is $(r - 4)/2$.

Proof. The r clique vertices of H_r are universal and have eccentricity one. Its other two vertices have eccentricity two and form the periphery. Apply Proposition 3.10; the edge count is $\binom{r}{2} + 2r + 3(r + 2)$. □

Thus L_5 has order 21, size 41, and $\gamma_2 = 14 > 27/2$. The graph L_6 has order 24, size 51, and $\gamma_2 = 16 > 15$, an integer-valued violation. The deficit is unbounded with r.

4 Triangle-free obstructions and the bipartite boundary of 394

The clique module in Corollary 3.3 already refutes 394. We now show that failure persists without triangles, while bipartiteness suffices for validity. These are extensions of the same refutation, not newly resolved numbered entries. Write $D_2 = \{v : d_G(v) = 2\}$, $\lambda_G(v) = \alpha(G[N_G(v)])$, and $M = \{v : \lambda_G(v) = \min_x \lambda_G(x)\}$. The proposed inequality is

$$\gamma_2(G) \leq \mu(G) + |E(G[D_2])| + |M|.$$

Here μ is matching number and every vertex outside a 2-dominating set must have at least two neighbors in that set.

4.1 Arbitrary internal graphs in the spoke construction

Let H be a connected graph on roots $c_1, \dots, c_r$, where $r \geq 1$. For each i, choose a graph F_i on $Q_i = \{q_{i1}, \dots, q_{ik_i}\}$ with minimum degree at least two. Add vertices $t_{i1}, \dots, t_{ik_i}$ and the edges $c_i t_{ij}, t_{ij} q_{ij}$. Apart from the edges of F_i and H, add only one further vertex p, adjacent to a designated root c_1. Call the resulting graph G, and let H^+ be H with this pendant vertex p attached. Put $K = \sum_i k_i$.

Theorem 4.1. *The construction satisfies*

$$|V(G)| = 2K + r + 1, \qquad \gamma_2(G) = K + r + 1, \qquad \mu(G) = K + \mu(H^+).$$

Moreover, $M = \{p\}$ and $D_2 = \bigcup_i \{t_{i1}, \dots, t_{ik_i}\}$ is independent. Thus the excess of the left side of 394 over its right side is exactly $r - \mu(H^+)$.

Proof. Consider a 2-dominating set D. If $c_i \in D$, every pair $\{t_{ij}, q_{ij}\}$ meets D, since otherwise t_{ij} has only one selected neighbor. This gives at least $k_i + 1$ selected vertices in that module. If $c_i \notin D$, all t_{ij} must belong to D. At least one q_{ij} must also be selected: otherwise each omitted q_{ij} has only its one selected neighbor t_{ij}. Again the module contributes at least $k_i + 1$. The pendant vertex p must be selected, proving the lower bound. Equality follows by selecting p, every root, and every vertex in every Q_i.

A matching contains at most k_i internal edges in the module of order $2k_i + 1$. Its remaining edges form a matching in H^+. Conversely, all K edges $t_{ij}q_{ij}$ are disjoint from a maximum matching of H^+. This proves the matching identity.

Each root has at least three pairwise nonadjacent neighbors among its t-vertices. Each t_{ij} has the two nonadjacent neighbors c_i, q_{ij}. Each q_{ij} has its neighbor t_{ij} and at least one neighbor in F_i, which are nonadjacent. Thus every vertex other than p has local independence at least two, whereas $\lambda(p) = 1$. Finally $d(q_{ij}) \geq 3$ and $d(c_i) \geq 3$, so exactly the t-vertices have degree two and no two of them are adjacent. □

This proof uses no clique assumption on F_i. Its counting and matching arguments are elementary; no claim of historical priority is made for the general construction.

Corollary 4.2. *Take every $F_i = C_4$, let H be a star on r roots, and attach p to its center. Then G is triangle-free and*

$$n = 9r + 1, \qquad \gamma_2(G) = 5r + 1, \qquad \mu(G) = 4r + 1.$$

Its deficit in 394 is $r - 1$, so every $r \geq 2$ is a counterexample. In particular $r = 2$ gives a 19-vertex triangle-free graph with $\gamma_2 = 11$ and proposed bound 10.

Proof. The graph H^+ is a star and has matching number one. No triangle uses a spoke, and both H and every F_i are triangle-free. Apply Theorem 4.1. □

4.2 A smaller triangle-free family

Let J have vertex set $\{0, 1, \dots, 6\}$, root 0, and edge set

$$E(J) = \{01, 03, 06, 12, 14, 23, 35, 45, 46\}.$$

This is a triangle-free graph with independent degree-two set $A = \{2, 5, 6\}$.

Lemma 4.3. *A set that 2-dominates every omitted vertex of J other than its root has at least four vertices. The set $\{0, 1, 3, 4\}$ 2-dominates J.*

Proof. Suppose at most three vertices are selected and let a be the number selected from A. An omitted vertex of A forces both its neighbors to be selected. If $a = 0$, their combined neighborhoods $\{0, 1, 3, 4\}$ force four selections. If $a = 1$, the two omitted vertices of A have at least three distinct neighbors, giving at least four selections together. If $a = 2$, the remaining omitted vertex forces two more selections. If $a = 3$, the selected set is exactly A, but the nonroot vertex 1 has only its neighbor 2 selected. These exhaust the possibilities. The stated four-set is checked directly: each omitted vertex $2, 5, 6$ has both its neighbors in it. □

Take r disjoint copies of J, connect their roots as a star centered at the first root, and attach one pendant vertex p to that root. Denote the resulting connected graph by G_r.

Theorem 4.4. *For every $r \geq 1$, G_r is triangle-free and*

$$n = 7r + 1, \quad \gamma_2(G_r) = 4r + 1, \quad \mu(G_r) = 3r + 1, \quad |M| = 1, \quad |E(G_r[D_2])| = 0.$$

Thus the deficit is $r - 1$, so 394 fails for every $r \geq 2$. At $r = 2$ this is a 15-vertex witness with $\gamma_2 = 9$ and proposed bound 8; no minimum-order claim is made.

Proof. Only roots receive external neighbors, so Lemma 4.3 forces four selections in every copy. The pendant p is mandatory. Select p and the displayed four-set in each copy to attain this bound. Every matching has at most three internal edges per copy and at most one external edge, since all external edges share the central root. Attain $3r + 1$ by matching that root to p, using $\{12, 35, 46\}$ in its copy, and using the same three edges in all other copies. The original degree-two vertices remain independent; all roots have degree at least four, all other nonpendant vertices have degree two or three, and the graph is triangle-free. Its local independence equals its degree, so $M = \{p\}$. □

4.3 Bipartite validity and an exact deletion bound

Let P be the pendant set of a connected graph of order greater than two. A set X is an odd-cycle transversal if $G - X$ is bipartite.

Proposition 4.5. *For every such X,*

$$\gamma_2(G) \leq |P \cup X| + \mu(G - (P \cup X)).$$

In particular 394 holds for every connected bipartite graph of order greater than two.

Proof. Put $I = P \cup X$. By König's theorem, the bipartite graph $G - I$ has a vertex cover C of size $\mu(G - I)$. Every vertex outside $I \cup C$ has all its neighbors in $I \cup C$; its original degree is at least two because it is not pendant. Hence $I \cup C$ is 2-dominating. If G is bipartite and has leaves, take $X = \varnothing$; then $\gamma_2 \leq |P| + \mu(G - P) \leq |P| + \mu(G)$ and $M = P$, since local independence equals degree. If there are no leaves, a minimum vertex cover itself is 2-dominating, giving $\gamma_2 \leq \mu(G)$. Both cases imply the proposed bound in 394. □

The proposition is an elementary application of the established bipartite matching–vertex-cover theorem, not a claimed new matching principle. Its value here is the exact comparison with the examples.

Proposition 4.6. *Both triangle-free families above have odd-cycle transversal number exactly r, and the bound in Proposition 4.5 is attained.*

Proof. For the C_4 spoke family, every module contains a five-cycle formed by an edge $q_{ij}q_{i\ell}$ and its two spokes. These r odd cycles are vertex-disjoint. Deleting all roots makes the graph bipartite, so the transversal number is r. Take those roots as X. After also deleting p, the edges $t_{ij}q_{ij}$ form a perfect matching of size $4r$. The deletion bound is therefore $(r+1)+4r = 5r+1$.

For G_r, every copy contains the five-cycle $1, 2, 3, 5, 4, 1$, giving the same lower bound r. Delete vertex 4 from each copy. The remaining copy is a four-cycle $0, 1, 2, 3, 0$ with leaves 5 and 6 attached to 3 and 0; it is bipartite. Its bipartition may be reversed independently in each copy so that all edges of the root star cross. The pendant p also receives the opposite color to its root. Thus these r deletions suffice. After also deleting p, the edges $06, 12, 35$ in every copy form a perfect matching of size $3r$. The bound is $(r+1)+3r = 4r+1$, as asserted. □

The standard-library checker `verification/extensions/module394/verify_extension394.py` enumerates every subset of the 15- and 19-vertex examples to compute their exact 2-domination numbers, and independently computes maximum matchings, local independence, and minimum odd-cycle transversals. These finite checks corroborate the proofs for the infinite families.

5 Total domination and distances between centers: statement 269

Graphs in this section are connected and have order at least two. A total dominating set D requires every vertex, including each vertex of D, to have a neighbor in D. Its minimum cardinality is $\gamma_t(G)$. For a center containing at least two vertices, put

$$\overline{d}_C(G) = \frac{1}{\binom{|C(G)|}{2}} \sum_{\{x,y\}\subseteq C(G)} d_G(x,y).$$

Distances are measured in G, and only pairs of distinct centers are averaged. For the reductions below we set this average to zero when the center is a singleton; all counterexamples have every vertex central and do not require this convention. Statement 269 proposes

$$\gamma_t(G) \geq \left\lceil 1 + \overline{d}_C(G) \right\rceil \quad \text{for connected } C_4\text{-free graphs.} \tag{11}$$

Here C_4-free excludes an ordinary four-cycle, whether induced or not, as in the original definition [2, 3].

We first consolidate the structural reduction from the earlier baseline and Round 2. It shows precisely why the counterexamples subsequently found in Round 5 occur at total domination number and radius four.

5.1 Equality in the radius bound

Lemma 5.1 (Spanning trees preserving a total dominating set)**.** *Let D be a total dominating set of a connected graph G. There is a spanning tree T in which D remains totally dominating. Every such tree satisfies* $\operatorname{diam}(T) \leq 2|D| - 1$. *In particular,* $\operatorname{rad}(G) \leq \gamma_t(G)$.

Proof. Choose a spanning tree of each component of $G[D]$, and attach each vertex outside D to one chosen neighbor in D. This gives a spanning forest. Add edges corresponding to a spanning tree of its connected quotient in G. Every member of D has a neighbor in its chosen component tree, since $G[D]$ has no isolated vertices, and every other vertex has its chosen neighbor. Thus D remains totally dominating.

For any path P in a tree, a vertex has at most two neighbors on P; if it is outside P, it has at most one. Count incidences from D to vertices of P. Every vertex of P has such an incidence, giving $|V(P)| \le 2|D|$. Hence $\operatorname{diam}(T) \le 2|D| - 1$, and the formula for the radius of a tree gives $\operatorname{rad}(G) \le \operatorname{rad}(T) \le |D|$. □

Proposition 5.2 (The equality quotient)**.** *Suppose G is connected and $\gamma_t(G) = \operatorname{rad}(G) = t$. For any minimum total dominating set D:*

1. *$t = 2k$ is even, and $G[D]$ consists of k disjoint edges;*
2. *every vertex outside D has neighbors in exactly one of these edges, possibly in both endpoints of that edge;*
3. *partition $V(G)$ into these k edges and their respective outside neighbors. The simple quotient Q of the partition is a path, a cycle, or a single vertex.*

Every spanning tree preserving total domination by D has diameter $2t - 1$. Both endpoints of its central edge are centers of G.

Proof. For any such tree, Lemma 5.1 and $t = \operatorname{rad}(G) \le \operatorname{rad}(T)$ force $\operatorname{diam}(T) = 2t - 1$. On a diametral path P there are exactly $2t$ vertices. Equality in the incidence count shows that every member of D is internal to P, has both path neighbors on P, and every vertex of P has exactly one neighbor in D. Starting at an endpoint, membership in D along P follows the repeating pattern $0, 1, 1, 0$. The other endpoint has value zero and its predecessor has value one, forcing $2t$ to be a multiple of four. Thus t is even.

If a component of $G[D]$ had at least three vertices, its chosen component tree could be taken to contain a three-vertex path in D. All its vertices lie on P, and uniqueness of tree paths puts that three-vertex path on P. Its middle vertex would have two neighbors in D, a contradiction. Hence $G[D]$ consists of disjoint edges.

If $x \notin D$ had neighbors d_i, d_j in distinct such edges, attach x to d_i and include the edge xd_j in the quotient spanning tree. This is possible because that quotient edge is not a loop and can be extended to a spanning tree. The resulting tree path between d_i and d_j passes through x. Since both endpoints lie on P, so does this path, and x has two D-neighbors on P, again a contradiction. This proves the second assertion and makes the partition in the third assertion unambiguous.

Every spanning tree of Q can now be realized by the construction of Lemma 5.1. Its diametral path contains all of D and therefore meets every part. Each part induces a connected subtree, so the path cannot leave a part and subsequently reenter it. Its projection is a Hamilton path in the chosen quotient tree, which must itself be a path. If Q had a vertex of degree at least three, the three incident edges could be extended to a spanning tree that was not a path. Thus the connected graph Q has maximum degree at most two, proving the classification.

Finally, the endpoints of the central edge of a tree of diameter $2t - 1$ have tree eccentricity t. Their eccentricities in G are at most t and at least $\operatorname{rad}(G) = t$, so they are centers of G. □

Lemma 5.3. *If G is connected and C_4-free, and $\gamma_t(G) = \operatorname{rad}(G) = 2$, then its center is a clique and $\overline{d}_C(G) \le 1$.*

Proof. Let $D = \{u, v\}$, where uv is an edge. Divide the outside vertices into A, adjacent to u only; B, adjacent to v only; and W, adjacent to both. The radius is not one, so both A and B are nonempty. A four-cycle forbids edges from A to B, forbids edges from W to $A \cup B$, and implies $|W| \le 1$. Therefore the distance from any member of A to any member of B is at least three. No member of $A \cup B$ is central. The vertices in $\{u, v\} \cup W$ all have eccentricity two and form a clique, proving the assertion. □

5.2 The radius-four reduction

Theorem 5.4 (The reduction recorded in Round 2)**.** *If a connected C_4-free graph violates* (11)*, then $\gamma_t(G) = \mathrm{rad}(G) = 4$.*

Proof. Write $t = \gamma_t(G)$ and $r = \mathrm{rad}(G)$. Distances between centers are at most r, so if $t > r$, then $\left\lceil 1 + \overline{d}_C(G) \right\rceil \le r + 1 \le t$. Thus a counterexample must have $t = r$. Proposition 5.2 makes t even, and Lemma 5.3 excludes $t = 2$. It remains to prove the inequality when $t = r = 2k$ with $k \ge 3$. Fix a minimum total dominating set with edges a_ib_i, and its quotient Q from Proposition 5.2.

Contacts at an internal part. Suppose a part has two different neighbors in Q. Choose one interpart edge to each of them, and choose a quotient spanning tree containing both. In a spanning tree preserving D, its diametral path contains both D-vertices of the part and traverses both chosen interpart edges. The two contacts must therefore attach to opposite endpoints of its D-edge. The choices of edges and attachments were arbitrary. Consequently all contacts toward one neighboring part are adjacent to exactly one fixed endpoint of the D-edge, and all contacts toward the other neighbor are adjacent to the opposite endpoint. No common neighbor of both endpoints can be a contact.

Case 1: Q is a cycle. Orient its parts cyclically, with incoming contacts at a_i and outgoing contacts at b_i. Choose one interpart edge at each adjacency. The D-edges, their contact attachment edges, and the chosen interpart edges form a simple cycle Z of length $4k$ containing all of D. Delete one interpart edge of Z, and attach each remaining vertex as a leaf to a D-neighbor. This gives a spanning tree preserving D. Its central edge lies on the remaining $4k$-vertex path. If k is odd, that edge is a D-edge; varying the deleted interpart edge proves that all $2k$ members of D are central. If k is even, the central edge is the opposite interpart edge. Because the chosen interpart edges were arbitrary, every endpoint of every interpart edge is central. In particular there are at least $2k$ centers in this case as well.

We use the following elementary metric fact. For m points, with repetitions allowed, on a circle of circumference L, the sum of their ordered pairwise circular distances is at most $Lm^2/4$. Indeed, parameterize antipodal cuts by a half-circle of length $L/2$. If a cut separates s points from the other $m - s$, the number of separated ordered pairs is $2s(m - s) \le m^2/2$. For a fixed pair, the measure of separating cuts is its circular distance. Integration proves the stated bound.

Project each center to a neighbor in D, leaving centers already in D fixed, and view the images on Z. Set $m = |C(G)|$ and $s = |C(G) \cap D|$. Travel to the images, along the shorter arc of Z, and back. The circular estimate and the attachment costs give

$$\overline{d}_C(G) \le \frac{km}{m-1} + 2 - \frac{2s}{m}. \tag{12}$$

For odd $k \ge 3$, we have $s = 2k$ and $m \ge 2k$, whence

$$\overline{d}_C(G) \le k + 2 + \frac{k}{m-1} - \frac{4k}{m} \le k + 2 \le 2k - 1.$$

For even $k \ge 4$, discard the last term of (12) and use $m \ge 2k$ to obtain

$$\overline{d}_C(G) \le k + 2 + \frac{k}{2k-1} \le 2k - 1.$$

Case 2: Q is a path on parts $1, \dots, k$. At internal parts, orient a_i toward the left and b_i toward the right as above. We first show that all contacts at the first part are adjacent to one fixed endpoint of its D-edge, and to no other. Otherwise the first-part contacts collectively meet both endpoints, possibly by a contact adjacent to both. The left contacts at part two are adjacent to

a_2. Thus a_2 is at distance at most three from both first-part D-vertices, and at most four from every vertex in the first part. Choose a path through parts $2, \dots, k$ using one interpart edge at each step. Assign positions $a_i = 4i - 3$, $b_i = 4i - 2$, with the contacts at the intervening integer positions. Its vertex at position $2k$ exists for $k \geq 3$. This vertex is within $2k - 5 + 4 = 2k - 1$ of every first-part vertex. For every other vertex, travel along the chosen path to a D-neighbor; the distance is at most

$$\max\{2k - 5 + 1, (4k - 2) - 2k + 1\} = 2k - 1.$$

This contradicts radius $2k$. Interchanging the endpoints of the first D-edge if necessary, all first-part contacts are consequently private neighbors of b_1. The symmetric argument makes all last-part contacts private neighbors of a_k.

Assign real coordinates to every vertex in part i as follows:

vertex type	a_i	b_i	private to a_i	private to b_i	common
coordinate	$4i - 3$	$4i - 2$	$4i - 4$	$4i - 1$	$4i - \frac{5}{2}$.

The C_4-free assumption forbids an edge between the two private classes, forbids an edge from a common neighbor to a private neighbor, and permits at most one common neighbor. All remaining within-part edges have coordinate difference at most one. Every interpart edge joins coordinates $4i - 1$ and $4i$, by the contact orientations. Thus the coordinate function changes by at most one along every edge.

There must be a private neighbor z of a_1, of coordinate zero. If there were none, choose one edge between each consecutive pair of parts and use the resulting path through all D-edges. Its vertex at coordinate $2k$ is at distance at most $2k - 1$ from a_1 and all other D-vertices. All outside vertices in the first part can be reached through b_1, while all other outside vertices can be reached through one of their D-neighbors, also within $2k - 1$. This again contradicts the radius. Symmetrically there is a private neighbor w of b_k, of coordinate $4k - 1$.

Every center has distance at most $2k$ from both z and w. The coordinate bound forces its coordinate into $[2k - 1, 2k]$. If k is odd, this interval contains only the middle D-edge and its possible common neighbor, a clique. Thus $\overline{d}_C(G) \leq 1$. If k is even, it contains only the outgoing private vertices of part $k/2$ and the incoming private vertices of part $k/2 + 1$. A center on the left must have an interpart neighbor on the right: a shortest path to w has the maximum allowed coordinate increase at every step, including its first step. The same holds with left and right reversed. Moreover both endpoints of every edge across this middle adjacency are central, by the central-edge assertion of Proposition 5.2, realized with that chosen edge. Centers on the same side have distance at most two through their common D-neighbor. If x and y are on opposite sides, take a cross-neighbor x' of x; the path from x to x' and then through the right-hand D-vertex to y has length at most three. Hence $\overline{d}_C(G) \leq 3 \leq 2k - 1$.

Both quotient cases give $\overline{d}_C(G) \leq t - 1$ for $t \geq 6$. Together with the initial reductions, this leaves only $t = r = 4$. □

5.3 An explicit family in the remaining case

Definition 5.5. For positive integers Y, Z, start with four vertices a, b, c, d and edges ab, cd. For $1 \leq i \leq Y$, add a path a, u_i, v_i, d. For $1 \leq j \leq Z$, add a path b, p_j, q_j, c, a vertex x_j adjacent to b, p_j, and a vertex y_j adjacent to c, q_j. There are no further edges. Denote the graph by $R(Y, Z)$. Its order and size are

$$n = 4 + 2Y + 4Z, \qquad m = 2 + 3Y + 7Z.$$

Lemma 5.6. *The graph $R(Y, Z)$ is connected and C_4-free, and $\gamma_t(R(Y, Z)) = 4$.*

Proof. Connectivity is immediate. Temporarily omit the x_j, y_j. Every cycle in the remaining graph either uses two length-three paths with the same endpoints and has length six, or uses one path of each kind together with ab, cd and has length eight. This base graph has neither triangles nor four-cycles. Reinsert the omitted vertices one at a time. Each is adjacent to the endpoints of a base edge, and no base edge receives two such vertices. A new four-cycle would require a previously existing length-two path between those endpoints. Its middle vertex cannot be a base vertex, as that would make a base triangle, or a previously added vertex, as that would repeat the same base edge. Therefore no four-cycle is created.

The set $\{a, b, c, d\}$ totally dominates $R(Y, Z)$. Conversely the four open neighborhoods

$$\{a, v_1\}, \quad \{d, u_1\}, \quad \{b, q_1, x_1\}, \quad \{c, p_1, y_1\}$$

of u_1, v_1, p_1, q_1, respectively, are pairwise disjoint. Every total dominating set meets all four, so its size is at least four. □

Lemma 5.7 (Distances after adding a vertex on an edge)**.** *Adding vertices each adjacent to the endpoints of an existing base edge does not change distances between base vertices. A new vertex with endpoint set A has distance $1 + \min_{a \in A} d_H(a, z)$ from a base vertex z, and two distinct new vertices with endpoint sets A, A' have distance $2 + \min_{a \in A, a' \in A'} d_H(a, a')$, where H is the base graph.*

Proof. A passage between base vertices through a new vertex can be replaced by the existing edge between its two neighbors. A shortest path therefore never needs a new internal vertex. Minimize over the first and last neighbor choices for paths with new endpoints. □

For completeness, the following tables determine every distance in $R(Y, Z)$. Among the hubs,

$$d(a, b) = d(c, d) = 1, \qquad d(a, d) = d(b, c) = 3, \qquad d(a, c) = d(b, d) = 4.$$

Distances to hubs, obtained from the length-three base paths and Lemma 5.7, are

	a	b	c	d
u_i	1	2	3	2
v_i	2	3	2	1
p_j	2	1	2	3
q_j	3	2	1	2
x_j	2	1	3	4
y_j	4	3	1	2

type pair	equal indices	unequal indices
p, q	1	3
p, x	1	2
p, y	2	3
q, x	2	3
q, y	1	2
x, y	3	4

Distinct vertices of any one of the six non-hub types are at distance two. Also $d(u_i, v_h)$ is one for $i = h$ and three otherwise, and

$$(d(u_i, p_j), d(u_i, q_j), d(u_i, x_j), d(u_i, y_j)) = (3, 4, 3, 4),$$
$$(d(v_i, p_j), d(v_i, q_j), d(v_i, x_j), d(v_i, y_j)) = (4, 3, 4, 3).$$

These formulas follow by choosing the closest hub endpoints in the base graph; Lemma 5.7 then handles every pair involving an added triangle vertex. They exhaust all pairs.

Proposition 5.8 (Center and exact distance distribution)**.** *Every vertex of $R(Y, Z)$ has eccentricity four. If N_j is the number of unordered vertex pairs at distance j, then*

$$N_1 = 2 + 3Y + 7Z,$$
$$N_2 = Y^2 + 3Y + 4Z^2 + 4Z,$$
$$N_4 = 4YZ + Z^2 + Z + 2,$$
$$N_3 = \binom{n}{2} - N_1 - N_2 - N_4.$$

The sum of unordered distances is

$$W = 5Y^2 + 28YZ + 21Z^2 + 12Y + 25Z + 16, \tag{13}$$

and its excess over $3\binom{n}{2}$ *is*

$$F(Y, Z) = -Y^2 + 4YZ - 3Z^2 - 9Y - 17Z - 2. \tag{14}$$

Proof. The tables give an upper bound of four on all distances. Partners at distance four are a, c; b, d; u_i, q_1; v_i, p_1; p_j, v_1; q_j, u_1; x_j, d; and y_j, a. Thus the radius and diameter are four and the center is all of V.

The edge count gives N_1. Distance-two pairs among the u, v types contribute $Y(Y-1)$, and between these types and hubs contribute $4Y$. Equal decorated types contribute $2Z(Z-1)$; unequal-index pairs of types p, x and q, y contribute another $2Z(Z-1)$; equal-index pairs of types p, y and q, x contribute $2Z$; and decorated-to-hub pairs contribute $6Z$. These sum to N_2. At distance four there are two hub pairs, $2Z$ decorated-to-hub pairs, $4YZ$ plain-to-decorated pairs, and $Z(Z-1)$ unequal-index pairs of type x, y. This gives N_4. All other pairs have distance three. Finally

$$W - 3\binom{n}{2} = N_4 - 2N_1 - N_2,$$

which expands to (14); adding $3\binom{n}{2}$ gives (13). □

Theorem 5.9 (Refutation of 269)**.** *There are infinitely many connected* C_4*-free graphs with* $\gamma_t = \mathrm{rad} = \mathrm{diam} = 4$ *for which* $\lceil 1 + \overline{d}_C \rceil = 5$*. In particular* $R(62, 35)$ *has order* 268*, size* 433*, and*

$$\overline{d}_C = \frac{17890}{5963} = 3 + \frac{1}{5963}.$$

No member of the family $R(Y, Z)$ *of smaller order violates statement 269.*

Proof. Take $Y = 7s$, $Z = 4s$. Then $n = 30s + 4$ and

$$F(7s, 4s) = 15s^2 - 131s - 2.$$

This equals $34 > 0$ at $s = 9$, and its increment when s increases by one is $30s - 116 > 0$ for $s \geq 9$. Hence the average distance exceeds three for all $s \geq 9$. It is strictly below four because there are edges, while all distances are at most four. Therefore the indicated ceiling is five, contradicting total domination number four. The average tends to $3 + 1/30$ along this subfamily.

For $Y = 62, Z = 35$, the exact distance counts are

$$(N_1, N_2, N_3, N_4) = (433, 9070, 16333, 9942).$$

There are 35778 unordered pairs and $W = 107340$, giving the displayed reduced fraction.

To prove minimality within this family, put $N = Y + 2Z = (n-4)/2$. Substitution in (14) and completion of a square give

$$F(Y, Z) = -15\left(Z - \frac{8N+1}{30}\right)^2 + \frac{4N^2 - 524N - 119}{60}.$$

The numerator in the last term is negative throughout $0 \leq N \leq 131$: it is a convex quadratic with value -119 at both endpoints. Consequently no such parameter pair has positive excess. All orders in the family are even; order below 268 has $N \leq 131$. The displayed order-268 member proves the claimed family minimum. This argument makes no claim about smaller counterexamples outside the family. □

6 Even-distance counts and complement neighborhoods: statement 302

For a vertex z of a connected graph, define

$$e_G(z) = |\{u : d_G(z,u) \text{ is even}\}|, \qquad e_{\min}(G) = \min_z e_G(z).$$

The vertex z itself is included, since its distance is zero. For a graph H, put

$$f_\lambda(H) = |\{z : \lambda_H(z) = \max_u \lambda_H(u)\}|, \qquad \lambda_H(z) = \alpha(H[N_H(z)]).$$

This frequency concerns local independence numbers, not a spectral multiplicity. Statement 302 proposes

$$\gamma_t(G) \le e_{\min}(G) + f_\lambda(\overline{G}). \tag{15}$$

These are the original invariant definitions [2, 6].

Definition 6.1. Let $r \ge 2$ and $q \ge \max\{r, 3\}$. Start with a clique $Q = \{a, b_1, \dots, b_q\}$ and a vertex v anticomplete to Q. For each $1 \le i \le r$, add vertices w_i, t_i and edges $vw_i, aw_i, w_it_i, t_ib_i$. For every $1 \le i \le r$ and $1 \le j \le q$, add a vertex ℓ_{ij} with exactly the two neighbors t_i, b_j. Denote this graph by $S_{r,q}$. It has

$$n = 2 + q + 2r + rq, \qquad m = \binom{q+1}{2} + 4r + 2rq.$$

Proposition 6.2. *For $S = S_{r,q}$,*

$$\gamma_t(S) = \min\{2r, q+2\}, \qquad e_{\min}(S) = r + 2, \qquad f_\lambda(\overline{S}) = 1.$$

The minimum even-distance count is attained uniquely at v. Moreover $\operatorname{rad}(S) = 2$ *and* $\operatorname{diam}(S) = 3$.

Proof. Let D be a total dominating set. If $t_i \notin D$ for some i, each ℓ_{ij} forces $b_j \in D$, so all q vertices b_j are selected. The vertex v also forces some $w_h \in D$. To dominate that selected vertex, at least one of v, a, t_h must be selected, giving $|D| \ge q + 2$. If every t_i is selected, their neighborhoods

$$N(t_i) = \{w_i, b_i\} \cup \{\ell_{ij} : 1 \le j \le q\}$$

are pairwise disjoint and avoid all the t_i. Each must meet D, so $|D| \ge 2r$. The two sets

$$\{t_1, \dots, t_r, b_1, w_2, \dots, w_r\}, \qquad \{b_1, \dots, b_q, a, w_1\}$$

are total dominating sets of sizes $2r$ and $q + 2$, respectively. Here $r \ge 2$ ensures that the first set dominates v. This proves the first formula.

All pairs are at distance at most three. For example, clique vertices connect all the t_i and ℓ_{ij} within three steps, w_i reaches such vertices through a, and the distance layers from v are

$$\{v\}, \quad \{w_i\}, \quad \{a\} \cup \{t_i\}, \quad \{b_j\} \cup \{\ell_{ij}\}.$$

The vertex a is within two of all vertices, whereas v is at distance three from every b_j. No vertex is universal. This proves the assertions about radius and diameter.

Thus $e_S(z)$ equals one plus the number of distance-two vertices. Enumerating them from the defining neighborhoods gives

z	$e_S(z)$
v	$r+2$
a	$rq+r+2$
w_i	$2q+r$
t_i	$q+r+1$
b_j	$rq+r+1-\mathbf{1}_{\{j\le r\}}$
ℓ_{ij}	$2q+r+\mathbf{1}_{\{j\le r,\ j\neq i\}}$.

(16)

For clarity, the distance-two vertices from w_i are the other $r-1$ vertices w_h, all q vertices b_j, and its row of q vertices ℓ_{ij}. From t_i they are v, a, the $q-1$ vertices b_j with $j\neq i$, and the $r-1$ vertices ℓ_{hi} with $h\neq i$. From b_j they are all w_i, all leaf-type vertices outside column j, and all t_i except t_j when $j\le r$. From ℓ_{ij} they are $Q\setminus\{b_j\}$, w_i, its other row and column vertices, and t_j precisely when $j\le r$ and $j\neq i$. The rows for v, a follow from their displayed layers and neighborhoods. These lists prove (16). Since $q\ge 3$, every entry other than the one for v is strictly greater than $r+2$.

Finally,

$$\lambda_{\overline{S}}(z)=\omega(S[V(S)\setminus N_S[z]]).$$

The clique Q has order $q+1\ge 4$ and is the unique clique of that order. Indeed $N(v)$ and $N(w_i)$ are independent; the only edge in $S[N(t_i)]$ is $b_i\ell_{ii}$; and every ℓ_{ij} has degree two. Any clique containing a vertex outside Q therefore has order at most three. The vertex v is the only vertex outside Q anticomplete to Q: every w_i sees a, every t_i sees b_i, and every ℓ_{ij} sees b_j. Consequently the maximum local independence number in the complement is $q+1$, attained only at v, proving $f_\lambda(\overline{S})=1$. □

Corollary 6.3 (Refutation of 302)**.** *For $r\ge 4$, take $q=2r-2$. Then $S_{r,2r-2}$ has*

$$n=2r(r+1),\quad m=6r^2-3r+1,\quad \gamma_t=2r,$$

whereas the right side of (15) is $r+3$. The deficit $r-3$ is unbounded. The first member $S_{4,6}$ has order 40, size 85, and $\gamma_t=8>7$.

Proof. Substitute the parameter choice into Proposition 6.2 and the order and size formulas. □

The related valid bound $\gamma_t(G)\le\omega(G)+f_\lambda(\overline{G})$ is an earlier theorem of DeLaViña, Liu, Pepper, Waller, and West [8, Theorem 10]. The present examples attain it: $\omega(S_{r,2r-2})=2r-1$ and $f_\lambda(\overline{S})=1$. This comparison identifies the loss caused by replacing the clique number with the minimum even-distance count; it does not count the earlier theorem as a new result.

7 Annihilation and cycles of length divisible by four: statement 298

Let G have n vertices, m edges, and degree sequence $d_1\le\cdots\le d_n$. Its annihilation number is

$$a(G)=\max\left\{k:\sum_{i=1}^{k}d_i\le m\right\}.$$

Equivalently, it is the maximum cardinality of a vertex set whose degree sum in G is at most m. For a connected nontrivial graph $a(G)<n$, and we write $d_{\mathrm{cut}}(G)=d_{a(G)+1}$ for its cutoff degree. The original statement 298, for connected graphs of order greater than two, is

$$\gamma_t(G)\le a(G)+\mathbf{1}_{\{G\text{ contains no ordinary }C_4\}}.\tag{17}$$

We prove the stronger assertion that the indicator can be omitted whenever G contains a cycle of any length divisible by four.

7.1 Credited antecedents and the classical input

The global bound $\gamma_t \le a + 1$ and the stronger bound $\gamma_t \le a$ when $d_{\mathrm{cut}} \ge 3$ are prior results credited in the Round 7 source to J. J. Gebendorfer's manuscript of 11 October 2026 [9]. We do not count those two prior results as new contributions of this consolidation. To expose the dependency of the present result, we reconstruct their required proof chain below, following the detailed proof audit included in the archive [10]. The prior PDF and text snapshots named in that audit were absent from the supplied ZIPs. During preparation of this revised edition, the primary manuscript archive was independently retrieved from the attachment to issue 7242. Its SHA-256 digest matched the recorded value, and its six-page written proof was read against the required dependency chain. The reconstruction below was checked against that primary text; every needed step is also proved here. No Lean rebuild is claimed.

The one classical theorem used without reproving it is the Chvátal–McDiarmid transversal bound: if a finite 3-uniform hypergraph has n_H vertices and m_H edges, then

$$4\tau(H) \le n_H + m_H. \tag{18}$$

Here τ is the minimum size of a set meeting every edge. Isolated vertices and repeated edges are allowed. The original theorem is in [11]; the exact formulation and multiset convention also appear explicitly in [12]. Alternatively, repeated edges can be deleted before applying the simple-hypergraph version, since this does not change τ and only decreases the right side.

Lemma 7.1 (Mixed transversal bound, credited antecedent)**.** *Let H be a finite hypergraph whose nonempty edges have sizes one, two, or three. If e_j counts the edges of size j, with multiplicity, and $n_H = |V(H)|$, then*

$$6\tau(H) \le 2n_H + 4e_1 + 2e_2 + e_3.$$

Proof. Suppose a counterexample exists and choose one minimizing $\Phi = 2n_H + 4e_1 + 2e_2 + e_3$. It has no isolated vertex, since such a vertex can be deleted. Write $d_j(v)$ for the number of incident j-edges. If $4d_1(v) + 2d_2(v) + d_3(v) \ge 4$, select v and delete it together with all incident edges. The potential drops by at least six, and a transversal of the smaller hypergraph together with v contradicts minimality. Hence this weighted incidence is at most three everywhere. There are no singleton edges; each vertex is in at most one 2-edge; and a vertex in a 2-edge is in at most one 3-edge.

Suppose uv is a 2-edge. If neither endpoint lies in a 3-edge, select one endpoint and delete both vertices and the 2-edge. The potential drops by six, again a contradiction. Otherwise assume u lies in a 3-edge f. Select u and delete u, uv, f, lowering the potential by five. If v is now isolated, delete it as well, giving a drop of seven. Otherwise v lies in exactly one remaining 3-edge $g \ne f$. Delete v and shrink g to the 2-edge $g \setminus \{v\}$. This lowers the potential by one more: deleting the vertex saves two while the new edge weight increases from one to two. The shrinking preserves the transversal number of the remaining hypergraph, since a transversal using v can replace it by any other member of g, and every transversal of the shrunken edge also meets the old edge. Adding the selected u therefore lifts a transversal with an additional cost of at most one, whereas the total potential drop is at least six. This is again impossible.

There are consequently no 2-edges. The hypergraph is 3-uniform with maximum degree at most three, so $m_H \le n_H$. The classical bound (18) now gives

$$6\tau(H) \le \frac{3}{2}(n_H + m_H) \le 2n_H + m_H = \Phi,$$

the final contradiction. □

Lemma 7.2 (Annihilation budget, credited antecedent)**.** *Let G be connected of order $n \geq 3$, with ℓ degree-one vertices and b degree-two vertices. If $d_{\mathrm{cut}}(G) \geq 3$, then*

$$a(G) \geq \left\lfloor \frac{n}{2} + \frac{\ell}{3} + \frac{b}{6} \right\rfloor.$$

Proof. All the vertices of degrees one and two belong to the affordable initial degree segment. Put $L = \ell + 2b$, let $r = n - \ell - b$ be the number of remaining vertices, and let E be their degree sum. Then $L \leq m$, $L + E = 2m$, and $E \geq L$, $E \geq 3r > 0$. Set

$$t = \left\lfloor \frac{r}{2} - \frac{rL}{2E} \right\rfloor.$$

This is an integer between zero and r. The sum of the t smallest degrees among the r high-degree vertices is at most $tE/r \leq (E - L)/2 = m - L$. Adjoining them to all the low-degree vertices gives an annihilation set. Therefore

$$\begin{aligned} a(G) &\geq \ell + b + t \\ &= \left\lfloor \frac{n}{2} + \frac{\ell}{2} + \frac{b}{2} - \frac{r(\ell + 2b)}{2E} \right\rfloor \\ &\geq \left\lfloor \frac{n}{2} + \frac{\ell}{3} + \frac{b}{6} \right\rfloor, \end{aligned}$$

where the last step uses $r/E \leq 1/3$. □

Theorem 7.3 (A degree-count bound, credited antecedent)**.** *Every connected graph G of order $n \geq 3$, with ℓ leaves and b degree-two vertices, satisfies*

$$6\gamma_t(G) \leq 3n + 2\ell + b.$$

Proof. We induct on n. For a star the total domination number is two; the inequality is immediate, including P_3, where the right side equals 14. Suppose G is not a star and has a strong support vertex v, meaning one with at least two leaf neighbors. Such a vertex has a nonleaf neighbor, by connectivity, so its degree is at least three. Delete one of its leaf neighbors to obtain H. The graph remains connected, its leaf count decreases by one, and its number of degree-two vertices increases by at most one. Every total dominating set of H contains v, forced by the remaining leaf, and is consequently also a total dominating set of G. The induction hypothesis gives

$$6\gamma_t(G) \leq 3(n - 1) + 2(\ell - 1) + (b + 1) \leq 3n + 2\ell + b.$$

It remains to consider graphs with no strong support. Let S be the support set and put $s = |S| = \ell$. Delete all leaves to form the connected core K, which has at least two vertices. Let S_2, S_3 consist of supports whose degrees in G are two and three, with sizes s_2, s_3. Put

$$T = N_K(S_2) \setminus S, \quad c = |T|, \quad D_0 = S \cup T, \quad t_0 = |N_K(D_0)|.$$

Each member of S_2 has just one core neighbor, so $c \leq s_2$. The disjoint sets T, S_2 both lie in $N_K(D_0)$, including when two supports are adjacent. Hence $t_0 \geq c + s_2$.

Let $U = V(K) \setminus N_K(D_0)$ be the vertices not yet dominated, and let $Q_0 = V(K) \setminus D_0$. For $u \in U$, all its core neighbors lie in Q_0, and it has at least two: a core vertex of degree one would be a member of S_2, already dominated. Build a hypergraph on Q_0 with one edge for each

$u \in U$, taking its whole core neighborhood if that neighborhood has size two and any three of its neighbors otherwise. There are

$$n_H = n - 2s - c, \qquad h = |U| = n - s - t_0$$

vertices and edges respectively. The number h_2 of 2-edges satisfies $h_2 \le b - s_2 + s_3$: a core-degree-two vertex is either a nonsupport of original degree two or a support of original degree three.

If X is a transversal of this hypergraph, $D_0 \cup X$ totally dominates G. Vertices in $N_K(D_0)$ see D_0, all remaining core vertices see X by their neighborhood rows, and leaves see their supports. This includes vertices selected into $D_0 \cup X$ themselves. The mixed transversal bound gives

$$\begin{aligned} 6\gamma_t(G) &\le 6(s+c) + 2(n-2s-c) + h + h_2 \\ &= 3n + s + 4c - t_0 + h_2 \\ &\le 3n + s + b + 3c - 2s_2 + s_3 \\ &\le 3n + s + b + s_2 + s_3 \\ &\le 3n + 2s + b = 3n + 2\ell + b. \end{aligned}$$

The argument also covers an empty row set or variable set; the empty transversal then has size zero. □

Corollary 7.4 (High-cutoff bound, credited antecedent)**.** *For a connected graph of order at least three with $d_{\mathrm{cut}}(G) \ge 3$, one has $\gamma_t(G) \le a(G)$.*

Proof. The integer $\gamma_t(G)$ is at most $\lfloor n/2 + \ell/3 + b/6 \rfloor$ by Theorem 7.3; this is at most $a(G)$ by Lemma 7.2. □

7.2 Low-cutoff transfer and incidence coloring

Lemma 7.5 (The cutoff-two formula)**.** *If G is connected of order at least three, then $d_{\mathrm{cut}}(G) \ge 2$. If equality holds and ℓ is its number of leaves, then*

$$a(G) = \left\lfloor \frac{m + \ell}{2} \right\rfloor.$$

Proof. Distinct leaves have distinct incident edges when $n \ge 3$. Thus $\ell \le m$ and all leaves form an annihilation set, which excludes cutoff one. At cutoff two, the first $a(G) + 1$ degrees consist of all ℓ ones followed by twos. The defining inequalities are $2a(G) - \ell \le m < 2(a(G) + 1) - \ell$, giving the formula. □

Lemma 7.6 (Annihilation under a suitable edge deletion)**.** *Suppose $d_{\mathrm{cut}}(G) = 2$, $uv \in E(G)$, and $d_G(v) \ge 3$. For $H = G - uv$ one has $a(H) \le a(G)$.*

Proof. Write $m = |E(G)|$ and choose an annihilation set A of H with $|A| = a(H)$. Its degree sum in H is at most $m - 1$, so restoring uv gives a sum in G of at most $m + 1$. If it is at most m, there is nothing to prove. Otherwise it is exactly $m + 1$, and both endpoints of the restored edge belong to A.

Let $L_2 = \{x : d_G(x) \le 2\}$. The first $a(G) + 1$ degrees lie in L_2 and sum to more than m, so the degree sum of L_2 is at least $m + 1$. The set A cannot contain all of L_2 as well as v: its degree sum would then be at least $m + 4$. Choose $w \in L_2 \setminus A$ and replace v by w. This preserves cardinality and decreases the degree sum by at least one, producing an annihilation set of G of size $a(H)$. □

Lemma 7.7 (A two-coloring of neighborhood incidences)**.** *Let G be a tree of order at least three, or a connected unicyclic graph whose cycle has length $4k$ for some $k \geq 1$. If S is its support set and ℓ its number of leaves, then*

$$\gamma_t(G) \leq \left\lfloor \frac{n + |S|}{2} \right\rfloor \leq \left\lfloor \frac{n + \ell}{2} \right\rfloor.$$

Proof. Put $Q = V(G) \setminus S$ and $U = V(G) \setminus N_G(S)$, where $N_G(S)$ is an open neighborhood union. For each $u \in U$, all its neighbors are in Q, and $d_G(u) \geq 2$: a leaf has a support neighbor and would lie in $N_G(S)$.

Build the bipartite incidence graph with row vertices indexed by U, variable vertices indexed by Q, and an edge for each adjacency in G. The graph G is bipartite. Its full row-variable neighborhood graph splits into two disjoint copies of G, one in each bipartition orientation. The restricted incidence graph therefore has only tree components or unicyclic components whose cycle has length $4k$. Every row has degree at least two.

Color variables with two colors so every row sees both colors. In a tree component, root at a variable. Each row has a parent variable and at least one child variable; give one child the color opposite its parent and color the other children arbitrarily, continuing outward. Isolated variables cause no difficulty. In a cyclic component, there are $2k$ variables around the cycle. Color them alternately, which is consistent because $2k$ is even. Every cycle row now sees both colors. Extend to branches by the same tree rule; branches issuing from a cycle row may start with arbitrary variable colors because that row is already satisfied.

Let Q_0, Q_1 be the color classes. Each set $S \cup Q_i$ totally dominates G: every vertex in $N_G(S)$ sees S, and every vertex of its complement U sees both variable colors. This reasoning also dominates selected vertices themselves. The two sizes sum to $n + |S|$, so the smaller is at most its half rounded down. Finally, distinct supports have disjoint nonempty sets of leaf neighbors, so $|S| \leq \ell$. □

The tree case of Lemma 7.7 is part of the credited antecedent argument. Its extension to a unicyclic $4k$-cycle is the additional incidence-coloring step used by the Round 7 result. The proof explains the divisibility condition: the variable vertices on each incidence cycle must have even number.

Theorem 7.8 (Global $a + 1$ bound, credited antecedent)**.** *Every connected graph of order at least two satisfies $\gamma_t(G) \leq a(G) + 1$.*

Proof. For K_2 this is $2 \leq 1 + 1$. For order at least three, Corollary 7.4 handles cutoff at least three, so only cutoff two need be considered. If G is a tree, $m = n - 1$ and Lemmas 7.7 and 7.5 give

$$\gamma_t(G) \leq \left\lfloor \frac{n + \ell}{2} \right\rfloor = \left\lfloor \frac{m + \ell + 1}{2} \right\rfloor \leq a(G) + 1.$$

For a cycle C_n, $a(C_n) = \lfloor n/2 \rfloor$. Label its vertices $0, \ldots, n - 1$ cyclically and write $n = 4q + s$ with $0 \leq s \leq 3$. Select $4j, 4j + 1$ for $0 \leq j < q$. For $s = 1$, also select $4q$; for $s = 2$ or 3, also select $4q, 4q + 1$. These are total dominating sets of size at most $\lfloor n/2 \rfloor + 1$, including C_3.

If a counterexample remained, choose one with the fewest edges. It is neither a tree nor a cycle, and has cutoff two. Some cycle has a vertex v of degree at least three: otherwise every vertex of any cycle would have degree two and connectivity would make G that cycle. Delete a cycle edge incident with v, obtaining a connected graph H. Minimality, monotonicity of total domination under edge addition, and Lemma 7.6 give

$$\gamma_t(G) \leq \gamma_t(H) \leq a(H) + 1 \leq a(G) + 1,$$

a contradiction. □

7.3 The prescribed-cycle theorem and the original inequality

Corollary 7.9. *If G is connected and unicyclic with cycle length divisible by four, then $\gamma_t(G) \leq a(G)$.*

Proof. At high cutoff use Corollary 7.4. At cutoff two, $m = n$, and Lemmas 7.5 and 7.7 both give the threshold $\lfloor (n + \ell)/2 \rfloor$. □

Lemma 7.10. *Let G be connected, let C be a cycle, and suppose G is not unicyclic. There is a nonbridge edge outside $E(C)$ with an endpoint of degree at least three.*

Proof. Take a different cycle C', which has an edge outside $E(C)$. If C' meets C, start at an intersection and traverse C' up to its first edge outside $E(C)$. Its initial vertex belongs to C and has that edge in addition to its two incident C-edges. If the cycles are vertex-disjoint, take a shortest path between them. Its endpoint on C' has a neighbor outside C', so has degree at least three. Choose either incident C'-edge there. In either case the chosen edge is outside C and lies on C', so is not a bridge. □

Theorem 7.11 (Prescribed cycles of length divisible by four)**.** *Every connected graph containing a cycle of length divisible by four satisfies $\gamma_t(G) \leq a(G)$.*

Proof. Choose a counterexample with the fewest edges and fix a cycle C of length $4k$ in it. Lemma 7.5 and Corollary 7.4 force cutoff two. Corollary 7.9 excludes the unicyclic case. By Lemma 7.10, delete a nonbridge edge outside C with a high-degree endpoint. The resulting graph H is connected, still contains C, and has fewer edges. No condition on its cutoff is needed. Minimality and the annihilation-transfer lemma yield

$$\gamma_t(G) \leq \gamma_t(H) \leq a(H) \leq a(G),$$

a contradiction. □

Corollary 7.12 (Statement 298)**.** *Every connected graph G of order greater than two satisfies* (17)*.*

Proof. If G contains an ordinary four-cycle, apply Theorem 7.11 with $k = 1$. Otherwise apply the credited global bound, Theorem 7.8. The four-cycle need not be induced. □

Example 7.13 (Sharpness of both branches). For $k \geq 1$, the cycle C_{4k} has $\gamma_t = a = 2k$. The annihilation value follows from its constant degree two. The lower bound for total domination follows by counting at most two dominated vertices per selected vertex, and the repeated pattern 1100 attains it.

For C_{4k+2}, the annihilation number is $2k + 1$ and the total domination number is $2k + 2$. To see the strict lower bound, suppose a total dominating set had $2k + 1$ vertices. Every nonempty run of selected vertices around the cycle has length at least two, and every intervening run of unselected vertices has length at most two. There would be at most k selected runs and thus at most $2k$ unselected vertices, whereas there are $2k + 1$. This is impossible. The construction in Theorem 7.8 attains $2k + 2$. These cycles are C_4-free, so the additive one in the four-cycle-free branch cannot be removed in general.

8 Counterexample families from Rounds 11–14

The four source statements treated in this section are mathematically refuted by explicit infinite families. Their exact definitions are retained, including the distinction between the periphery and arbitrary boundary vertices, the largest-output interpretation of Maxine, and the half-coefficient open-neighbourhood version of 354. These are source-qualified conclusions; archived open flags are not assertions of current curator status or worldwide publication priority.

8.1 Conjecture 171: peripheral averages and even-distance counts

8.1.1 Statement and scope

The archived original row 171 states, for a simple connected graph G,

$$L_s(G) \geq \frac{\max_{v \in V(G)} \operatorname{dist}_{\text{even}}(v) - 1}{\operatorname{dist}_{\text{avg}}(B)}.$$

Here L_s is the maximum number of leaves in a spanning tree. The set B is the periphery: vertices of maximum eccentricity. The denominator is the average of all nonzero distances **between pairs of vertices of** B, not the average between B and all vertices and not the average distance to the set B. The even-distance count includes distance zero. Our principal counterexample remains a strict counterexample if distance zero is omitted.

The source glossary defines the historical word *boundary* as the maximum-eccentricity periphery, not all leaves of a tree; the archived row and definitions are recorded in [2]. The construction below refutes that precise formulation.

8.1.2 The 407-vertex witness

Construct a tree T as follows. All paths mentioned have no vertices in common except their prescribed common center c.

1. From c, draw a path of length 14 ending at a vertex a; attach 27 new pendant vertices $p_1, \ldots, p_{27}$ to a.
2. From c, draw a path of length 15 ending at a vertex q.
3. From c, draw 25 further paths, each of length 14, ending at vertices $s_1, \ldots, s_{25}$.

These operations attach trees at a single existing vertex, so T is a finite simple connected tree. Its order is

$$n = 1 + 14 + 27 + 15 + 25 \cdot 14 = 407.$$

Maximum spanning-tree leaves A tree has itself as its unique spanning tree. Its leaves are precisely the 27 vertices p_i, the vertex q, and the 25 vertices s_j. Hence

$$L_s(T) = 27 + 1 + 25 = 53.$$

Periphery and its average distance For every i,

$$d(p_i, q) = 15 + 15 = 30.$$

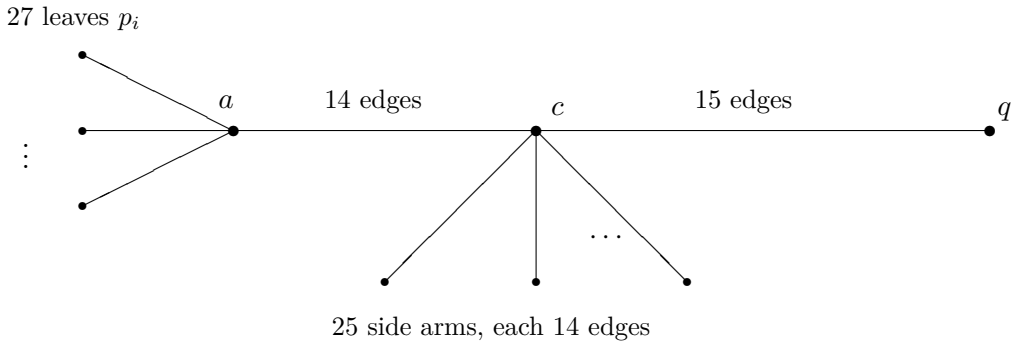


Figure 1: Compressed construction: degree-two vertices are suppressed. Three representative leaves or arms stand for the indicated multiplicities. Only the 27 left leaves and q are peripheral.

Every vertex has distance at most 15 from c, so the diameter is 30. Moreover, p_i and q have eccentricity 30. Every other vertex x has $d(c,x) \le 14$, and consequently

$$d(x,y) \le d(x,c) + d(c,y) \le 14 + 15 = 29 \qquad \text{for all } y.$$

Thus no other vertex is peripheral, and

$$B = \{p_1, \dots, p_{27}, q\}, \qquad |B| = 28.$$

Two distinct p_i, p_j have distance 2, while $d(p_i, q) = 30$. Therefore the average over unordered distinct pairs (equivalently over ordered nonzero pairs) is

$$\mathrm{dist}_{\mathrm{avg}}(B) = \frac{2\binom{27}{2} + 30 \cdot 27}{\binom{28}{2}} = \frac{702 + 810}{378} = 4.$$

Maximum even-distance count In a tree, the vertices at even distance from a given root are precisely the bipartition class containing that root. Count the two bipartition classes using c as the initial root. The class containing c has

$$1 + 7 + 7 + 25 \cdot 7 = 190$$

vertices: c, seven vertices on the length-14 left stem, seven vertices on the length-15 right path, and seven vertices on each further length-14 path. None of the p_i belong to this class, since their distance from c is 15. The other class therefore has $407 - 190 = 217$ vertices. Both classes are nonempty, so

$$\max_v \mathrm{dist}_{\mathrm{even}}(v) = 217.$$

For example, any neighbor of c attains this maximum.

Strict contradiction The right side of Conjecture 171 is

$$\frac{217 - 1}{4} = 54,$$

whereas $L_s(T) = 53$. Hence the conjectured inequality is false:

$$\boxed{53 < 54.}$$

If distance zero were excluded from the even-distance count, the right side would instead be $215/4 = 53.75$, still strictly larger than 53.

8.1.3 An infinite counterexample family

Keep the first two parts of the construction fixed and use k additional length-14 paths instead of 25. For every integer $k \geq 24$, the same argument gives

$$n = 57 + 14k, \qquad L_s = k + 28, \qquad |B| = 28, \qquad \mathrm{dist}_{\mathrm{avg}}(B) = 4.$$

The bipartition classes have orders $15 + 7k$ and $42 + 7k$, respectively. Thus the proposed right side exceeds the left side by

$$\frac{41 + 7k}{4} - (k + 28) = \frac{3k - 71}{4} > 0.$$

In particular, $k = 24$ gives a 393-vertex counterexample; the $k = 25$ witness was selected above because its claimed lower bound is the integer 54 and the violation is independent of the distance-zero convention.

8.1.4 No positive constant-factor repair

The same construction gives a stronger consequence. For every odd integer $R \geq 19$, use a length-$(R-1)$ left stem with $m = 2R - 3$ pendant leaves, a length-R right arm, and $k = R$ further length-$(R-1)$ arms. Exactly the m left leaves and the right endpoint are peripheral. The same distance and parity counts give

$$n = R^2 + 3R - 3, \quad L_s = 3R - 2, \quad \mathrm{dist}_{\mathrm{avg}}(B) = 4, \quad \max_v \mathrm{dist}_{\mathrm{even}}(v) = \frac{R^2 + 5R - 6}{2}.$$

The proposed right side exceeds L_s by

$$\frac{R^2 - 19R + 8}{8} > 0.$$

For an integer-valued parametrization set $R = 8t + 19$, $t \geq 0$. Then

$$n = 64t^2 + 328t + 415, \qquad L_s = 24t + 55,$$

$$\text{proposed right side} = 8t^2 + 43t + 56, \qquad \text{excess} = 8t^2 + 19t + 1.$$

Its ratio to L_s tends to infinity. Therefore there is no fixed constant $c > 0$ for which

$$L_s(G) \geq c\,\frac{\max_v \mathrm{dist}_{\mathrm{even}}(v) - 1}{\mathrm{dist}_{\mathrm{avg}}(B)}$$

holds for all trees. This is a consequence of the same refutation, not an additional numbered conjecture closure. No global minimum-order claim is made for any displayed witness.

8.2 Conjecture 382e: Maxine and two-domination

8.2.1 Statement, definitions, and prior construction

All graphs are finite and simple. A set D is 2-dominating if every vertex outside D has at least two neighbors in D. Its minimum size is $\gamma_2(G)$. Ordinary domination requires one such neighbor and has minimum size $\gamma(G)$. A *Maxine run* repeatedly deletes a vertex of maximum degree in the current induced graph until no edges remain. We use $\mathrm{Maxine}(G)$ for the largest possible number of remaining vertices over all tie-breaking choices. Original WOWII row 382e asserts

$$\gamma_2(G) \le \mathrm{Maxine}(G) + \gamma(G) \tag{19}$$

for every connected graph of order greater than two [2]. Our examples have equal output size for *every* run, so no choice-dependent interpretation can remove the violation.

DeLaViña, Larson, Pepper, and Waller prove the related valid inequality $\gamma_2 \le \gamma + \alpha$ as a special case of their Theorem 6 and Corollary 7 [7]; this is not the Maxine inequality. The forced-deletion mechanism has prior art: Cao and Eppstein's 2018 comments describe retaining the edges of a triangle-free cubic graph and adding a degree-two vertex on each edge. Minimum-degree deletion removes the new vertices first [13]. We credit that mechanism. Here the private vertices instead have *three* prescribed neighbors, including a common distinguished vertex. The changed neighborhoods and a universal-vertex construction give the domination obstruction below. We do not claim a smallest counterexample or first-ever worldwide discovery; source-priority conclusions must be limited to the accompanying bounded audit.

Example 8.1 (The attributed peeling precursor). Let $k \ge 3$, retain the edges of $K_{k,k}$, and for each core edge ab add a private vertex adjacent exactly to a, b. Call the result Q_k and put $P_k = \overline{Q_k}$. If i denotes the minimum order of a maximal independent set, then

$$i(P_k) = \alpha(P_k) = 3, \qquad \mathrm{Maxine}(P_k) = 2.$$

For $J = K_1 \vee P_k$ with new vertex u,

$$\gamma(J) = 1, \quad \{u\} \text{ is its unique minimum dominating set}, \quad i(J-u) = 3 > \mathrm{Maxine}(J) = 2,$$

but $\gamma_2(J) = 3 = \mathrm{Maxine}(J) + \gamma(J)$. Thus these examples invalidate the corresponding independent-domination bridges while still satisfying 382e. The construction and its peeling behavior are attributed to Cao and Eppstein [13].

Proof. Every maximal clique of Q_k is a core edge together with its private vertex. Consequently every maximal independent set of P_k has order three. In Q_k, private vertices have degree two, while each core vertex retains at least its k original core neighbors. Every minimum-degree deletion therefore removes all private vertices first. From $K_{k,k}$, minimum-degree deletion alternates between the two sides whenever their sizes differ, eventually reaching $K_{1,1}$. Complementation exchanges minimum-degree and maximum-degree deletion, and a clique in Q_k is an independent set in P_k. Every Maxine run thus ends with two vertices.

No vertex of P_k is universal, since every vertex of Q_k has at least two neighbors. The new u is therefore both the unique dominating singleton of J and its unique maximum-degree vertex. It is deleted first by every Maxine run, proving the asserted residual values. For two-domination, choose u and the private vertices belonging to two disjoint core edges. Every remaining core vertex is adjacent in P_k to at least one of the two private vertices, and every other private vertex is adjacent to both. This is a 2-dominating set of order three. Every old vertex has at least two old nonneighbors, so a set of two vertices containing an old vertex x leaves some old nonneighbor of x outside the set; that vertex has at most one selected neighbor. Hence no two-element 2-dominating set exists. □

8.2.2 The base graph

Fix $k \geq 4$. Let

$$A = \{a_0, a_1, \ldots, a_{k-1}\}, \qquad B = \{b_0, b_1, \ldots, b_{k-1}\}, \qquad X = \{x_{ij} : 1 \leq i \leq k-1,\ 0 \leq j \leq k-1\}.$$

Define H_k as follows: retain every edge between A and B; for each x_{ij} add exactly the three edges to a_0, a_i, b_j; add no other edges. Put $F_k = \overline{H_k}$. Thus $|F_k| = 2k + k(k-1) = k^2 + k$. Equivalently, A, B, and X each induce a clique in F_k; there are no A–B edges, and x_{ij} is adjacent to every core vertex except a_0, a_i, b_j. In particular,

$$N_{F_k}(a_0) = A \setminus \{a_0\}. \tag{20}$$

Lemma 8.2. *Every Maxine run on F_k leaves exactly two vertices, one in A and one in B.*

Proof. On any fixed surviving vertex set, maximum-degree deletion in a graph is minimum-degree deletion in its complement. As long as a vertex of X remains in H_k, it has degree three, while every core vertex has degree at least $k \geq 4$, because the retained $K_{k,k}$ is still present. Consequently every run deletes all of X before any core vertex. The remaining graph in F_k is $K_k \sqcup K_k$. Deleting positive-degree vertices cannot delete the last, isolated vertex of either clique. The terminal set therefore has one vertex from each clique. □

Lemma 8.3. *For every $k \geq 4$, $\gamma(F_k) = 3$ and $\gamma_2(F_k) = 4$.*

Proof. First, $\{a_0, x_{10}, x_{11}\}$ dominates F_k: a_0 dominates A, each vertex of B is adjacent to at least one of the two selected private vertices, and every other private vertex is adjacent to both. No pair dominates. A pair in X misses a_0. A pair in the same core part misses the other core part. A core pair a_i, b_j misses x_{ij} if $i \neq 0$, and misses x_{1j} if $i = 0$. A pair a_i, x_{pq} misses b_q, while a pair b_j, x_{pq} misses a_0. These cases exhaust all pairs. Since a dominating singleton could be enlarged to a dominating pair, $\gamma(F_k) = 3$.

The four vertices $\{a_1, a_2, b_0, b_1\}$ form a 2-dominating set. An unselected core vertex has two selected neighbors in its own part. A private vertex misses at most one selected vertex in A and at most one in B, so it has at least two selected neighbors.

Suppose, for a contradiction, that a three-element set D is 2-dominating. If $a_0 \notin D$, equation (20) forces at least two vertices of D into $A \setminus \{a_0\}$. The one possible remaining vertex cannot supply two selected neighbors to any unselected vertex of B. Such a vertex exists, so this is impossible. Thus $a_0 \in D$. The possibilities for the other two vertices are listed below; in each row the indicated unselected vertex has fewer than two neighbors in D.

Locations of $D \setminus \{a_0\}$	Witness to failure
A, A	any vertex of B
A, B	a vertex of $B \setminus D$
B, B	a vertex of $A \setminus \{a_0\}$
$A,\ x_{ij}$	b_j
$B,\ x_{ij}$	a_i
$x_{ij},\ x_{pq}$	b_j

The needed unselected core vertices exist because $k \geq 4$. No smaller 2-dominating set exists either, since adjoining vertices preserves 2-domination. Hence $\gamma_2(F_k) = 4$. □

8.2.3 Connected counterexamples and amplification

Let rF_k denote the disjoint union of r copies of F_k. Form $J_{k,r} = K_1 \vee (rF_k)$ by adding a vertex u adjacent to every vertex in those copies.

Theorem 8.4. *For every $k \geq 4$ and $r \geq 1$, $J_{k,r}$ is connected and satisfies*

$$|J_{k,r}| = 1 + r(k^2 + k), \qquad \gamma(J_{k,r}) = 1, \qquad \mathrm{Maxine}(J_{k,r}) = 2r, \qquad \gamma_2(J_{k,r}) = 3r + 1.$$

Every Maxine run has the stated output size. Consequently,

$$\gamma_2(J_{k,r}) - \big(\mathrm{Maxine}(J_{k,r}) + \gamma(J_{k,r})\big) = r > 0,$$

so WOWII 382e is false, with unbounded additive defect.

Proof. The vertex u is universal. No vertex in a copy of F_k is universal, since it has a nonneighbor already in that copy. Hence u is the unique maximum-degree vertex and is deleted first by every Maxine run. The remaining graph is rF_k. Whenever a vertex in one component is deleted, it has maximum degree within that component; ignoring the interleaving therefore gives a valid local Maxine run. Lemma 8.2 shows that each component contributes exactly two terminal vertices. Thus $\mathrm{Maxine}(J_{k,r}) = 2r$, independently of ties.

Clearly $\gamma(J_{k,r}) = 1$. If a 2-dominating set contains u, its intersection with each copy of F_k must dominate that copy, and conversely this condition suffices. Its minimum possible size is therefore $1 + 3r$. If the set omits u, it must 2-dominate each copy separately, requiring at least $4r$ vertices. Conversely, the union of minimum 2-dominating sets in the copies also gives at least two neighbors to u. It follows that

$$\gamma_2(J_{k,r}) = \min\{1 + 3r, 4r\} = 3r + 1,$$

using $r \geq 1$ and Lemma 8.3. □

8.3 Conjecture 154: the ratio of radial-circle orders

8.3.1 Exact statement and scope

For a simple connected graph G, let $L_s(G)$ be the maximum number of leaves in a spanning tree. Let r be its radius. A radial circle is the set of vertices at distance r from a central vertex. Write $R_{\min}$ and $R_{\max}$ for the smallest and largest orders of these radial circles. Original WOWII 154 asserts

$$L_s(G) \geq \frac{1 + R_{\max}}{R_{\min}}.$$

The construction below has $L_s = 13$, $R_{\min} = 1$ and $R_{\max} = 13$. Thus it refutes the original assertion by the strict inequality $13 < 14$. It does **not** refute WOWII 155, whose different right side counts distinct radial-circle orders.

The distinct original formulations of 154 and 155 are retained from the archived rows [2]. The following proof applies to the stated nontrivial connected-graph formulation of 154.

8.3.2 Construction

In fact there is an infinite family, indexed by an integer $t \geq 2$.

Start with seven distinct vertices

$$v, w, a, u, b_1, b_2, c$$

and the six edges

$$vw,\ va,\ au,\ wb_1,\ wb_2,\ ac.$$

Thus these seven vertices induce a tree. For each stem $x \in \{b_1, b_2, c\}$ and each index $j \in \{1, \dots, t\}$, introduce two new vertices $p_{x,j}, z_{x,j}$ and the four edges

$$xp_{x,j}, \quad xz_{x,j}, \quad p_{x,j}z_{x,j}, \quad uz_{x,j}.$$

Finally introduce a vertex q with the single edge uq. There are no other edges. Denote the resulting graph by G_t. It is visibly simple and connected, with

$$n = 8 + 6t.$$

The principal witness is G_2, of order 20 and size 31.

8.3.3 Connected domination number is exactly seven

Let γ_c denote minimum connected dominating set cardinality. The original seven core vertices form a connected dominating set, so $\gamma_c(G_t) \le 7$.

We prove the reverse bound. Let D be any connected dominating set. Necessarily $u \in D$: otherwise the pendant vertex q must belong to D to be dominated, but then it has no neighbor in D and cannot be part of a connected dominating set of this graph.

For a stem x, let its branch be

$$X_x = \{x\} \cup \{p_{x,j}, z_{x,j} : 1 \le j \le t\}.$$

Its parent is w when $x = b_1, b_2$, and a when $x = c$. Two observations give a branch-cost lower bound.

- If $x \notin D$, then domination of each $p_{x,j}$ requires D to meet $\{p_{x,j}, z_{x,j}\}$. These pairs are disjoint and $t \ge 2$, so $|D \cap X_x| \ge 2$.
- If $x \in D$ but its parent is absent from D, then selecting no further vertex of X_x would leave x isolated in $G[D]$, because its only neighbors outside X_x are its parent. Since $u \in D$ and $u \ne x$, this is impossible. Again $|D \cap X_x| \ge 2$.

Consequently a branch costs at least one selected vertex if its parent is in D, and at least two otherwise.

Write $W = \mathbf{1}_{w \in D}$, $A = \mathbf{1}_{a \in D}$ and $V = \mathbf{1}_{v \in D}$. Summing the three disjoint branch costs and the required core selections gives

$$|D| \ge 1 + (4 - 2W) + (2 - A) + W + A + V = 7 - W + V.$$

If $|D| \le 6$, the only possibility is $W = 1$, $V = 0$, $|D| = 6$, with equality in every cost estimate. In particular, each of the branches X_{b_1}, X_{b_2} contains exactly one selected vertex. It must be its stem b_i, since excluding that stem requires at least two selected vertices. Thus neither branch contains

any selected p- or z-vertex. Since $v \notin D$, the set $\{w, b_1, b_2\}$ is then a connected component of $G[D]$ not containing the required vertex u. This contradicts connectivity.

Therefore

$$\boxed{\gamma_c(G_t) = 7.}$$

For a connected graph of order at least three, $L_s = n - \gamma_c$. Indeed the internal vertices of any spanning tree form a connected dominating set; conversely a connected dominating set can be spanned by a tree and all remaining vertices attached as leaves. Hence

$$\boxed{L_s(G_t) = 6t + 1.}$$

8.3.4 Radius and a radial circle of order one

The distance layers from v are exactly

$$V_0 = \{v\}, \qquad V_1 = \{w, a\}, \quad V_2 = \{u, b_1, b_2, c\},$$
$$V_3 = \{q\} \cup \{p_{x,j}, z_{x,j} : x \in \{b_1, b_2, c\}, 1 \le j \le t\}.$$

Thus $\mathrm{ecc}(v) = 3$ and its third sphere has order $6t + 1$.

There is no vertex of eccentricity at most two. Any such vertex would have to be within distance two of the pendant vertex q, and hence belong to

$$\{q, u, a\} \cup \{z_{x,j} : x \in \{b_1, b_2, c\}, 1 \le j \le t\}.$$

Each candidate fails:

- $d(u, w) = 3$ and $d(q, w) = 4$;
- $d(a, p_{b_1,1}) = 3$;
- for any $z_{x,j}$, choose a different stem $y \neq x$. The vertices $z_{x,j}$ and $p_{y,1}$ are neither adjacent nor have a common neighbor, so their distance is at least three.

Since v has eccentricity three, the radius is exactly three.

The vertex u also has eccentricity three, and its unique vertex at distance three is w. Indeed u is adjacent to a, every $z_{x,j}$ and q; it reaches v through a, every stem through either a or a z-vertex, and every p-vertex through its z-vertex, all within two steps. Its distance to w is three. Therefore

$$\boxed{R_{\min} = 1.}$$

8.3.5 Maximum radial-circle order and contradiction

The radial circle at v has order $6t + 1 = L_s(G_t)$. Every radial circle has order at most L_s: in a BFS spanning tree rooted at its center, all vertices of the final distance layer are leaves. Thus

$$\boxed{R_{\max} = 6t + 1.}$$

Conjecture 154 would require

$$6t+1 \geq \frac{1+(6t+1)}{1} = 6t+2,$$

which is false for every integer $t \geq 2$. At $t = 2$, the explicit contradiction is

$$\boxed{13 < 14.}$$

This is a full counterexample family to the original numbered statement 154, not merely to an auxiliary strengthening.

8.4 Conjecture 354 in the historical half-coefficient, open-neighbourhood form

8.4.1 Exact statement and source qualification

Write $\gamma_t(T)$ for the total domination number, $C(T)$ for the full center, and $D_2(T) = \{x : \deg_T(x) = 2\}$. For a set A, let $N(A) = \bigcup_{a\in A} N(a)$ be the union of open neighborhoods. The notation $c(H)$ counts all connected components of H, including isolated vertices.

The pinned original WOWII row #354, dated 18 February 2009, states that every tree T of order greater than two satisfies

$$\gamma_t(T) \geq \frac{|C(T)|}{2}\, c\big(T[N(D_2(T))]\big). \tag{21}$$

This neighborhood is neither the external neighborhood $N(A) \setminus A$ nor the closed neighborhood $N[A]$.

The literal half coefficient and the union of open neighborhoods are retained from the archived HTML formula [2]. The source audit found that an ASCII export dropped the half glyph; this is not evidence for changing the target formula. In particular, a theorem about a coefficient-one or closed-neighborhood variant does not decide (21). The bounded historical audit did not complete the comparison with the full Desormeaux–Henning 2016 article, so this section makes no publication-priority claim.

8.4.2 Feit's earlier construction and the modification

The eleven-vertex branch used below is Feit's earlier construction. His 34-vertex example for WoWII 240/346 consists of a root with three copies of this branch. Feit's example and an unbounded-gap extension are credited in DeLaViña, Larson, Pepper and Waller [14, p. 7 of the author manuscript]. The branch is therefore an attributed input to the bicentral modification below.

Feit's tree has total domination number 14 and 15 open-neighborhood components, but only one center: its radius is five and diameter ten. Its right-hand side in (21) is therefore 15/2, so that older example does not refute the half-coefficient formula.

Our principal 36-vertex tree is a *bicentral modification of Feit's construction*: split the common root into adjacent vertices, distribute the three branches one on one side and two on the other, and add a leaf to the one-branch side. The last operation prevents either central vertex from having degree two. The contribution asserted here is this source-qualified modification/application, not the original branch or the earlier component-bound counterexample.

8.4.3 The branch and the family

Let B have eleven distinct vertices

$$a, b, c, d, e, f, g, h, i, j, \ell$$

and edges

$$ab,\ ac,\ bd,\ de,\ df,\ fg,\ ch,\ hi,\ hj,\ j\ell.$$

The vertex a is the attachment vertex. When a is joined to a new vertex r, its branch vertices have the following distances from r:

Distance	Vertices
1	a
2	b, c
3	d, h
4	e, f, i, j
5	g, ℓ

The branch vertices of degree two are exactly b, c, f, j. Vertices a, d, h have degree three; vertices e, g, i, ℓ are leaves. Consequently the open union neighborhood contributed by the four degree-two vertices is

$$\{a, d, h, g, \ell\}. \tag{22}$$

These five vertices are pairwise nonadjacent.

Fix $k \geq 2$. Begin with an edge uv. Attach one disjoint copy of B to u and k disjoint copies to v, joining each copy's vertex a to its designated endpoint. Add a new leaf z adjacent to u. There are no other edges. Denote the resulting tree by T_k.

Each branch is attached by one edge, so T_k is a tree with

$$|V(T_k)| = 2 + 1 + 11(k+1) = 11k + 14, \qquad |E(T_k)| = 11k + 13.$$

The principal example T_2 has 36 vertices and 35 edges.

8.4.4 Degree-two neighborhood components

The degrees of u and v are respectively 3 and $k+1$, neither of which is two. The new vertex z has degree one. Therefore each branch contributes exactly b, c, f, j to $D_2(T_k)$, and there are no other degree-two vertices.

By (22), the set $N(D_2(T_k))$ consists of five vertices in each branch. It contains none of u, v, z. Within each branch the five vertices are independent, and different branches have no edges between them. Hence

$$c(T_k[N(D_2(T_k))]) = 5(k+1). \tag{23}$$

This calculation uses the open-union neighborhood directly, without replacing it by an external or closed neighborhood.

8.4.5 The center

Each branch has maximum depth five from its designated endpoint, and has vertices at that depth. Depth-five vertices on opposite sides of the edge uv are at distance 11. Vertices on one side

are at distance at most 10, and vertices on opposite sides are at distance at most $5 + 1 + 5 = 11$. Thus the diameter is 11.

The eccentricities of u and v are both six. Any other vertex x has positive depth q from the endpoint on its side, so its distance to a depth-five vertex on the opposite side is $q + 1 + 5 \geq 7$. This includes the extra leaf z. Consequently,

$$C(T_k) = \{u, v\}, \qquad \mathrm{rad}(T_k) = 6. \tag{24}$$

8.4.6 Exact total domination number

Let D contain u, v and the four vertices d, f, h, j from each branch. Its order is

$$|D| = 2 + 4(k + 1) = 4k + 6.$$

The centers dominate each other, and u dominates z. Within a branch, a is dominated by its center, b by d, and c by h. The adjacent pair d, f dominates itself and the leaves e, g; the adjacent pair h, j dominates itself and the leaves i, ℓ. Thus D is a total dominating set, proving $\gamma_t(T_k) \leq 4k + 6$.

Conversely, the four leaves e, g, i, ℓ of each branch force their distinct neighbors d, f, h, j into every total dominating set. The leaf z forces u. These give $4(k + 1) + 1$ forced vertices. None of them is a neighbor of u: the neighbors of u are v, the vertex a of its single branch, and z. Total domination of u therefore requires at least one additional vertex. It follows that

$$\gamma_t(T_k) = 4k + 6. \tag{25}$$

Only the upper certificate is necessary to refute (21); the lower argument establishes the exact value without an exhaustive search.

8.4.7 The contradiction

By (23), (24), and (25), the right-hand side of (21) is $5(k+1)$, whereas its left-hand side is $4k+6$. Thus

$$\frac{|C(T_k)|}{2} c(T_k[N(D_2(T_k))]) - \gamma_t(T_k) = 5k + 5 - (4k + 6) = k - 1 > 0.$$

Every T_k with $k \geq 2$ is therefore a counterexample to the pinned historical formulation. The additive error is unbounded. In particular, T_2 gives

$$\boxed{\gamma_t(T_2) = 14 < 15 = \tfrac{1}{2}|C(T_2)|c(T_2[N(D_2(T_2))]).}$$

No minimum-order claim is made.

9 Leaf inequalities 177 and 180 on claw-free graphs

9.1 Scope

Two subclass theorems, not full resolutions of the original conjectures. Both original inequalities allow arbitrary connected graphs. This section proves them for connected claw-free graphs. No numbered full closure or auxiliary-lemma priority claim is made.

Graphs are finite and simple. Write $n = |V(G)|$, L for the maximum number of leaves in a spanning tree, b for the largest order of an induced bipartite subgraph, $a = \alpha(G)$, and γ_c for

connected domination number. Let δ be minimum degree and $\sigma = d_2$ the second-smallest degree, with multiplicity. For a root v, let $O(v)$ be its odd-distance vertices and $E(v)$ its even-distance vertices; distance zero counts in $E(v)$.

The following theorem proves the two source inequalities on the specified class.

Theorem 9.1. *Every nontrivial connected claw-free graph satisfies both*

$$L + b \geq 2a + \sigma \qquad \text{and} \qquad L + b \geq 1 + a + \max_{v \in V(G)} |E(v)|.$$

These are, respectively, the inequalities in 177 and 180.

For $n \geq 3$, $L = n - \gamma_c$. Put $\eta = b - \gamma_c - 1$, so $L + b = n + 1 + \eta$. The published DeLaViña–Waller construction gives $\eta \geq 0$. The two inequalities are therefore equivalent to

$$n + 1 + \eta \geq 2a + \sigma, \qquad \eta \geq a - \min_{v} |O(v)|.$$

The graph K_2 satisfies both directly. We henceforth assume $n \geq 3$.

9.2 Prior construction and the exact refinements used

The construction below is the published method of DeLaViña and Waller [15], with its prescribed-root retention and the odd-path modification made explicit. It is not claimed as a new foundational inequality.

9.2.1 Rooted connected domination

For every prescribed vertex r of a nontrivial connected graph, there is a connected dominating set containing r of size at most $b - 1$.

Start with a maximal connected induced bipartite set B_0 containing r, grown from r and one neighbor. As long as the union of the selected pieces does not dominate, choose a vertex x_j at distance two from that union and a connector c_j adjacent to x_j and to an earlier piece. In the graph remaining after deleting all earlier pieces and their neighborhoods, grow a maximal connected induced bipartite set B_j containing x_j. Let the final pieces be $B_0, \ldots, B_k$. They are pairwise anticomplete, so their union B is induced bipartite and $|B| \leq b$; there are at most k distinct connectors.

Every uncolored vertex has neighbors of both colors in the first piece it meets. Indeed, at the construction stage of that piece it had no neighbor in an earlier piece and was eligible to be added; if it met only one color, it could have been added without destroying bipartiteness. In particular, a singleton piece is never the first piece met by an uncolored vertex. Connectors stay uncolored because they neighbor an earlier piece.

Choose a spanning tree of each piece, rooted at r in B_0 and at x_j in B_j. Delete one leaf different from its root from every nonsingleton piece; delete the sole vertex from any singleton later piece. Retain all connectors. The retained initial piece is nonempty and contains r. Every nonempty later piece retains its root and connects through c_j to an earlier retained piece: the connector retains a neighbor in its first piece because it had neighbors in both colors. Singleton pieces cannot interrupt this attachment, since none is a connector's first piece.

The retained set dominates: each uncolored vertex keeps a neighbor in its first piece; deleted tree leaves have retained neighbors; a deleted singleton root is adjacent to its connector. Its order is at most $|B| + k - (k + 1) \leq b - 1$.

9.2.2 Zero slack excludes simplicial vertices except in complete graphs

When $\eta = 0$, the rooted bound shows every vertex lies in a minimum connected dominating set. Suppose a simplicial vertex v lies in such a set D. If $D = \{v\}$, then v is universal and simplicial, so G is complete. Otherwise choose a neighbor $u \in D$ of v. Deleting v preserves connectedness of $G[D]$, because its neighborhood is a clique. It also preserves domination: u dominates v and every vertex whose only neighbor in D was v. This contradicts minimality. Thus a noncomplete zero-slack graph has no simplicial vertex and, in particular, $\delta \geq 2$.

9.2.3 A maximal odd bipartite path forces positive slack

If an inclusion-maximal connected induced bipartite subgraph is an odd-order path of order at least three, then $\gamma_c \leq b - 2$.

Use that path as B_0 in the preceding construction. Instead of deleting one initial tree leaf, delete both path endpoints. They have the same color, so the retained nonempty path contains the entire opposite color class. Every uncolored vertex meeting the initial path retains a neighbor there. All connectivity and domination arguments above remain valid, while one additional vertex was deleted. The resulting connected dominating set has order at most $|B| + k - (k + 2) \leq b - 2$.

9.2.4 The published weaker form of 177

For completeness, $L + b \geq 2a + 1$ follows by the following short argument, also published in [15]. Fix a maximum independent set I and let $C = V \setminus I$ have m induced connected components. Selecting one vertex from each component together with I gives an induced bipartite set of order $a + m$, so $b \geq a + m$. The set C dominates G, and at most $m - 1$ vertices of I suffice to connect its components. Hence $\gamma_c \leq |C| + m - 1$ and $L \geq a - m + 1$. Adding gives the assertion.

9.3 Independence core and odd layers in claw-free graphs

9.3.1 Odd-layer injection

For every independent set I in a connected claw-free graph and every root v,

$$|I| \leq |O(v)| + \mathbf{1}_{v \in I}.$$

For each member $x \in I$ at positive even distance $2k$, choose a neighbor $p(x)$ in layer $2k - 1$. Two independent vertices cannot choose the same predecessor p: a preceding neighbor of p in layer $2k - 2$ is nonadjacent to both of them, producing a claw. Map each odd-distance member of I to itself. Its image cannot coincide with a predecessor image, because that predecessor is adjacent to a member of I. This injects $I \setminus \{v\}$ into $O(v)$.

9.3.2 An essential independent vertex forces an odd maximal bipartite path

The exchanges used here are part of the classical alternating-path approach to independent sets in claw-free graphs; see Minty [16]. We prove the precise assertion required below.

Suppose v belongs to every maximum independent set of a connected claw-free graph. Fix one maximum independent set I, and choose a longest induced path P containing v, alternating between I and its complement, with both endpoints in I. Initially the singleton $\{v\}$ is allowed.

Every internal non-I vertex of P already has two neighbors in I on the path. Claw-freeness therefore prohibits further neighbors in I outside the path.

The path has order at least three. A neighbor w of v lies outside I. If v were its only neighbor in I, replacing v by w would yield a maximum independent set avoiding v. Thus w has another neighbor $z \in I$, giving the three-vertex alternating induced path.

Suppose an outside vertex w could extend P while preserving connectedness and induced bipartiteness. Every connected bipartite claw-free graph has maximum degree at most two. Thus w either meets one endpoint alone, or meets exactly the two endpoints, closing an even cycle.

If it meets one endpoint and has a further neighbor $z \in I \setminus P$, the vertex z has no neighbor in P: not among its independent vertices, nor among its internal non-I vertices, which already have two independent neighbors. Appending w, z extends the alternating odd path, a contradiction. If it has no such further neighbor, replace all I vertices of P by its non-I vertices together with w. The two sets have equal size; the replacement is independent and has no neighbors in $I \setminus P$. It yields a maximum independent set avoiding v, again a contradiction.

If w meets both endpoints, it already has two neighbors in I and therefore no neighbor in $I \setminus P$. The identical exchange around the even cycle yields the same contradiction.

No vertex can extend P bipartitely. Any larger connected induced bipartite supergraph would supply a first outside neighbor of P that could be added. Thus P is inclusion-maximal.

9.3.3 Consequence at zero slack

In a claw-free graph with $\eta = 0$, the preceding lemma and Section 1.3 rule out every essential independent vertex. Consequently, each vertex can be avoided by a maximum independent set. The odd-layer injection then gives

$$a \le |O(v)| \quad \text{for every } v.$$

Equivalently, the intersection of all maximum independent sets—the independence core—is empty.

9.4 WOWII 180 on the claw-free class

If $\eta \ge 1$, the injection gives $a \le |O(v)| + 1 \le |O(v)| + \eta$. If $\eta = 0$, Section 9.3.3 gives $a \le |O(v)|$. Taking the minimum odd-distance count proves 180 for every connected claw-free graph.

This does not treat graphs containing induced claws.

9.5 WOWII 177 on the claw-free class

Complete graphs satisfy the inequality with equality. For $\sigma = 1$, Section 9.2.4 proves it. Assume henceforth $a \ge 2$ and $\sigma \ge 2$.

For a maximum independent set I, every outside vertex has at most two neighbors in I. At most one vertex of I has degree below σ, so

$$(a-1)\sigma + \delta \le e(I, V \setminus I) \le 2(n-a). \tag{26}$$

9.5.1 A general degree bound

Every noncomplete connected claw-free graph with $\sigma \geq 2$ satisfies

$$2a + \sigma \leq n + 2. \tag{27}$$

If $\sigma = 2$, (26) gives $n \geq 2a - 1 + \delta/2$, hence $n \geq 2a$ by integrality. If $a = 2$, a noncomplete graph has two nonuniversal vertices, so $\sigma \leq n - 2$. If $a = 3$ and $\sigma \geq 3$, (26) gives $n \geq \sigma + 3 + \delta/2$, hence $n \geq \sigma + 4$. Finally, if $a \geq 4$ and $\sigma \geq 3$, (26) gives

$$n - (2a + \sigma - 2) \geq \frac{(a-3)\sigma + \delta - 2a + 4}{2} \geq \frac{a-4}{2} \geq 0.$$

Thus $\eta \geq 1$ already proves 177.

9.5.2 The zero-slack equality profiles

Assume $\eta = 0$. Section 9.2.2 gives $\delta \geq 2$ and no simplicial vertex. When $a \geq 4$ and $\sigma \geq 3$, the final lower bound above improves to $(a-3)/2 > 0$, so integrality yields $2a+\sigma \leq n+1$. It remains to exclude equality in (27) in the following three profiles.

Profile A: $\sigma = 2$, $n = 2a$. Equality in (26) forces every vertex of I to have degree two and every vertex of $J = V \setminus I$ to have exactly two neighbors in I. If two vertices of J are adjacent, they must have a common neighbor in I; otherwise one endpoint and its two independent neighbors form a claw. The common neighbor then has degree two with adjacent neighbors and is simplicial. Hence there are no edges inside J. The graph is connected and 2-regular bipartite, so is an even cycle. But an even cycle has $b = n$, $\gamma_c = n - 2$, and $\eta = 1$, contradiction.

Profile B: $a = 3$, $\sigma \geq 3$, $n = \sigma + 4$. Equality in (26) forces $\delta = 2$, one vertex $x \in I$ of degree two, the other two vertices $y, z \in I$ of degree σ, and exactly two I-neighbors at every vertex of J. Write $N(x) = \{u, v\}$ and $W = J \setminus \{u, v\}$, where $|W| = \sigma - 1$. Every member of W meets both y, z. The degree counts force u to meet y but not z, and v to meet z but not y, after relabeling. Nonsimpliciality of x gives $uv \notin E(G)$. Since u has degree at least $\sigma \geq 3$, it has a neighbor $w \in W$. The set $\{u, w, z\}$ is connected and dominating: it covers x, y through u, v and all of W through z. Meanwhile y, u, x, v, z induce a five-vertex path. Thus $\gamma_c \leq 3$ and $b \geq 5$, giving $\eta \geq 1$, contradiction.

Profile C: $a = 2$, $\sigma = n - 2$. A universal vertex gives $\gamma_c = 1$ and $b \geq a + 1 = 3$, again positive slack. Without a universal vertex, the complement has minimum degree at least one and at most one vertex of degree greater than one. It is a disjoint union of a star and matching edges. It cannot be a single spanning star, since that would disconnect G. Two complement edges from different components induce a four-cycle in G, so $b \geq 4$. Choose a vertex u of degree $n - 2$, its unique nonneighbor v, and a neighbor w of v. Then uw is an edge and $\{u, w\}$ is connected dominating: u dominates every vertex except v, which is dominated by w. Hence $\gamma_c \leq 2$, again giving positive slack.

All equality profiles are impossible at $\eta = 0$. Therefore $2a + \sigma \leq n + 1 = L + b$, completing the claw-free proof of 177.

10 Induced forests, degree sequences, and matching structure

The results of Rounds 10, 12, 16, 17, 19, and 20 are presented here with one common residue framework. The induced-matching candidate is left open in general. Forest equality and a

smallest-counterexample restriction come first, followed by split-composition inheritance and unigraphs. The first triangle-free residue equality then gives the diameter-at-most-six case of 61. The final two subclasses use different matching hypotheses and yield the stronger half-diameter term: graphs with one feedback vertex use acyclic matchings, while the matching-geodesic case uses an ordinary maximum matching on a diametral path. These hypotheses are retained separately.

10.1 Residue, induced matchings, and equality for forests

10.1.1 Scope, notation, and the unresolved candidate

All graphs are finite, simple, and undirected. We allow empty and disconnected graphs except where connectedness is explicitly assumed. Write $n(G)$ for order, $\alpha(G)$ for independence number, $\nu(G)$ for ordinary matching number, and $f(G)$ for the largest order of an induced forest. An *induced matching* is a matching whose endpoints induce exactly its matching edges; its maximum cardinality is $\mathrm{im}(G)$. A feedback vertex is a vertex whose deletion leaves a forest. A nontrivial star will mean $K_{1,d}$ with $d \geq 2$, keeping isolated vertices and edges separate.

For a nonincreasing graphical degree list $D = (d_1, \ldots, d_n)$ with $d_1 > 0$, the Havel–Hakimi step removes d_1, subtracts one from the next d_1 entries, and sorts again. The Havel–Hakimi theorem guarantees that the result is graphical [17]. Repeating until only zeros remain defines the *residue* $\mathrm{res}(D)$ as the number of those zeros; set $\mathrm{res}(\varnothing) = 0$ and $\mathrm{res}(G) = \mathrm{res}(D(G))$. The elimination sequence lists the removed positive heads, followed by the final zeros. It has length n and total sum $|E(G)|$.

Conjecture 10.1. *For every finite simple graph G,*

$$f(G) \geq \mathrm{res}(G) + \mathrm{im}(G). \tag{28}$$

Status distinction. Candidate 10.1 is left unresolved in this paper. Our proved results are the forest equality characterization in Theorem 10.11 and the conditional smallest-counterexample restriction in Theorem 10.13. No worldwide novelty claim for these auxiliary statements is made on the basis of a limited literature search.

WOWII Conjecture 61 asks, for a connected nontrivial graph of diameter D, whether

$$f(G) \geq \mathrm{res}(G) + \lceil D/3 \rceil; \tag{29}$$

see the recorded formulation [18]. A diametral geodesic is induced. Taking its edges with indices $0, 3, 6, \ldots$ gives an induced matching of size $\lceil D/3 \rceil$. Thus a general proof of (28) would prove (29) directly. The unproved part is (28) itself, not this implication.

10.1.2 Classical residue dependencies

For lists of equal length and equal sum, sorted nonincreasingly, write $A \succeq B$ when every prefix sum of A is at least the corresponding prefix sum of B. We use the following established result, explicitly credited rather than reproved here.

Theorem 10.2 (Graphical elimination majorization; Triesch)**.** *If graphical degree lists A, B have equal length and equal total degree, and $A \succeq B$, then their Havel–Hakimi elimination sequences satisfy the same majorization relation.*

This is Triesch's theorem [19], stated with attribution as Theorem 1.6 in Hiller's research manuscript [20]. The equal-sum and graphical hypotheses are essential to the way it is used here.

The proof does not assume that individual Havel–Hakimi successor lists preserve an arbitrary majorization relation.

Lemma 10.3 (Residue order)**.** *Under the hypotheses of Theorem 10.2,* $\operatorname{res}(A) \geq \operatorname{res}(B)$.

Proof. The two elimination sequences have the same total sum. If the second has t positive entries, its first t entries already sum to that total. Majorization forces the first sequence's first t entries to equal the total as well, so it has at most t positive entries. Its zero count is therefore at least that of the second sequence. □

Lemma 10.4 (Largest-entry decrements)**.** *Fix a nonnegative list and an integer d no larger than its number of positive entries. Among the lists obtained by decrementing d distinct positive entries by one, decrementing the d largest entries produces a majorization-minimal sorted list.*

Proof. If a smaller entry is decremented and a larger entry is not, exchange these choices. For original entries $a \geq b$, this changes the surviving pair $(a, b-1)$ to $(a-1, b)$. When $a = b$ the change is only a permutation; otherwise it is a unit balancing transfer and cannot increase sorted-prefix sums. Repeated exchanges reach the largest-entry choice. □

Lemma 10.5 (Positive maximum-degree deletion)**.** *If $\Delta(G) > 0$ and $d_G(v) = \Delta(G)$, then* $\operatorname{res}(G - v) \geq \operatorname{res}(G)$.

Proof. Remove the common maximum degree head $d = d_G(v)$. The canonical Havel–Hakimi step decrements the d largest tail entries; actual vertex deletion decrements the d neighbor entries. By Lemma 10.4, the latter list majorizes the former. Both are graphical lists of the same order and sum. Lemma 10.3 gives $\operatorname{res}(G - v) \geq \operatorname{res}(\mathrm{HH}(D(G))) = \operatorname{res}(G)$. □

The positive-degree hypothesis cannot be omitted: deleting a vertex from an edgeless graph decreases residue by one.

Corollary 10.6 (Classical residue bound)**.** *For every graph,* $\operatorname{res}(G) \leq \alpha(G)$.

Proof. Induct on order. For an edgeless graph both sides equal its order. Otherwise choose a maximum-degree vertex v. By Lemma 10.5 and induction, $\operatorname{res}(G) \leq \operatorname{res}(G - v) \leq \alpha(G - v) \leq \alpha(G)$. □

This is the established Havel–Hakimi residue bound, proved in the classical literature, including Griggs and Kleitman [21]; the derivation above records our exact dependency chain.

Lemma 10.7 (Classical disjoint-union subadditivity)**.** *For any two disjoint graphs G, H,*

$$\operatorname{res}(G \sqcup H) \leq \operatorname{res}(G) + \operatorname{res}(H).$$

Proof. Induct on total order, writing A, B for the component degree lists. If both are zero lists, equality is immediate. Otherwise choose a largest pooled positive head, say in A, of value d. Let C be the global canonical reduction of $A \cup B$. Instead decrementing the d largest remaining entries of A alone gives $S = \mathrm{HH}(A) \cup B$. This is feasible because A is graphical. Both C and S are graphical; they have the same length and sum. By Lemmas 10.4 and 10.3, $S \succeq C$ and $\operatorname{res}(C) \leq \operatorname{res}(S)$. Induction on the shorter total length now gives

$$\operatorname{res}(A \cup B) = \operatorname{res}(C) \leq \operatorname{res}(S) \leq \operatorname{res}(\mathrm{HH}(A)) + \operatorname{res}(B) = \operatorname{res}(A) + \operatorname{res}(B).$$

□

Lemma 10.7 is the $k = 1$ case of the established Disjoint Union Lemma, Theorem 11 of Amos, Davila and Pepper [22]. Their Lemma 8 gives the maximum-degree residual comparison, and their Theorem 16 combines the two facts at a maximum-degree cut vertex. The proof here makes the dependencies transparent; it is not a claim to a new residue theorem.

10.1.3 Weak majorization by matching padding

The preceding majorization theorem requires equal degree sums. The following extension, extracted from the Round 10 auxiliary note, removes that requirement on graphical inputs. It remains a sequence theorem; it does not supply the graph-level forest augmentation needed for a general proof of 61.

For a nonincreasing list A, let $P_k(A)$ be its first k entries' sum, with zero padding when necessary. Write $E(A)$ for its canonical Havel–Hakimi elimination sequence, including terminal zeros. This sequence is nonincreasing, and its total is one half the degree sum.

Lemma 10.8 (Matching-padding identity)**.** *For a graphical list B and an integer $m \geq 0$,*

$$E(B \cup 1^{2m}) = \operatorname{sort}\big(E(B) \cup 1^m \cup 0^m\big). \tag{30}$$

Adding isolated vertices likewise appends only zeros to the elimination sequence.

Proof. Run the canonical reductions on the original B portion first. At a positive head h of that portion, graphicality guarantees at least h positive entries after the head: a realization has h distinct neighbors of the head vertex. Each entry selected by the B-only Havel–Hakimi step is therefore at least one. The appended ones can tie with these entries but cannot strictly displace them. Resolve ties in favor of the B entries, leaving every appended one unchanged. This is a valid canonical reduction of the combined multiset.

When the B portion reaches zeros, the $2m$ appended ones produce exactly m further eliminated heads equal to one and m additional terminal zeros. A tie-breaking choice cannot change the reduced degree multiset or the later head values. This proves (30). Appended zeros cannot affect a positive head or a decrement, proving the final assertion. □

Theorem 10.9 (Weak prefix order for graphical elimination)**.** *Let A, B be graphical degree lists, possibly of different lengths. If $P_k(A) \geq P_k(B)$ for every k, then*

$$P_k(E(A)) \geq P_k(E(B)) \qquad \text{for every } k.$$

Equality of the total degree sums is not required.

Proof. Pad A, B with zeros to a common length n. Their nonnegative degree-sum difference is even; write it as $2m$. Form sorted graphical lists

$$A' = A \cup 0^{2m}, \qquad B' = B \cup 1^{2m}.$$

They have equal lengths and sums: the first is realized by adjoining $2m$ isolates to a realization of A, and the second by adjoining m disjoint edges to a realization of B.

Let p be the number of positive entries in A. If $k \leq p$, a selection of k entries from B' uses some $q \leq k$ entries from B and $k - q$ appended ones. Its sum is at most

$$P_q(B) + (k - q) \leq P_q(A) + (k - q) \leq P_k(A),$$

because the first k entries of A are positive integers. In particular, this bounds the maximum-sum selection and gives $P_k(B') \leq P_k(A')$. For $k > p$, the latter prefix is already the entire sum of A', equal to the sum of B'. Thus $A' \succeq B'$.

Triesch's theorem, Theorem 10.2, now gives $P_k(E(A')) \geq P_k(E(B'))$. Adding isolates does not change the left prefix sum. By Lemma 10.8, the multiset defining $E(B')$ contains every entry of $E(B)$ and extra nonnegative entries. Therefore its largest k entries sum to at least $P_k(E(B))$. Together these facts give the asserted weak prefix order. □

Corollary 10.10 (A residual degree-sum bound)**.** *Under the hypotheses of Theorem 10.9, let A_k, B_k be the residual lists after k canonical steps, padding a completed history by zero heads. Then*

$$\sum A_k - \sum B_k \leq \sum A - \sum B.$$

Proof. Each eliminated head h removes $2h$ degree-sum units. Hence

$$\sum A_k - \sum B_k = \sum A - \sum B - 2(P_k(E(A)) - P_k(E(B))),$$

and the theorem bounds the parenthesized expression below by zero. □

This is not a claim that the residual lists themselves continue to majorize each other at every step. Nor is a statement about arbitrary nongraphical input lists being made. The proof depends on the credited equal-sum graphical theorem and the exact padding identity.

10.1.4 Equality for forests

We use the forest identity $\alpha(F) = |V(F)| - \nu(F)$. For completeness, a forest's minimum vertex-cover number equals its matching number: at a leaf x with neighbor y, some minimum vertex cover contains y rather than x, and some maximum matching contains xy. Both parameters therefore obey the recurrence 1 plus the corresponding parameter on $F - \{x, y\}$, with edgeless base zero. Independence number is order minus vertex-cover number.

Theorem 10.11. *For every finite forest F,*

$$|V(F)| \geq \mathrm{res}(F) + \mathrm{im}(F). \tag{31}$$

Equality holds if and only if every component is an isolated vertex or an edge, except possibly one component that is a star $K_{1,d}$ with $d \geq 2$.

Proof. The inequality follows immediately from

$$\mathrm{res}(F) \leq \alpha(F) = |V(F)| - \nu(F) \leq |V(F)| - \mathrm{im}(F). \tag{32}$$

We determine when it can be tight.

Step 1: connected nonstar trees are strict. If $\nu(T) > \mathrm{im}(T)$, strictness follows already from (32). Otherwise let $a = \nu(T) = \mathrm{im}(T)$, and fix an induced matching M of size a. It is also a maximum ordinary matching.

Every edge $xy \in M$ has a leaf endpoint. If both endpoints had other neighbors u, w, inducedness of M would put u, w outside its endpoint set. They are distinct because T has no triangle. Replacing xy by ux, yw would augment M, a contradiction.

Choose the nonleaf endpoint of each matching edge as its support, choosing either endpoint if the edge is a component on its own. The set A of supports is independent, has cardinality a, and each support has a private leaf partner. It is a vertex cover: unmatched vertices are independent, and a leaf partner is incident only with its matching edge. Thus every edge meets A.

Since T is connected and not a star, $a \geq 2$, and two distinct supports x, y have a common neighbor z on a path between supports. Let x', y' be their private leaf partners. Perform the degree-preserving switch

$$xx',\ yy' \quad \longmapsto \quad xy,\ x'y'.$$

The new edges were absent, because A is independent and x', y' are private leaves. The resulting simple graph T' has the same degree sequence and residue as T. It contains, on mutually disjoint vertex sets, the triangle xyz, the edge $x'y'$, and the other $a-2$ edges of M. Every vertex cover of T' therefore has size at least $2+1+(a-2) = a+1$, regardless of additional cross edges. Corollary 10.6 gives

$$\operatorname{res}(T) = \operatorname{res}(T') \leq \alpha(T') \leq |V(T)| - a - 1.$$

Hence $\operatorname{res}(T) + \operatorname{im}(T) \leq |V(T)| - 1$, as required.

Step 2: any nonstar component forces strictness. Induced matching number is additive across components. If a forest has a nonstar component, Lemma 10.7 and Step 1 give

$$\operatorname{res}(F) + \operatorname{im}(F) \leq \sum_{T \text{ component}} (\operatorname{res}(T) + \operatorname{im}(T)) < \sum_{T} |V(T)| = |V(F)|.$$

We do not assert $\operatorname{res}(T) < \alpha(T)$ for every nonstar tree: for example P_4 has $\operatorname{res}(P_4) = \alpha(P_4) = 2$, and its strictness instead comes from $\nu > \operatorname{im}$.

Step 3: two nontrivial star components force strictness. Suppose all components are stars, edges, or isolates, and two star centers x, y each have at least two leaves. Choose distinct leaves x_1, x_2 at x and y_1, y_2 at y, and switch

$$xx_1,\ yy_1 \quad \longmapsto \quad xy,\ x_1y_1.$$

In this degree-equivalent realization F', the three edges xx_2, yy_2, x_1y_1, together with one edge from every other nontrivial component, form a matching of size $\nu(F)+1$. In every graph, a vertex cover has size at least that of any matching. Consequently

$$\operatorname{res}(F) = \operatorname{res}(F') \leq \alpha(F') \leq |V(F)| - \nu(F) - 1.$$

For a union of stars, $\nu(F) = \operatorname{im}(F)$, so (31) is strict.

Step 4: the remaining forests attain equality. For $K_{1,d} \sqcup eK_2 \sqcup iK_1$ with $d \geq 2$, the degree list is $(d, 1^{d+2e}, 0^i)$. Its first Havel–Hakimi step leaves $2e$ ones and $d+i$ zeros. Thus

$$\operatorname{res} = d + e + i, \qquad \operatorname{im} = 1 + e, \qquad n = d + 1 + 2e + i,$$

giving equality. If there is no nontrivial star, $eK_2 \sqcup iK_1$ has $\operatorname{res} = e + i$, $\operatorname{im} = e$, and $n = 2e + i$. This includes the empty forest. □

10.1.5 A smallest-counterexample restriction

The assertions in this section are conditional on a counterexample to Candidate 10.1 existing. They do not assume its truth on graphs of the same or larger order.

Lemma 10.12. *Let G have minimum order among all counterexamples to Candidate 10.1. Then G is connected and is not a forest. For every maximum-degree vertex v, with $H = G - v$, one has*

$$\operatorname{res}(H) = \operatorname{res}(G), \tag{33}$$

$$\operatorname{im}(H) = \operatorname{im}(G) - 1, \tag{34}$$

$$f(G) = f(H) = \operatorname{res}(G) + \operatorname{im}(G) - 1. \tag{35}$$

Moreover, H is candidate-tight, residue is additive across its components, and every component is candidate-tight.

Proof. If G were disconnected, minimality on its components, additivity of f and im, and Lemma 10.7 would prove the candidate for G. Theorem 10.11 excludes forests. In particular $\Delta(G) > 0$.

Deleting one vertex destroys at most one edge of any induced matching. Minimality and Lemma 10.5 therefore give

$$f(G) \geq f(H) \geq \mathrm{res}(H) + \mathrm{im}(H) \geq \mathrm{res}(G) + \mathrm{im}(G) - 1.$$

Since G fails the integer-valued inequality (28), every inequality in this chain is equality. This proves (33)–(35). Applying the same component sum to H shows that neither a strict residue-sum inequality nor a surplus in any component is possible. □

Theorem 10.13. *If G is a smallest counterexample to Candidate 10.1, then $G - v$ contains a cycle for every maximum-degree vertex v.*

Proof. Suppose $H = G - v$ were a forest. By Lemma 10.12, it is tight in (31), so Theorem 10.11 applies. Equation (34) says that a maximum induced matching of G consists of an edge vu and $\mathrm{im}(H)$ edges in H.

The vertex u must be isolated in H. If it belonged to an edge component, that component's one matching edge could not be retained. If it belonged to the possible star component, every star edge would meet or have an endpoint adjacent to u, so the star's one matching edge could not be retained either. There is no way to compensate elsewhere, because the other retained edges already attain the sum of component induced matching numbers.

There is also no edge component in H: retaining its sole edge would force v to have no neighbor in that component, contradicting connectedness of G. If there were no nontrivial star component either, G would be a star, contrary to Lemma 10.12. Hence H consists of a star $K_{1,d}$ with center w, together with $p \geq 1$ isolates.

To retain a star edge in the induced matching, v is nonadjacent to w and misses at least one star leaf. Connectedness forces it to meet some star leaf. Let it meet exactly a of them. All p isolates of H are neighbors of v. Since G is not a forest, $a \geq 2$; also $a \leq d - 1$, so $d \geq 3$. Maximum degree of v implies $a + p \geq d$.

The degree list of G is thus

$$(a + p,\ d,\ 2^a,\ 1^{d-a+p}), \qquad a + p \geq d \geq 3.$$

Choose the maximum head $a + p$ in the first Havel–Hakimi step. It decrements the degree d, all a entries equal to two, and $p - 1$ ones. The residual list is

$$(d - 1,\ 1^{d+1},\ 0^{p-1}).$$

The next positive head $d - 1$ leaves exactly two ones; the following head removes their edge. There are exactly three positive heads in total. Hence

$$\mathrm{res}(G) = n(G) - 3 = d + p - 1, \qquad \mathrm{res}(H) = d + p,$$

contradicting (33). □

Remark 10.14 (Exact scope). Theorem 10.13 is a restriction on a *smallest counterexample* to the unproved candidate. It is not a proof of the candidate for every graph of feedback vertex number one: a feedback vertex in such a graph need not have maximum degree. It does not resolve original WOWII 61.

10.1.6 Why a simpler maximum-degree induction does not suffice

A tempting replacement for the unresolved step would assert that some maximum-degree vertex can be avoided by a maximum induced matching whenever the graph has no leaves. This is false even at arbitrarily large minimum degree.

Proposition 10.15. *For integers $m \geq 2$ and $k \geq 2m$, let $G = I_k \vee (mK_2)$, the join of an independent set of size k and m disjoint edges. Every maximum-degree vertex belongs to every maximum induced matching, and $\delta(G) = 2m$.*

Proof. The matching endpoints have degree $k + 1 > 2m$, while independent-set vertices have degree $2m$, so the endpoints are exactly the maximum-degree vertices. The original m edges form an induced matching. Any induced matching using a join edge has size only one: its independent-set endpoint is adjacent to every matching endpoint, and there are no edges inside the independent set. Since $m \geq 2$, the unique maximum induced matching is therefore the set of all m original edges. □

For clarity, this family does satisfy Candidate 10.1. Label its matching edges a_jb_j for $1 \leq j \leq m$. For each $j < m$, choose two new, distinct vertices x_j, y_j of I_k, and switch $b_jx_j, a_{j+1}y_j$ to b_ja_{j+1}, x_jy_j. This is possible since $k \geq 2m$. In the degree-equivalent realization, the matching endpoints induce a path on $2m$ vertices, and I_k induces $m - 1$ disjoint edges and isolates. Each vertex has lost at most one join incidence, with distinct missing partners. An independent set meeting both sides has size at most two; one within the path has size at most m, and one within the other side has size at most $k - m + 1$. Hence $\mathrm{res}(G) \leq k - m + 1$. The original graph contains a star on $k + 1$ vertices, so $\mathrm{res}(G) + \mathrm{im}(G) \leq k + 1 \leq f(G)$.

The family shows that a minimum-degree threshold cannot justify the proposed deletion shortcut, even though the target inequality itself holds on the family. The remaining mathematical gap is to rule out a smallest counterexample whose every maximum-degree deletion leaves tight cyclic components as in Lemma 10.12.

10.2 Inheritance under split composition and the unigraph class

10.2.1 Definitions and scope

A *unigraph* is a graph determined up to isomorphism by its degree sequence. We retain the residue, forest number, and induced-matching notation from Section 10.1, including the unresolved candidate (28). Recall the classical bound

$$\mathrm{res}(G) \leq \alpha(G) \tag{36}$$

from Corollary 10.6; Favaron, Mahéo and Saclé [23] are an earlier primary source. The subclass result is the following.

Theorem 10.16. *Inequality (28) holds for every unigraph. It also holds for every graph obtained from a unigraph by any finite sequence of split compositions; the outer split factors need not be unigraphs.*

A *split graph* has a specified partition $A \dot\cup B$ into an independent set A and a clique B. For a nonempty graph J, its split composition $S(A, B) \circ J$ is formed from disjoint copies of S and J by adding every edge between B and $V(J)$, and no edge between A and $V(J)$. Either A or B may be empty.

We explicitly credit two inputs from Barrus [24]:

$$\operatorname{res}(S) = \alpha(S) \qquad \text{for split graphs } S \qquad [24, \text{Proposition 2.2}], \tag{37}$$

$$\operatorname{res}(S(A,B) \circ J) = |A| + \operatorname{res}(J) \qquad \text{for nonempty } J \qquad [24, \text{Corollary 3.3}]. \tag{38}$$

These residue identities, and the canonical classification used below, are existing results, not contributions claimed by this section.

10.2.2 Inheritance under split composition

For any graph with an edge,

$$f(G) \geq \alpha(G) + 1. \tag{39}$$

Indeed, adjoining any vertex outside a maximum independent set produces an induced star together with isolated vertices. A split graph with an edge has $\operatorname{im}(G) = 1$, since every edge meets its clique part. Equations (37) and (39) therefore prove (28) for split graphs with an edge. For edgeless graphs, $f = \operatorname{res} = |V|$ and $\operatorname{im} = 0$.

Proposition 10.17. *If a nonempty graph J satisfies (28), then so does $S(A,B) \circ J$ for every split graph with specified partition $A \dot\cup B$.*

Proof. If J is edgeless, the composition is split and has already been handled. Suppose that J has an edge, and put $G = S(A,B) \circ J$. Deleting B leaves J and $|A|$ isolates, so

$$f(G) \geq |A| + f(J).$$

Every induced matching of size at least two lies entirely in J. An edge not lying in J has an endpoint in B, and that endpoint is adjacent to every vertex of J and every other vertex of B; it therefore conflicts with every other possible matching edge. Since J has an edge, $\operatorname{im}(G) = \operatorname{im}(J)$. By (38),

$$f(G) \geq |A| + \operatorname{res}(J) + \operatorname{im}(J) = \operatorname{res}(G) + \operatorname{im}(G).$$

□

Corollary 10.18. *A smallest counterexample to (28), if one exists, is indecomposable under nontrivial split composition.*

Proof. In a nontrivial representation $S(A,B) \circ J$, the inner graph J is smaller. Minimality and Proposition 10.17 would close the inequality. The edgeless-inner case is split and is also covered. □

10.2.3 The established unigraph classification

Tyshkevich's canonical decomposition [25], in the form used by Barrus [24, Theorems 4.1–4.2], expresses every unigraph as a sequence of split compositions over an indecomposable unigraph core. If that core is nonsplit, then, up to complementation, it is one of

$$C_5, \qquad mK_2 \ (m \geq 2), \qquad mK_2 \dot\cup K_{1,n} \ (m \geq 1,\ n \geq 2), \qquad U_m \ (m \geq 1). \tag{40}$$

Here U_m consists of a chordless four-cycle and m triangles sharing one common vertex and otherwise vertex-disjoint. The classification is an external theorem; we do not purport to reprove it. By Proposition 10.17, it remains to check these cores and their complements. Split cores were handled above.

Lemma 10.19. *A nonempty graph has residue one if and only if it is complete.*

Proof. The complete-graph direction is immediate. For the converse, induct on the order N. The one-vertex case is immediate. If $N > 1$ and the residue is one, the first HH head is positive and its residual list has residue one, hence is the complete-graph list of length $N-1$ by induction. If the first head d were less than $N-1$, a decremented tail entry would originally have degree $N-1 > d$, contradicting maximality of the head. Thus $d = N-1$ and every original entry is $N-1$. □

The five-cycle and the star forests For C_5, the values are res $= 2$, im $= 1$ and $f = 4$. For the two forest families in (40), direct computation gives

$$(f, \mathrm{res}, \mathrm{im})(mK_2) = (2m, m, m),$$
$$(f, \mathrm{res}, \mathrm{im})(mK_2 \dot\cup K_{1,n}) = (2m+n+1, m+n, m+1).$$

The residue calculation removes the star head, when present, and then pairs off the remaining ones. Both forest families satisfy (28) with equality.

The complement of either listed forest has independence two. It has no induced $2K_2$, since that would correspond to an induced four-cycle in the forest. Thus its induced matching number is one. By (36) and Lemma 10.19, its residue is two, and (39) gives $f \geq 3$. This proves the required inequality for both complementary families as well.

The shared-cycle graph and its complement The degree list of U_m is

$$(2m+2, 2^{\,2m+3}),$$

where exponents indicate multiplicities. One HH step leaves $(2, 1^{2m+2})$, the degree list of $mK_2 \dot\cup P_3$. Hence $\mathrm{res}(U_m) = m+2$. Deleting the common vertex leaves a forest on $2m+3$ vertices; since the original graph contains a cycle, this is maximum. An induced matching avoiding that vertex takes one edge from each of the m triangles and one edge from the residual P_3, for $m+1$ edges. If the common vertex is used, no second edge can coexist: its only nonneighbor is the opposite four-cycle vertex. Therefore

$$f(U_m) = 2m+3 = \mathrm{res}(U_m) + \mathrm{im}(U_m), \qquad \mathrm{im}(U_m) = m+1.$$

The complementary degree list is

$$((2m+1)^{\,2m+3}, 1).$$

For $m \geq 1$, two HH steps produce

$$((2m-1)^{\,2m+1}, 1),$$

the same list with parameter $m-1$. At $m = 0$ this is 1^4, of residue two. Consequently $\mathrm{res}(\overline{U_m}) = 2$. The original four-cycle induces $2K_2$ in the complement, giving both an induced matching of size two and an induced forest of order four. There cannot be an induced matching of size three: its six endpoints would induce minimum degree four in U_m, although every vertex of U_m except the common vertex has total degree two. It follows that

$$\mathrm{im}(\overline{U_m}) = 2, \qquad f(\overline{U_m}) \geq 4 = \mathrm{res}(\overline{U_m}) + \mathrm{im}(\overline{U_m}).$$

All cores in the credited classification are now covered. Repeated application of Proposition 10.17 proves Theorem 10.16.

10.2.4 Relation to the original diameter conjecture

As explained in Section 10.1, a general proof of (28) would imply original 61. Theorem 10.16 establishes only the specified subclass.

More importantly, connected unigraphs already have diameter at most three. Each connected nonsplit core in (40) or its complement has diameter at most three. In a connected nontrivial split composition, the outer clique is nonempty, every outer independent vertex has a neighbor in it, and every inner vertex is adjacent to it. Two outer independent vertices are therefore at distance at most three, and all other pairs at distance at most two. Connected split graphs have the same diameter bound. Thus (29) on unigraphs is already the elementary inequality (39) combined with (36). It is not a new original-conjecture advance.

The actual auxiliary strengthening here is the induced-matching term, which is unbounded on the family U_m. This is an auxiliary strengthening, not a full resolution of a numbered WoWII conjecture.

10.3 The first residue equality in triangle-free graphs

10.3.1 Definitions and scope

Retain the notation of Section 10.1. The residue depends only on the degree sequence, and the classical bound is

$$\operatorname{res}(H) \le \alpha(H). \tag{41}$$

No residue result beyond Corollary 10.6 is needed here.

For every connected graph with an edge,

$$f(G) \ge \alpha(G) + 1 \ge \operatorname{res}(G) + 1. \tag{42}$$

Indeed, a maximum independent set together with one vertex outside it induces a star and isolated vertices. Our main result identifies equality in the last bound for bipartite graphs.

Theorem 10.20. *Let G be a connected bipartite graph with at least one edge. Then*

$$f(G) = \operatorname{res}(G) + 1 \quad \Longleftrightarrow \quad G \text{ is a star or } C_4.$$

Thus every other connected bipartite graph satisfies $f(G) \ge \operatorname{res}(G) + 2$.

Here a star includes $K_{1,1}$. The single-vertex graph is excluded because its forest order and residue are both one. Our proof is a classification of this first equality case, not a proof of the unrestricted candidate $f \ge \operatorname{res} + \operatorname{im}$.

An attributed preliminary observation. Gunderson, Meagher, Morris and Pantangi already observed that when $f(G) = \alpha(G) + 1$, relative to any maximum independent set I, adjacent vertices outside I share a neighbor in I, and nonadjacent ones share at least two [26, Section 5]. In particular, the triangle-free implication $f = \alpha + 1 \Rightarrow G$ bipartite follows from that prior observation. Its short proof is retained below. The subsequent residue-equality argument is a separate step.

10.3.2 Maximum independent sets

Lemma 10.21. *Let G be connected and bipartite, with an edge and bipartition (A, B). If $f(G) = \alpha(G) + 1$, every maximum independent set is either A or B.*

Proof. Let I be a maximum independent set. Suppose it omits both a vertex $a \in A$ and a vertex $b \in B$. Their neighborhoods within I are disjoint, since $N(a) \cap I \subseteq B$ and $N(b) \cap I \subseteq A$. The induced graph on $I \cup \{a, b\}$ is therefore two stars, possibly joined by the edge ab, together with isolated vertices. It is a forest of order $\alpha(G) + 2$, a contradiction. Hence I contains an entire bipartition class. Every vertex has a neighbor, so independence prevents adding any vertex from the other class. Thus $I = A$ or $I = B$. □

We next prove a residue bound under this independent-set restriction, without assuming anything about f.

Theorem 10.22. *Let G be connected and bipartite, with an edge. Suppose every maximum independent set is a bipartition class. If G is neither a star nor C_4, then*

$$\operatorname{res}(G) \leq \alpha(G) - 1.$$

10.3.3 The unequal-side case

Proof of Theorem 10.22 when the sides are unequal. Write $|A| = p > |B| = q$. The hypothesis implies that A is the unique maximum independent set and $\alpha(G) = p$. For every nonempty $X \subseteq B$, the independent set $X \cup (A \setminus N(X))$ differs from A, so has order at most $p - 1$. Thus

$$|N(X)| \geq |X| + 1 \qquad (\varnothing \neq X \subseteq B). \tag{43}$$

If $q = 1$, connectivity makes G a star, contrary to the assumption. Hence $q \geq 2$. Connectivity gives a vertex $w \in A$ with two distinct neighbors $u, v \in B$; otherwise no path could join two vertices of B. By (43) and Hall's theorem, $G - w$ has a matching M saturating B. Let $au, bv \in M$. The vertices a, b, w are distinct.

Delete the edges au, bv and insert ab, uv, obtaining G'. Both new edges were absent, because they join vertices in the same bipartition class. Thus this is a valid degree-preserving switch. The graph G' contains the following vertex-disjoint subgraphs: the triangle on $\{w, u, v\}$, the edge ab, and the $q - 2$ other matching edges of M. Every vertex cover of G' has at least

$$2 + 1 + (q - 2) = q + 1$$

vertices. Since a set is a vertex cover exactly when its complement is independent, $\alpha(G') \leq p - 1$. Equality of degree sequences and (41) now give

$$\operatorname{res}(G) = \operatorname{res}(G') \leq \alpha(G') \leq p - 1,$$

as required. □

10.3.4 The balanced case and conformal cycles

An even cycle is called *conformal* here if deleting its vertices leaves a graph with a perfect matching. The cycle need not be induced. A balanced bipartite graph whose only maximum independent sets are its two sides is classically called *bistable*. The equivalence of this condition with strict Hall inequalities (and with connectedness and every edge belonging to a perfect matching) is established in Levit–Mandrescu [27, Theorem 2.5]. We include the short direction needed here and then give the explicit conformal-cycle and degree-switch constructions.

Lemma 10.23. *Let G be bipartite with $|A| = |B| = q \geq 3$. Suppose*

$$|N(X)| \geq |X| + 1 \qquad (\varnothing \neq X \subsetneq A). \tag{44}$$

Then G contains a conformal even cycle of length at least six.

Proof. The displayed inequalities also ensure $N(A) = B$: applying one of them to a set of size $q-1$ already gives all q vertices of B. Hall's theorem therefore supplies a perfect matching $M = \{a_i b_i : 1 \le i \le q\}$. Define a directed graph D on the matching indices, with an arc $i \to j$ for $i \neq j$ whenever $a_i b_j$ is an edge of G. Every nonempty proper index set has an outgoing arc by (44). Hence D is strongly connected.

If D has a directed cycle on $k \ge 3$ indices, its arcs together with the corresponding k matching edges form an even cycle of length $2k$. The other matching edges show that this cycle is conformal.

Otherwise D has no directed cycle of length at least three. For every arc $i \to j$, a shortest directed return path from j to i has length one: a longer return path would give such a forbidden directed cycle. Thus every arc has its reverse. The underlying undirected graph is connected and acyclic, because an undirected cycle could be oriented to give a directed cycle of length at least three. It is a tree. Since $q \ge 3$, it contains a three-vertex path $i - j - k$. The six distinct vertices of those matching pairs form the cycle

$$a_i, b_i, a_j, b_k, a_k, b_j, a_i.$$

Again the other edges of M certify conformality. □

Proof of Theorem 10.22 when the sides are equal. Write $|A| = |B| = q = \alpha(G)$. For a nonempty proper subset X of A, the independent set $X \cup (B \setminus N(X))$ equals neither A nor B. It therefore has order at most $q-1$, proving (44). If $q = 1$, $G = K_2$ is a star. If $q = 2$, the strict Hall inequalities force all four possible edges, so $G = C_4$. These cases are excluded, hence $q \ge 3$.

By Lemma 10.23, choose a conformal cycle with consecutive vertices $c_0, c_1, \dots, c_{2k-1}$, where $k \ge 3$. Delete c_0c_1, c_3c_4 and insert c_0c_4, c_1c_3. The inserted edges were absent because their endpoints lie in the same bipartition class. Degrees are preserved. The resulting graph G' contains two vertex-disjoint odd cycles:

$$c_1, c_2, c_3, c_1, \qquad c_4, c_5, \dots, c_{2k-1}, c_0, c_4.$$

Their orders are 3 and $2k-3$, respectively, so their vertex covers require at least $2+(k-1) = k+1$ vertices in total. A perfect matching outside the original cycle supplies $q-k$ further disjoint edges. Every vertex cover of G' consequently has size at least $q+1$. Therefore

$$\operatorname{res}(G) = \operatorname{res}(G') \le \alpha(G') \le q - 1,$$

which completes the proof of Theorem 10.22. □

10.3.5 Equality and carefully limited consequences

Proof of Theorem 10.20. If $f(G) = \operatorname{res}(G) + 1$, then (42) forces $f(G) = \alpha(G) + 1$ and $\alpha(G) = \operatorname{res}(G)$. Lemma 10.21 and Theorem 10.22 exclude every graph except a star or C_4. Conversely, a star with q leaves has forest order $q+1$ and residue q; C_4 has forest order three and residue two. □

Corollary 10.24. *Let G be a connected triangle-free graph with an edge. Then $f(G) = \operatorname{res}(G)+1$ if and only if G is a star or C_4. Every other such graph satisfies $f(G) \ge \operatorname{res}(G) + 2$.*

Proof. Equality in (42) forces $f(G) = \alpha(G) + 1$. Let I be a maximum independent set. If $V(G) \setminus I$ contains an edge uv, triangle-freeness gives $N(u) \cap N(v) \cap I = \varnothing$. Then $G[I \cup \{u, v\}]$ is two stars joined by uv, together with isolates, and is a forest of order $\alpha(G) + 2$, a contradiction. Thus $V(G) \setminus I$ is independent, so G is bipartite and Theorem 10.20 applies. The converse is already proved. □

Original WOWII Conjecture 61 asks for

$$f(G) \geq \mathrm{res}(G) + \lceil \mathrm{diam}(G)/3 \rceil \tag{45}$$

in connected graphs. For diameter at most three this already follows from (42), with the one-vertex case checked directly. Corollary 10.24 additionally proves (45) for connected triangle-free graphs of diameter four, five, or six: none is a star or C_4, and the required ceiling is two. Thus the whole triangle-free subclass with diameter at most six is covered, but diameter at least seven is not.

The same candidate (28) holds for connected triangle-free graphs with $\mathrm{im}(G) \leq 2$. The case $\mathrm{im} = 1$ follows from (42); when $\mathrm{im} = 2$, the exceptions in Corollary 10.24 are impossible. The one-vertex case is immediate. No assertion is made here for larger induced matching number or for general graphs containing triangles. In particular, no full original WOWII conjecture is counted as closed by these subclass results. These are mathematical subclass consequences, not a claim of established publication novelty.

10.4 Acyclic matchings and triangle-free graphs with one feedback vertex

10.4.1 Definitions, credited input, and exact scope

All graphs are finite and simple. Write $n(G) = |V(G)|$, $\alpha(G)$ for the independence number, $\nu(G)$ for the ordinary matching number, and $f(G)$ for the maximum order of an *induced* forest. The feedback-vertex number is $n(G) - f(G)$. Thus a graph has feedback-vertex number at most one if it is a forest or becomes a forest after deleting one vertex. No maximum-degree condition on that vertex is imposed.

A matching is *acyclic* if its endpoints induce a forest. Its maximum possible size is denoted by $\nu_{\mathrm{ac}}(G)$. This is the established acyclic matching notion introduced in the subgraph-restricted matching framework [28]; see also [29]. It is stronger than requiring that the matching be the unique perfect matching on its endpoints. Always

$$\nu_{\mathrm{ac}}(G) \leq \nu(G). \tag{46}$$

We again use the classical residue bound (Corollary 10.6),

$$\mathrm{res}(H) \leq \alpha(H) \tag{47}$$

for every graph H. Also,

$$\alpha(H) \leq n(H) - \nu(H), \tag{48}$$

since every vertex cover meets each edge of a maximum matching and the complement of an independent set is a vertex cover.

The use of degree-equivalent realizations to upper-bound residue through independence, and the two-edge-switch method itself, are classical; Fajtlowicz discusses both in the July 2004 *Written on the Wall* notes [30, p. 37]. They are not new ingredients claimed here.

10.4.2 An unmatched nonleaf forces strictness

Lemma 10.25. *Let G be triangle-free, with n vertices. If some maximum matching leaves a vertex of degree at least two unmatched, then*

$$\mathrm{res}(G) \leq n - \nu(G) - 1.$$

Proof. Let M be such a maximum matching, let x be the unmatched vertex, and choose distinct neighbors u, v of x. Two unmatched vertices cannot be adjacent, since their edge could be added to M. Thus u and v have partners a and b in M, respectively. Triangle-freeness implies that uv is absent, so ua and vb are distinct matching edges. In particular, x, u, v, a, b are five distinct vertices.

First suppose $ab \in E(G)$. Then

$$x, u, a, b, v, x$$

is a five-cycle. It is vertex-disjoint from the $\nu(G) - 2$ remaining edges of M. Every vertex cover of G therefore has at least

$$3 + (\nu(G) - 2) = \nu(G) + 1$$

vertices. Hence $\alpha(G) \leq n - \nu(G) - 1$, and (47) proves the assertion.

Now suppose $ab \notin E(G)$. Delete ua, vb and insert uv, ab, producing G'. The added edges were absent and their four endpoints are distinct, so this is a valid degree-preserving switch. The graph G' contains the mutually vertex-disjoint triangle on $\{x, u, v\}$, edge ab, and the $\nu(G) - 2$ other edges of M. Every vertex cover of G' has at least

$$2 + 1 + (\nu(G) - 2) = \nu(G) + 1$$

vertices. Consequently

$$\operatorname{res}(G) = \operatorname{res}(G') \leq \alpha(G') \leq n - \nu(G) - 1.$$

The two cases exhaust all possibilities. □

Corollary 10.26. *If a triangle-free graph satisfies* $\operatorname{res}(G) = n(G) - \nu(G)$*, every vertex left unmatched by any maximum matching has degree at most one.*

The unmatched vertices of an acyclic ordinary maximum matching form an independent feedback vertex set, as explicitly observed by Fürst–Rautenbach [31, Conclusion]. We recall that standard observation in the proof below; the strict residue conclusion additionally uses Lemma 10.25.

Corollary 10.27. *If G is triangle-free and contains a cycle, and $\nu_{\mathrm{ac}}(G) = \nu(G)$, then*

$$\operatorname{res}(G) \leq n(G) - \nu(G) - 1.$$

Proof. Choose an acyclic matching M of size $\nu(G)$. Its endpoint-induced graph is a forest. The unmatched set is independent, because M is a maximum matching. A cycle of G cannot be wholly contained in the endpoint-induced forest, so it contains an unmatched vertex x. This vertex has at least two neighbors, and Lemma 10.25 applies. □

The hypothesis in Corollary 10.27 is that an *ordinary maximum* matching can be chosen acyclic. It does not assume that all maximum matchings are acyclic.

10.4.3 Triangle-free apex forests

Theorem 10.28. *If G is triangle-free and has feedback-vertex number at most one, then*

$$f(G) \geq \operatorname{res}(G) + \nu_{\mathrm{ac}}(G). \tag{49}$$

Proof. If G is a forest, every matching is acyclic and $f(G) = n(G)$. Combining (47) and (48) gives

$$f(G) = n(G) \geq \mathrm{res}(G) + \nu(G) = \mathrm{res}(G) + \nu_{\mathrm{ac}}(G).$$

Otherwise G is cyclic and its feedback-vertex number is exactly one, so $f(G) = n(G) - 1$. If $\nu_{\mathrm{ac}}(G) \leq \nu(G) - 1$, then

$$\mathrm{res}(G) + \nu_{\mathrm{ac}}(G) \leq (n(G) - \nu(G)) + (\nu(G) - 1) = n(G) - 1 = f(G).$$

If $\nu_{\mathrm{ac}}(G) = \nu(G)$, Corollary 10.27 instead yields

$$\mathrm{res}(G) + \nu_{\mathrm{ac}}(G) \leq (n(G) - \nu(G) - 1) + \nu(G) = n(G) - 1 = f(G).$$

These cases are exhaustive by (46). □

The theorem allows arbitrarily many cycles through the feedback vertex; it is not limited to unicyclic graphs. Its forest case is simply the classical independence bound and is not counted as a separate new result.

Corollary 10.29. *Let G be a connected triangle-free graph with feedback-vertex number at most one, and let $D = \mathrm{diam}(G)$. Then*

$$f(G) \geq \mathrm{res}(G) + \lceil D/2 \rceil. \tag{50}$$

In particular, original WOWII Conjecture 61 holds for every graph in this subclass, without a diameter restriction.

Proof. A diametral path is induced and has $D + 1$ vertices. A maximum matching of that path has size

$$\left\lfloor \frac{D+1}{2} \right\rfloor = \left\lceil \frac{D}{2} \right\rceil.$$

The path matching size is standard; see also [28, Theorem 3]. Its endpoints induce a subgraph of the induced path, hence a forest. Thus $\nu_{\mathrm{ac}}(G) \geq \lceil D/2 \rceil$, and Theorem 10.28 gives (50). Original 61 asks for $f(G) \geq \mathrm{res}(G) + \lceil D/3 \rceil$, which is weaker. The one-vertex case has $D = 0$ and $f = \mathrm{res} = 1$. □

10.4.4 Limitations of the stronger inequality

No proof is claimed here for arbitrary triangle-free graphs of feedback-vertex number at least two. Neither the general half-diameter bound nor the general acyclic-matching bound is established by this paper. In particular, this is not a full closure of original 61.

The triangle-free hypothesis cannot simply be dropped. A triangle with a pendant leaf at each of two distinct triangle vertices has $f = 4$, $\mathrm{res} = 3$, $\nu_{\mathrm{ac}} = 2$, and diameter three. It violates both (49) and (50). Ordinary matching number also cannot replace acyclic matching number, even within the feedback-vertex-one subclass: C_4 has $f = 3$, $\mathrm{res} = 2$, $\nu = 2$, and $\nu_{\mathrm{ac}} = 1$.

10.5 When a diametral path attains the matching number

10.5.1 Definitions, credited input, and precise scope

All graphs are finite and simple. Write $n(G) = |V(G)|$, $\alpha(G)$ for the independence number, $\nu(G)$ for the *ordinary* matching number, and $f(G)$ for the largest number of vertices inducing a forest. A feedback-vertex set is a set whose deletion leaves a forest; its minimum size is

$$\mathrm{fvs}(G) = n(G) - f(G).$$

A vertex cover meets every edge, so its minimum size is $n(G) - \alpha(G)$. The feedback-vertex and vertex-cover parameters are different.

The residue depends only on the degree sequence. Our residue input is again Corollary 10.6:

$$\operatorname{res}(H) \le \alpha(H) \qquad \text{for every finite simple graph } H. \tag{51}$$

The degree-equivalent-realization viewpoint and two-edge-switch method are classical, and are explicitly discussed in Fajtlowicz's notes [30, p. 37]. They are not claimed as new methods here.

A diametral path of length D contains a matching of size $\lfloor (D+1)/2 \rfloor = \lceil D/2 \rceil$, giving the elementary lower bound $\nu(G) \ge \lceil D/2 \rceil$ for a connected graph. We treat precisely the equality case.

Theorem 10.30. *Let G be a finite simple connected triangle-free graph, and put $D = \operatorname{diam}(G)$ and $k = \lceil D/2 \rceil$. If $\nu(G) = k$, then*

$$f(G) \ge \operatorname{res}(G) + k. \tag{52}$$

Original WOWII Conjecture 61 asks for $f(G) \ge \operatorname{res}(G) + \lceil \operatorname{diam}(G)/3 \rceil$ for every simple connected graph. Consequently Theorem 10.30 establishes that inequality only on its stated subclass, with a stronger diameter term. No conclusion for triangle-free graphs with $\nu(G) > \lceil D/2 \rceil$ is obtained below. The proof is combinatorial, with (51) as its credited input; finite computations are supporting controls rather than its basis.

10.5.2 The structure forced by a diametral path

For the remainder of the proof assume $D \ge 1$ and fix a diametral path

$$P = p_0 p_1 \cdots p_D.$$

Every subpath is a shortest path between its ends; in particular, P is induced. Set $R = V(G) \setminus V(P)$.

Lemma 10.31. *The set R is independent. Every $x \in R$ has either one neighbor on P, or exactly two neighbors p_i, p_{i+2}. It has no other neighbors.*

Proof. A matching of k edges in P is maximum in G. Any edge inside R could be added to it, which is impossible. Thus R is independent. Connectedness ensures that each $x \in R$ has a neighbor on P. If $p_i, p_j \in N(x)$ with $i < j$, the two-edge path $p_i x p_j$ shows $j - i \le 2$, since otherwise it would shorten the corresponding geodesic subpath. Consecutive neighbor indices would form a triangle with x. Thus the neighborhood is a singleton or a pair at index difference two. □

We now choose k distinct path vertices called *supports* and pair each with a distinct *partner* in a fixed k-edge path matching M. The supports and partners are disjoint sets. In both parity cases all partners will form an independent set.

Even diameter: one support chain Suppose $D = 2k$. For each $j \in \{0, \dots, k\}$ there is a k-edge matching in P exposing precisely p_{2j}: match the two even-order subpaths on its sides. If p_{2j} had a neighbor in R, that edge would extend the matching. Therefore outside vertices have only odd-indexed neighbors on P.

For $1 \le i \le k$ take

$$u_i = p_{2i-1}, \qquad a_i = p_{2i-2}, \qquad M = \{u_i a_i : 1 \le i \le k\}.$$

The supports are $u_1, \dots, u_k$, their partners $a_1, \dots, a_k$ are independent, and p_{2k} is unmatched. Consecutive supports u_i, u_{i+1} are joined by one or more internally disjoint two-edge paths: the path through p_{2i}, and any paths through outside common neighbors. All other vertices are pendant at a support. This includes the two path endpoints. There are no other edges, by Lemma 10.31 and the inducedness of P.

Odd diameter: two support chains and an optional bridge Suppose $D = 2k - 1$, and fix

$$M = \{p_{2j}p_{2j+1} : 0 \le j < k\}.$$

Let E and O be respectively the even and odd path indices having a neighbor in R. There are no $e \in E$, $o \in O$ with $e < o$. Indeed, choose outside neighbors x of p_e and y of p_o. They are distinct by Lemma 10.31, since an outside vertex has neighbors of only one parity. The path

$$x, p_e, p_{e+1}, \dots, p_o, y$$

is an M-augmenting path: its end edges are outside M, and the path segment starts and ends with edges of M. Exchanging alternating edges would increase the matching size, a contradiction.

It follows that all odd outside attachment indices precede all even ones. Choose $h \in \{0, \dots, k\}$ so that every odd attachment index is less than $2h$, and every even one is at least $2h$. If $O \neq \varnothing$, one may take $h = (\max O + 1)/2$; if $O = \varnothing$, take $h = 0$. For each $0 \le j < k$ designate the support and its partner by

$$(u_j, a_j) = \begin{cases} (p_{2j+1}, p_{2j}), & j < h, \\ (p_{2j}, p_{2j+1}), & j \ge h. \end{cases}$$

Every outside vertex is adjacent only to supports. The odd supports on the left form one chain, and the even supports on the right form another. Within either chain, consecutive supports are joined by one or more internally disjoint two-edge paths. All remaining outside vertices are pendant at a support.

If both chains are nonempty, their sole connection is the path edge $p_{2h-1}p_{2h}$. This edge is a bridge: there are no path chords, no edges within R, and no outside vertex adjacent to both parities of supports. If $h = 0$ or $h = k$, one chain is empty and this bridge is absent. All partners are independent. Partners within one chain have the same parity; across the boundary the nearest two have indices $2h - 2$ and $2h + 1$, whose difference is three. The inducedness of P rules out all such partner edges.

Remark 10.32. This is a necessary description relative to the fixed diametral path, not a converse assertion that arbitrary pendant additions preserve its diameter. Empty attachment sets, a chain with one support, and an empty chain are allowed. If R is empty, the description reduces to P itself.

10.5.3 The exact feedback-vertex number

Call the collection of two-edge paths between two consecutive supports in the same chain a *link*. Its internal vertices are precisely the common neighbors used by these paths, and it induces $K_{2,c}$ for some $c \ge 1$. Exactly one of its internal vertices belongs to P; the others belong to R. Each internal vertex belongs to exactly one link. A link is *bad* if $c \ge 2$.

Let Q be the graph on all k supports whose edges are the bad links. The optional bridge between the two odd-case chains is not an edge of Q. Thus Q is a disjoint union of paths, including isolated vertices. Put $t = \nu(Q)$.

Lemma 10.33. *With the notation above,* $\mathrm{fvs}(G) = t$ *and* $f(G) = n(G) - t$.

Proof. Every cycle of G lies in one link. To see this, the support-chain skeleton, including the optional bridge, is a path; a support separating two links is a cut vertex separating their internal vertices. A simple cycle cannot cross such a support and return through it, nor can it use a bridge. Pendant vertices do not lie on cycles. A link $K_{2,c}$ is cyclic exactly when $c \geq 2$.

A vertex cover of Q is consequently a feedback-vertex set of G. For a path with ℓ edges, a matching and a vertex cover of size $\lceil \ell/2 \rceil$ are obtained by alternating edges and alternating vertices, respectively. No vertex cover can be smaller than a matching. Summing over path components, Q therefore has a vertex cover of size t, giving $\mathrm{fvs}(G) \leq t$.

Conversely, choose a matching of t bad links. In each link choose two of its internal vertices. They and its two supports form a four-cycle. The resulting t four-cycles are vertex-disjoint: their support pairs are disjoint, and internal vertices belong to distinct links. Every feedback-vertex set meets all these cycles, so $\mathrm{fvs}(G) \geq t$. □

10.5.4 Simultaneous degree-preserving switches

Proof of Theorem 10.30. When $D = 0$, connectedness gives $G = K_1$, so $f(G) = \mathrm{res}(G) = 1$ and $k = 0$. Assume $D \geq 1$ and use the structure and matching M above. Write $n = n(G)$, and choose a matching L of t bad links in Q.

For each $uv \in L$, choose an extra common neighbor $x \notin V(P)$. Such a vertex exists because the link has at least two internal vertices and exactly one of them is on P. Let a, b be the partners of u, v in M. The five vertices x, u, v, a, b are distinct. Replace the two edges ua, vb by the two edges uv, ab.

Both new edges were absent: u, v have path indices differing by two, and all partners are independent. The selected support pairs are disjoint, their partners are distinct, and the outside witnesses are distinct because each belongs to only one link. Supports, partners and outside witnesses are disjoint classes. Hence these switches have pairwise disjoint five-vertex domains and may be performed simultaneously, producing a finite simple graph G' with exactly the same degree at every vertex as G.

In G' the following subgraphs are pairwise vertex-disjoint:

- t triangles xuv, one for each selected link;
- the t new partner edges ab;
- the $k - 2t$ unchanged edges of M on the remaining supports.

Their edges survive all the switches. Additional edges between these subgraphs are harmless: they are not asserted to be induced components. Since L is a matching on k supports, $2t \leq k$. Every vertex cover of G' must therefore contain at least

$$2t + t + (k - 2t) = k + t$$

vertices. By (51) and Lemma 10.33,

$$\mathrm{res}(G) = \mathrm{res}(G') \leq \alpha(G') \leq n - k - t = f(G) - k.$$

This proves (52). When $t = 0$, no switch is made and the unchanged matching alone is the required cover certificate. The switched graph need not remain triangle-free, connected, or have the same diameter; none of these extra conditions is needed for (51). □

10.5.5 Boundary cases and an unbounded family

The small diameter cases are included: $D = 1$ gives K_2, and $D = 2$ under the matching hypothesis gives a star. Neither case has a bad link. Both odd split endpoints $h = 0, k$, missing attachment

sets, and $2t = k$ are included in the argument. Link multiplicities may exceed two; a selected bad link always has an outside witness, and only two of its internal vertices are needed for the disjoint-cycle certificate.

Proposition 10.34. *For each $k \geq 1$ there is a graph satisfying the theorem with $n = 3k$, $D = 2k$, $\nu = k$, and* $\mathrm{fvs} = \lfloor k/2 \rfloor$.

Proof. Take k supports in path order, replace each consecutive-support link by two internally disjoint two-edge paths, and attach one pendant vertex to each end support. For $k = 1$, attach two distinct pendant vertices to the single support. The graph is triangle-free. It has $k + 2(k-1) + 2 = 3k$ vertices. The two pendant vertices have distance $2k$, and every other vertex lies between their levels in the chain, so the diameter is $2k$. The supports form a vertex cover of size k, while the diametral path has a matching of size k, proving $\nu = k$. Every link is bad, so Q is the path on k vertices and Lemma 10.33 gives $\mathrm{fvs} = \lfloor k/2 \rfloor$. □

In particular, the theorem is not restricted to feedback-vertex number at most one. The proposition states structural parameters, not equality in (52) for every member.

Scope and priority. The unrestricted half-diameter bound for triangle-free graphs remains unproved here, as does original 61 for arbitrary connected graphs. The classical residue bound, path matching observation, vertex-cover certificate, and degree-switch methodology are credited or elementary inputs. The source checkpoint is a bounded literature search, not a certification of worldwide novelty or publication priority.

11 Conjecture 155 when spanning trees have at most three leaves

For a nontrivial connected graph G, let $L(G)$ be the maximum number of leaves of a spanning tree, let $r = \mathrm{rad}(G)$, and let $C(G)$ be the full center. Define

$$R(v) = |\{x : d(v,x) = r\}| \quad (v \in C(G)), \qquad q(G) = |\{R(v) : v \in C(G)\}|.$$

Thus q counts the *distinct orders* of radial circles. Original statement 155 is

$$L(G) \geq 1 + q(G). \tag{53}$$

It contains no ratio of radial-circle orders; that ratio belongs to 154.

Theorem 11.1. *For every nontrivial connected graph with $L(G) \leq 3$, inequality (53) holds. More precisely, $L = 2$ implies $q = 1$, and $L = 3$ implies $q \leq 2$.*

11.1 A full-star criterion and the resulting families

Lemma 11.2. *For a connected graph of maximum degree at most three, the following are equivalent:*

1. *$L(G) \leq 3$;*
2. *for every two distinct degree-three vertices, the union of their full incident-edge stars contains a cycle;*
3. *every adjacent pair of degree-three vertices has a common neighbor, and every nonadjacent pair has at least two common neighbors.*

Proof. Every forest in a connected graph extends to a spanning tree. If two full stars form a forest, an extension retains degree three at both centers. A subcubic tree has two more leaves than degree-three vertices, so the extension has at least four leaves. Conversely, a spanning tree with at least four leaves has two degree-three vertices, and its edge set contains their full stars, which form a forest. Two adjacent stars contain a cycle exactly when their centers have a common neighbor; two nonadjacent stars contain a cycle exactly when their centers have at least two common neighbors. These observations prove all equivalences. □

If $L \le 3$, then $\Delta(G) \le 3$: the star at any vertex extends to a spanning tree, and a tree has at least as many leaves as the degree of each of its vertices. Let B be the set of degree-three vertices.

If B is independent, fix $x \in B$. Its three neighbors have degree at most two, so together they have at most three edges to $B \setminus \{x\}$. Every further vertex of B requires at least two such edges by Lemma 11.2. Consequently $|B| \le 2$.

Suppose $|B| \ge 3$. There is an edge ab with both endpoints in B. They have a common neighbor c. If $c \in B$, there is a triangle in B. Otherwise c has degree two and is adjacent only to a, b. A third vertex $z \in B$ cannot miss both a and b: to share two neighbors with a, it would have to be adjacent to b or c, which is then impossible. Suppose, by symmetry, that za is an edge. The adjacent pair a, z needs a common neighbor; of the other neighbors b, c of a, only b can meet z, because c is saturated. Hence a, b, z form a triangle in B.

Fix a triangle abc in B. Every further $z \in B$ must meet the triangle, since otherwise it could share at most the one remaining neighbor of a with a. If za is an edge, the adjacent-star condition forces z to meet b or c as well. Each additional vertex therefore uses at least two of the triangle's three free incidences. At most one is possible, proving $|B| \le 4$.

Every component outside B is a path, unless B is empty and the whole graph is a cycle. Its ends attach to free incidences of vertices in B or terminate as leaves. A *tail* means a positive-length pendant path. A *theta graph* has two endpoints joined by three internally disjoint paths of the specified lengths. The exhaustive list is:

1. $|B| = 0$: a path or a cycle;
2. $|B| = 1$: a three-leaf tree, or a cycle with one tail;
3. $|B| = 2$ with the vertices adjacent: a triangle with tails at two vertices, or a theta graph of lengths $1, 2, \ell$, with $\ell \ge 2$;
4. $|B| = 2$ with the vertices nonadjacent: a four-cycle with tails at opposite vertices, or a theta graph of lengths $2, 2, \ell$, with $\ell \ge 2$;
5. $|B| = 3$: a triangle with three tails, or a theta graph of lengths $1, 2, \ell$, with a tail at the internal vertex of the length-two path;
6. $|B| = 4$: K_4, a graph obtained by subdividing exactly one edge of K_4, or a diamond $K_4 - e$ with tails at its two degree-two vertices.

For the last case, the fourth vertex meets two or three vertices of the triangle. Three gives K_4. Two gives a diamond; its remaining two incidences either join by a path or terminate in separate tails. For $|B| = 3$, the triangle's three free incidences are either all tails, or two join by a path and the third is a tail. For $|B| = 2$, the one or two required common neighbors give exactly the listed alternatives; three common neighbors is the theta graph with lengths $2, 2, 2$. This proves the classification directly from the full-star criterion.

11.2 Paths, cycles, trees, and a cycle with one tail

If $L = 2$, then $\Delta \le 2$, so G is a path or cycle. A cycle is vertex-transitive, and all its radial circles have the same order. A path has one or two centers; the radial-circle order is two at a

unique center, or one at each of two centers. Thus $q = 1$. A tree has at most two centers, the middle vertex or vertices of a diametral path, so a three-leaf tree has $q \le 2$.

Now attach a tail of length $t \ge 1$ at a vertex c of a cycle of length m, and put $s = \lfloor m/2 \rfloor$. At a cycle vertex x,

$$\mathrm{ecc}(x) = \max\{s, d_{C_m}(x, c) + t\}.$$

At depth j on the tail, eccentricity is $\max\{t - j, s + j\}$. If $t > s$, the center lies on the tail, and minimizing the maximum of these two opposite affine functions gives at most two centers. If $t \le s$, the radius is s, no positive-depth tail vertex is central, and the central cycle vertices satisfy $d(x, c) \le s - t$. Each cycle vertex has ϵ farthest cycle vertices, where $\epsilon = 1$ for even m and $\epsilon = 2$ for odd m. At a central cycle vertex the tail endpoint is the only possible additional farthest vertex, occurring exactly when $d(x, c) + t = s$. Thus all radial orders are in $\{\epsilon, \epsilon + 1\}$, and $q \le 2$.

11.3 A triangle with tails

Let $A \ge B \ge C \ge 0$ be tail lengths at the triangle vertices a, b, c; zero means no tail, and at least two lengths are positive. The eccentricity of a is $\max\{A, B + 1\}$, and those of b, c are $A + 1$. If $A = B$, all three triangle vertices are central with radius $A + 1$, and no positive-depth tail vertex is central. A radial circle consists of the endpoints of maximum-length tails at the other triangle vertices, so has order one or two. If $A > B$, the eccentricity of a is at most A, whereas every vertex outside its tail has eccentricity at least $A + 1$. The center lies on that tail, including a, and at depth j eccentricity is $\max\{A - j, B + 1 + j\}$. There are at most two minimizing vertices. Therefore $q \le 2$ in either case.

11.4 A four-cycle or diamond with opposite tails

Let the nonadjacent attachment vertices c, d have tail lengths $A \ge B \ge 1$, and let a, b be the other two vertices of the four-cycle. The edge ab is optional, producing the diamond when present. In both cases,

$$\mathrm{ecc}(c) = \max\{A, B + 2\}, \quad \mathrm{ecc}(d) = A + 2, \quad \mathrm{ecc}(a) = \mathrm{ecc}(b) = A + 1.$$

If $A = B$, the center is exactly $\{a, b\}$. If $A = B + 1$, it is exactly $\{a, b, c\}$, with radius $A + 1 \ge 3$. The vertices a, b have the endpoint of the A-tail as their unique farthest vertex; c has the endpoint of the B-tail as its unique farthest vertex. All radial orders are one. If $A \ge B + 2$, the center lies on the c-tail, including c, where the eccentricity at depth j is $\max\{A - j, B + 2 + j\}$. There are at most two centers. The exclusion of other centers follows because a positive-depth tail vertex reaches the opposite tail endpoint through a core path of length two. Thus $q \le 2$ throughout.

11.5 A theta graph of lengths 1, 2, and an arbitrary third length

Let a, b be the theta endpoints, let $a - c - b$ be the length-two path, and let the third path have length $\ell \ge 2$. Allow a tail of length $t \ge 0$ at c. The edge ab and the long path form a cycle Q of length $m = \ell + 1$, and distances within Q are unchanged by c and the tail. Put $s = \lfloor m/2 \rfloor$ and

$$h(x) = d(x, c) = 1 + \min\{d_Q(x, a), d_Q(x, b)\} \quad (x \in V(Q)).$$

Then

$$\mathrm{ecc}(x) = \max\{s, h(x) + t\}. \tag{54}$$

At depth j on the tail, including $j = 0$ at c, eccentricity is

$$\max\{t - j, H + j\}, \qquad H = \max_{x \in V(Q)} h(x). \tag{55}$$

If $m = 2s$, each value $1, \dots, s$ of h occurs twice and $H = s$. For $t < s$, the radius is s. The centers are c and the cycle vertices with $h(x) \le s - t$. Each such cycle vertex has one cycle antipode and gains the tail endpoint exactly when $h(x) + t = s$; the vertex c has its two farthest cycle vertices. All radial orders are one or two. For $t = s$, the sole center is c, with radial order three. For $t > s$, every cycle vertex has eccentricity at least $t + 1$, whereas c has eccentricity t. The center is on the tail, and (55) gives at most two centers.

If $m = 2s + 1$, each value $1, \dots, s$ occurs twice and $s + 1$ occurs once, so $H = s + 1$. For $t < s$, the radius is s, and the centers are exactly the cycle vertices with $h(x) \le s - t$. Each has two cycle antipodes and gains the tail endpoint precisely when $h(x) + t = s$. All radial orders are two or three. For $t = s$, the center is $\{a, b, c\}$, with radius $s + 1$: a, b have only the tail endpoint at that distance, and c has only the unique cycle vertex with $h = s + 1$. Every radial order is one. For $t \ge s + 1$, every cycle vertex has eccentricity at least $t + 1$ and c has eccentricity t; again the center consists of at most two tail vertices. These cases include $t = 0$, and always give $q \le 2$.

11.6 The remaining theta graphs and subdivisions of one edge of a clique

For a theta graph of lengths $2, 2, \ell$, $\ell \ge 2$, form a cycle Q from the long path and one length-two path. Its length is $m = \ell + 2 \ge 4$; let c be the internal vertex of its length-two path. The other length-two path adds a vertex d with the same neighbors as c. Optionally add cd. With that edge, the graph is obtained by subdividing one edge of K_4; without it, it is the given theta graph.

Distances within Q are unchanged. For $x \in V(Q) \setminus \{c\}$, $d(x, d) = d_Q(x, c)$ whether or not cd is present. The distance between c, d is two without that edge and one with it. Put $s = \lfloor m/2 \rfloor \ge 2$. Every vertex has eccentricity s. Each vertex of Q has ϵ cycle antipodes, with $\epsilon = 1$ for even m and $\epsilon = 2$ for odd m. A vertex other than c gains d as a farthest vertex only when its distance to c is s. The vertices c, d likewise have either ϵ or $\epsilon + 1$ farthest vertices, according as their mutual distance equals s. Hence all radial orders lie in $\{\epsilon, \epsilon + 1\}$. The unsubdivided K_4 has every radial-circle order equal to three.

Conclusion of the proof of Theorem 11.1. The classification above is exhaustive for $L \le 3$. The case $L = 2$ has $q = 1$, and all families with $L = 3$ have $q \le 2$. Therefore $L \ge 1 + q$ throughout the claimed class. □

This proves an arbitrary-order subclass theorem. It does not establish 155 for $L \ge 4$ or the separate, unproved radial-order lemma discussed in the source notes.

12 Degree frequencies and quartile sufficient conditions for 389a

12.1 The source convention and the scope of the reduction

The original statement 389a asks whether every connected graph of order $n > 2$ satisfies

$$\gamma_2(G) \le \mu(q + a) - 1, \tag{56}$$

where a is the number of vertices of degree *at most* two, μ is the size of the mode-min degree class, and q is the first-quartile degree. All modal classes have the same size, so μ is simply the largest frequency of any degree. We use μ here to distinguish this frequency from the induced-forest number $f(G)$ used in the preceding sections. The condition on a is different from the exactly-degree-two set in the formulation of 354.

The archived definition orders the degrees as $d_1 \le \cdots \le d_n$ and specifies the entry at rank $n/4$, without a rounding convention when $4 \nmid n$ [2]. Accordingly, throughout this section, for $n \ge 4$ we

assume that q is a positive integer satisfying

$$\left|\{v : d(v) \le q\}\right| \ge \lfloor n/4 \rfloor. \tag{57}$$

This includes both the lower-rank and upper-rank choices, and the literal source definition when $4 \mid n$. No interpolation convention is assumed. For $n = 3$, the connected graphs are P_3 and K_3; each has $\gamma_2 = 2$ and $a = 3$, with $\mu = 2$ and 3, respectively, so (56) holds for every positive integer q.

Connectedness puts all degrees in $\{1, \dots, n-1\}$, and hence

$$\mu \ge 2, \qquad a \le 2\mu. \tag{58}$$

If $q \ge 3$, counting the degree classes in (57) gives

$$\lfloor n/4 \rfloor \le a + \mu(q-2), \qquad n \le 4a + 4\mu(q-2) + 3. \tag{59}$$

12.2 Two upper bounds for two-domination

Lemma 12.1 (A cut bound)**.** *For any graph with n vertices, of which a have degree at most two,*

$$\gamma_2(G) \le \left\lfloor \frac{n+a}{2} \right\rfloor.$$

Proof. Take a cut locally maximal under moving a single vertex to the other side. Every vertex of degree at least three has at least two neighbors across the cut, since otherwise moving it would increase the cut. One side contains at least $\lceil (n-a)/2 \rceil$ vertices of degree at least three. Omit exactly those vertices and select all remaining vertices. Each omitted vertex has at least two selected neighbors across the cut, so the selected set is 2-dominating and has the required size. □

Lemma 12.2 (A frequency-sensitive alteration bound)**.** *Let G have no isolated vertices, let a count its vertices of degree at most two, and let μ be its largest degree frequency. For $0 < p < 1$,*

$$\gamma_2(G) \le pn + (1-p)a + \mu(1-p)^3\left(1 + \frac{2}{p}\right). \tag{60}$$

In particular,

$$\gamma_2(G) \le \frac{n}{6} + \frac{5a}{6} + \frac{1625\mu}{216}. \tag{61}$$

Proof. Select each vertex independently with probability p, producing a set S. Add every unselected vertex having fewer than two neighbors in S. The resulting set is 2-dominating. Put $z = 1 - p$. A vertex of degree d is added in the second step with probability

$$z^{d+1} + dpz^d.$$

For $d = 1$ this equals z, and for $d = 2$ it equals $z^2(1+p) \le z$. At most μ vertices have any given degree $d \ge 3$. The expected size of the final set is therefore at most

$$pn + za + \mu \sum_{d=3}^{\infty} (z^{d+1} + dpz^d).$$

Using the geometric series and its derivative gives

$$\sum_{d=3}^{\infty} (z^{d+1} + dpz^d) = 3z^3 + \frac{2z^4}{p} = z^3\left(1 + \frac{2}{p}\right).$$

Some outcome has size at most its expectation, proving (60). Substituting $p = 1/6$ proves (61). □

These are elementary applications of the usual cut and alteration methods; the methods themselves are not claimed as new.

12.3 Explicit sufficient conditions

Theorem 12.3. *Under* (57)*, suppose* $q \geq 3$*. Inequality* (56) *holds if either of the following is true:*

$$\mu(a + 4 - q) > \frac{5a + 3}{2}, \tag{62}$$

$$\mu\left(\frac{q}{3} - \frac{1337}{216}\right) + a\left(\mu - \frac{3}{2}\right) > \frac{1}{2}. \tag{63}$$

Proof. Lemma 12.1 and (59) imply

$$\gamma_2(G) \leq \frac{5a + 4\mu(q - 2) + 3}{2}.$$

Condition (62) says precisely that this upper bound is strictly less than the integer $\mu(q + a)$. Integrality then yields (56). Similarly, (61) and (59) give

$$\gamma_2(G) \leq \frac{2\mu q}{3} + \frac{3a}{2} + \frac{1337\mu}{216} + \frac{1}{2}. \tag{64}$$

Condition (63) makes this bound strictly smaller than $\mu(q + a)$, giving the same conclusion. □

Corollary 12.4. *Under the stated quartile convention,* 389a *holds whenever* $q \geq 20$*.*

Proof. The left side of (63) minus $1/2$, using $\mu \geq 2$ and $a \geq 0$, is at least

$$2\left(\frac{20}{3} - \frac{1337}{216}\right) - \frac{1}{2} = \frac{49}{108} > 0.$$

□

Corollary 12.5. *Inequality* (56) *holds for* $q = 2$*, and for* $q = 3$ *with* $\mu \geq 3$*. It also holds for* $q = 3$*,* $\mu = 2$*,* $a = 0$*.*

Proof. If $q = 2$, then (57) gives $n \leq 4a + 3$ and $a \geq 1$. The cut bound becomes $\gamma_2 \leq \lfloor (5a + 3)/2 \rfloor$. For $a = 1, 2, 3, 4$, these upper bounds are $4, 6, 9, 11$, while $\mu(a + 2) - 1 \geq 2(a + 2) - 1$ gives $5, 7, 9, 11$. For $a \geq 5$, use $\mu \geq a/2$ and

$$\frac{a(a + 2)}{2} - 1 - \frac{5a + 3}{2} = \frac{a^2 - 3a - 5}{2} > 0.$$

For $q = 3$, condition (62) reduces to $\mu(a + 1) > (5a + 3)/2$, which holds when $\mu \geq 3$, and also when $\mu = 2$, $a = 0$. □

Corollary 12.6. *Under the stated quartile convention,* 389a *holds whenever* $a \geq 7$*.*

Proof. First let $q \geq 3$. Since $a \leq 2\mu$, we have $\mu \geq 4$. Define

$$U = a + 4 - \frac{5a + 3}{2\mu}, \qquad L = \frac{1337}{72} - 3a + \frac{9a + 3}{2\mu}.$$

Conditions (62) and (63) are respectively $q < U$ and $q > L$. Their intervals overlap, since

$$U - L = 4a - \frac{1049}{72} - \frac{7a + 3}{\mu} \geq 28 - \frac{1049}{72} - 13 = \frac{31}{72} > 0.$$

For the inequality, the expression increases with μ, and for $\mu \geq 4$ its coefficient of a is $4 - 7/\mu > 0$. Thus every real q satisfies at least one sufficient condition. The case $q = 2$ is Corollary 12.5.

For $q = 1$, let ℓ be the number of leaves. Then $n \leq 4\ell + 3$, $\ell \geq 1$, and $\mu \geq \ell$. A connected graph of order greater than two has a vertex of degree at least two. Omitting that one vertex gives a 2-dominating set, so $\gamma_2 \leq n - 1 \leq 4\ell + 2$. Since $a \geq 7$,

$$\mu(a + 1) - 1 \geq \ell(a + 1) - 1 \geq 4\ell + 2,$$

as required. □

12.4 Residual parameter profiles

Any counterexample under (57) must have $a \leq 6$ and $q \leq 19$, avoid the cases in Corollary 12.5, and, when $q \geq 3$, satisfy both

$$\begin{aligned} \mu(a + 4 - q) &\leq \frac{5a + 3}{2}, \\ \mu\left(a + \frac{q}{3} - \frac{1337}{216}\right) &\leq \frac{3a + 1}{2}. \end{aligned} \tag{65}$$

If either coefficient of μ is positive, these inequalities give a finite bound on μ, after which (59) bounds n. For $q \geq 3$ both coefficients can be nonpositive only in the following ranges:

a	possibly unbounded degree frequency
0	$4 \leq q \leq 18$
1	$5 \leq q \leq 15$
2	$6 \leq q \leq 12$
3	$7 \leq q \leq 9$.

These are necessary parameter restrictions, not a resolution of 389a. They do not assert that all graphs realizing the remaining profiles have been tested. The accompanying exact-rational checker verifies the displayed algebra on its declared parameter grid; it does not enumerate graphs. The case $q = 1$ outside the proved $a \geq 7$ range remains subject to its separate rank condition and is not covered by (65).

13 Local independence and induced-path density: Conjecture 133

For this part, let $p(G)$ be the maximum *vertex order* of an induced path, $r(G) = \mathrm{rad}(G)$, and

$$\lambda_G(v) = \alpha(G[N_G(v)]), \qquad \Lambda(G) = \sum_v \lambda_G(v), \qquad \ell(G) = \Lambda(G)/|V(G)|.$$

The average ℓ is never rounded. A graph is called C_4-free here only when it has no four-cycle as an ordinary subgraph, including one with chords; P_j-free means no induced path on j vertices. The archived formulation of Conjecture 133 is

$$p(G) \geq r(G) + \ell(G)^{\chi_{C_4}(G)}, \tag{66}$$

where $\chi_{C_4} = 1$ exactly in the C_4-free case [2, 3]. If a four-cycle occurs, a geodesic gives $p \geq r + 1$; the substantive branch is $p \geq r + \ell$. In a C_4-free graph every neighborhood is a matching plus isolates. Writing $t(v)$ for the number of triangles through v, and t for the number of triangles, therefore gives

$$\lambda(v) = d(v) - t(v), \quad \Lambda = 2m - 3t, \quad \Lambda(G) - \Lambda(G - v) = 2\lambda(v) - t(v). \tag{67}$$

The next results retain the sharper deficit and equality information from the earlier rounds before proving the stronger global P_7-free bound. A clique cutset is a clique whose removal disconnects a connected graph, and an atom is a connected graph with no clique cutset.

13.1 An elementary P_6-free bound

Theorem 13.1. *If G is connected, C_4-free, and P_6-free, then $\ell(G) \le 3$, with equality exactly for the Petersen graph P. More precisely, for $D(G) = 3n - 2m + 3t$ one has*

$$D(K_1) = 3, \quad D(P) = 0, \quad D(P - v) = 3,$$

and $D(G) \ge 4$ for every other such graph. If G contains a triangle, then $D(G) \ge 6$.

The general atom structure studied by Brandstädt–Hoàng [32], and restated in [33, Theorem 9], motivated the following specialization. Under our stronger four-cycle exclusion the needed contact and domination arguments are elementary; those external structure theorems are not dependencies of Theorem 13.1.

Lemma 13.2. *A C_4-free, P_6-free atom is one of the graphs in Table 2.*

For the table, label a five-cycle $c_0, \dots, c_4$ modulo five. Given an index set I, add a_i adjacent to c_i for each $i \in I$, and join a_i, a_j exactly when $i - j = \pm 2 \pmod 5$. The nonempty permitted sets I induce an edge, P_3, P_4, or C_5 in the pentagram on the indices. Write the resulting graph as $Q_{|I|}$. Thus $Q_4 = P - v$ and $Q_5 = P$. The graph T consists of the five-cycle, a vertex a adjacent to c_3, and a vertex b adjacent to c_0, c_1, a.

Table 2: The atoms needed for the P_6-free deficit bound.

Graph	n	m	t	$D = 3n - 2m + 3t$
K_1	1	0	0	3
K_2	2	1	0	4
K_3	3	3	1	6
C_6	6	6	0	6
C_5	5	5	0	5
Q_2	7	8	0	5
Q_3	8	10	0	4
$Q_4 = P - v$	9	12	0	3
$Q_5 = P$	10	15	0	0
T	7	9	1	6

Proof of Lemma 13.2. Suppose first that the atom H contains an induced five-cycle as above. An exterior vertex cannot contact two cycle vertices at cycle-distance two, since that creates a C_4. Thus its nonempty contact set is either $\{c_i\}$ or $\{c_i, c_{i+1}\}$. The cycle is dominating. Otherwise choose adjacent exterior vertices y, x with y having no cycle contact and x having a contact. Append to y, x four consecutive cycle vertices, starting at a contact of x and moving away from its other contact if present. This is an induced P_6.

Let A_i and B_i be the singleton-contact and adjacent-pair-contact classes, respectively. All indices in this paragraph are modulo five. The following restrictions account for every possible adjacency.

1. $|B_i| \le 1$, since two members would have two common neighbors. Each A_i induces a matching, being contained in $N(c_i)$.

2. A_i is anticomplete to $A_{i\pm1}$, since an edge would form a C_4 with the corresponding cycle edge. It is complete to $A_{i\pm2}$: a nonedge xy, with $x \in A_i, y \in A_{i+2}$, gives the induced path $x, c_i, c_{i-1}, c_{i-2}, c_{i+2}, y$.
3. B_i and B_{i+1} cannot both be present. Their vertices, if adjacent, form a C_4 with c_i, c_{i+1}; if nonadjacent they are the ends of the induced path $b_i, c_i, c_{i-1}, c_{i-2}, c_{i+2}, b_{i+1}$. Edges between B_i and $B_{i\pm2}$ are also forbidden by a C_4 using the cycle edge between their closest contacts. Thus there are no edges between distinct B-classes.
4. A vertex of A_i can contact a B-class only at B_{i+2}; every other such edge creates a C_4 using a cycle edge or two-edge segment. At most one member of A_i can contact the vertex in B_{i+2}, since otherwise that vertex and c_i have two common neighbors.

Every $x \in A_i$ must have a neighbor outside $A_i \cup \{c_i\}$. Otherwise $N(x)$ consists of c_i and possibly its matching partner in A_i, a clique whose deletion separates x from the remaining cycle. If $|A_i| \geq 2$, restriction 2 and C_4-freeness force $A_{i-2} = A_{i+2} = \varnothing$. The only possible exterior neighbor is then the unique vertex of B_{i+2}, and at most one member of A_i can contact it. This contradicts the required exterior neighbor for every member. Hence $|A_i| \leq 1$.

If B_i is present, its vertex must contact A_{i-2}; otherwise its neighborhood is the clique $\{c_i, c_{i+1}\}$, a clique cutset. This edge excludes A_i, A_{i+1} by C_4-freeness. Applying the same observation to other B-classes excludes $B_{i\pm2}$, while restriction 3 excludes $B_{i\pm1}$. Thus B_i is the only B-class present. The remaining possible classes A_{i-1}, A_{i+2} must either both be absent or both be present, because each needs an exterior neighbor. If both are present,

$$b_i, c_i, c_{i-1}, a_{i-1}, a_{i+2}, c_{i+2}$$

is an induced P_6. Hence both are absent and $H = T$.

If no B-class is present, every present A_i must have a neighbor in $A_{i-2} \cup A_{i+2}$. The selected indices therefore induce a subgraph of the pentagram without isolated vertices. The possibilities are the empty set, an edge, P_3, P_4, and C_5, giving the stated five triangle-free graphs.

Finally suppose H contains no induced C_5. If H is chordal, it is a clique: a noncomplete connected chordal graph has a clique cutset. Its order is at most three. Otherwise an induced cycle must be C_6, since C_4, C_5 are excluded and any longer induced cycle contains a P_6. If there is an exterior vertex, connectedness supplies one with a cycle contact. It has at most two contacts, which if two are adjacent or opposite. One contact or two adjacent contacts produce an induced P_6 by appending five successive cycle vertices in a suitable direction. Opposite contacts produce an induced C_5. Thus $H = C_6$. The counts in the table are immediate. □

Lemma 13.3. *Neither P nor $P - v$ admits a one-vertex extension with a nonempty clique as its contact set while remaining P_6-free.*

Proof. Represent $P - v$ by the six-cycle $0, 1, \ldots, 5$ and three vertices a_0, a_1, a_2, where a_i contacts $i, i + 3$. Adding a vertex z adjacent to the three a_i gives P. Rotation of the six-cycle covers all contact types in the following table. Each sequence is an induced P_5 beginning at one contact and avoiding the other when there are two.

Contact set, up to rotation	Induced P_5
$\{0\}$	$0, 1, a_1, 4, 3$
$\{a_0\}$	$a_0, 0, 1, a_1, 4$
$\{z\}$, in P only	$z, a_0, 0, 1, 2$
$\{0, 1\}$	$0, a_0, 3, 2, a_2$
$\{0, a_0\}$	$0, 1, a_1, 4, 3$
$\{z, a_0\}$, in P only	$z, a_1, 1, 0, 5$

The graphs are triangle-free, so these are all nonempty clique-contact types. Prepending the added vertex gives an induced P_6. □

Proof of Theorem 13.1. Prove the deficit assertions simultaneously by induction on n. Atoms are covered by Table 2. Otherwise let K be a clique cutset, partition the components of $G - K$ into two nonempty unions X_1, X_2, and put $G_i = G[K \cup X_i]$. Both sides are connected proper induced subgraphs with at least two vertices. Neither can be P or $P - v$: an outside vertex adjacent to the separator would contradict Lemma 13.3. Thus $D(G_i) \geq 4$ by induction. Inclusion–exclusion for vertices, edges, and triangles gives

$$D(G) = D(G_1) + D(G_2) - D(K). \tag{68}$$

For $|K| = 1, 2$, the subtracted values are $3, 4$, giving $D(G) \geq 5, 4$. For $|K| = 3$, both sides have a triangle, so their deficits are at least six, and subtraction of $D(K_3) = 6$ leaves at least six. If G has a triangle and $|K| \leq 2$, one side has deficit at least six and the other at least four; this gives at least seven or six. The induction is complete. Equation (67) gives $D = n(3 - \ell)$.

For sharpness, realize P as $KG(5, 2)$: its vertices are the two-element subsets of a five-element set, adjacent when disjoint. It is cubic and triangle-free; adjacent vertices have no common neighbor, and nonadjacent vertices have exactly one. In particular it is C_4-free. In a hypothetical induced P_6, its six pairs of vertices at path-distance at least three would require distinct common neighbors outside the path. An exterior vertex contacts at most two path vertices, whose indices must differ by at least three, so it can witness at most one of those pairs. Only four exterior vertices are available. This contradiction proves P_6-freeness, and $\ell(P) = 3$. □

13.2 A P_7-free consequence of a published structure theorem

Theorem 13.4. *If G is connected, C_4-free, and P_7-free, then $\ell(G) < 4$. More precisely, $F(G) = 4n - \Lambda(G)$ is at least eight when G is noncomplete and at least nine when G contains a triangle. The complete cases K_1, K_2, K_3 have $F = 4, 6, 9$, respectively.*

The required prior result is Chen–Wu–Xu [34, Theorem 1.3]: a connected graph with no induced P_7, C_4, or gem, and no clique cutset, is a clique blow-up of P or has a *bisimplicial* vertex, whose neighborhood is covered by two cliques. The journal version appeared in 2024 [35]. A gem contains a four-cycle as a non-induced subgraph, so this theorem applies under our hypotheses.

Proof. Induct simultaneously on the two deficit assertions. Complete graphs are immediate. First let G be a noncomplete atom. A clique blow-up of P can be C_4-free only when every bag is a singleton: two vertices in a bag and one vertex from each of two neighboring bags would form a C_4. Thus this case is P, with $F(P) = 10$ and no triangle.

Otherwise take a bisimplicial vertex v, so $\lambda(v) \leq 2$. Using (67), with $q = t(v)$, gives

$$F(G) - F(G - v) = 4 - 2d(v) + 3q = 4 - 2\lambda(v) + q \geq 0. \tag{69}$$

An atom has no cutvertex, so $H = G - v$ is connected. A noncomplete atom has at least four vertices; hence H has at least three and $F(H) \geq 8$ by induction, including its possible complete case K_3. This proves $F(G) \geq 8$. If a triangle survives in H, then $F(H) \geq 9$. If G has a triangle but H does not, then $q \geq 1$, making the increment in (69) at least one. Either way $F(G) \geq 9$.

For a clique cutset decomposition as before,

$$F(G) = F(G_1) + F(G_2) - F(K).$$

If $|K| = 1$, each side has at least two vertices and deficit at least six, so subtraction of four leaves at least eight. If $|K| = 2$, each side has at least three vertices and deficit at least eight, so subtraction of six leaves at least ten. If $|K| = 3$, both sides have a triangle and deficit at least nine, leaving at least nine after subtraction. When a triangle occurs and $|K| \le 2$, at least one side has deficit at least nine, giving at least eleven in either case. This proves the induction and $\Lambda(G) < 4n$. □

13.3 The stronger global bound without an induced seven-vertex path

Theorem 13.5. *Every C_4-subgraph-free, P_7-free graph satisfies $\Lambda(G) \le 3n$.*

It suffices to treat connected graphs, since the two sides are additive over components. The proof proceeds through dominating and nondominating five-cycles, the branch with neither a five- nor a seven-cycle, and finally seven-cycle contacts.

We begin with two five-cycle statements, whose complete proofs are included below.

Theorem 13.6. *Every connected C_4-subgraph-free, P_7-free graph containing an induced C_5 and no induced C_7 satisfies $\Lambda(G) \le 3n$.*

Theorem 13.7. *A C_4-subgraph-free, P_7-free graph with a dominating induced C_5 satisfies $\Lambda(G) \le 3n$.*

The stronger deficit estimate in the dominating case will also be needed. The first five-cycle argument is independent of any structural classification theorem.

13.4 A dominating five-cycle: contact classes

Fix a dominating induced cycle $c_0 \dots c_4 c_0$, with subscripts modulo five. Every exterior contact set is a singleton or an adjacent pair: a distance-two pair gives a C_4, and any three cycle vertices contain such a pair. Let A_i contain the vertices with contact set $\{c_i\}$, and let B_i contain those with contact set $\{c_i, c_{i+1}\}$. Four-cycle exclusion gives the following properties.

(i) Each $G[A_i]$ is a matching plus isolated vertices. A three-vertex path in $N(c_i)$ would complete a C_4.
(ii) $|B_i| \le 1$, and there are no edges between different B classes. Two vertices with the same pair of contacts give a C_4; an edge between different pair-contact classes completes a C_4 with suitable cycle contacts.
(iii) For $i \ne j$, an A_i–A_j edge is possible only if $j = i \pm 2$. Edges between any fixed two classes form a matching, since two neighbors in A_j give a C_4 with c_j.
(iv) An A_i–B_j edge is possible only if $i = j + 3$. Each B_j vertex has at most one such neighbor, by the same common-neighbor test.

Thus every B vertex has degree two or three and belongs to exactly one triangle, on its two cycle contacts.

In any C_4-subgraph-free graph each neighborhood is a matching plus isolated vertices. Consequently

$$\Lambda(G) = 2m - 3t. \tag{70}$$

Deleting a B vertex decreases Λ by either one or three. If $G_0 = G - B$ satisfies $\Lambda(G_0) \le 3|V(G_0)|$, reinstating the B vertices gives $\Lambda(G) \le \Lambda(G_0) + 3|B| \le 3n$. We may therefore assume that all exterior vertices are in $A = \bigcup_i A_i$.

13.5 Diagonal components

Let $a = |A|$, let q count the edges within the individual A_i, and let D be the graph on A containing only edges between different contact classes. Write $d = |E(D)|$. The maximum degree of D is two. Along a nonbacktracking path of D, contact indices progress consistently by $+2$ or by -2 modulo five: reversing direction would give a vertex two neighbors in one contact class, and hence a C_4.

Lemma 13.8. *Every component of D is P_1, P_2, P_3, P_4, or C_5.*

Proof. Take five consecutive vertices $x_0, \dots, x_4$ of a diagonal path. Their contacts, in order, are $c_i, c_{i+2}, c_{i+4}, c_{i+1}, c_{i+3}$. If x_0x_4 is absent, then

$$c_i, x_0, x_1, c_{i+2}, c_{i+3}, x_4, x_3$$

is an induced P_7. No within-class matching edge can be a chord, since the five exterior vertices have distinct contact classes. The only possible additional diagonal chord is x_0x_4. If that edge is present, the vertices form an entire C_5 component, since each already has diagonal degree two. Every other component has at most four vertices, which proves the assertion. □

For a component Q of D, put $e(Q) = 2|E(Q)| - |V(Q)|$. Its values on P_1, P_2, P_3, P_4, C_5 are respectively $-1, 0, 1, 2, 5$. The only triangles of G arise from the q within-class matching edges and their cycle contacts. Hence (70) gives

$$3n - \Lambda(G) = 5 + a + q - 2d = 5 + q - \sum_{Q} e(Q). \tag{71}$$

It suffices to bound the sum of the positive excesses by $5 + q$.

13.6 Rooted paths force matching edges

A component is *positive* if it is P_3, P_4, or C_5. For such a component Q, let $T(Q)$ be the set of contact indices of endpoints of its induced three-vertex paths. Then

$$|T(P_3)| = 2, \qquad |T(P_4)| = 4, \qquad T(C_5) = \mathbb{Z}/5\mathbb{Z}.$$

Lemma 13.9. *If two distinct positive components have intersecting T sets, some within-class matching edge joins them. Moreover a C_5 component requires at least two such edges to each P_3 component, four to each P_4 component, and five to each other C_5 component.*

Proof. For the first assertion, choose a three-vertex path in each component ending in the same class A_i, and join the paths through c_i. The seven vertices are distinct. Each three-vertex path uses three distinct contact classes, so c_i has no other neighbor on either path. No diagonal edge joins the two components. Thus a matching edge must join the two paths, or their concatenation is an induced P_7.

For the stronger assertion, let $x \in A_i$ be an endpoint of a three-vertex path in the other component. The C_5 component contains one vertex $y \in A_i$, with rooted three-vertex paths in both circular orientations. Choose the orientation opposite to that of the path rooted at x. The two sets of non-root contact indices are disjoint. Therefore the only possible matching chord in the joined seven-vertex path is xy, and that edge must be present. A P_3 has two possible root vertices, a P_4 has four, and another C_5 has five. This forces the stated numbers of distinct edges. Edges required by distinct component pairs are also distinct. □

13.7 Counting the excess

Let h, k, s be the numbers of C_5, P_4, and P_3 components of D. If $h \geq 1$, Lemma 13.9 gives

$$q \geq 5\binom{h}{2} + h(4k + 2s).$$

Consequently

$$5h + 2k + s - q \leq 5h - 5\binom{h}{2} + (2 - 4h)k + (1 - 2h)s \leq 5.$$

Here $h - \binom{h}{2} \leq 1$ for every positive integer h, and the remaining coefficients are nonpositive.

Now suppose $h = 0$. A four-element contact set intersects every other positive contact set, which has size two or four. Thus $q \geq \binom{k}{2} + ks$. If $k \geq 1$, then

$$2k + s - q \leq 2k - \binom{k}{2} + (1 - k)s \leq 3.$$

If $k = 0$, form the intersection graph of the s two-element sets from the P_3 components. Distinct connected components of this intersection graph use disjoint collections of contact indices. Each uses at least two of the five indices, so there are at most two components. The intersection graph has at least $s - 2$ edges, each requiring a distinct matching edge by Lemma 13.9. Therefore $s - q \leq 2$; the case $s = 0$ is immediate.

In all cases, the positive excess sum is at most $5 + q$. The P_1 and P_2 components have nonpositive excess, so (71) proves $\Lambda(G) \leq 3n$. Reinstating the B vertices proves Theorem 13.7.

13.8 The nondominating layers

We prove Theorem 13.6. Fix an induced cycle $c_0 \dots c_4 c_0$, and use the contact classes A_i, B_i defined above, now without assuming that this cycle dominates. Let R be the set of vertices with no cycle contact. Every first-layer vertex x starts an induced five-vertex path using x and four consecutive cycle vertices; for adjacent-pair contacts, omit the unused contact.

Every component Q of $G[R]$ is complete to each first-layer vertex that touches it. Otherwise an edge ab in Q with xa present and xb absent gives an induced P_7, starting b, a, x and continuing along the four cycle vertices. The neighborhood of x is a matching plus isolated vertices, so connectedness forces Q to be K_1 or K_2. Every component touches the first layer because G is connected. In particular there are no vertices at distance at least three from the cycle.

A K_2 component has exactly one first-layer neighbor: two would give its two vertices two common neighbors and hence a C_4. It is a pendant triangle. Removing it decreases Λ by three and n by two, so it is safe for proving $\Lambda \leq 3n$.

Lemma 13.10. *Every singleton component of $G[R]$ has degree at most two. Its possible pairs of neighbors have types*

$$(A_i, A_{i\pm2}), \qquad (A_i, B_{i+2}), \qquad (B_i, B_{i\pm2}).$$

In the first pair the neighbors are adjacent; in the last pair they are nonadjacent.

Proof. Let z be the singleton. Two neighbors x, y cannot share a cycle contact, or they give a C_4 with z. For $x \in A_0, y \in A_1$, the sequence

$$c_2, c_3, c_4, c_0, x, z, y$$

is an induced P_7. For $x \in A_0, y \in A_2$ with xy absent, the cycle

$$x, c_0, c_4, c_3, c_2, y, z, x$$

is an induced C_7. The same cycle excludes an A_0, B_1 pair; in that case xy is already forbidden by the C_4 on x, c_0, c_1, y. Reflection excludes the A_0, B_3 pair, while B_0, B_4 share a contact with A_0. Two B types must have disjoint cycle edges, so their indices differ by ± 2. Their mutual edge is excluded by C_4-freeness. These exhaust all pairs up to rotation and reflection.

No three types are pairwise compatible. The compatibility graphs among A types alone and among B types alone are five-cycles. Moreover an A type is compatible with only one B type, and conversely. Repeating a type shares a cycle contact. Thus $d(z) \leq 2$. □

A singleton leaf has deletion cost $\Delta\Lambda = 2$ for one vertex. A degree-two singleton whose neighbors are adjacent has $\Delta\Lambda = 1$. Remove these safe vertices and all the K_2 components. Every remaining R vertex has degree two with nonadjacent neighbors and subdivides A_i–B_{i+2} or B_i–$B_{i\pm 2}$. There is at most one subdivision vertex for each fixed endpoint pair, by C_4 exclusion.

13.9 Removing isolated marked vertices

Call an A vertex *marked* if it is an endpoint of a remaining A–B subdivision. Each A_i vertex supports at most one such subdivision, since its only possible B endpoint is the unique B_{i+2}, and repeating the pair creates a C_4.

Let D again denote the diagonal-edge graph on A. Suppose a marked vertex x has degree zero in D, with subdivision vertex z. Its neighbors are its cycle contact, z, and possibly its within-class mate. It cannot have a direct B neighbor, since the only possible one is the nonadjacent endpoint of the subdivision. Deleting x, z therefore removes three edges and no triangle, or four edges and one triangle. Its cost $\Delta\Lambda$ is six or five for two vertices. Perform all such safe pair deletions. Every remaining marked vertex now has a diagonal neighbor. The fixed cycle and all remaining contact vertices stay connected.

13.10 Restrictions imposed by a subdivision

Let z subdivide B_j–x, where $x \in A_{j+3}$, and call B_j *active*. All subscripts in the remainder are modulo five.

At most two marks at one active vertex. Suppose x has a diagonal neighbor $y \in A_j$; the A_{j+1} case is symmetric. If another mark x' at B_j is not adjacent to x, with subdivision vertex z', then

$$c_{j+1}, c_j, y, x, c_{j+3}, x', z'$$

is an induced P_7. In particular, yx' is impossible because y already has its unique neighbor x in that contact class. Thus every other mark must be the within-class mate of x. There are at most two marks at B_j, and if there are two, they are matched.

Neighboring double-contact types are absent. If B_{j+1} existed, then

$$c_j, B_j, z, x, c_{j+3}, c_{j+2}, B_{j+1}$$

would be an induced P_7. Reflection excludes B_{j-1}. Consequently active indices form an independent set in a five-cycle, so there are at most two. One active B leaves at most three B vertices in total; two active B vertices leave exactly those two.

Other double-contact vertices have no direct single-contact neighbor. The only other possible types are B_{j+2}, B_{j+3}. If B_{j+2} has its permitted direct neighbor $y \in A_j$, then

$$c_{j+1}, c_j, y, B_{j+2}, c_{j+3}, x, z$$

is an induced P_7. The potential chord xy would give the C_4 $x, c_{j+3}, B_{j+2}, y, x$. Reflection handles B_{j+3}. Thus, with one active B, at most that vertex has a direct A neighbor. With two active vertices, neither can have one.

A subdivision between double-contact vertices leaves only two such vertices. Suppose z subdivides B_0–B_2. The other three types give these induced P_7 sequences:

$$\begin{aligned} B_1 &: \quad c_4, c_0, B_0, z, B_2, c_2, B_1, \\ B_3 &: \quad c_2, B_2, z, B_0, c_0, c_4, B_3, \\ B_4 &: \quad c_1, B_0, z, B_2, c_3, c_4, B_4. \end{aligned}$$

Rotation and reflection cover every possible pair. Hence a B–B subdivision forces exactly two B vertices, and there is at most one such subdivision in total.

13.11 The residual deficit

Let b be the number of B vertices, e the number of direct A–B edges, s the number of A–B subdivisions, and u the number of B–B subdivisions. Put

$$G_A = G[V(C) \cup A], \qquad D_A = 3|V(G_A)| - \Lambda(G_A).$$

Adding the B vertices creates $2b + e$ edges and b triangles. Each remaining subdivision vertex creates two edges and no triangle. Hence

$$3n - \Lambda(G) = D_A + 2b - 2e - s - u. \tag{72}$$

The component proof above gives $D_A \geq 2$ if D has no C_5 component: in that case the positive diagonal excess minus the matching count is at most three, and (71) starts with five.

The restrictions just established give

$$2b - 2e - s - u \geq -2. \tag{73}$$

Here is the complete case check. With no active vertex, $s = 0$; use $e \leq b$ if $u = 0$, or $b = 2, e \leq 2$ if $u = 1$, obtaining a bound of zero or -1. With one active vertex, $s \leq 2$ and $e \leq 1$. For $b = 1$ we have $u = 0$ and the bound is -2; for $b = 2$, $u \leq 1$ gives -1; for $b = 3$, $u = 0$ gives two. With two active vertices, $b = 2, e = 0, s \leq 4, u \leq 1$, giving -1. Thus (72) and (73) finish the case in which D has no five-cycle component.

13.12 The Petersen alternative

Suppose D has a C_5 component, labelled $u_i \in A_i$ with edges $u_i u_{i\pm2}$. Together with the original cycle these vertices induce the Petersen graph. No other component of D can have an edge: an edge xy with $x \in A_i, y \in A_{i+2}$ outside this component gives

$$c_i, x, y, c_{i+2}, c_{i+3}, u_{i+3}, u_{i+1}$$

as an induced P_7. The only possible within-class mates of x, y have indices $i, i+2$, neither of which occurs among the two displayed u vertices.

Every remaining mark therefore belongs to the displayed component, since marks with diagonal degree zero were removed. A subdivision z between B_j and u_{j+3} gives the induced P_7

$$c_j, B_j, z, u_{j+3}, c_{j+3}, c_{j+2}, u_{j+2}.$$

Thus there is no A–B subdivision. A subdivision z between B_0, B_2 gives the induced P_7

$$c_0, c_1, u_1, u_3, c_3, B_2, z.$$

There is no B–B subdivision either. The fixed cycle now dominates the reduced graph, and Theorem 13.7 applies. Reversing all safe deletions completes the proof of Theorem 13.6.

13.13 The branch with neither a five-cycle nor a seven-cycle

Assume that G has no induced C_5 or C_7. If it also has no induced C_6, it is chordal: a longer hole contains P_7, and a four-cycle is excluded. A chordal graph has a simplicial vertex; here its degree is at most two, since a clique of order four contains a four-cycle. Deleting a simplicial vertex from a connected graph leaves a connected graph, unless it was the only vertex. Reinstating a leaf adds two to Λ, while reinstating a degree-two simplicial vertex adds one, by (70). Induction gives $\Lambda \le 2(n-1)$.

We may therefore fix an induced cycle $C = c_0 \dots c_5 c_0$, with indices modulo six. Each exterior cycle-contact set is a singleton, an adjacent pair, or an opposite pair. Indeed, a distance-two pair gives a four-cycle, and every three cycle vertices contain such a pair. Denote the singleton classes by A_i, the adjacent-pair classes by B_i, and the opposite-pair classes by Z_i, with contact sets respectively

$$\{c_i\}, \qquad \{c_i, c_{i+1}\}, \qquad \{c_i, c_{i+3}\}.$$

Here $Z_i = Z_{i+3}$. Each B_i and Z_i has at most one vertex, by the common-neighbor test. We also use B_i, Z_i for those vertices when present. There are at most three Z vertices, and they are independent: an edge between two different opposite-contact types completes a four-cycle with adjacent cycle contacts.

Let R consist of vertices with no cycle contact. Every A or B vertex starts an induced P_6 using five consecutive cycle vertices. An edge from it to R would therefore produce an induced P_7. Thus only the Z vertices can have neighbors in R.

13.13.1 Removing first-layer vertices at nonnegative deficit cost

Every B vertex has degree two. To see this, four-cycle exclusion permits a B_0–A edge only to A_3 or A_4, and a B_0–B edge only to B_3. Such edges give an induced five-cycle; for example, with $x \in A_3$ the cycle is $B_0, x, c_3, c_2, c_1, B_0$, and for the second case it is $B_0, c_1, c_2, c_3, B_3, B_0$. All B–Z edges give a four-cycle. Edges to R were excluded above. Therefore each B is a degree-two simplicial vertex, belonging to one triangle, and deleting it decreases Λ by one. Remove all B vertices.

Edges among distinct A classes can join only opposite classes: adjacent classes give a four-cycle, and distance-two classes give an induced five-cycle. These opposite edges form a matching, since every vertex has at most one neighbor in a fixed other contact class. Within each A_i there is a matching plus isolated vertices.

An A_i vertex can meet only Z_i among the Z vertices; another type would complete a four-cycle. If $x \in A_i$ is adjacent to Z_i, it cannot have a within-class mate y, because x, y, c_i, Z_i, x would be a four-cycle. Nor can it have an opposite-class neighbor y, because x, y, c_{i+3}, Z_i, x would be a

four-cycle. It has no neighbor in R. Thus x is another degree-two simplicial vertex. Delete all such vertices, at cost one in Λ per vertex.

The remaining A vertices meet only the cycle and other A vertices. Let a be their number, q their within-class matching count, and e the number of opposite-class edges. Reinstating them adds

$$\Delta\Lambda = 2a + 2e - q, \qquad \Delta(3n - \Lambda) = a - 2e + q \geq 0,$$

since $2e \leq a$. We may therefore remove them while retaining this nonnegative contribution to the deficit.

13.13.2 Components attached to opposite-contact vertices

The reduced graph consists of C, at most three Z vertices, and R. Every component Q of $G[R]$ touches at least one Z vertex. No vertex of R can meet two distinct Z vertices: their two opposite pairs have adjacent cycle contacts, and these, together with the common neighbor, give an induced C_5.

For each Z vertex z, the graph $G[R \cup \{z\}]$ has no induced four-vertex path with endpoint z. Otherwise append the three cycle vertices of a rooted induced P_4 starting at z, giving P_7. Fix a component Q touching z, and put

$$X = N(z) \cap Q, \qquad Y = Q \setminus X.$$

Every vertex of Q is within distance two of z, by the forbidden rooted path. Each $y \in Y$ has a unique neighbor in X; two would give a four-cycle through z. Call that neighbor its parent. Both $G[X]$ and each set of children of one parent induce matchings plus isolated vertices. No edge joins children with different parents, since then z, x, y, y' would be an induced four-vertex path. Connectedness of Q now implies that X is K_1 or K_2. Thus Q consists of one or two adjacent parent vertices and their private child sets.

If Q touches only z, then $Q \cup \{z\}$ is a block graph with all blocks edges or triangles. For $q = |Q|$ and τ its triangle count, including triangles through z, the attachment has $q + \tau$ edges. It adds $2q - \tau \leq 2q$ to Λ, and hence adds nonnegative deficit.

Suppose instead that Q touches another Z vertex z'. Its neighbor y in Q is a child of some $x \in X$, since no vertex meets two Z vertices. If X contained another vertex x', then z', y, x, x' would be an induced four-vertex path rooted at z'. Hence $X = \{x\}$. There is at most one child adjacent to z', by the common-neighbor condition applied to x, z'. Every other child w must be adjacent to y, or z', y, x, w would be the forbidden rooted path. Since the child set induces a matching, there is at most one such w. Consequently Q is K_2 or K_3, and its different boundary Z vertices attach to different vertices of Q. The only contributions are

attachment	$\Delta(3n - \Lambda)$
K_2 with two Z neighbors	$6 - 2(1 + 2) = 0$
K_3 with two Z neighbors	$9 - [2(3 + 2) - 3] = 2$
K_3 with three Z neighbors	$9 - [2(3 + 3) - 3] = 0.$

Thus every component adds nonnegative deficit. Finally the base C_6 with $k \leq 3$ opposite-contact vertices has

$$n = 6 + k, \qquad m = 6 + 2k, \qquad t = 0, \qquad 3n - \Lambda = 6 - k \geq 3.$$

Reinstating all deleted vertices proves $\Lambda \leq 3n$ in this branch. Together with Theorem 13.6, this settles every graph without an induced seven-cycle.

13.14 Seven-cycle contacts and a connected deletion

Fix an induced $C_7 = c_0 \dots c_6 c_0$, with indices modulo seven. Every nonempty exterior cycle-contact set is of one of the forms

$$X_i = \{c_i, c_{i+3}\}, \qquad Y_i = \{c_i, c_{i+3}, c_{i+4}\}.$$

Indeed, distance-two contacts give a four-cycle. A singleton or adjacent pair produces P_7 by appending six consecutive cycle vertices, omitting the unused contact in the adjacent-pair case. Four contacts include a distance-two pair, and the three-contact sets without such a pair are exactly the displayed Y_i. No type is repeated, since that would give two common cycle neighbors.

All exterior vertices having a cycle contact are independent. For an edge xy between two such vertices, no cycle edge may run from a contact of x to a distinct contact of y, since that edge and xy complete a four-cycle. If x has type X_0, the contacts of y must therefore lie in $\{c_0, c_3, c_5\}$; the only admissible type contained there is X_0, already excluded by repetition. If x has type Y_0, the permitted set is just $\{c_0\}$, which contains no admissible type.

The following compatibility restrictions are necessary:

$$X_i, X_{i\pm2} \text{ cannot coexist}, \tag{74}$$
$$X_i, Y_j \text{ coexist only if } j - i \in \{4, 5, 6\}, \tag{75}$$
$$Y_i, Y_j \text{ coexist only if } j - i \in \{2, 5\}. \tag{76}$$

The common-two-contact cases give four-cycles. All remaining exclusions follow by rotating or reflecting these induced P_7 witnesses, with x, y denoting the two indicated exterior types:

$$\begin{array}{r|l} X_0, X_2 & c_1, c_0, x, c_3, c_4, c_5, y \\ X_0, Y_1 & c_2, c_3, x, c_0, c_6, c_5, y \\ X_0, Y_2 & c_1, c_0, x, c_3, c_4, c_5, y \\ Y_0, Y_1 & c_2, c_3, x, c_0, c_6, c_5, y. \end{array}$$

The exact contact sets and the just-proved independence verify every chord condition. The standard X/Y partition and several of these exclusions also occur in Chen–Wu–Xu [35, proof of Theorem 1.1, claims (1)–(6)]; the exclusion of a four-cycle as a subgraph is used here throughout.

If there is no Y, the X indices form an independent set in the seven-cycle with steps ±2, and there are at most three. If Y_0 occurs, the only possible X indices are $1, 2, 3$, with X_1, X_3 incompatible. The only other possible Y indices are $2, 5$, which are incompatible with each other. If Y_2 also occurs, only X_3 remains permitted, and the reflected case is the same. Thus the entire first layer has at most three vertices.

Proof of Theorem 13.5. Induct on the order of a connected graph. All graphs without induced C_7 have been settled. Otherwise use the contact description above. If a Y occurs, take it to be Y_0. The only other possible Y types are Y_2, Y_5, neither of which touches c_3 or c_4. The vertex c_3 has degree three unless X_3 occurs, and c_4 has degree three unless X_1 occurs. Since these two X types cannot coexist, one of c_3, c_4 has degree exactly three and lies in exactly one triangle. Delete that vertex v. Its two cycle neighbors are connected by the remaining cycle path, and its third neighbor Y_0 retains two cycle contacts. Thus $G - v$ is connected. Since

$$\Lambda(G) - \Lambda(G - v) = 2d(v) - 3t(v) = 3,$$

induction proves $\Lambda(G) \le 3(n-1) + 3 = 3n$.

If there is no Y, there are two adjacent cycle vertices untouched by all X types. With at most one X this is immediate. For two X indices, up to rotation and reflection the pairs are

$\{0,1\}$ or $\{0,3\}$, leaving respectively c_5, c_6 or c_1, c_2 untouched. Every permitted three-index set is a rotation of $\{0,1,4\}$, again leaving c_5, c_6 untouched. Delete this adjacent degree-two pair. The remaining cycle path is connected, and no exterior vertex touched the deleted pair, so the remaining graph is connected. Exactly three edges and no triangles are lost. Therefore induction gives $\Lambda(G) \le 3(n-2)+6 = 3n$. This completes the proof. □

13.15 A sharp elementary bipartite P_8-free bound

Theorem 13.11. *A connected bipartite C_4-free, P_8-free graph has average degree at most three, with equality exactly for the Heawood graph.*

Proof. Trees have average degree $2 - 2/n < 2$. Otherwise the girth is six or eight: a shortest cycle is induced and even, and a longer one contains P_8. If the girth is eight, fix a shortest C_8. An exterior vertex has at most one cycle contact: two contacts have even cycle-distance, and yield a C_4 or C_6. A single contact followed by seven successive cycle vertices gives an induced P_8. Connectedness therefore forces $G = C_8$.

Now fix an induced $C_6 = c_0c_1\cdots c_5c_0$, with indices modulo six. Let A, B be the vertices at distance one and two from this cycle. Every vertex in A has exactly one cycle contact. Its contacts must have the same parity on the cycle, and two would be at distance two, creating a C_4. Put

$$A_i = \{x \in A : N(x) \cap V(C_6) = \{c_i\}\}.$$

There are no vertices at distance at least three: a shortest path z, y, x, c_0 to the cycle, followed by c_1, c_2, c_3, c_4, would be an induced P_8. Thus $V(G) = V(C_6) \cup A \cup B$.

Every edge in $G[A]$ joins A_i to A_{i+3}. Bipartiteness requires opposite parity, and adjacent contact indices would yield a C_4 using the cycle edge. Moreover $G[A]$ is a matching: two neighbors of $x \in A_i$ would both lie in A_{i+3}, forming a C_4 with c_{i+3}. Write $k = |E(G[A])|$. The set B is independent. If bb' were an edge and $x \in A_i$ a neighbor of b, bipartiteness excludes $b'x$, and

$$b', b, x, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}$$

would be an induced P_8.

A vertex $b \in B$ has at most one neighbor in any A_i, by C_4-freeness, and can contact classes of only one parity. We claim that if it contacts A_i, it is complete to A_{i+2} and A_{i+4}. Indeed, for $x \in A_i$ adjacent to b and $y \in A_{i+4}$ nonadjacent to b,

$$b, x, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, y$$

is an induced P_8. Here xy is absent because exterior edges only join opposite contact classes. Reverse the cycle direction to treat A_{i+2}.

Let $A^0 = A_0 \cup A_2 \cup A_4$ and $A^1 = A_1 \cup A_3 \cup A_5$, and let B^j consist of vertices whose neighbors lie in A^j. Write

$$a_j = |A^j|, \quad b_j = |B^j|, \quad e_j = |E(A^j, B^j)|,$$

and let q_j count the nonempty contact classes defining A^j. Call parity j special if $q_j \ge 2$ and $b_j > 0$. For a special parity, the claim forces any $b \in B^j$ to contact every nonempty class. Applying the claim from a second class also makes it complete to its original class. Each class is therefore a singleton. Every member of B^j contacts all these vertices, so C_4-freeness gives $b_j = 1$. Consequently

$$a_j = q_j \in \{2,3\}, \quad b_j = 1, \quad e_j = q_j, \quad R_j := a_j + 3b_j - 2e_j = 3 - q_j \ge 0. \tag{77}$$

For a nonspecial parity, either $b_j = 0$ or $q_j \le 1$; in the latter case every vertex of B^j has degree one. Thus

$$R_j \ge a_j \ge 0. \tag{78}$$

Counting the cycle edges, cycle-to-A edges, matching edges, and A–B edges gives

$$3n - 2m = 6 + R_0 + R_1 - 2k. \tag{79}$$

If neither parity is special, $R_0 + R_1 \ge a_0 + a_1 \ge 2k$, giving a positive deficit of at least six. If exactly one is special, its A-side has at most three vertices, so $k \le 3$, while the other side contributes $R \ge k$. The deficit is then at least $6 - k \ge 3$. If both are special,

$$3n - 2m = 12 - q_0 - q_1 - 2k \ge 0,$$

since $q_0, q_1 \le 3$ and $k \le \min(q_0, q_1)$. Equality requires $q_0 = q_1 = k = 3$.

The equality graph has vertices $c_0, \dots, c_5$, $a_0, \dots, a_5$, z_0, z_1, and exactly the edges

$$c_i c_{i+1}, \quad c_i a_i \ (0 \le i < 6), \quad a_i a_{i+3} \ (0 \le i < 3), \quad z_0 a_i \ (i = 0, 2, 4), \quad z_1 a_i \ (i = 1, 3, 5).$$

This is the Heawood graph, equivalently the incidence graph of the Fano plane. The displayed graph is cubic, bipartite, and C_4-free. Every pair in the same bipartition has exactly one common neighbor, as is seen directly, or from C_4-freeness and $7\binom{3}{2} = \binom{7}{2}$. In an induced path $v_0, \dots, v_7$, the pairs v_6, v_0 and v_6, v_2 would need distinct common neighbors outside the path: they cannot coincide because v_0, v_2 already share v_1, and an internal contact would be a chord. Together with v_5, v_7, these give v_6 four neighbors, a contradiction. Hence the equality graph is indeed P_8-free. □

Remark 13.12. The same layer argument gives a useful simpler refinement: a cyclic connected bipartite C_4-free, P_7-free graph satisfies $3n - 2m \ge 6$. A shortest cycle is C_6. A distance-two vertex would give P_7 by appending five cycle vertices, so the cycle is dominating. Its exterior vertices have unique contacts and induce a matching. With k matching edges and s exterior isolated vertices, $n = 6 + 2k + s$, $m = 6 + 3k + s$, and $3n - 2m = 6 + s$.

14 Radius classifications for Conjecture 133

The classical induced-path radius inequality used in this section is

$$p(G) \ge 2r(G) - 1. \tag{80}$$

It is recorded by Erdős–Saks–Sós with credit to Fan Chung [36, Section II, pp. 64–65]. We first prove an arbitrary-cycle attachment theorem, then use connected domination to obtain complete radius-four and radius-five classifications. These are statements about all finite graphs with the stated hypotheses, not bounded-order searches.

14.1 Dominating even cycles

Theorem 14.1. *Fix $k \ge 3$. Let G be a connected C_4-subgraph-free graph with a dominating induced cycle $C = c_0 \dots c_{2k-1} c_0$. The graph has radius $k + 1$ and is P_{2k+2}-free if and only if it has the following form:*

(i) *For every i, add a nonempty set A_i, join all its vertices to c_i, and place a matching plus isolated vertices on A_i.*
(ii) *For each cycle edge $c_i c_{i+1}$, optionally add one vertex b_i adjacent to its endpoints.*

(iii) *Add no other vertices or edges.*

Every graph in the classification satisfies

$$p(G) = 2k + 1 = 2r(G) - 1, \qquad \Lambda(G) = 2n - t \le 2n,$$

where $n = |V(G)|$ *and* t *counts its triangles.*

14.2 Contact sets and radius witnesses

Put $N = 2k$, and take all indices modulo N. Suppose an exterior vertex x has two cycle contacts at circular distance $d \ge 3$, choosing the shorter arc so that $d \le k$. Every cycle vertex is within

$$1 + \left\lfloor \frac{2k - d}{2} \right\rfloor \le k - 1$$

of x. Since C dominates, x has eccentricity at most k, contrary to the assumed radius. A pair of contacts at distance two gives a four-cycle. Thus every exterior contact set is a singleton or an adjacent pair; three contacts would contain a prohibited pair. Denote the singleton class by A_i and the adjacent-pair vertex by b_i, with contacts c_i, c_{i+1}. There is at most one vertex of the latter type, by the common-neighbor condition for four-cycle exclusion. Each $G[A_i]$ is a matching plus isolated vertices, since a three-vertex path in $N(c_i)$ would complete a four-cycle.

The only possible cross-class edges are

$$\begin{aligned} A_i\text{–}A_j: &\quad \text{circular distance at least two,} \\ A_i\text{–}b_j: &\quad j - i \in \{2, \dots, N-3\}, \\ b_i\text{–}b_j: &\quad j - i \in \{3, \dots, N-3\}. \end{aligned}$$

The excluded cases give a four-cycle using the two exterior endpoints and an appropriate cycle edge, or the shared cycle contact and its neighbor.

The cycle vertex c_{i+k} is within distance k of every vertex except possibly those in A_i. Since its eccentricity is at least $k + 1$, there is a vertex $z_i \in A_i$ with

$$\operatorname{dist}(c_{i+k}, z_i) = k + 1. \tag{81}$$

In particular every A_i is nonempty. The vertex z_i has no exterior neighbor outside A_i. Indeed, the cycle contact of any allowed A_j neighbor is within cycle distance $k - 2$ of c_{i+k}, giving a route of length at most k to z_i. The same argument applies to either contact of an allowed b_j neighbor. Both contradict (81). Thus the only possible exterior neighbor of z_i is one mate in A_i.

14.3 Excluding every cross-class edge

Every displayed sequence in this section has exactly $N + 2$ distinct vertices. Consecutive indices in an ellipsis follow the indicated ordinary cyclic direction. The omitted cycle vertices separate the two cycle segments. The lack of cross-class neighbors of the z_i excludes all further exterior chords.

14.3.1 Two distant single-contact classes

Suppose $x \in A_0$ and $y \in A_d$ are adjacent, where $3 \le d \le N - 3$. Then

$$z_{d-2}, c_{d-2}, c_{d-3}, \dots, c_0, x, y, \\ c_d, c_{d+1}, \dots, c_{N-2}, z_{N-2}$$

is an induced P_{N+2}. The omitted cycle vertices are c_{d-1} and c_{N-1}, and the two witness classes differ from those of x, y. This is impossible.

14.3.2 Single-contact classes at distance two

Let $x \in A_0$ and $y \in A_2$ be adjacent. The witness z_0 differs from x, since x has a cross-class neighbor. If z_0x were absent, then

$$z_{N-2}, c_{N-2}, c_{N-3}, \ldots, c_2, y, x, c_0, z_0$$

would be induced P_{N+2}. Hence z_0x is an edge. Reflection gives z_2y as an edge. But now

$$z_3, c_3, c_4, \ldots, c_{N-1}, c_0, x, y, z_2$$

is induced P_{N+2}. The omitted cycle vertices are c_1, c_2; z_2 has only its mate y among the exterior path vertices. Rotation and reflection cover every distance-two pair.

14.3.3 Single-contact and double-contact classes

Suppose $x \in A_0$ is adjacent to b_d, where $2 \le d \le N-3$. The sequence

$$z_{d-1}, c_{d-1}, c_{d-2}, \ldots, c_0, x, b_d, \\ c_{d+1}, c_{d+2}, \ldots, c_{N-2}, z_{N-2}$$

is induced P_{N+2}. Here c_d, c_{N-1} are omitted, so the unused contact c_d of b_d is absent. Both witness classes differ from A_0.

14.3.4 Two double-contact classes

For an edge b_0b_d, with $3 \le d \le N-3$, use

$$z_{d-1}, c_{d-1}, c_{d-2}, \ldots, c_1, b_0, b_d, \\ c_{d+1}, c_{d+2}, \ldots, c_{N-1}, z_{N-1}.$$

The omitted cycle vertices c_0, c_d are precisely the unused contacts of the two double-contact vertices. This is again induced P_{N+2}.

The list exhausts all possible cross-class edges. None can therefore occur, proving the asserted structure.

14.4 Converse and the local-independence sum

Consider a graph constructed as in Theorem 14.1. An A attachment lies in a pendant edge or triangle. A four-cycle containing b_i would require a second length-two path between c_i and c_{i+1}, and the construction has no such path. The remaining core cycle has length at least six. Thus the graph is C_4-subgraph-free.

Every exterior vertex is simplicial: it has one neighbor or two adjacent neighbors. It can therefore occur in an induced path only as an endpoint. An induced path uses at most $N-1$ cycle vertices and two exterior endpoints, giving $p \le N+1$. Since all A_i are nonempty, an induced path from A_0 through $c_0, c_1, \ldots, c_{N-2}$ to A_{N-2} has $N+1$ vertices. Hence $p = N+1$.

Every cycle vertex has eccentricity $k+1$, witnessed by the opposite A class. Every A vertex has eccentricity $k+2$, also witnessed by an opposite-class vertex. Each b_i has eccentricity $k+1$, witnessed by either farthest A class. Thus the radius is exactly $k+1$.

Put $a = \sum_i |A_i|$, $q = \sum_i |E(G[A_i])|$, and let b count the vertices b_i. Then

$$n = N + a + b, \qquad m = N + a + q + 2b, \qquad t = q + b.$$

Every neighborhood in a C_4-subgraph-free graph is a matching plus isolated vertices. If $t(v)$ counts triangles through v, then $\lambda(v) = d(v) - t(v)$. Summing yields

$$\Lambda = 2m - 3t = 2N + 2a - q + b = 2n - (q + b) = 2n - t.$$

This finishes the proof of Theorem 14.1.

Corollary 14.2. *For $k \geq 3$, Conjecture 133 holds for every connected graph with a dominating induced C_{2k}, radius $k + 1$, and $\ell \leq k + 1$.*

Proof. The four-cycle-present case follows from a geodesic. Otherwise (80) settles $\ell \leq k$. If $k < \ell \leq k + 1$, Theorem 14.1 excludes P_{2k+2}-freeness, since its classified graphs have $\ell \leq 2 < k$. Hence $p \geq 2k + 2 \geq (k + 1) + \ell$. □

14.5 Radius four: the classification used in the corollary

Write dist, ecc, and rad for distance, eccentricity, and radius.

Theorem 14.3. *A connected C_4-subgraph-free graph has radius four and is P_8-free if and only if it is C_8, or has the following form:*

(i) *Start with an induced cycle $c_0c_1 \dots c_5c_0$.*
(ii) *For each i, add a nonempty set A_i, join every vertex of A_i to c_i, and put a matching plus isolated vertices on A_i.*
(iii) *On each cycle edge c_ic_{i+1} optionally add one vertex b_i adjacent to its endpoints.*
(iv) *Add no other edges.*

The six-cycle family satisfies $\Lambda(G) = 2n - t$. In particular all graphs in the classification have $\ell(G) \leq 2$.

14.5.1 Reduction to a dominating cycle

We use the classical induced-path bound

$$p(G) \geq 2\,\mathrm{rad}(G) - 1, \tag{82}$$

recorded by Erdős–Saks–Sós with credit to Fan Chung [36, Section II, pp. 64–65].

Lemma 14.4. *A connected C_4-subgraph-free P_6-free graph of radius three is C_6.*

Proof. Theorem 6 of van 't Hof–Paulusma [37] supplies a dominating induced six-cycle or a dominating complete bipartite subgraph. In a C_4-subgraph-free graph the latter is a star, whose center has eccentricity at most two.

Fix a dominating induced cycle $c_0 \dots c_5c_0$. Every exterior vertex has at most two cycle neighbors, since three vertices on a six-cycle include a pair at circular distance two, which would complete a C_4. Two contacts at distance two are likewise forbidden. The remaining contact sets are singletons, adjacent pairs, and opposite pairs. A singleton or adjacent pair gives an induced P_6: take the exterior vertex followed by five consecutive cycle vertices, leaving out the unused contact in the adjacent-pair case. Thus every exterior vertex has an opposite pair of contacts.

If an exterior vertex has contacts c_0, c_3, then c_0 is within distance two of the whole cycle. Every opposite pair contains one of c_0, c_1, c_5, so c_0 is also within distance two of every exterior vertex. This contradicts radius three. Hence there is no exterior vertex. □

Lemma 14.5. *Every connected C_4-subgraph-free P_8-free graph of radius four is C_8 or has a dominating induced C_6.*

Proof. Theorem 5 of Camby–Schaudt [38], in the pinned arXiv version, supplies a connected dominating induced subgraph H that is P_6-free or is C_8. Since H dominates, $\mathrm{rad}(G) \le \mathrm{rad}(H)+1$. In the P_6-free alternative, (82) gives $\mathrm{rad}(H) \le 3$, so $\mathrm{rad}(H) = 3$ and Lemma 14.4 yields $H = C_6$.

Suppose instead that $H = C_8$. The cycle-contact set of each exterior vertex contains a pair at circular distance three or four. Indeed, distance-two contacts create a C_4; without a distance-three/four pair, the contacts are a singleton or adjacent pair, which yields an induced P_8 by appending seven consecutive cycle vertices. For each cycle vertex, its three cycle vertices at distance at least three are consecutive. They cannot contain such a contact set. Thus every cycle vertex is within distance three of every exterior vertex. If an exterior vertex has two contacts at distance four, it shortens the distance between those antipodal vertices to two. If its contacts have distance three, it gives a path of length three from a contact to that contact's antipode. In either case one cycle vertex is now within distance three of the entire cycle and all exterior vertices. This contradicts radius four. Therefore the C_8 alternative has no exterior vertices. □

Completion of the radius-four classification. Lemma 14.5 leaves C_8 or a dominating induced C_6. The latter is precisely the $k = 3$ case of Theorem 14.1. That theorem supplies both the attachment restrictions and their converse, including $\Lambda = 2n - t$. The conclusions for C_8 follow directly. This proves Theorem 14.3. □

14.6 The radius-five classification

In this subsection, a cycle is *distance-two dominating* if every vertex of the graph has distance at most two from the cycle. This distance condition is distinct from the two-neighbor condition in the parameter γ_2. For an even cycle $C = c_0 \dots c_{N-1}c_0$, a *one-level decoration* has the following form. Each cycle vertex c_i receives a nonempty set A_i of neighbors; the graph on A_i is a matching plus isolates. For each edge c_ic_{i+1} there may also be one vertex b_i adjacent to its two endpoints. There are no other vertices or edges. All indices are read modulo N.

A *two-level decoration* is obtained in the same way, allowing some A_i initially to be empty, and then adding a set of children to any first-layer vertex in an A_i or of type b_i. Each child has exactly one first-layer neighbor, its parent. The children of each parent induce a matching plus isolates, and different child classes are anticomplete. We require, for every i, that some vertex of A_i has a child. No other vertices or edges are permitted.

Theorem 14.6. *A connected C_4-subgraph-free graph has radius five and is P_{10}-free if and only if it is one of the following:*

(i) *C_{10};*
(ii) *a one-level decoration of C_8;*
(iii) *a two-level decoration of C_6.*

Every graph in the list satisfies $p = 9$ and $\Lambda = 2n - t \le 2n$.

Corollary 14.7. *Original WOWII133 holds for every connected graph with $r(G) = 5$ and $\ell(G) \le 5$.*

14.7 Previously established inputs and the structural reduction

We use the following two classification results, already proved in Theorems 14.3 and 14.1. The remaining argument classifies all possible second-layer attachments.

Proposition 14.8. (a) *A connected C_4-subgraph-free, P_8-free graph of radius four is either C_8 or a one-level decoration of C_6.*

(b) *For $k \geq 3$, a connected C_4-subgraph-free graph with a dominating induced C_{2k}, radius $k+1$, and no induced P_{2k+2} is a one-level decoration of that cycle. Conversely every such decoration has those properties, with $p = 2k+1$ and $\Lambda = 2n - t$.*

The induced-path radius bound, due to Fan Chung as credited in Erdős–Saks–Sós [36, Section II, pp. 64–65], is

$$p(G) \geq 2r(G) - 1. \tag{83}$$

Camby–Schaudt [38, Theorem 5, arXiv version 1] shows that every connected P_{10}-free graph has a connected dominating induced subgraph H which is P_8-free or is an induced C_{10}. If $r(G) = 5$ and H is P_8-free, then

$$5 \leq r(H) + 1, \qquad 2r(H) - 1 \leq p(H) \leq 7,$$

so $r(H) = 4$. Proposition 14.8(a) gives a dominating C_8 in G, or a fixed induced C_6 whose distance from every vertex of G is at most two.

If the dominating subgraph is C_{10}, there can be no exterior vertex. Indeed, a singleton or adjacent-pair contact set yields an induced P_{10} by appending nine consecutive cycle vertices, and contacts at cycle distance two yield a C_4. Hence an exterior vertex has two contacts at cycle distance at least three. For any cycle vertex c, the cycle vertices at distance at least four from c form three consecutive vertices. This set cannot contain an admissible contact set: two contacts in it would be adjacent or at distance two. Consequently c is within distance four of every exterior vertex. If an exterior vertex has contacts at cycle distance $d \in \{3, 4, 5\}$, one contact reaches its antipode in $2 + (5 - d) \leq 4$ edges through that exterior vertex. This contact is therefore within distance four of the whole graph, a contradiction. Thus $G = C_{10}$.

The dominating-C_8 branch follows from Proposition 14.8(b) with $k = 4$. It remains to prove the following lemma.

Lemma 14.9. *Let G be connected, C_4-subgraph-free, P_{10}-free, and of radius five. If an induced C_6 is distance-two dominating, then G is a two-level decoration of that cycle.*

14.8 Contacts and clean radius witnesses

Fix the cycle $C = c_0 \dots c_5 c_0$ in Lemma 14.9. Indices in this proof are modulo six. Three cycle contacts include a pair at distance two, which produces a C_4. Hence nonempty contact sets are singletons, adjacent pairs, or opposite pairs. A vertex with opposite contacts is within distance two of the cycle, and hence within distance four of all of G, contrary to the radius. Thus the first layer has classes A_i with contact set $\{c_i\}$ and classes B_i with contact set $\{c_i, c_{i+1}\}$. Four-cycle exclusion gives $|B_i| \leq 1$; write its vertex, when present, as b_i. Let R be the set with no cycle contact. Every vertex of R has a first-layer neighbor.

Each A_i induces a matching plus isolates. The only possible first-layer cross edges have types

$$A_iA_j \ (j - i \in \{2, 3, 4\}), \qquad A_ib_j \ (j - i \in \{2, 3\}), \qquad b_ib_{i+3}. \tag{84}$$

All other types immediately give a four-cycle.

Every cycle vertex has eccentricity at most five by distance-two domination, and at least five by the radius. A distance-five witness for c_{i+3} must therefore belong to R, and all its first-layer neighbors must belong to A_i: any other contact gives a route of length at most four. Two neighbors in A_i would give a four-cycle through c_i. We may therefore choose six arms

$$c_i - a_i - z_i, \quad a_i \in A_i, \ z_i \in R, \quad N(z_i) \cap (\text{first layer}) = \{a_i\}, \quad \operatorname{dist}(c_{i+3}, z_i) = 5. \tag{85}$$

Every a_i has no first-layer neighbor outside A_i. Indeed, any permitted cross edge in (84) shortens the prescribed distance in (85) to at most four. Within A_i, the vertex a_i has at most one mate.

We next prove that each z_i is simplicial, of degree one or two. Rotate to $i = 0$, and suppose z_0w is an edge with $w \in R$. A first-layer neighbor x of w can only have type A_0, A_1, A_5, B_0, B_5, since all other types put z_0 within distance four of c_3. The following alternatives give induced ten-vertex paths:

$$\begin{aligned} x \in A_0 \setminus \{a_0\}: &\quad z_2, a_2, c_2, c_3, c_4, c_5, c_0, x, w, z_0; \\ x \in A_1: &\quad z_0, w, x, c_1, c_0, c_5, c_4, c_3, a_3, z_3; \\ x = b_0: &\quad z_0, w, b_0, c_0, c_5, c_4, c_3, c_2, a_2, z_2. \end{aligned}$$

Reflect the latter two for A_5, B_5. In the first row, the possible chords wa_2, wz_2, z_0z_2 shorten one of the prescribed witness distances to at most four. The same argument excludes wa_3, wz_3, z_0z_3 in the second row and wa_2, wz_2, z_0z_2 in the third. Exact cycle contacts, the absence of cross edges at clean parents a_i, and the unique first-layer neighbor of z_0 exclude all other chords. The unused contact c_1 of b_0 is omitted in the third row.

Thus every first-layer neighbor of w is a_0. At least one exists, so wa_0 is an edge. Two such vertices w would have the two common neighbors a_0, z_0, giving a four-cycle. This proves the assertion for z_0, and by rotation for all z_i.

14.9 Eliminating first-layer cross edges

Every displayed path in this section has ten vertices. Clean parents have no cross-class neighbors, clean terminals are simplicial, and the cycle contacts are exact; these properties exclude the chords. An edge xy with $x \in A_0, y \in A_3$ gives

$$z_1, a_1, c_1, c_0, x, y, c_3, c_4, a_4, z_4.$$

An edge from $x \in A_0$ to b_2 or b_3 gives, respectively,

$$\begin{aligned} &z_1, a_1, c_1, c_0, x, b_2, c_3, c_4, a_4, z_4; \\ &z_5, a_5, c_5, c_0, x, b_3, c_3, c_2, a_2, z_2. \end{aligned}$$

An edge b_0b_3 gives

$$z_2, a_2, c_2, c_1, b_0, b_3, c_4, c_5, a_5, z_5.$$

Finally, suppose $x \in A_0, y \in A_2$ are adjacent. Necessarily $x \neq a_0$, since a_0 has no cross edge. If a_0x is absent, then

$$z_4, a_4, c_4, c_3, c_2, y, x, c_0, a_0, z_0$$

is induced. Hence a_0x is present, and reflection gives a_2y. But now

$$z_3, a_3, c_3, c_4, c_5, c_0, x, y, a_2, z_2$$

is induced. Rotations and reversals exhaust (84). Therefore the first layer contains only the individual A_i matchings; every b_i has no first-layer neighbor.

14.10 Unique parenthood in the second layer

Suppose $w \in R$ has at least two first-layer neighbors. It has at most one neighbor in each A_i, by four-cycle exclusion through c_i. It is nonadjacent to every z_i, because a neighbor of z_i in R has only a_i as its first-layer neighbor. We repeatedly use

$$x \in A_i,\ wx \in E(G) \quad \Longrightarrow \quad wa_{i+3} \notin E(G). \tag{86}$$

Otherwise $c_i, x, w, a_{i+3}, z_{i+3}$ contradicts the prescribed distance five in (85).

14.10.1 Two neighbors of type A

First suppose $x \in A_0, y \in A_2$ are neighbors of w. According to the relation between x and a_0, use one of the following paths:

$$\begin{aligned} x = a_0 :&\quad z_0, x, w, y, c_2, c_3, c_4, c_5, a_5, z_5; \\ x \neq a_0,\ a_0x \in E :&\quad a_0, x, w, y, c_2, c_3, c_4, c_5, a_5, z_5; \\ x \neq a_0,\ a_0x \notin E :&\quad z_0, a_0, c_0, x, w, y, c_2, c_3, a_3, z_3. \end{aligned}$$

In the latter two cases wa_0 would create a four-cycle through c_0, x. The endpoint-parent chord wa_5 or wa_3 is excluded by (86). All first-layer cross edges have already been eliminated, so these are induced P_{10}s. Thus two A contacts cannot have indices at distance two.

For $x \in A_0, y \in A_1$, if wa_2 is absent, use

$$z_2, a_2, c_2, c_3, c_4, c_5, c_0, x, w, y.$$

If wa_2 is present, the distance-two case applies to x, a_2. Thus adjacent A indices are also impossible. For $x \in A_0, y \in A_3$, the preceding cases exclude every A neighbor of w in the other four classes, and

$$a_1, c_1, c_0, x, w, y, c_3, c_4, a_4, z_4$$

is induced. Hence w has at most one neighbor of type A.

14.10.2 One neighbor of each type

Let $x \in A_0$ be a neighbor of w. Types b_0, b_5 give a four-cycle through c_0. Every clean parent outside A_0 is a nonneighbor of w by the preceding exclusion of two A neighbors. For a second neighbor b_2, use

$$a_1, c_1, c_0, x, w, b_2, c_3, c_4, a_4, z_4;$$

the b_3 case is its reflection. For b_1, use

$$\begin{aligned} x = a_0 :&\quad a_2, c_2, b_1, w, x, c_0, c_5, c_4, a_4, z_4; \\ x \neq a_0,\ a_0x \notin E :&\quad a_0, c_0, x, w, b_1, c_2, c_3, c_4, a_4, z_4; \\ x \neq a_0,\ a_0x \in E :&\quad a_0, x, w, b_1, c_2, c_3, c_4, c_5, a_5, z_5. \end{aligned}$$

Reflect for b_4. Whenever needed, wa_0 is excluded by a four-cycle. The unused contact of the displayed B vertex is omitted, and all other chord exclusions were established above. Thus no mixed pair is possible.

14.10.3 Two neighbors of type B

In this case w has no A neighbor, by the preceding subsection. Two B types sharing a cycle contact yield a four-cycle. Up to rotation and reflection the remaining pairs are b_0, b_2 and b_0, b_3, giving respectively

$$\begin{aligned} &a_1, c_1, b_0, w, b_2, c_3, c_4, c_5, a_5, z_5; \\ &a_2, c_2, c_1, b_0, w, b_3, c_4, c_5, a_5, z_5. \end{aligned}$$

Both are induced P_{10}s. Every vertex of R therefore has exactly one first-layer neighbor, called its parent.

14.11 Child classes, converse, and counting

Suppose uv is an edge in R with distinct parents x, y. If one parent is of type A, orient so $x \in A_i$ and choose $s \in \{i+2, i-2\}$ with $y \neq a_s$. Take the five-vertex cycle arc from c_s to c_i that omits the other neighbor of c_i. Then

$$z_s, a_s, \ [\text{five cycle vertices from } c_s \text{ to } c_i], \ x, u, v$$

is an induced P_{10}. Distinct parents exclude xv, unique parenthood excludes the other first-layer chords, and the choice of s together with the simplicial terminal property excludes chords to the clean arm. If both parents have type B, rotate so $x = b_0$ and use

$$z_2, a_2, c_2, c_3, c_4, c_5, c_0, b_0, u, v.$$

The unused contact c_1 is omitted. Hence distinct child classes are anticomplete. Each child class has maximum degree one by four-cycle exclusion in its parent's neighborhood. The arms in (85) ensure the required nonempty depth-two class at each cycle vertex. This proves Lemma 14.9.

Conversely, consider any stated two-level decoration of C_6. The A attachments and child matchings introduce only pendant edges or triangles. A four-cycle through b_i would require a second length-two path between c_i, c_{i+1}, which the construction does not contain. Thus the graph is C_4-subgraph-free. An induced path meeting the core meets it in one contiguous induced cycle path, of at most five vertices, with at most two exterior vertices at each endpoint. Traversing a matching edge and then the common parent would produce a triangular chord. A path missing the core lies in one attachment component and has at most four vertices. Therefore $p \leq 9$, and depth-two vertices at the ends of a five-vertex cycle arc give $p = 9$.

Cycle, A, and B vertices have eccentricities five, six, and five, respectively. A child of an A vertex has eccentricity seven, while a child of a B vertex has eccentricity six. Opposite or farthest attachment classes supply these distances, and matching edges do not shorten routes between classes. In particular the radius is five. If a, b, s count the A vertices, the B vertices, and their children, and q, h count the matching edges on the A and child classes, then

$$n = 6 + a + b + s, \quad m = 6 + a + 2b + s + q + h, \quad t = b + q + h.$$

Equation (67) gives $\Lambda = 2n - t$. The C_{10} and decorated-C_8 cases have the same conclusions, by direct calculation and Proposition 14.8(b). The structural reduction now proves Theorem 14.6.

Proof of Corollary 14.7. If G contains a four-cycle, a geodesic supplies $p \geq r + 1 = 6$, as required by the original exponent-zero case. Otherwise (83) gives $p \geq 9$, proving $p \geq 5 + \ell$ for $\ell \leq 4$. If $4 < \ell \leq 5$, Theorem 14.6 rules out P_{10}-freeness, since every classified graph has $\ell \leq 2$. Hence $p \geq 10 \geq 5 + \ell$, proving the remaining range. □

Antecedents. Equal-length pendant-arm cores are classical ciliates. Relevant antecedents include Fajtlowicz's characterization of radius-critical graphs and the results on induced ciliates and radius equality discussed by DeLaViña, Larson, Pepper, and Waller [39, 40]. The attachment proofs above use the explicitly stated connected-domination theorems.

14.12 The unrounded average range up to four

Corollary 14.10. *Conjecture 133 holds for every connected graph with $\ell(G) \leq 4$.*

Lemma 14.11. *Every connected C_4-subgraph-free graph with $\ell \leq 3$ satisfies $p \geq r + \ell$.*

Proof. First, a connected P_5-free, C_4-subgraph-free graph has $\Lambda \le 2n$. If it has an induced C_5, any exterior neighbor of that cycle has a singleton or adjacent-pair contact set, producing P_5 with four consecutive cycle vertices. Connectedness therefore implies that the whole graph is C_5. Without C_5, the graph is chordal, since every longer hole contains P_5; the simplicial deletion argument above gives $\Lambda \le 2(n-1)$.

Also every connected P_4-free, C_4-subgraph-free graph has a universal vertex. If it is not complete, choose an induced u, w, v path. Any vertex x nonadjacent to w cannot meet u or v: such a contact would give P_4 or C_4 on these four vertices. A connected P_4-free graph has diameter at most two, so take a path x, y, w. The vertex y cannot meet both u, v, by four-cycle exclusion. Choosing the one it does not meet gives an induced P_4 starting x, y, w, a contradiction. Thus w is universal.

For radius one, a universal vertex has a matching plus isolates on its neighborhood. If that matching has q edges, then $n = 1 + d$, $m = d + q$, $t = q$, and $\Lambda = 2d - q < 2n$. A noncomplete connected graph has $p \ge 3 > 1 + \ell$ in this case. A complete graph has $p = 2$, $r = \ell = 1$, except for the trivial one-vertex case, which is immediate.

At radius two, $p \ge 4$ by the universal-vertex observation. This is sufficient for $\ell \le 2$. If $2 < \ell \le 3$, the P_5-free density bound forces $p \ge 5 \ge 2 + \ell$. At radius three, (80) gives $p \ge 5$, sufficient for $\ell \le 2$. For $2 < \ell \le 3$, Lemma 14.4 excludes P_6-freeness: its only radius-three possibility is C_6, with $\ell = 2$. Hence $p \ge 6 \ge 3 + \ell$. Finally, for $r \ge 4$, (80) gives $p \ge 2r - 1 \ge r + 3 \ge r + \ell$. □

Proof of Corollary 14.10. If a four-cycle occurs, (66) asks only for $p \ge r + 1$, which follows from a geodesic. Assume henceforth that G is C_4-subgraph-free. The range $\ell \le 3$ is Lemma 14.11. Suppose $3 < \ell \le 4$. Radius one is impossible by the calculation in that lemma. For $r = 2$ or 3, Theorem 13.5 forces $p \ge 7 \ge r + \ell$. For $r = 4$, Theorem 14.3 has $\ell \le 2$, so $p \ge 8 \ge 4 + \ell$. For $r \ge 5$, (80) gives $p \ge 2r - 1 \ge r + 4 \ge r + \ell$. □

15 Closed core certificates and the final range for Conjecture 133

The finite objects in this section are proof certificates: every permitted extension is accounted for, and the extension lemmas give them force at arbitrary graph order. Explicit paths and adjacency-preserving permutations are checked directly. The certificate files and independent verifiers are included in the accompanying reproducibility package.

15.1 Four cores without an induced eight-vertex path

Let H be the graph obtained from the Heawood graph by deleting one vertex and joining its three neighbors pairwise. This is the classical Y–Δ construction; neither the operation nor the graph is claimed as new. It is the graph labeled C_{13} in the Heawood family (not the thirteen-cycle); see [41, Figure 1.3], where its ΔY successor C_{14} is identified as the Heawood graph. We use the following adjacency table, in which the unique triangle is $T = \{0, 1, 2\}$:

Vertex	Neighbors	Vertex	Neighbors
0	1, 2, 3, 9	7	3, 4, 5
1	0, 2, 4, 12	8	5, 9, 12
2	0, 1, 5, 10	9	0, 8, 11
3	0, 6, 7	10	2, 6, 11
4	1, 7, 11	11	4, 9, 10
5	2, 7, 8	12	1, 6, 8
6	3, 10, 12		

Its graph6 representation is `L{O__wC_HC@BOg`. It has $(n, m, t) = (13, 21, 1)$ and $\lambda(v) = 3$ at every vertex. The certificate verification also checks that it is C_4-subgraph-free and P_8-free.

Theorem 15.1. *Let G be connected, C_4-subgraph-free, and P_8-free. If $\lambda_G(v) \geq 3$ at every vertex, then G is exactly one of:*

(i) *the Petersen graph;*
(ii) *the Heawood graph;*
(iii) *the Heawood graph with one edge contracted;*
(iv) *H.*

The triangle-free cases (i)–(iii) *are the theorem of Chudnovsky–Stacho [42, Theorem 1]; the proof below establishes the triangle-containing case by a closed finite certificate and an arbitrary-order extension argument.*

Corollary 15.2. *Every nonempty C_4-subgraph-free P_8-free graph satisfies $\Lambda(G) < 4n$.*

15.2 A complete one-vertex extension rule

Lemma 15.3. *Let F be an induced subgraph of a C_4-subgraph-free graph G with $\lambda_G(v) \geq 3$ at every vertex. If $u \in V(F)$ satisfies $\lambda_F(u) < 3$, then some $x \in V(G) \setminus V(F)$ is adjacent to u and anticomplete to $N_F(u)$.*

Proof. Both neighborhoods of u induce matchings plus isolates. Suppose every new neighbor of u had a neighbor in $N_F(u)$. A new neighbor cannot meet two old neighbors, and an old neighbor cannot meet two new ones. Nor can any new neighbor paired with an old neighbor have an additional neighbor inside $N_G(u)$. Thus every new neighbor merely pairs with an old isolated neighbor. Replacing an isolate by a matched pair leaves the independence number unchanged. This contradicts $\lambda_G(u) \geq 3 > \lambda_F(u)$. □

The contact set $S = N_G(x) \cap V(F)$ of this new vertex satisfies

$$u \in S, \qquad S \cap N_F(u) = \varnothing, \qquad N_F(v) \cap N_F(w) = \varnothing \quad (v, w \in S,\ v \neq w). \tag{87}$$

Conversely, the last condition exactly excludes a four-cycle through x in the extension of F. Therefore the required contact sets are precisely the cliques containing u in the compatibility graph whose edges join pairs with no common neighbor, subject to the middle restriction in (87). Any deficient vertex u may be chosen; the choice need not be invariant under graph isomorphism.

15.3 The closed finite certificate

The supplemental file `p8_core_certificate.json.gz` is a finite directed certificate with the labeled triangle as its root. Each state specifies a graph by integer adjacency bitmasks. A

nonterminal state specifies a vertex u with $\lambda_F(u) < 3$ and contains one row for *every* contact set satisfying (87). Each extension $F + x$ is accompanied by either:

(a) eight distinct vertices inducing P_8; or
(b) a target state of order $|V(F)| + 1$, with an explicit adjacency-preserving permutation from $F + x$ to that state.

Every terminal state has local independence at least three at every vertex and comes with an explicit isomorphism to the displayed H.

Certificate quantity	Count
States	192
Maximum state order	25
Required one-vertex extensions	93,895
Explicit induced-P_8 rejections	93,608
Explicit retained-extension isomorphisms	287
Terminal states	1
Unaccounted extensions or cutoff frontier	0

The maximum order 25 is an observed property of this *closed* certificate, not an assumption on the order of a target graph. The builder uses no order cutoff. Canonical labeling is used only to build a compact certificate; it is not trusted by the verifier.

The separate program `verify_p8_core_certificate.py` uses only the Python standard library. It independently enumerates all contact sets in (87), requires exact equality with the listed rows, checks every induced-path witness on all vertex pairs, and checks every retained-extension permutation on all vertex pairs. It also checks the state graphs, the root, reachability, increasing order along retained arcs, and the terminal isomorphism. No graph-isomorphism or canonical-labeling package is a verification dependency. The compressed certificate has SHA256

`d7be0e33ac427e6ed0aabeef4e69efd50be0ebd7a05063a30bee4aa42fe31982.`

Lemma 15.4. *Every C_4-subgraph-free, P_8-free graph containing a triangle and having $\lambda_G(v) \geq 3$ everywhere contains an induced H.*

Proof from the certificate. Embed the root triangle in G. If an embedded state F is nonterminal, apply Lemma 15.3 at its designated vertex. The resulting new vertex has a contact set present in the certificate. Its row cannot be an induced-P_8 rejection, since that path would also be induced in G. The retained-extension permutation therefore gives an induced embedding of the specified next state in G. This increases the order by one. A finite certificate with all nonterminal extensions accounted for cannot support an infinite sequence of increasing orders. Thus a terminal state is reached, and its isomorphism gives an induced H. No bound on $|V(G)|$ is assumed. □

15.4 Attachment restrictions at the terminal graph

A second, small finite certificate is included in `weighted_core_contact_certificates.json`. The compatibility graph on the 13 vertices of H has 30 edges and no triangle; the latter is checked on all $\binom{13}{3} = 286$ triples. Consequently a nonempty contact set of a new vertex in a C_4-subgraph-free extension has size at most two. Of its $13 + 30 = 43$ possibilities, 40 have an explicit induced-P_8 witness. The only survivors are $\{0\}, \{1\}, \{2\}$; the three single-leaf extensions are also checked to be P_8-free. The same standard-library verifier checks all these facts and the rooted paths used below.

Lemma 15.5. *Every connected C_4-subgraph-free P_8-free graph containing an induced H consists of H together with, at each vertex of $T = \{0, 1, 2\}$, a possibly empty matching plus isolates whose vertices are all adjacent to that vertex of T. There are no other vertices or edges. Conversely every graph of this form is C_4-subgraph-free and P_8-free.*

Proof. The finite contact certificate shows that every exterior vertex with a neighbor in H has exactly one such neighbor, belonging to T. If y had distance two from H, choose a path $y - x - i$ with $i \in T$. The following induced six-vertex paths start at the respective roots:

i	path in H
0	$0, 1, 4, 7, 5, 8$
1	$1, 0, 3, 6, 10, 11$
2	$2, 0, 3, 6, 12, 8$

Prepending y, x gives an induced P_8, since y has no H contact and x has only i. Thus there are no distance-two vertices, and connectedness excludes further layers. Let A_i be the exterior vertices with contact $i \in T$. An edge between A_i, A_j for $i \neq j$ forms a four-cycle through ij. A three-vertex path $x - y - z$ within A_i forms the four-cycle $i - x - y - z - i$. Hence each A_i is a matching plus isolates and the classes are anticomplete.

Conversely, the attachments add only pendant edges or triangles, so there is no four-cycle. An induced path meeting H uses at most one attachment vertex at either end: traversing a matching edge and then the common root creates a chord. If both exterior endpoints have distinct roots, those roots are adjacent, so their segment in H has only that edge. If exactly one endpoint is exterior, an induced P_8 would give a rooted seven-vertex path in H, contrary to the checked single-leaf extension. Paths entirely outside H have at most two vertices. Thus the graph is P_8-free. □

Every added vertex in a proper extension from Lemma 15.5 has local independence one. Hence a graph with local independence at least three everywhere cannot properly contain H. This proves the triangle-containing case of Theorem 15.1, using Lemma 15.4. In the triangle-free case $\lambda(v) = d(v)$, and the remaining three possibilities follow from Chudnovsky–Stacho [42, Theorem 1].

Proof of Corollary 15.2. Induct on the order, allowing disconnected graphs. If some vertex has $\lambda(v) \leq 2$, equation (67) gives

$$\Lambda(G) - \Lambda(G - v) = 2\lambda(v) - t(v) \leq 4.$$

The induction hypothesis proves the strict inequality; the one-vertex base case is immediate. Otherwise every connected component is one of the four graphs in Theorem 15.1. Their pairs (n, Λ) are, respectively,

$$(10, 30), \qquad (14, 42), \qquad (13, 40), \qquad (13, 39).$$

Each satisfies $\Lambda < 4n$, and summing over components completes the proof. □

15.5 Eleven cores without an induced nine-vertex path

Table 3 specifies eleven graphs. The final column counts induced five-cycles. All tuples (n, m, t, c_5) are distinct, and therefore the models are pairwise nonisomorphic. For completeness, their graph6 strings use the following convention: the first character has code $n + 63$; the remaining characters, after subtracting 63 and writing each as six binary digits, encode the adjacency bits in the order $(0, 1), (0, 2), (1, 2), (0, 3), (1, 3), (2, 3), \ldots$, with trailing zero padding. The certificate also stores explicit adjacency arrays.

Model	graph6	n	m	t	Λ	c_5
M_0	`L{OcA?_F?b@S@o`	13	21	1	39	12
M_1	`M{OcA?_F?b@S@a@P?`	14	24	1	45	21
M_2	`O{OcA?_F?a@P@_@OaDOPI`	16	30	4	48	24
M_3	`N{OcA?_E?g@_@sAiAK_`	15	27	1	51	30
M_4	`IheA@GUAo`	10	15	0	30	12
M_5	`MheA@GUA_QCA?a@@_`	14	23	0	46	17
M_6	`LheA@GUA_QCA?J`	13	21	0	42	18
M_7	`KheA@GQA?K_M`	12	18	0	36	9
M_8	`KheA@GQA?H_Y`	12	18	0	36	8
M_9	`LheA@GQA?H?YOE`	13	20	0	40	8
M_{10}	`MhEKA?_C_Q?c?i?T?`	14	21	0	42	0

Table 3: The eleven explicitly specified models.

Theorem 15.6. *A connected C_4-subgraph-free, P_9-free graph has $\lambda_G(v) \geq 3$ at every vertex if and only if it is isomorphic to one of the eleven models in Table 3.*

Corollary 15.7. *Every nonempty C_4-subgraph-free P_9-free graph satisfies $\Lambda(G) < 4n$.*

The classification proof uses the closed finite certificate described below as a proof dependency, not merely as supplementary evidence.

15.6 Two exhaustive extension rules

The deficient-vertex extension rule is Lemma 15.3, whose proof uses only four-cycle exclusion. The new vertex's contact set $S = N_G(x) \cap V(F)$ therefore satisfies

$$u \in S, \qquad S \cap N_F(u) = \varnothing, \qquad N_F(v) \cap N_F(w) = \varnothing \quad (v, w \in S,\ v \neq w). \tag{88}$$

The last condition is necessary and sufficient to exclude a new four-cycle. If the target G is triangle-free, we additionally require S to be independent in F.

There is a separate rule when F already has local independence at least three. If F is proper in a connected target G, connectedness supplies an exterior vertex with a *nonempty* contact set in F. We must then consider every nonempty set satisfying the four-cycle condition, and the independence condition in triangle-free mode. There is no designated deficient vertex or neighborhood-anticompleteness condition in this second rule. This is why an accepting finite core cannot simply terminate the supergraph analysis.

15.7 Initial states and contact partitions

15.7.1 Six roots

A target containing a triangle starts with an induced C_3. A triangle-free target has minimum degree at least three, so it contains a cycle. A shortest cycle is induced, has length at least five, and has length at most nine: an induced cycle on at least ten vertices contains an induced P_9. Consequently the six roots are C_3 in unrestricted mode, and C_5, C_6, C_7, C_8, C_9 in triangle-free mode. The latter mode preserves triangle-freeness by requiring independent contact sets.

15.7.2 A region of possible contacts

For a state F, define

$$\mathrm{Compat}(v) = \{w \neq v : N_F(v) \cap N_F(w) = \varnothing\}.$$

In triangle-free mode, remove neighbors of v from this set as well. A node of a contact-partition tree represents disjoint sets (S, R): S contains the contacts already selected, and R contains undecided vertices still available. Selected contacts are pairwise compatible, and every vertex of R is compatible with S.

At a deficient state, with chosen vertex u, start with

$$S = \{u\}, \qquad R = \mathrm{Compat}(u) \setminus (N_F(u) \cup \{u\}).$$

At a state without a deficient vertex, start with $S = \varnothing$, $R = V(F)$; the empty final contact set will be discarded.

A binary split chooses any $v \in R$ and has exactly the children

$$\begin{aligned} &\text{include: } (S \cup \{v\},\ (R \setminus \{v\}) \cap \mathrm{Compat}(v)),\\ &\text{exclude: } (S,\ R \setminus \{v\}). \end{aligned}$$

These two regions partition all admissible completions. Incompatible contacts in the include branch are forced absent, not omitted without justification.

15.7.3 Three leaf types

There are precisely three permitted leaf types.

(i) **Forbidden-path box.** An explicit induced P_9 is supplied in the extension $F + x$ with current contact set S. If x lies on the path, no vertex of R lies on it. Thus no further contact can add a chord to that path: additional edges at x only reach vertices outside the displayed path. If x is absent, the path is entirely in F and cannot change. The entire region is therefore forbidden.

(ii) **Retained extension.** Here $R = \varnothing$ and $S \neq \varnothing$. An explicit adjacency-preserving permutation maps $F + x$ to a target state of order $|V(F)| + 1$ in the same mode.

(iii) **Empty contacts.** Here $S = R = \varnothing$, at a state with no deficient vertex. This is not a connected extension and is discarded.

Finding a path in a partial extension is not, by itself, enough to reject that region. The explicit no-repair condition in leaf type (i) is essential and is checked by the verifier.

15.8 The finite certificate and its arbitrary-order force

Each graph state either specifies a vertex of local independence at most two or is an *accepting state*. An accepting state has local independence at least three everywhere and includes an explicit isomorphism to one of the eleven models. It also includes a full contact-partition tree covering every possible proper connected extension. Thus accepting states are not unexamined endpoints.

The closed certificate has the following counts:

Quantity	Count
Graph states	13,930
Maximum state order	45
Initial roots	6
Accepting model states	11
Contact-tree nodes	442,204
Binary split nodes	214,137
Forbidden-P_9 boxes	207,299
Retained-extension permutations	20,757
Empty-contact leaves	11
Unaccounted branches or cutoff frontier	0

The builder has no graph-order or state cutoff. The maximum order 45 is an observed feature of the closed certificate, not a bound assumed for target graphs. Every canonical merge used in construction is accompanied by an exact isomorphism; no canonical-labeling package is trusted by verification.

The file `p9_core_certificate.json.gz` has SHA256

`9d76bd6ba46c60a43175f792e0e67517c4d19b22b068917eeee5e34327544e1b.`

The standard-library-only verifier checks the state graphs, deficient vertices, accepting-model maps, exact include/exclude regions, forbidden paths and their no-repair conditions, retained-extension permutations, roots, modes, and reachability. It directly verifies each model's connectivity, four-cycle exclusion, P_9-freeness, local independence, potential Λ, and induced-five-cycle count. A separate table check verifies the graph6 transcriptions against the certificate's adjacency arrays.

Proof of Theorem 15.6 from the certificate. Choose a root cycle in G as above and embed its state as an induced subgraph. Suppose a state F has been embedded. If $F = G$, then F has no deficient vertex, so its accepting-model map identifies G with a listed model.

Otherwise, if F is deficient, Lemma 15.3 supplies a new vertex with a legal contact set. If F is accepting, connectedness supplies an exterior neighbor of F with a nonempty legal contact set. Follow this exact set through the binary contact partition. A forbidden-path box is impossible: its displayed path persists under those contacts and would be induced in G. An empty-contact leaf is also impossible. Therefore a retained leaf gives an induced embedding of a state with one more vertex.

This process cannot continue indefinitely, because the certificate has finitely many states and all retained arcs increase order. Since all branches are accounted for, the process must reach $F = G$ at an accepting state. Thus G is isomorphic to a listed model, with no restriction on its initial order. Conversely, all required properties of each model are verified directly. Distinct (n, m, t, c_5) tuples establish pairwise nonisomorphism. □

15.9 Density and the original conjecture

Proof of Corollary 15.7. Induct on the order, allowing disconnected graphs. If some vertex has $\lambda(v) \le 2$, equation (67) gives

$$\Lambda(G) - \Lambda(G - v) = 2\lambda(v) - t(v) \le 4.$$

For $n > 1$ the induction hypothesis gives the strict bound $\Lambda(G) < 4n$; the one-vertex case is immediate. Otherwise every connected component is one of the eleven models, each satisfying $\Lambda < 4n$ by Table 3. Summing over components completes the proof. □

15.10 Final consolidated consequence for the original conjecture

Theorem 15.8. *The original, unrounded Conjecture 133 holds for every connected graph with $\ell(G) \le 5$. It also holds in the additional ranges $r = 2$, $\ell \le 7$, and $r = 3$, $\ell \le 6$.*

Proof. A graph containing a four-cycle is covered by the geodesic bound. Assume C_4-freeness. Corollary 14.10 settles $\ell \le 4$. For $4 < \ell \le 5$, Corollary 15.7 forces $p \ge 9$, sufficient when $r \le 4$. At radius five, Theorem 14.6 excludes P_{10}-freeness because all its graphs have $\ell \le 2$. Thus $p \ge 10 \ge 5 + \ell$. For $r \ge 6$, inequality (80) gives $p \ge 2r - 1 \ge r + 5 \ge r + \ell$. More generally, $\ell > 4$ forces $p \ge 9$, which also covers the stated radius-two and radius-three extensions. Their lower-average endpoints are already included in Corollary 14.10. □

The unrestricted inequality remains outside the conclusions of this paper. The eleven-core proof does not need an external graph classification theorem. The preceding four-core result credits the triangle-free theorem of Chudnovsky–Stacho, and its Y–Δ–Heawood graph is a classical member of the Heawood family [42, 41]. Individual known graphs and earlier parameter ranges are not counted as additional resolutions of an original catalogue item.

16 Triangle diversity and 2-domination: the remaining case of WoWII 384b

For a vertex v of a finite simple graph, write $\theta_G(v)$ for the number of triangles containing v, and put

$$T(G) = \{\theta_G(v) : v \in V(G)\}, \qquad t(G) = |T(G)|.$$

Thus $t(G)$ counts distinct triangle-incidence values; it does not count triangles. A set D is *2-dominating* if every vertex outside D has at least two neighbors in D. Its minimum cardinality is $\gamma_2(G)$. The archived statement of WoWII 384b is

$$\gamma_2(G) \le \left\lfloor n - \frac{t(G)}{2} \right\rfloor \qquad \text{for every connected graph of order } n > 2. \tag{89}$$

The results consolidated here leave this statement unresolved. They do, however, reduce any possible failure to a sharply constrained finite range. The distinction between a bound on a representative and a bound on every counterexample is essential.

Theorem 16.1 (Combined reduction for 384b)**.** *Let G be connected of order greater than two, and let h be the number of its vertices of degree at least three.*

1. *Inequality (89) holds if $h \le 10$ or $h \ge 34$.*
2. *If a counterexample exists, then a connected counterexample G' exists with $|V(G')| \le 37$ and $11 \le h(G') \le 33$. More specifically, every counterexample has such a representative with the same number of vertices of degree at least three.*
3. *In every counterexample, putting $L = \{v : d_G(v) \le 2\}$ and $H = G[V(G) \setminus L]$, one has*

$$n - \gamma_2(G) = \lceil h/2 \rceil, \qquad \gamma_2(G) = |L| + \lfloor h/2 \rfloor, \qquad \mu(H) = \lfloor h/2 \rfloor,$$

where μ denotes the matching number. For odd h, all high-degree triangle counts are distinct and at least two, and $t = h + 2$. For even h, at least $h - 1$ high vertices have distinct counts at least two, and $t \in \{h + 1, h + 2\}$.

The proof occupies the rest of this section. Universal structural arguments are given explicitly. The finite parts are integer or rational certificates with specified input coverage; they are not

numerical optimization conclusions. The verification files and their hashes are specified in Section 16.13. The terminal exclusion of $h = 35$ uses a different sampler from the preceding exclusions and cannot be obtained by treating its matching edges as independent.

16.1 Omission sets, the critical profile, and matching

Call $S \subseteq V(G)$ *omittable* if

$$d_G(v) - |N_G(v) \cap S| \geq 2 \qquad (v \in S). \tag{90}$$

Its complement is 2-dominating. We use $k = n - \gamma_2(G)$, the maximum size of an omittable set, and consistently use H for the induced high-degree graph and $h = |V(H)|$. Since n and $\gamma_2(G)$ are integers, (89) is equivalent to $t \leq 2k$.

Lemma 16.2 (Critical profile)**.** *A counterexample satisfies the assertions about k, γ_2, and triangle counts in Theorem 16.1.*

Proof. In a maximum cut, every vertex has at least half its neighbors across the cut. Every high vertex consequently has at least two crossing neighbors. The high vertices on either side are omittable, whence $k \geq \lceil h/2 \rceil$. A low vertex has triangle count zero or one, so $t \leq h + 2$. In a counterexample $2k < t \leq h + 2$. For $h = 2r$ this forces $k = r$ and $t = h + 1$ or $h + 2$; at least $t - 2 \geq h - 1$ distinct values at least two are realized by high vertices. For $h = 2r - 1$ it forces $k = r$ and $t = h + 2$; every high vertex supplies a different value at least two, and low vertices supply both zero and one. Subtracting k from $n = |L| + h$ gives the claimed formula for γ_2. □

We use the following published bound of DeLaViña, Goddard, Henning, Pepper, and Vaughan [43, Theorem 3]. We give its specialized proof to make clear that connectivity is not required when it is later applied after deleting edges.

Lemma 16.3 (Matching bound)**.** *For any finite simple graph, with H and L defined by the same degree threshold three,*

$$\gamma_2(G) \leq |L| + \mu(H). \tag{91}$$

Consequently a critical counterexample has a perfect matching in H when h is even and a matching missing exactly one vertex when h is odd.

Proof. Take a maximum cut of H and let B be its bipartite graph of crossing edges. Every high vertex has at least $\lceil d_H(v)/2 \rceil$ neighbors in B. By the bipartite matching–vertex-cover theorem, B has a vertex cover C with $|C| = \mu(B) \leq \mu(H)$. Every $v \in V(H) \setminus C$ has all its crossing neighbors in C, and therefore has at least

$$|N_G(v) \cap L| + \lceil d_H(v)/2 \rceil \geq 2$$

neighbors in $L \cup C$; the inequality follows from $d_G(v) \geq 3$. Thus $L \cup C$ is 2-dominating. In a counterexample, Lemma 16.2 and $\mu(H) \leq \lfloor h/2 \rfloor$ force equality. □

For comparison, another established result is [43, Theorem 5]: if no two vertices of degree less than two are adjacent, then $\gamma_2(G) \leq (n + \alpha(G[L]))/2$. Its hypothesis holds for connected $n > 2$. In the present notation it gives $k \geq \lceil (h + |L| - \alpha(G[L]))/2 \rceil$. The sharper structural statements below are proved directly; this published bound is not a substitute for their proofs.

16.2 Stability at equality in the maximum-cut bound

For this subsection suppose $h > 0$ and $k = \lceil h/2 \rceil$. A degree-two vertex is *good* in a cut if both its neighbors are across the cut. Define a multigraph J on $V(H)$ by giving one edge ab for each degree-two vertex with high neighbors a, b. The pair ab need not be an edge of H. At this point parallel edges are allowed in the definition.

Lemma 16.4 (Even stability)**.** *If $h = 2k$, then $G[L]$ is independent, every degree-two vertex has two high neighbors, and J is a simple matching. Every high vertex has strictly more crossing than noncrossing neighbors in every maximum cut.*

Proof. Each high side of a maximum cut is omittable, so both sides have k high vertices. A good degree-two vertex could be added to the high vertices on its own side, contradicting maximality of k. Every degree-two vertex thus has exactly one crossing neighbor; flipping it preserves maximality of the cut.

A degree-two vertex cannot be adjacent to a leaf: the leaf edge crosses, and the flip would give the leaf no crossing neighbor in a maximum cut. If two degree-two vertices were adjacent, flipping one would give the other zero or two crossing neighbors, both impossible. Two leaves cannot be adjacent in a connected graph of order greater than two. Hence $G[L]$ is independent. The high neighbors of a degree-two vertex are on opposite sides; if they were on the same side, maximizing its two incident edges would make it good.

If a high vertex a supported two such vertices x, y, replace a in its k-element high side by x, y. Both new vertices have their neighbors selected, and no retained high vertex on that side is adjacent to x or y. This yields $k + 1$ omissions, a contradiction. Thus J is a simple matching. Finally, a high vertex with equally many neighbors on each side could be flipped while preserving the maximum cut, giving a high side of size $k + 1$. This proves the strict majority assertion. □

Lemma 16.5 (Odd low-degree structure)**.** *If $h = 2k - 1$, then $G[L]$ has at most one edge. The multigraph J is simple and consists of a matching together with at most one component isomorphic to P_3. If $G[L]$ has an edge, J is a matching.*

Proof. In a maximum cut let its high sides be A, B with $|A| = k$ and $|B| = k - 1$. No good degree-two vertex lies on the side of A, and at most one lies on the side of B: good vertices on one side are independent and can be added to that high side.

Every low component is a path or isolated vertex. A low cycle would be an entire component of G, contrary to $h > 0$. A low path with an edge is either a chain of at least two degree-two vertices between high endpoints, possibly equal, or a chain of degree-two vertices ending in a leaf and attached at its other end to a high vertex. Keep the high endpoints fixed and maximize crossing edges along a chain. If a chain between high endpoints has at least three internal degree-two vertices, all its edges can cross or all but one can cross, with the possible noncrossing edge placed at an end. In either case two consecutive internal degree-two vertices are good on opposite sides, which is impossible. A pendant chain with at least two internal degree-two vertices has the same obstruction, since all its edges can cross. Thus every nontrivial low component is a single edge.

For a path $a - x - y - b$ with x, y low, opposite high colors would make both internal vertices good. High endpoints both on the side of B would permit a maximum two-crossing-edge orientation with a good vertex on the side of A. Therefore both endpoints are on the side of A, and the low edge can be oriented to give a good vertex on the side of B. A leaf–degree-two edge similarly supplies a good degree-two vertex on that side. Two low edges could be oriented independently to supply two good vertices there, a contradiction. This proves the assertion about $G[L]$.

There is no J edge inside B, and at most one inside A. Among crossing J edges, an A vertex has incidence at most one: replacing it in A by two incident low vertices would increase the omission

set. A B vertex has incidence at most two, and two B vertices cannot both have incidence two: replacing the two by their four incident low vertices gives $(k-1)-2+4=k+1$ omissions. Those four vertices are distinct by the incidence restriction on A.

If there is no edge inside A, these conditions give a matching plus at most one P_3. If there is one, its good low vertex can first be added to B. Now any B vertex with two crossing attachments would enlarge this k-element omission set. Thus crossing J edges form a matching. Adding the single edge inside A can create only a matching, a P_3, or a P_4 with high-side pattern B, A, A, B. The last case is impossible: replace the two middle A vertices by the three low vertices on its edges to get $k-2+3=k+1$ omissions.

Finally, if a low edge is present, use its good low vertex on the side of B. This rules out an edge inside A and forces crossing incidence at most one also in B, giving a matching. All replacements used here preserve domination at retained omitted high vertices: newly omitted low vertices are adjacent, within the old high omission set, only to vertices explicitly removed from that set. □

Proposition 16.6 (Cubic vertices in the even case)**.** *An even-h counterexample has between one and three vertices of full degree three. In every maximum cut, at most six high vertices have at least two neighbors outside the opposite high side. More precisely, the number of such vertices on one side is at most twice the number of cubic vertices on the other side.*

Proof. Among $h-1$ high vertices with distinct triangle counts at least two, at most two can be cubic, since a cubic count is at most three. There is at most one exceptional high vertex, proving the upper bound three.

Suppose all high degrees were at least four. In a maximum cut the strict-majority assertion of Lemma 16.4 gives at least three crossing neighbors at every high vertex. Write A, B for the two k-element high sides. If $v \in B$ had at least two neighbors outside A, then $A \cup \{v\}$ would be omittable: v is supported and each retained vertex of A loses at most one of its at least three selected neighbors. Therefore each high vertex has at most one neighbor that is low or on its own high side. In particular $H[A]$ and $H[B]$ are matchings.

For a high vertex with a same-side mate, triangles using that mate number at most k, and the remaining triangles use an edge of the opposite-side matching, numbering at most $\lfloor k/2 \rfloor$. Such a vertex has no low neighbor. A high vertex without a same-side mate has at most $\lfloor k/2 \rfloor$ triangles using two high neighbors and at most one further triangle using a low neighbor. Thus all triangle counts are at most $k + \lfloor k/2 \rfloor$, and $t \le k + \lfloor k/2 \rfloor + 1 \le 2k$ for $k \ge 2$. When $k = 1$, all counts are at most one, with the same conclusion. This contradiction proves existence of a cubic vertex.

For the final assertion, let $v \in B$ have at least two neighbors outside A. The attempted omission set $A \cup \{v\}$ must fail at some $u \in A$ adjacent to v with exactly two neighbors outside A. Strict crossing majority forces u to be cubic. Each such u can block at most two candidates v. Apply this argument on both sides and the already proved bound of three cubic vertices. Only one new high vertex is added at a time; no simultaneous two-vertex augmentation is asserted. □

16.3 Triangle kernels and the small high-degree orders

In a connected graph with $h > 0$, every triangle using low vertices has one of two forms. A *pendant triangle* has a high root and two adjacent degree-two vertices. An *edge attachment* has one degree-two vertex attached to both ends of an edge of H. Let P, Q count these two kinds of triangles.

Lemma 16.7 (Finite kernel reduction)**.** *In a critical counterexample $P + Q \le k$. Let K be the graph induced by the high vertices and all low vertices belonging to triangles. Then*

$|V(K)| = h + 2P + Q \le h + 2k$. *Set*

$$b(v) = \begin{cases} \max(3, d_K(v)), & v \in V(H), \\ 2, & v \in V(K) \setminus V(H). \end{cases}$$

Every $S \subseteq V(K)$ satisfying $b(v) - |N_K(v) \cap S| \ge 2$ for all $v \in S$ is omittable in G. For even h the edge attachments form a matching and $P = 0$. For odd h they form a matching with at most one P_3 component; if $P = 1$ they form a matching, and $P \le 1$.

Proof. Omit every edge-attachment vertex and one private vertex from every pendant triangle. These choices are independent and retain both neighbors selected, so their number is at most k. Every high vertex has full degree at least $b(v)$, every omitted vertex lies in K, and any additional vertex is kept selected. Thus the displayed inequalities imply (90). The attachment restrictions are immediate from Lemmas 16.4 and 16.5. This is a one-way safe reduction; we do not assert that every enumerated kernel is realizable as a counterexample. □

For an odd critical core, write $a_v = \theta_H(v)$, let p_v be the number of pendant triangles at v, and let Q also denote the graph of edge attachments. Then

$$\theta_G(v) = a_v + p_v + d_Q(v). \tag{92}$$

Each offset is zero or one except that at most one can be two. For the elementary $h \le 7$ checks it is harmless to enumerate the larger class of all attachment multisets with $P + Q \le k$.

Lemma 16.8 (Safe sorted-count filters)**.** *For odd h, sort the core counts $a_1 \le \cdots \le a_h$ and set $b_0 = 1$, $b_i = \max(a_i, b_{i-1} + 1)$. If offsets of the form just described can make distinct final counts at least two, then $b_i - a_i \le 2$ for every i and at most one difference equals two. For even h, after deleting a suitable one of the core counts, the same greedy construction on $h - 1$ counts must have all differences at most one.*

Proof. Consider a feasible assignment of final counts and an inversion $a_i \le a_j$ but $x_i > x_j$. If $a_i = a_j$, interchange x_i, x_j, which only interchanges offsets. If $a_i < a_j$, after interchanging both offsets are nonnegative and at most one: $x_j - a_i \le x_i - a_i - 1$ and $x_i - a_j \le x_i - a_i - 1$. Thus removing inversions does not create an extra offset two. The resulting increasing assignment dominates the greedy sequence termwise, proving the odd assertion. In the even case choose the $h - 1$ distinct high counts at least two from Lemma 16.2; all their attachment offsets are at most one, and the same interchange argument applies. □

Here is the explicit witness used in the larger small-core checks. For a permitted attachment pattern set $d_v^* = \max(3, d_H(v) + 2p_v + d_Q(v))$. Test every $R \subseteq V(H)$ with $d_v^* - |N_H(v) \cap R| \ge 2$ for $v \in R$. With R, omit one private vertex of each pendant triangle whose root is outside R, and every edge-attachment vertex whose two roots are outside R. The resulting omission set has size

$$|R| + \sum_{v \notin R} p_v + |\{uv \in E(Q) : u, v \notin R\}|. \tag{93}$$

It is safe because the new omitted low vertices are adjacent to no retained vertex of R. For a high-only witness, use $d_v^* = \max(3, d_H(v))$, which is still a lower bound on full degree.

Proposition 16.9 (Verified small-core theorem)**.** *Every connected graph with $h \le 10$ satisfies* (89)*.*

Proof. For $h = 0$, the graph is a path or cycle. All triangle counts are equal, so $t = 1$, and a degree-two vertex can be omitted. Suppose $h > 0$ and that a counterexample exists. Lemmas 16.2–16.7 give the finite verification described next.

Encode a labeled core on r vertices by the bits of its unordered pairs in lexicographic order. Triangle counts are calculated by counting edges inside each neighborhood. For $1 \le h \le 6$, enumerate all $2^{\binom{h}{2}}$ masks. For odd h, enumerate attachment multisets with $P + Q \le \lceil h/2 \rceil$; the necessary inequality

$$\sum_{v \in H} \theta_H(v) + 2k \ge h(h+3)/2$$

can first discard a core, because each attachment raises the high count sum by at most two. For even h, enumerate only matching edge attachments. Retain only patterns for which the set of high counts, together with zero and with one if a triangle attachment is present, has more than $2k$ values. Adding a value not realized by the graph enlarges this set and is a safe relaxation. In each retained kernel enumerate $(k+1)$-subsets and test the inequalities in Lemma 16.7.

For $h = 7$ use the 1,044 supplied seven-vertex masks and the same attachment procedure. Coverage of these masks is separately verified: all 7! relabelings of all masks mark every one of the 2^{21} labeled masks. For $h = 8$, extend each of these seven-vertex masks by each of the 2^7 possible neighborhoods of an eighth vertex, then enumerate matching attachments. Every eight-vertex graph is isomorphic to one of these extensions; duplicates cause no difficulty.

For $h = 9$, use the 12,346 supplied eight-vertex masks. Their coverage is certified by an explicit permutation for each of the preceding 133,632 extensions, mapping it to a listed mask. Extend by all 2^8 neighborhoods. Apply Lemma 16.8, search first for a high-only omission set of size six, and, if needed, enumerate the two attachment families in Lemma 16.7 and all 2^9 choices in (93).

For $h = 10$, Proposition 16.6 supplies a full cubic vertex. Relabel it as the tenth vertex. Its core neighborhood has at most three vertices, so only $\sum_{j=0}^{3} \binom{9}{j} = 130$ extensions per nine-vertex mask are needed. The supplied list of 274,668 nine-vertex masks is covered by explicit permutations for all $12,346 \cdot 2^8$ extensions of the eight-vertex list. Check the even filter of Lemma 16.8 for each possible exceptional vertex. Every surviving extension already has six high-only omissions using the lower degree bound $\max(3, d_H(v))$; attachments do not need to be enumerated at this step.

The complete enumerations give the following counts. “Critical” means that the triangle-count filter allows failure, not that a counterexample exists.

h	Core masks or extensions	Attachment patterns	Critical patterns	Failures
1	1	2	0	0
2	2	3	0	0
3	8	0	0	0
4	64	304	0	0
5	1,024	24,416	1,800	0
6	32,768	708,608	25,920	0
7	1,044	2,446,446	61,253	0
8	133,632	17,150,528	857,204	0
9	3,160,576	20,064,253	296,450	0
10	35,706,840	0	0	0

For $h = 7$, 170 cores pass the triangle-sum filter. For $h = 9$, 342,889 extensions pass the sorted-offset filter and 340,059 already have a high-only witness. For $h = 10$, all 3,971,627 offset-compatible extensions have a high-only witness. The last row’s zero critical attachment patterns records that no attachment search was needed. All coverage and witness checks are exact and were rerun for this consolidation. Their algorithms and hashes are recorded in Section 16.13. A failure would require an unresolved kernel, and none occurs. □

16.4 A component restriction at the lower boundary

Lemma 16.10. *In an odd-h counterexample, every component of H has at least five vertices. A component with only matching edge attachments, and no pendant triangle or fork, has at least seven vertices. In particular, if $h = 11$, then H is connected.*

Proof. All high counts are distinct and at least two. A one-vertex core component has at most one pendant increment and cannot reach two. A two-vertex component has at most one edge attachment; even a pendant increment leaves one endpoint with count at most one. A three-vertex component has all core counts zero unless it is a triangle, in which case all are one. Offsets at most one, with at most one two, cannot give three distinct counts at least two. In order four, a core count three forces K_4, where only the three values $3, 4, 5$ are possible after offsets. Otherwise core counts are at most two and there are only the values $2, 3$ and possibly one four. Thus every component has order at least five.

Consider now a component K of order five with only matching increments. Distinct corrected counts at least two have sum at least 20. If z is the number of triangles of K and M its attachment matching, then $3z + 2|M| \geq 20$. Since $|M| \leq 2$, $z \geq 6$. Deleting two edges from K_5 destroys at least five triangles, so K is K_5 or $K_5 - e$. Their count multisets are respectively 6^5 and $3^2, 5^3$; adding zero or one cannot separate a triple of equal counts.

For order six, suppose first that the largest core count is at most six. The six distinct corrected counts must then be $2, 3, 4, 5, 6, 7$, of sum 27. The equality $3z + 2|M| = 27$ gives $|M| = 0$ or 3. The first cannot produce seven, so M is perfect and the base counts are exactly $1, 2, 3, 4, 5, 6$. A vertex with base count six and degree four belongs to a K_5, forcing repeated counts at least six. Thus it is universal. Its five-vertex neighborhood W has six edges and exactly one triangle, since the total graph has seven triangles. The other five counts are $d_W(w) + \theta_W(w) = 1, 2, 3, 4, 5$. Count five forces degree four there (degree at most three plus at most one triangle is at most four), hence a universal vertex of that neighborhood. A universal vertex in a five-vertex six-edge graph belongs to two triangles, a contradiction.

If the largest count is at least seven, its vertex is universal and the six-vertex component has at least twelve edges. Its complement has at most three edges. Exhausting the possible complements gives

Complement	Triangle-count multiset in K
$6K_1$	10^6
$K_2 + 4K_1$	$6^2, 9^4$
$2K_2 + 2K_1$	$5^4, 8^2$
$P_3 + 3K_1$	$3, 6^2, 8^3$
$3K_2$	4^6
$P_3 + K_2 + K_1$	$2, 4^2, 5^2, 7$
$P_4 + 2K_1$	$3^2, 5^2, 7^2$
$K_{1,3} + 2K_1$	$1, 6^3, 7^2$
$K_3 + 3K_1$	$3^3, 7^3$

A triple of equal values cannot be separated by binary increments. In the $P_3 + K_2 + K_1$ row, four base values in $\{4, 5\}$ have only three possible corrected values. In the $P_4 + 2K_1$ row, separating all three pairs requires exactly three matched vertices, impossible. This excludes all components of orders five and six without an exceptional attachment.

There is at most one exceptional attachment component globally: either one pendant triangle, with the other attachments a matching, or one fork and no pendant triangle. An exceptional component needs at least five vertices and every other component at least seven. Two components

would therefore require at least twelve vertices. This proves connectivity when $h = 11$; it does not exclude that order. □

16.5 Triangle rank and basic probabilistic alteration

We next turn triangle diversity into degree lower bounds. Throughout, an out-of-range binomial coefficient is zero.

Lemma 16.11 (Untouched-neighbor rank bound)**.** *Let v have degree d and triangle count u in a graph. At least $2u - d(d-2)$ neighbors of v have triangle count at least u. Consequently, in a critical counterexample, the ith of the increasing distinct high counts at least two has full degree at least*

$$a_i(h) = \min\{d \geq 3 : \ \tbinom{d}{2} \geq i+1, \ d(d-2) \geq 3i+2-h\}. \tag{94}$$

For odd h these indices are $1, \dots, h$; for even h they are $1, \dots, h-1$, with one additional high vertex of degree at least three.

Proof. There are $r = \binom{d}{2} - u$ missing edges in the graph induced by $N(v)$. At most $2r$ neighbors are endpoints of missing edges. Every untouched neighbor w is adjacent to the other $d-1$ neighbors of v. The $d-1$ triangles using vw, together with the neighborhood edges not incident with w, give at least $(d-1) + (u - (d-1)) = u$ triangles at w. There are at least $d - 2r = 2u - d(d-2)$ such neighbors.

For the vertex of rank i, $u \geq i+1$. Every untouched neighbor has count at least two and is therefore high. Among the other high vertices at most $h - i$ can have count at least u: these are the later ranked vertices and, in the even case, possibly the one unranked vertex. Hence $2u - d(d-2) \leq h - i$, which gives (94). The binomial inequality is immediate. □

The simpler profile $\rho(j) = \min\{d \geq 3 : \binom{d}{2} \geq j\}$ suffices for the first range of exclusions. Select all low vertices initially, select each high vertex independently with probability p, and add each unselected high vertex having fewer than two selected neighbors. For $z = 1 - p$, the probability of adding a high vertex of full degree d is at most

$$g_d = z^{d+1} + dpz^d, \qquad g_d - g_{d+1} = dp^2 z^d > 0. \tag{95}$$

Indeed, replacing a preselected low neighbor by an independently selected artificial neighbor can only increase the bad event; with d independent neighbors the two terms are the probabilities of zero or one selected neighbor while the vertex itself is unselected. The resulting 2-dominating set proves

$$k \geq (1-p)h - \sum_{v \in H} g_{d_G(v)}. \tag{96}$$

Proposition 16.12. *Every connected graph with $h \geq 44$ satisfies (89).*

Proof. First put $p = 3/8$ and define $A_h = \sum_{j=2}^{h+1} g_{\rho(j)}$ and $B_h = g_3 + A_{h-1}$. The critical odd and even profiles bound the repair sums by A_h and B_h, respectively. To contradict the critical value of k, it suffices that $h/8 - A_h - 1/2 > 0$ in odd order or $h/8 - B_h > 0$ in even order. Exact evaluation gives

$$\frac{52}{8} - B_{52} = \frac{3607919267}{34359738368}, \qquad \frac{53}{8} - A_{53} - \frac{1}{2} = \frac{3547612629}{34359738368}.$$

When either order is increased by two, the margin increases by at least

$$\frac{1}{4} - 2g_{11} = \frac{3367232921}{17179869184} > 0,$$

because the two newly required triangle counts have degree lower bound at least eleven. This is a parity induction proving the result for every $h \geq 52$, not a finite parameter extrapolation.

For $44 \leq h \leq 51$, select a uniformly random s-subset of the high vertices. If a high vertex has full degree at least d, its repair probability is at most

$$R_0(h, s, d) = \frac{\binom{h-1-e}{s} + e\binom{h-1-e}{s-1}}{\binom{h}{s}}, \qquad e = \min(d, h-1). \tag{97}$$

To see this when some neighbors are low, fill the shortfall in actual high neighbors by distinct other high vertices. The preselected low neighbors provide at least as much support as the chosen filler vertices. If no filler position remains, the preselected support already makes the event impossible, so the displayed upper bound remains safe. Counting samples that omit the vertex and select zero or one of its e artificial neighbors gives (97).

The expected selected-high count after alteration is bounded by $s + \sum R_0(h, s, \rho(j))$, with the odd or even critical profile. The exact margins below the forbidden threshold $\lfloor h/2 \rfloor$ are

h	s	Margin	h	s	Margin
44	16	1745160/12198197	45	16	1763081/12198197
46	17	36171337/109383285	47	17	479277388/1339779509
48	17	16780745963/32154708216	49	18	337323149/617148728
50	18	75018443/101504725	51	18	78679171/101504725

Every margin is positive, completing the finite remaining range. □

16.6 Preselected low neighbors and the orders 41–43

If at least one low neighbor is already selected, the corresponding one-neighbor repair bound is

$$R_1(h, s, d) = \frac{\binom{h-1-e}{s}}{\binom{h}{s}}, \qquad e = \min(d-1, h-1), \tag{98}$$

and with two preselected low neighbors repair is unnecessary. These bounds follow from the same filling argument, reserving the credited low neighbors before filling. For any independent set I of degree-two vertices, omit $x \in I$ if all its high neighbors are initially selected; any low neighbors are kept. These omissions are simultaneous, and each costs at least

$$q(h, s) = \frac{s(s-1)}{h(h-1)} \tag{99}$$

in expected selected size. A pendant triangle permits one private vertex to be omitted when its root is sampled, with probability $p(h, s) = s/h$, and its root needs no repair while both private vertices remain selected. When the private vertex is omitted its root was initially selected, so there is no conflict between these two savings.

Lemma 16.13. *There is no counterexample with $h = 43$, or with $h = 42$ and a degree-two vertex. For $h = 41$, a counterexample would have exactly one degree-two vertex.*

Proof. For a fixed critical ρ-profile put $u_i = R_0(h, s, d_i) - R_1(h, s, d_i)$. An ordinary degree-two vertex, with two high neighbors, saves at least the sum of the two smallest u_i, in addition to its omission credit $q(h, s)$. A pendant triangle saves at least the smallest $R_0(h, s, d_i)$, together with $p(h, s)$.

Odd critical order requires a low vertex of triangle count one, hence an ordinary triangle attachment or a pendant triangle. For $h = 43$, take $s = 16$. The exact margins below 21 are

respectively 1057200/7762489 and 2799123/7762489. For $h = 42$, every degree-two vertex is ordinary by even stability; the same sample size gives margin 75065/541569 below 21.

For $h = 41$, take $s = 15$. Two nonadjacent degree-two vertices give two simultaneous credits (99); even without crediting repair reductions, the exact margin below 20 is 162283/3197836. If there are at least two degree-two vertices but none are nonadjacent, there are exactly two: three pairwise adjacent degree-two vertices would be a separate triangle component. The required low triangle count one then makes the two vertices a pendant triangle, whose margin is 28995/155992. There must be a degree-two vertex to realize the low count one. Thus only the exactly-one case remains. All four evaluations are substitutions into (97)–(99) with the specified profiles and were reproduced exactly. □

Lemma 16.14 (The leaf-only case at $h = 42$)**.** *There is no counterexample with $h = 42$ and no degree-two vertex.*

Proof. All low vertices are leaves. Criticality forces them to be present, and forces all 42 high triangle counts to be distinct and positive. These are also their core counts. In increasing order, $u_i \geq i$. Applying the untouched-neighbor argument inside H gives $d_H(v_i)(d_H(v_i) - 2) \geq 3i - 42$. Apart from a possible core-degree-two vertex at rank one, a lower profile is

$$(3^3, 4^3, 5^4, 6^5, 7^6, 8^7, 9^7, 10^5, 11^2).$$

The exception can only have count one and must have a leaf neighbor in order to be high in G.

For a uniform 16-subset of the core define

$$F_0(d) = \frac{\binom{41-d}{16} + d\binom{41-d}{15}}{\binom{42}{16}}, \qquad F_1(d) = \frac{\binom{41-d}{16}}{\binom{42}{16}}.$$

For the displayed profile the baseline is $16 + \sum F_0 = 705171683/33577278$. The least saving from one leaf neighbor is $c = \min(F_0(d) - F_1(d)) = 10120/987567$. If all core degrees are at least three, at least one leaf supplies this saving. If the rank-one exception occurs, its actual contribution is at most $F_1(2)$, and its saving against the nominal $F_0(3)$ is $F_0(3) - F_1(2) = 160/861 > c$. In either case the expected high cost is at most $21 - 295235/33577278 < 21$, contradicting criticality. □

Lemma 16.15 (The one-attachment case at $h = 41$)**.** *There is no counterexample with $h = 41$ and exactly one degree-two vertex.*

Proof. Write x for that vertex. It supplies the required low triangle count one, so its high neighbors a, b are adjacent. The low count zero forces a leaf z at a high root c. Ignore any other leaves in the repair test. Order the 41 distinct high counts increasingly as t_i, so $t_i \geq i + 1$, and put $\epsilon_i = 1$ at the two ranks a, b and zero elsewhere. The core count at rank i is $t_i - \epsilon_i$. The untouched-neighbor argument inside H now gives

$$\binom{d_i}{2} \geq i + 1 - \epsilon_i, \quad d_i(d_i - 2) \geq 3i + 2 - 41 - 3\epsilon_i, \quad d_i \geq 3 - \epsilon_i.$$

For completeness, when $\epsilon_i = 1$, a neighbor with core count at least $t_i - 1$ has full count at least that value, and at most $41 - i + 1$ other ranks can qualify; this is the extra one that produces the coefficient $-3\epsilon_i$. Let ℓ_i^ϵ be the least integer satisfying these three inequalities.

Choose a uniform 15-subset S of the core and put

$$F_r(d) = \frac{\binom{40-d}{15} + \mathbf{1}_{r=0} d\binom{40-d}{14}}{\binom{41}{15}} \quad (r = 0, 1), \qquad F_r(d) = 0 \quad (r \geq 2).$$

For each placement of the ranks $a < b$ and c, the expected high cost, after omitting x when both its roots are sampled, is at most

$$U(a,b,c) = 15 + \sum_{i=1}^{41} F_{\epsilon_i + \mathbf{1}_{i=c}}(\ell_i^{\epsilon_i}) - \frac{15 \cdot 14}{41 \cdot 40}. \tag{100}$$

Looping over the $\binom{41}{2} \cdot 41 = 33,620$ placements with integer binomial coefficients gives

$$\max U = \frac{127990267}{6395672} = 20 + \eta, \qquad \eta = \frac{76827}{6395672},$$
$$\max_{\min(a,b,c)\le 34} U = \frac{127854567}{6395672} < 20,$$
$$F_0(3) - F_0(4) = \frac{3150}{28823} > \eta, \qquad F_0(9) - F_0(10) = \frac{108675}{6395672} > \eta.$$

The first two maxima are attained at $(40, 41, 41)$ and $(28, 40, 40)$. These formulas and the finite loop completely specify the arithmetic certificate; there are only 22 placements with $U \ge 20$.

Thus a counterexample requires $a, b, c \ge 35$. The ranks at most 34 have no distinguished low support. Since $\ell_1^0 = \ell_2^0 = 3$, the first saving inequality forces their core degrees to be exactly three. Their full counts are distinct and at least two, so $t_1 = 2$ and $t_2 = 3$. Since $\ell_i^0 \le 9$ for $i \le 34$ and $\ell_{34}^0 = 9$, the second saving inequality forces all these degrees to be at most nine and the degree at rank 34 to equal nine. The exact untouched-neighbor inequality at that vertex gives $7 \ge 2t_{34} - 9(9-2)$, so $t_{34} = 35$. Its neighborhood is K_9 minus one edge. The seven untouched neighbors are precisely ranks 35 through 41, and they induce a clique.

The neighborhood of rank one, whose degree is three and count is two, is a P_3. At least one of its two nonadjacent endpoints w is outside the seven-vertex clique. Then $3 \le d_H(w) \le 9$, w and v_1 have exactly one common core neighbor, and neither is a distinguished boundary root. On the event that neither lies in S and each has exactly one neighbor in S, both are initially bad. Add w but do not add v_1. The latter now has its original selected neighbor and the added neighbor w. All other bad vertices are still added; all other vertices left outside already had two initially selected neighbors. Thus this modification is valid and saves one. Optional omission of x affects only roots in S.

For $d = d_H(w)$ the event has probability

$$\frac{\binom{39-d}{14} + (d-2)\binom{39-d}{13}}{\binom{41}{15}} \ge \frac{198375}{12791344} \qquad (3 \le d \le 9).$$

The minimum is at $d = 9$; all seven evaluations are in the exact certificate. The improved expectation is at most $20 + (153654 - 198375)/12791344 < 20$, a contradiction. □

Together, Proposition 16.12 and Lemmas 16.13–16.15 exclude every $h \ge 41$.

16.7 Matching-coupled repair with a fixed-size sample

The next improvement uses the actual perfect or near-perfect matching forced by Lemma 16.3. We first give the common support and repair argument, which is used with several sampling laws.

Lemma 16.16 (Artificial support and pair repair)**.** *Suppose a high vertex v has a degree lower bound $a(v)$, and that $r(v)$ distinct low neighbors are credited as initially selected. If $1 \le a(v) - r(v) \le h - 1$, an artificial set $A_v \subseteq V(H) \setminus \{v\}$ of size $a(v) - r(v)$ can be chosen such that*

$$|N_G(v) \cap (S \cup L)| \ge r(v) + |A_v \cap S| \qquad (S \subseteq V(H)). \tag{101}$$

When v has a partner in a fixed matching, A_v can include that actual partner. The sets need not be symmetric.

Call an endpoint outside S bad if its artificial support is at most one. Add bad endpoints, except that when both endpoints of a matching edge are bad and at least one has support one, add only one: add the zero-support endpoint if there is one, and otherwise either endpoint. Repair an unmatched bad vertex individually. The resulting set is 2-dominating.

Proof. Take actual high neighbors first, including the partner if present, and fill any shortfall with distinct other high vertices. Since $d_G(v) \geq a(v)$, the number of fillers is at most the number of uncredited low neighbors. For every S, those permanently selected neighbors provide at least as much support as the selected fillers. This proves (101). If actual high neighbors exceed the required size, choose a suitable subset instead.

When one bad endpoint is left outside, it had one initial selected neighbor and its actual partner is now added. Thus it has two selected neighbors. All other vertices left outside were already supported, and additions on other matching edges only help. Since matching edges are disjoint, all repairs are simultaneously valid. □

For a uniform s-subset of the high vertices, first take $r(v) = 0$. Let a, b be the artificial degrees of a matched pair and let c be the intersection size of their artificial neighborhoods outside the pair. Define the integer numerators

$$F_{h,s}(a) = \binom{h-1-a}{s} + a\binom{h-1-a}{s-1}, \tag{102}$$

$$\begin{aligned} Q_{h,s}(a,b,c) = {} & (a+b-2-c)\binom{h-a-b+c}{s-1} \\ & + (a-1-c)(b-1-c)\binom{h-a-b+c}{s-2}. \end{aligned} \tag{103}$$

The probability that v is bad is $F_{h,s}(a)/\binom{h}{s}$. The numerator $Q_{h,s}$ counts the event that both endpoints are outside S, each has support at most one, and at least one has support one. One selected vertex in the union of the outside neighborhoods gives the first term. Two selected vertices, one in each exclusive part, give the second. Consequently the expected pair repair cost is at most $(F_{h,s}(a) + F_{h,s}(b) - Q_{h,s}(a,b,c))/\binom{h}{s}$.

Choose rational weights β_a satisfying

$$\beta_a + \beta_b \leq Q_{h,s}(a,b,c) \quad \text{for } \max(0, a+b-h) \leq c < \min(a,b). \tag{104}$$

For a perfect matching put $\zeta = 0$; for a near-perfect matching choose $\zeta \geq \max_a \beta_a$. Summing over the actual matching, including the unmatched correction, yields the universal bound

$$\mathbb{E}(|D| - |L|) \leq s + \frac{\sum_a n_a(F_{h,s}(a) - \beta_a) + \zeta}{\binom{h}{s}}. \tag{105}$$

Weights are allowed to be negative. Their role is solely the displayed inequality, verified by exact substitution for every allowable overlap.

Proposition 16.17 (The order $h = 40$)**.** *There is no counterexample with exactly forty high vertices.*

Proof. The rank profile (94), including the one exceptional type three, has multiplicities

$$(n_3, \ldots, n_{10}) = (3, 3, 4, 5, 6, 7, 6, 6).$$

Take $s = 14$ and $D = \binom{40}{14} = 23206929840$. The following integer weights satisfy (104):

a	β_a	a	β_a
3	923276120	7	236242200
4	775290440	8	49774300
5	571290100	9	−137436500
6	413052400	10	−389725340

There are 192 overlap inequalities over 36 unordered type pairs. Their exact binomial evaluation is the finite certificate. The checker also compares the formula with 1,297 direct small-set controls. Equation (105) gives

$$\mathbb{E}(|D| - |L|) \le \frac{92365610053}{4641385968} = 20 - \frac{462109307}{4641385968} < 20,$$

contrary to the critical value $|L| + 20$. □

16.8 Credited boundary support at $h = 39$

At odd critical order the mandatory low count one supplies either a pendant triangle (case P) or an edge attachment (case Q). The mandatory low count zero supplies extra boundary support in case Q.

Proposition 16.18 (The order $h = 39$)**.** *There is no counterexample with exactly thirty-nine high vertices.*

Proof. The rank-profile multiplicities are $(2, 3, 4, 5, 6, 7, 6, 6)$ for degrees $3, \dots, 10$, and H has a near-perfect matching. If a pendant triangle exists, credit its two private selected neighbors at its root. Otherwise choose a private degree-two vertex x with adjacent high roots u, v. A low vertex of count zero exists; a path from its low component to H supplies a low vertex $z \neq x$ adjacent to a high root w. Credit x at u, v and z at w. Thus the credited support is either one at three distinct roots, or two at one of u, v and one at the other.

Choose a uniform 14-subset and use Lemma 16.16 with types (a, r), artificial-set size $a - r$. Write $D = \binom{39}{14} = 15084504396$ and

$$F_r(a) = \begin{cases} \binom{38-a}{14} + a\binom{38-a}{13}, & r = 0, \\ \binom{39-a}{14}, & r = 1, \\ 0, & r = 2. \end{cases}$$

For a pair $(a, r), (b, t)$, set $d = a - r$, $e = b - t$. If their outside artificial sets have intersection c, the pair saving has numerator

$$Q(a, r; b, t; c) = \sum_{\substack{i,j,\ell \in \{0,1\} \\ r+i+j \le 1,\ t+j+\ell \le 1 \\ r+t+i+\ell+2j \ge 1}} \binom{d-1-c}{i}\binom{c}{j}\binom{e-1-c}{\ell}\binom{39-d-e+c}{14-i-j-\ell}. \tag{106}$$

Here i, j, ℓ count selected vertices in the first exclusive part, the common part, and the second exclusive part. The constraints are exactly that both endpoints are bad and at least one has support one. This derives the formula, including its use with credited support.

In P omit one private vertex if its root is initially sampled, saving $14/39 = \binom{38}{13}/D$. In Q omit x if both roots are initially sampled, saving $\binom{37}{12}/D$. There is one additional possible exchange in Q. Suppose $w = u$, so u has support credit two, and its matching partner q is an ordinary type $(a, 0)$. On the event

$$u, q \notin S, \quad v \in S, \quad |A_q \cap S| = 1,$$

the ordinary repair would add q. Add u instead. Now q has its one initial neighbor and u, while both roots of x are selected; omit x to save one extra vertex. If $v \in A_q$ this event has numerator $\binom{38-a}{13}$, and otherwise numerator $(a-1)\binom{37-a}{12}$. Thus its numerator is at least

$$E(a) = \min\{\binom{38-a}{13}, (a-1)\binom{37-a}{12}\}. \tag{107}$$

The event is disjoint from the preceding ordinary omission event. There is at most one support-two root, so this exchange is never counted on more than one matching edge. The symmetric case $w = v$ is identical; no extra credit is used if the support-two root is unmatched or matched to the other credited root.

Use the following integer weights, all scaled by D:

a	$\beta_{a,0}$	$\beta_{a,1}$
3	383804427	0
4	364489027	0
5	291476815	0
6	203279727	0
7	104890051	−101591575
8	41134373	−212045947
9	−62663615	−284255827
10	−161677327	−323629527

Set $\zeta = 383804427$. In Q take $\beta_{a,2} = 0$; in P take $\beta_{a,2} = -\zeta$. All weights are at most ζ. Add $E(b)$ to (106) when $(r,t) = (2,0)$ in Q, and add $E(a)$ when $(r,t) = (0,2)$; call the resulting numerator $\widehat{Q}$. For every applicable pair and overlap the exact inequalities are

$$\beta_{a,r} + \beta_{b,t} \le \widehat{Q}(a,r;b,t;c).$$

There are 1,249 Q inequalities and 565 P inequalities. They are all checked by directly evaluating the finite sums of binomial coefficients.

Summing over the near-perfect matching, and writing $n_{a,r}$ for the credited-type multiplicities, yields the upper bound

$$14 + \frac{\sum_{a,r} n_{a,r}(F_r(a) - \beta_{a,r}) + \zeta - O}{D}, \qquad O = \begin{cases} \binom{37}{12}, & \text{Q}, \\ \binom{38}{13}, & \text{P}. \end{cases} \tag{108}$$

All allocations of the credited roots to the profile require 351 Q type placements and eight P type placements. Repeated types at distinct roots and coincidence of w with a Q root are included. The exact maxima of (108) are

$$\frac{5617583945}{295774596} \quad (\text{Q}), \qquad \frac{16662763795}{887323788} \quad (\text{P}),$$

attained at three distinct type-ten roots and at a type-ten pendant root, respectively. Both are below 19; the smaller strict margin is $2133379/295774596$. All other low vertices remain selected, and each low omission affects only roots that are selected after the specified repair. Hence this is a valid 2-dominating construction, contradicting the critical value $|L| + 19$. □

16.9 Independent choices on matching edges

A fixed-size vertex sample has negative dependence across every pair of vertices. A different useful distribution makes choices independent on different matching edges, while permitting at

most one selected endpoint on an edge. The support indicators within an artificial neighborhood are still not independent, and their joint distribution must be evaluated.

Use Lemma 16.16 with no explicitly credited support. For a focal matched pair let X, Y be its outside artificial support counts, capped at two, and define

$$g(x,y) = \begin{cases} 2, & x = y = 0, \\ 0, & x = y = 2, \\ 1, & \text{otherwise.} \end{cases} \tag{109}$$

If the focal pair is inactive, this is its pair-repair cost. If just its first endpoint is selected, the second already has its partner selected and needs repair exactly when $Y = 0$; the other orientation is symmetric.

We give an explicit finite classification that will also be used for the dependent sampler in Section 16.10. Color every outside vertex according to membership in the two outside artificial neighborhoods by

$$0 = (0,0), \quad A = (1,0), \quad B = (0,1), \quad C = (1,1).$$

If the artificial types are a, b and their outside intersection is c, the nonzero color populations are

$$n_A = a - 1 - c, \quad n_B = b - 1 - c, \quad n_C = c, \qquad 0 \le c < \min(a,b). \tag{110}$$

If there is an unmatched singleton, assign it any available color and remove it from that color population. The external matched edges between nonzero colors are specified by six nonnegative integers

$$m_{AA}, m_{AB}, m_{AC}, m_{BB}, m_{BC}, m_{CC}, \tag{111}$$

subject to

$$\begin{aligned} 2m_{AA} + m_{AB} + m_{AC} &\le n_A, \\ 2m_{BB} + m_{AB} + m_{BC} &\le n_B, \\ 2m_{CC} + m_{AC} + m_{BC} &\le n_C. \end{aligned}$$

Every remaining nonzero-colored vertex is paired with a zero. These six counts, the common count, and the singleton color specify all information needed for the capped distribution of (X, Y). Allowing unlimited zeros is a safe enlargement when edges are sampled independently, since zero–zero edges have no effect on that distribution. Later, when the number of active edges is fixed, this relaxation will no longer be permissible.

Proposition 16.19 (The even orders 34, 36, and 38)**.** *There is no counterexample with* $h \in \{34, 36, 38\}$.

Proof. The critical core has a perfect matching. Independently on each edge, choose neither endpoint with probability $1/5$ and either specified endpoint with probability $2/5$. Then $\mathbb{E}|S| = 2h/5$, and the focal pair's expected repair cost is

$$W = \frac{1}{5}\mathbb{E}\left[g(X,Y) + 2\mathbf{1}_{X=0} + 2\mathbf{1}_{Y=0}\right]. \tag{112}$$

For an external edge with endpoint color vectors $(i,j), (i',j')$, its probability-generating factor is

$$\frac{1 + 2x^i y^j + 2x^{i'} y^{j'}}{5}. \tag{113}$$

Multiply these factors and cap each exponent at two. This gives an exact nine-state distribution, without making independent-neighbor assumptions.

For $3 \le a \le b \le 10$, enumeration of all common counts and all six-tuples (111) gives 12,556 patterns over 36 unordered type pairs. Evaluating (113) and (112) on every pattern proves $W \le \lambda_a + \lambda_b$ for the rational potentials

a	$10^9\lambda_a$	a	$10^9\lambda_a$
3	295986688	7	64396800
4	206041600	8	43481600
5	140654080	9	30569984
6	95296000	10	25344512

The finite procedure uses integer factor coefficients and exact fractions. In addition, 234 small feasible pairings and their 27,324 sampling outcomes were directly checked against the generating functions. These controls test implementation; the six-count classification is what proves coverage.

The rank-profile multiplicities for $a = 3, \dots, 10$ and the resulting strict bounds are

h	Profile multiplicities	Expected high cost
34	$(3, 3, 4, 5, 6, 6, 5, 2)$	33195294/1953125
36	$(3, 3, 4, 5, 6, 7, 5, 3)$	6978444/390625
38	$(3, 3, 4, 5, 6, 7, 6, 4)$	36563928/1953125

Each bound is $2h/5 + \sum n_a\lambda_a$. Their margins below $h/2$ are respectively 7831/1953125, 52806/390625, and 545447/1953125, all positive. Criticality is contradicted. □

Lemma 16.20 (Odd independent-edge certificate)**.** *For an odd critical core whose rank-profile types lie in* $\{3, \dots, 10\}$*, independently choose neither endpoint of a near-perfect matching edge with probability* $1/4$ *or either specified endpoint with probability* $q = 3/8$*, and choose the unmatched vertex independently with probability* q*. Pair repair has expected total high cost at most*

$$B_h = \frac{3h}{8} + \sum_a n_a\lambda_a + \zeta, \tag{114}$$

where

a	$10^9\lambda_a$	a	$10^9\lambda_a$
3	*341385154*	*7*	*88287832*
4	*248550156*	*8*	*61517419*
5	*177279951*	*9*	*43816963*
6	*125316621*	*10*	*32792315*

and $\zeta = 107833599/10^9$*.*

Proof. For a focal pair,

$$W = \frac{1}{8}\mathbb{E}[2g(X, Y) + 3\mathbf{1}_{X=0} + 3\mathbf{1}_{Y=0}]. \tag{115}$$

The external-edge factor is $(2 + 3x^iy^j + 3x^{i'}y^{j'})/8$, and the singleton factor for color (i, j) is $(5 + 3x^iy^j)/8$. Enumerate all singleton colors and the patterns (111), allowing unlimited zero fillers. There are 37,022 patterns, again covering all 36 unordered type pairs. Exact multiplication and (115) verify $W \le \lambda_a + \lambda_b$ on every pattern.

For the unmatched vertex of type a, suppose j matching edges contain two artificial neighbors. The probability-generating function of its selected support is

$$\left(\frac{2 + 6x}{8}\right)^j \left(\frac{5 + 3x}{8}\right)^{a-2j}, \qquad 0 \le j \le \lfloor a/2 \rfloor .$$

The repair probability is $5/8$ times the sum of coefficients of degrees zero and one. Exact evaluation verifies it is at most $\lambda_a + \zeta$ for every a, j. Summing the pair inequalities and this one unmatched inequality proves (114). As controls, 333 small pairings and all their 34,938 sampling outcomes were compared with the generating-function values. □

The following special configuration supplies the extra saving needed at odd orders. It also explains why an essential edge cannot simply be treated as if its endpoints were sampled on distinct matching edges.

Lemma 16.21 (Essential triangle edge and cubic repair)**.** *Suppose a critical odd-h counterexample has a degree-two vertex x with adjacent high roots u, v. If every maximum matching of H uses uv, then some root, say u, has full degree three. Its third neighbor w is high, both uw, vw are edges, and $\theta_G(u) = 2$. In particular u has rank-profile type three.*

For a sampler that never selects both endpoints of a matching edge, the usual repair on the matched pair uv can be replaced by a repair of net cost one exactly when neither root nor w is initially selected, and net cost zero otherwise. Here net cost counts added high vertices minus the possible omission of x.

Proof. If both roots had full degree at least four, deleting uv would preserve the high set. It would reduce $\mu(H)$ because every maximum matching uses that edge. Since deleting edges cannot reduce γ_2, Lemma 16.3 would imply

$$\gamma_2(G) \le \gamma_2(G - uv) \le |L| + \mu(H - uv) < |L| + \mu(H) = \gamma_2(G),$$

a contradiction. Thus one root is cubic. Its other neighbors are x, v, w; x is adjacent only to u, v. Having at least two triangles forces $vw \in E(G)$ and gives exactly two triangles at u.

If w were low, then x, w would be two degree-two vertices with the same high roots, contrary to Lemma 16.5. There is also a direct cut argument: u has no neighbor outside this diamond, so with v fixed a maximum cut puts u, v together and x, w opposite, with four local crossing edges instead of three. The side of x, w has at least $k - 1$ high vertices and can omit both x, w, exceeding k (or already exceeds k using one if its high side has size k). Hence w is high.

If neither root is selected and w is selected, keep x and add neither root; both roots are supported by x, w. If neither root nor w is selected, add both roots and omit x, with net cost one. If exactly one root is selected, add the other and omit x, with net cost zero. In every case an omitted x has both roots selected; other pair repairs remain valid. Distinct high counts make u the rank-one vertex, and (94) gives type three. □

Proposition 16.22 (The order $h = 37$)**.** *There is no counterexample with exactly thirty-seven high vertices.*

Proof. The rank profile is $(2, 3, 4, 5, 6, 7, 5, 5)$, so Lemma 16.20 gives $B_{37} = 18.090352599$; the decimal is exact, with denominator 10^9. A mandatory low triangle exists. If it is pendant, omit one private vertex when its root is sampled, giving $B_{37} - 3/8 = 17.715352599 < 18$.

Otherwise take an edge attachment x on uv. If a maximum matching avoids uv, use it: its two roots lie on distinct matching edges or one is unmatched, and their selections are independent with probability $q = 3/8$. Omit x when both are selected, giving $B_{37} - 9/64 = 17.949727599 < 18$.

If uv belongs to every maximum matching, use Lemma 16.21. Focal inactivity has probability $1/4$, and w, lying on another edge or unmatched, is independently unselected with probability $5/8$. The special net repair costs $5/32$ in expectation. Replacing the generic pair potential gives

$$B_{37} - \lambda_3 - \min_a \lambda_a + 5/32 = 17.87242513 < 18.$$

Each case contradicts the critical value $|L| + 18$. □

16.10 Exactly thirteen active edges: the exclusion of $h = 35$

The independent sampler does not exclude $h = 35$ without an order assumption. The last round replaces it by a finite-capacity sampler with an exact number of active matching edges. We spell out its dependence rather than importing the previous probability calculation.

Proposition 16.23 (The order $h = 35$)**.** *Every connected graph with exactly thirty-five high vertices satisfies* (89)*.*

Proof. Assume a counterexample. Its core has a near-perfect matching of 17 edges and one unmatched vertex, and its rank profile is

$$(n_3, \dots, n_{10}) = (2, 3, 4, 5, 6, 7, 5, 3).$$

Choose exactly 13 of the 17 matching edges uniformly, and on each active edge choose one endpoint independently and uniformly. Select the unmatched vertex independently with probability $\vartheta = 13/32$. Then matched vertices have marginal probability 13/34, the singleton has marginal 13/32, and $\mathbb{E}|S| = 13 + 13/32$.

Use the artificial sets and pair repair of Lemma 16.16. For a focal matching edge, inactivity has probability 4/17 and leaves exactly 13 active edges among the 16 external edges. Activity has probability 13/17 and leaves exactly 12 active external edges. Its expected repair cost is

$$W = \frac{4}{17}\mathbb{E}_{13} g(X, Y) + \frac{13}{34}\big(\Pr_{12}(X = 0) + \Pr_{12}(Y = 0)\big). \tag{116}$$

The subscripts are conditioning on the number of active external edges. These two distributions are different.

Use colors (110), remove the chosen singleton color, and enumerate the six counts (111). Pair unused nonzero colors with zeros. If these nonzero external edges number E, impose $E \le 16$ and set $Z = 16 - E$. The remaining Z edges are zero–zero edges. Unlimited extra zeros would change the conditional sampling law and are not allowed here.

For the E nonzero edges form the integer polynomial

$$P(z, x, y) = \prod_{\{u,v\}} \left[2 + z(x^{i_u} y^{j_u} + x^{i_v} y^{j_v})\right], \tag{117}$$

capping the x, y exponents at two. Write $P_{j,x,y}$ for its coefficients. Conditional on r active external edges, the capped support distribution before the singleton is

$$\Pr_r(X = x, Y = y) = \frac{\sum_j P_{j,x,y} \binom{Z}{r-j}}{2^E \binom{16}{r}}. \tag{118}$$

To derive this, assign two possible orientations also to inactive edges, so all activation subsets and orientations have the common denominator $2^{16}\binom{16}{r}$. Each zero–zero edge contributes $2(1 + z)$; cancelling their factor 2^Z gives precisely (118). Finally mix the singleton color with weights 19/32 and 13/32, capping again after its possible shift.

For all $3 \le a \le b \le 10$, all common counts, all available singleton colors, and all feasible six-count patterns, exact evaluation of (116)–(118) gives $W \le \lambda_a + \lambda_b$, with

a	$10^6\lambda_a$	a	$10^6\lambda_a$
3	333547	7	71002
4	234704	8	45945
5	161174	9	30585
6	108489	10	22040

There are 37,014 patterns and 36 type pairs. Exactly eight patterns from the unlimited-zero classification exceed the finite capacity and are excluded. This is an exact coverage restriction, not a probabilistic approximation.

For an unmatched vertex of type a, let j matching edges have two marked artificial neighbors. The 17 edges comprise j doubly marked, $a-2j$ singly marked, and $17-a+j$ zero–zero edges. Apply the same polynomial method conditional on 13 active edges among all 17. Multiplying its support-zero-or-one probability by $19/32$ gives a repair cost at most $\lambda_a+\zeta$, where $\zeta = 89937/10^6$. All $0 \le j \le \lfloor a/2 \rfloor$ are checked. Summing over the near-perfect matching gives

$$B = 13 + \frac{13}{32} + \sum_a n_a \lambda_a + \zeta = \frac{8510603}{500000} = 17.021206. \tag{119}$$

The finite certificate also compares 333 small feasible patterns with direct enumeration of 17,664 fixed-active outcomes.

It remains to use the mandatory low triangle. In case P, omit one private vertex when the root is initially selected. Its selection probability is at least $13/34$, including an unmatched root, and

$$B - 13/34 = 141430251/8500000 < 17.$$

In case Q choose a private vertex x on an edge uv. If a maximum matching avoids uv, use that matching. When both roots are matched on distinct edges their joint selection probability is

$$\frac{13 \cdot 12}{17 \cdot 16} \cdot \frac{1}{4} = \frac{39}{272}.$$

If one root is unmatched, it is $(13/32)(13/34)$, which is larger. Omitting x when both roots are selected therefore gives

$$B - 39/272 = 143461501/8500000 < 17.$$

If every maximum matching contains uv, invoke Lemma 16.21. Conditional on focal inactivity, a third vertex w on another matching edge is selected with probability $13/(2 \cdot 16) = 13/32$. If w is unmatched, its independently chosen probability is the same, by design. The special net repair cost is therefore $(4/17)(19/32) = 19/136$. Replacing the generic focal-pair potential in (119) gives

$$B - \lambda_3 - \min_a \lambda_a + 19/136 = 285690523/17000000 < 17.$$

Every branch constructs a 2-dominating set with expected size below $|L| + 17$, contradicting criticality. No order bound on G was used. □

Propositions 16.12, 16.17, 16.18, 16.19, 16.22, and 16.23, together with the exclusions of 41–43, now exclude every $h \ge 34$.

16.11 Leaf trimming and matching resilience

Lemma 16.24 (One-leaf representative)**.** *Every counterexample has a connected counterexample representative with the same high-degree order and the same set of triangle counts, and with at most one leaf. If n_2 is the number of degree-two vertices in that representative, then*

$$n_2 \le \lceil h/2 \rceil + 1, \qquad n \le h + \lceil h/2 \rceil + 2. \tag{120}$$

Proof. Deleting a leaf cannot increase k: every omission set in the smaller graph lifts to the larger by leaving the deleted leaf selected. Therefore a leaf deletion preserving the high set and $T(G)$ preserves counterexample status. The cut bound and critical equality also force k to remain unchanged.

Every high vertex with triangle count at least two has at least three nonleaf neighbors. If h is odd, all high vertices have this property, so their incident leaves may be deleted freely. There is at most one leaf adjacent to a degree-two vertex, because $G[L]$ has at most one edge. Retain such a leaf if it exists. Otherwise retain one high-adjacent leaf only when needed to preserve the triangle count zero. This leaves at most one leaf, preserves every triangle count, and does not disconnect the remaining graph.

For even h, there is at most one exceptional high vertex v without the count-at-least-two guarantee. By even stability every leaf is high-adjacent. Delete leaves at the other high vertices, retaining one if needed for the value zero. At v, retain only enough leaves to keep its degree at least three; any retained leaf there already preserves zero. Since $h \geq 11$ by Proposition 16.9, connectedness gives v at least one nonleaf neighbor. The only possible need for two retained leaves is that v has exactly one nonleaf neighbor. Then v lies in no triangle. Delete v and its two leaves. All remaining $h-1$ high vertices still have their at least two triangles and remain high. A maximum cut in the remaining graph, which need not be connected, gives an omission set of at least $\lceil (h-1)/2 \rceil = h/2$ high vertices. Lift it and also omit v, retaining its two leaves. Existing omitted vertices retain their supporting neighbors from the smaller graph, and v has the two selected leaves. This gives $h/2+1 > k$ omissions, a contradiction. Thus at most one leaf is needed in even order too.

At least $n_2 - 1$ degree-two vertices are independent, by stability of $G[L]$. An independent set of degree-two vertices is omittable, so $n_2 - 1 \leq k = \lceil h/2 \rceil$. Adding at most one leaf proves (120). □

Lemma 16.25 (Matching resilience)**.** *In a critical counterexample, let $F \subseteq E(H)$ satisfy $d_G(v) - d_F(v) \geq 3$ for every high vertex. Then $\mu(H - F) = \mu(H)$.*

Proof. The high and low sets of $G - F$ are unchanged. Edge deletion cannot decrease the 2-domination number. Using the equality in Lemma 16.3,

$$|L| + \mu(H) = \gamma_2(G) \leq \gamma_2(G - F) \leq |L| + \mu(H - F) \leq |L| + \mu(H).$$

All inequalities must be equalities. The matching bound applies even when deletion disconnects the graph. □

16.12 Simultaneous low omissions and the order-37 representative

It remains to prove the representative bound. Apply Lemma 16.24 and work in the resulting connected counterexample. The completed high-order exclusions give $11 \leq h \leq 33$. If $h \leq 23$, then (120) already gives $n \leq 37$. Assume for contradiction that $n \geq 38$; since there is at most one leaf,

$$n_2 \geq 37 - h. \tag{121}$$

Lemma 16.26 (Uniform-sample exclusion for $24 \leq h \leq 32$)**.** *A trimmed counterexample of order at least 38 cannot have $24 \leq h \leq 32$.*

Proof. Use the full-degree rank profile (94), the fixed-size paired alteration (105) with $s = 13$, and the exact weights in the supplied `R21/order37_uniform_input.json`. They are rational weights for each degree type; an odd-order row has its unmatched correction as the last entry. For $h = 24, 25$ all weights are zero. For every other row they are the indicated exact weights reused from the earlier order-39 certificate. Verify each inequality (104) for every type pair and overlap; there are 1,270 such inequalities over the nine orders.

Put $q_h = 156/[h(h-1)]$ and $p_h = 13/h$. If $G[L]$ is independent, omit every degree-two vertex when both its roots are initially sampled, earning $n_2 q_h$ in expectation. If $G[L]$ has one edge,

choose a degree-two endpoint x and keep its low neighbor selected. Omit x when its one high neighbor is sampled, earning p_h. At most two degree-two vertices are on that low edge, and all others are ordinary and supply q_h each. Since this case has odd $h \geq 25$, $p_h \geq 2q_h$. Therefore the total credit is at least $n_2 q_h$ in both cases. All omissions affect only high roots that were initially selected, so they can be taken simultaneously after the pair repairs.

Using (121), subtract $(37-h)q_h$ from the high bound. The exact positive margins below $\lfloor h/2 \rfloor$ are

h	Margin	h	Margin
24	1968779/1248072	25	1274117/1300075
26	10702053/10400600	27	11190077/20058300
28	235339/367080	29	2013666/7540435
30	2452616/6653325	31	2038426/29464725
32	57723899/347373600		

For example the $h = 24$ high bound is 17593393/1248072, and its low credit is 169/46. The checker reconstructs every rank profile, evaluates all pair inequalities, and compares every displayed margin as a fraction; it does not require a graph catalogue or an optimizer. Every row contradicts criticality. □

Only $h = 33$ remains. An ordinary degree-two vertex means one with two high neighbors; these neighbors need not be adjacent. Let Q be the graph of those endpoint pairs which are edges of H. By odd stability, Q is a simple matching with at most one two-edge fork. If an ordinary vertex's endpoint pair is not in the chosen matching, the odd sampler of Lemma 16.20 omits it with probability $q^2 = 9/64$ when both roots are initially selected. The purpose of resilience is to choose one matching for all these credits at once.

Lemma 16.27 (A fork with two preselected neighbors)**.** *In the odd independent-edge sampler, suppose a full-degree-four high vertex u is incident to two edges of Q. Its profile type is three, and its two private low neighbors allow the high cost bound B_h to be reduced by at least*

$$\Delta = 321442183/10^9.$$

At least one of those two private vertices simultaneously provides an additional omission credit $q^2 = 9/64$.

Proof. The four neighbors of u are its two private degree-two vertices and two distinct high roots. Its triangle count is two or three, so distinctness puts it at rank one or two and gives profile type three for the orders under consideration. While u is unselected, both private low neighbors remain selected, and no repair at u is necessary.

If u is matched to a vertex of type a, let Y be the partner's outside artificial support. The focal pair costs at most $\frac{1}{4}\Pr(Y \leq 1) + \frac{3}{8}\Pr(Y = 0)$. Among its $a-1$ outside artificial neighbors, let j external matching edges contain two marked vertices and $m = a-1-2j$ contain a single marked vertex; a possible marked singleton has the same selection probability $q = 3/8$ as a singly marked edge. With $p = 3/4$ put

$$P_0 = (1-p)^j(1-q)^m, \quad P_1 = P_0\left(\frac{jp}{1-p} + \frac{mq}{1-q}\right),$$

$$C_a = \max_{0 \leq j \leq \lfloor (a-1)/2 \rfloor} \left[\tfrac{1}{4}(P_0+P_1) + \tfrac{3}{8}P_0\right].$$

These are exact support-zero and support-one probabilities. Direct evaluation for $3 \leq a \leq 10$ proves $\lambda_3 + \lambda_a - C_a \geq \Delta$. If u is unmatched, its whole repair potential $\lambda_3 + \zeta$ can be removed,

an even larger saving. At most one of its two incident Q edges belongs to the matching, so at least one private vertex has roots on distinct sampling units and earns q^2. It is omitted only when u was initially selected, so this omission is compatible with the use of both low neighbors to support an unselected u. □

Lemma 16.28 (The last order case)**.** *A trimmed counterexample of order at least 38 cannot have $h = 33$.*

Proof. The odd independent-edge certificate gives the exact high bound $B_{33} = 16.430458235$. If the configuration in Lemma 16.27 occurs, its upper bound becomes

$$B_{33} - \Delta - q^2 = 15.968391052 < 16,$$

which already contradicts criticality.

Suppose there is no such full-degree-four fork root. A cubic vertex incident with a Q edge has exactly two triangles, by the same neighborhood argument as in Lemma 16.21. There is at most one such cubic vertex, and it is incident with only one Q edge $e = uv$. Indeed a second incident private attachment would force two low vertices with the same endpoint pair, contrary to stability. If no such cubic vertex exists, there is no exceptional edge e.

Delete every edge of Q except the possible e. An endpoint losing one edge has full degree at least four. A fork center losing two edges has full degree at least five, because its degree-four case was already excluded. All high vertices remain high. By Lemma 16.25, a near-perfect matching exists that avoids all these deleted edges.

If this matching also avoids e, every ordinary degree-two vertex earns q^2. If a low–low edge exists, omit one of its degree-two endpoints when its high neighbor is sampled, earning q. At most two degree-two vertices lie on the edge, and $q \geq 2q^2$. Thus the simultaneous total low credit is at least $n_2 q^2$, including the low-edge case. Using $n_2 \geq 4$ gives

$$B_{33} - 4q^2 = 15.867958235 < 16.$$

If the matching contains e, use the cubic repair of Lemma 16.21. The lemma's local repair only requires the cubic marked configuration; its essential-edge hypothesis was used to force that configuration, which is already known here. Its expected net cost is $5/32$. Replace the pair potential and do not count its private vertex again. All other ordinary vertices still earn q^2, and the possible low edge is handled as above. Other low omissions affect only initially selected high roots, so they are compatible with the special repair. The resulting bound is

$$B_{33} - \lambda_3 - \min_a \lambda_a + 5/32 - 3q^2 = 15.790655766 < 16.$$

All branches contradict the critical 2-domination number. □

Completion of Theorem 16.1. The small-core certificate excludes $h \leq 10$. The independent alteration, boundary rank arguments, uniform matching repair, independent-edge sampling, and fixed-active sampling exclude every $h \geq 34$. Thus a counterexample has $11 \leq h \leq 33$ and satisfies the critical equalities. Trim it using Lemma 16.24. For $h \leq 23$ its order is already at most 37. If its order exceeded 37, Lemmas 16.26 and 16.28 would exclude every remaining possible h. The trimmed graph is therefore the required connected representative, with unchanged h and triangle-count set. □

Remark 16.29 (What has and has not been closed). The quantified conclusion is that existence of any counterexample implies existence of a connected counterexample on at most 37 vertices. The argument does not bound the order of every counterexample. It does not enumerate all graphs

on at most 37 vertices, and it does not prove (89) in the remaining range $11 \le h \le 33$. The connectivity conclusion at $h = 11$ and the even cubic filters are additional necessary conditions within that range.

16.13 Certificate structure, reproduction, and provenance

The companion directory is `verification/paired_384b/`; its subdirectories `R08`, `R09`, and so on retain the original round numbers solely to identify provenance [44]. The proofs above form one dependency chain; a theorem is not counted again when a later round strengthens its numerical boundary. The main sources are the 384b manuscripts of Rounds 8–11 and 14–22, together with the initial Round-5 reduction and the Round-10 cubic supplement. The consolidated audit JSON records the exact source paths.

The small-core data are masks and relabeling certificates. The seven- vertex catalogue is certified by enumerating all permutations and marking all 2^{21} masks. The eight-vertex coverage file assigns an explicit relabeling to each seven-mask extension, and the nine- vertex coverage file does the same for each eight-mask extension. The eight-vertex records use four-byte little-endian words with three bits per vertex image; the nine-vertex records use eight-byte little-endian words with four bits per vertex image. A coverage verifier checks that each record encodes a permutation, reconstructs the relabeled edge mask, and tests membership in the target list. It also checks record counts and exhaustion of the file. Coverage therefore does not depend on trusting a graph-isomorphism or canonical-labeling package. Duplicates in the extension families are permitted and counted.

The probability certificates have three distinct forms. The rank certificates evaluate finite binomial sums and all explicitly listed rank placements. The fixed-size pair certificates check every type-pair and overlap inequality and then sum the resulting rational weights. The matching-edge certificates enumerate the six color-pair counts and, in odd order, all singleton colors; they multiply capped integer generating functions. The fixed-active certificate also imposes the actual 16-edge external capacity and conditions on the proper active-edge count. The extra physical controls verify selected repair implementations on sampled finite graphs, but are not used as exhaustive graph coverage. In particular, the 600 controls and 46,800 weighted outcomes in `R21/check_joint_low_repairs.py` supplement the universal compatibility proofs above.

For this consolidation 21 checkers were rerun, all with exit status zero. They include all finite-kernel checks through $h = 10$, all three coverage checks, both rank exclusions, the uniform matching certificates, all three matching-edge distributions, the order-37 certificate, and the simultaneous-repair controls. The runner `run_verification.py` compiles the C++ verifiers with `g++-O3-std=c++17`, runs Python standard-library checkers, and saves each command, exit status, elapsed time, and output log in `rerun_results.json` and `rerun_logs/`. These runs use copies and do not alter archival source files. They reproduce exact finite certificates; discovery sweeps and optimization searches are not verification dependencies.

The original series successively obtained larger representative bounds, including 217, 67, 64, 43, 42, 40, and 39 before the bound 37. Those numerical consequences are subsumed by Theorem 16.1. Their reusable arguments—leaf trimming, low omission credit, rank constraints, paired repair, and matching resilience—are proved here once. The order-37 certificate also contains its original $h = 35$ calculation, with the three conditional bounds 16.993517969, 16.9162155, and 16.812700786; these exclude $h = 35$ representatives of order at least 38. The stronger Proposition 16.23 now excludes that high order without a total-order assumption.

The following full SHA-256 digests identify the data needed for catalogue coverage and the only external weight input to the final order reduction. The verifier digests and the manifest binding the remaining certificate outputs are listed immediately afterward. The complete source manifest

also records byte sizes and archival paths, while the consolidated audit records which statements are finite certificates and which are universal proofs.

Table 4: 384b certificate input and verifier digests. Paths are relative to `verification/paired_384b/`. Each two-line hexadecimal string is one complete SHA-256 digest.

File	SHA-256
`R11/core7_masks.txt`	`de8140ae11b1d575d725febccb6140ae` `faea77bbce7880d78283bb63b9e5486a`
`R11/core8_masks.txt`	`865fd27c2bb07b21c11f57a5d71d0e20` `ebcdd6beea937a043751f70cc89f456f`
`R11/core9_masks.txt`	`018a48bfc1548601c0c71c539d4fe25d` `809e05c6280ee9d3c91c1f50ada24d01`
`R11/core8_cover_permutations.bin`	`ec105831fab7638d77fae72b4d9d2e18` `1246ba3d73e4f852924e9e6521a1c781`
`R11/core9_cover_permutations.bin`	`2c6f017c01a45645f93b71fa8f13dca8` `c4eccc5fd22cca76b20e10b3b3cf7549`
`R21/order37_uniform_input.json`	`f97ee255bb05f749a9f2aeef7a83ae6c` `a99b9114f3e97355c7adc311c387cd81`
`R08/check_384b_alteration.py`	`11d05965d783e89fd2d15fce3488486d` `64b53db3c001aa38b1627a46ee16dac6`
`R08/check_384b_hypergeometric.py`	`796f885ac58b0e52e6c2db05f13b895f` `ca5276096029fdfecea0c6c51542438a`
`R08/check_384b_kernels.py`	`92dafc0831fd9f5cb3aae50739593b0d` `9ea19dd4da6e61a6441b76cbc32f5017`
`R08/verify_384b_h7.py`	`74d894388f8fe04a1478d07a6fc33427` `51a654e902f432c772c34bd8762eaf51`
`R08/verify_384b_h8.cpp`	`d674b73d21333315aefb6602c5c4f948` `4d60355dbfb1060f8c8a02651b1a07f6`
`R09/verify_384b_h9.cpp`	`d5bb333c69cd65d778744da1ac0c176d` `10b9a1714f350da6a8eb2832ea16c3a7`
`R10/check_low_omission_credit.py`	`ebe2632c4606e40093dbe11e2bac3725` `239cb24ae1eeaa517f4f42bf2ba1d95f`
`R10/check_order43.py`	`578320b635bc69f4d7cd4a0f1791db08` `fafcf0c4229860fdfcf56e69dd18a33b`
`R11/certify_core7_coverage.cpp`	`6824967f10e2d451189561b79c02ed86` `0c224ed209c741136cbfe3b296973f43`
`R11/verify_384b_h10.cpp`	`5fc5a41e4ff03ffde85bff00d1cffaa5` `08064274de7ea95735184ae36f8e68a6`
`R11/verify_core8_cover.py`	`4aae049fd7cc29b83ffef992b2e9bc34` `571736ef2b18f2633f0254bf023aec36`
`R11/verify_core9_cover.cpp`	`72cc50a60d005a531f82c888be066d98` `a238bff750f11b93de385492a1693415`
`R15/check_rank_h42.py`	`f441532e3e0cfafc9b8ac7991339dc86` `7a49e9e6a39e0309eb733645ba44ea5e`
`R16/check_rank_h41.py`	`04fc4d94918fa4065e8fcb476c3c277f` `9ce5a63cf72c26cdb8ec5337e703b0bf`
`R17/check_matching_h40.py`	`4c53641c6efce77949522a5052eea980` `883933ce4feff214965a5aba0c80198e`
`R18/check_matching_h39.py`	`b462f1b60db5d65594178c9ee2e33fb9` `125434c1be96d9b32a84df6d67c0009d`
`R19/check_paired_sampling_even.py`	`69c7bb48ec2a368f0249149211df33ca` `a7ab8ba1a509d17f1c256ed69934e05e`
`R20/check_paired_sampling_h37.py`	`de00f742fd056f53e3130759243f15c7` `687885fa0557e8eab20bcb43a09e1d78`
`R21/check_joint_low_repairs.py`	`249a18c5a7f1819bf3f6614922058bf8` `41ed5e722b22057fd88341c0c4227112`
`R21/check_order37.py`	`65770c371aa08d4182fffe66312e3131` `6a2c68d5efc036242aa836968cd79e70`
`R22/check_fixed_active_h35.py`	`3501301c351b1e86732ee7f902502c4e` `b701d65dcd9d049f2300016557d76baa`

File	SHA-256
`artifact_manifest.json`	`9cf90736538db5a592aaf9ac3ff27612 3415bb521ed089ec954b27bf6dc95b0a`
`source_manifest.json`	`f57cca0e02f5956b406969c91418bbb3 045a8c300ec35ca483571c92a7ab65a5`
`rerun_results.json`	`928bf4bf7366956b19c2558827229320 9aba2bf23d8f5660f6205fb8d80e5c27`
`run_verification.py`	`b04bb7f1e1480cde9411ffb490faf2dd db68e2cd8ae16dca4a25ffb53ca2207f`

The artifact manifest identifies every bundled source, data, and exact output JSON by full digest. The source manifest separately identifies the unchanged archival inputs, so a regenerated output can be compared with its source without conflating the two. Both are bound by their full digests in Table 4. The audit file `paired_384b_audit.json` maps each theorem to its mathematical and finite dependencies, and explicitly distinguishes this rerun from independent-audit records already present in the supplied archive.

17 Traceability in split graphs: Conjecture 199

This and the following two sections consolidate the split-graph results of Rounds 17–22. All path conclusions refer to a Hamiltonian path. The maximum induced-tree order is written τ here to distinguish it from the maximum induced-forest order used elsewhere.

17.1 Definitions and the precise scope

All graphs are finite and simple. A graph is *traceable* if it has a Hamiltonian path, that is, a path containing every vertex. Write $\alpha(G)$ for the independence number and $\kappa(G)$ for vertex connectivity. The order of a graph is its number of vertices. In particular, $\tau(G)$ is the maximum *vertex order* of a connected acyclic *induced* subgraph; it is not an induced-forest parameter.

A *split graph* has a partition $V(G) = C \dot\cup I$ in which C is a clique and I is independent. WOWII Conjecture 199, dated January 12, 2006 in the archived list [2], asserts that every connected graph of order greater than one satisfying

$$\tau(G) - 2 \le \kappa(G) \tag{122}$$

is traceable. We prove the following restricted statement.

Theorem 17.1. *Let G be a connected split graph of order greater than one satisfying (122). Then G is traceable in either of these cases:*

1. *$\kappa(G) \le 3$;*
2. *G is noncomplete, $k = \kappa(G) \ge 2$, and some split partition $V(G) = C \dot\cup I$ with I maximum satisfies*

$$k \ge 2m - 2, \qquad m = \max_{c \in C} |N(c) \cap I|.$$

Complete graphs are traceable without either additional restriction.

The proof of the second part is in Section 17.3. The first part follows from Sections 17.4–17.6. Neither part asserts that an arbitrary capacity-two factor is acyclic or can be patched into a spanning path.

17.2 Split partitions and elementary tools

Lemma 17.2. *A split graph has a split partition $C \dot\cup I$ with I a maximum independent set. In such a partition every $c \in C$ has a neighbor in I. If the graph is connected and noncomplete, then*

$$\kappa(G) = \min_{i \in I} d_C(i).$$

Writing $k = \kappa(G)$ and $A_c = N(c) \cap I$, condition (122) implies

$$d_C(i) \geq k, \qquad 1 \leq |A_c| \leq k+1, \qquad |A_c \triangle A_d| \leq k \quad (c \neq d). \tag{123}$$

Proof. Start with any split partition. If a clique vertex has no neighbor in the independent side, move it to that side. Every remaining clique vertex is adjacent to the moved vertex, so no further move is needed. When each clique vertex has an independent-side neighbor, an independent set using one clique vertex omits at least one independent-side vertex; it therefore has size at most $|I|$. An independent set uses at most one clique vertex, so I is maximum.

For a connected noncomplete graph, $|I| \geq 2$. Put $k_0 = \min_{i \in I} d_C(i)$. Removing the neighbors of a vertex attaining this minimum isolates it from another vertex of I, so $\kappa(G) \leq k_0$. Deleting fewer than k_0 vertices leaves a nonempty clique, and every remaining independent vertex retains a neighbor in it. The remaining graph is connected, proving equality.

The induced star on $\{c\} \cup A_c$ has order $1 + |A_c|$. For distinct $c, d \in C$, the induced graph on $\{c, d\} \cup (A_c \triangle A_d)$ is a double star of order $2 + |A_c \triangle A_d|$. Common independent neighbors are omitted. Applying (122) to these two trees proves (123). □

We use the following precise classical input. Chvátal and Erdős [45, Theorem 2, p. 112] proved that an s-connected graph with no independent set of size $s+2$ has a Hamiltonian path. Thus

$$\alpha(G) \leq \kappa(G) + 1 \quad \Longrightarrow \quad G \text{ is traceable.} \tag{124}$$

Their proof adjoins a universal vertex, applies their Hamiltonian-cycle theorem, and then deletes that vertex. We invoke (124) only when the size of our maximum independent side is explicitly at most $k+1$.

Lemma 17.3. *Let $C \dot\cup I$ be a split partition with I maximum. The following path operations are valid.*

1. *A path containing every vertex of I can be extended to a spanning path by inserting the unused clique vertices.*
2. *If a path has an edge ac, with $a \in I$, $c \in C$, and $x \in I$ is outside the path, let p, q be distinct unused neighbors of x in C. If ap is an edge, replace ac by a, p, x, q, c.*
3. *If the path has an edge cd within C, and p, q are distinct unused clique neighbors of a vertex $x \in I$ outside the path, replace cd by c, p, x, q, d. The resulting path still has an edge within C.*

Proof. For the first operation, an unused $d \in C$ has a neighbor $a \in I$ already on the path. Replace an incident path edge ac, $c \in C$, by a, d, c. The paths to which we apply this operation have an edge incident with each independent vertex. If the path initially consists of a single independent vertex, first append one of its clique neighbors. The other operations follow by checking the displayed incidences and the clique edges. All inserted vertices are unused, so simplicity is preserved. The edge cp in the third operation is still a clique edge. □

Lemma 17.4 (Classical path closure)**.** *If u, v are nonadjacent vertices of an n-vertex graph F and $d_F(u) + d_F(v) \geq n - 1$, then F is traceable if and only if $F + uv$ is traceable.*

Proof. Only the reverse implication needs proof. Add a universal vertex w to F, obtaining H of order $N = n + 1$. A spanning path of $F + uv$ gives a Hamiltonian cycle of $H + uv$, and $d_H(u) + d_H(v) \geq N$. If this cycle does not use uv, delete w and finish. Otherwise deleting uv gives a spanning path $u = v_1, v_2, \ldots, v_N = v$ in H. Among the $N - 1$ indices j, the sets specified by $uv_{j+1} \in E(H)$ and $v_j v \in E(H)$ have total size at least N. Some index belongs to both. Then

$$v_1, \ldots, v_j, v_N, v_{N-1}, \ldots, v_{j+1}, v_1$$

is a Hamiltonian cycle in H. Removing w gives a spanning path in F. This is the standard Bondy–Chvátal closure argument [46]. □

17.3 Small maximum independent neighborhoods

Theorem 17.5. *Under the hypotheses and notation of Lemma 17.2, suppose $k \geq 2$ and $k \geq 2m-2$, where $m = \max_{c \in C} |A_c|$. Then G is traceable.*

Proof. If $m = 1$, the clique-neighbor sets of different independent vertices are disjoint. Choose two neighbors of each independent vertex, form disjoint three-vertex paths, and concatenate them and the unused clique vertices using clique edges. Hence assume $m \geq 2$.

Choose c_0 with $A = A_{c_0}$ of size m, and put $X = I \setminus A$. Maximality of m and (123) give

$$2|A_c \setminus A| = |A_c \triangle A| + |A_c| - |A| \leq k.$$

Put $r = \lfloor k/2 \rfloor \geq 1$. For $J \subseteq X$, count incidences between J and its clique neighborhood to obtain

$$k|J| \leq r|N_C(J)|, \qquad |N_C(J)| \geq 2|J|.$$

Hall's theorem [47], applied to two copies of each vertex of X, therefore supplies two distinct clique neighbors per vertex of X, with all chosen clique vertices distinct. Indeed, a subset of the copied side with underlying set J has at most $2|J|$ vertices and neighborhood $N_C(J)$, so it satisfies Hall's condition.

Form these disjoint three-vertex paths and add every unselected clique vertex as a singleton path. This covers $C \cup X$ by paths whose endpoints are in C. The vertex c_0 is a singleton, since it has no neighbors in X. List $A = \{a_1, \ldots, a_m\}$.

Choose a neighbor $p \neq c_0$ of a_1, which exists because $d(a_1) \geq k \geq 2$. Join the singleton c_0 and the component containing p through a_1. At most one clique vertex becomes internal: p might become internal, but c_0, previously isolated, becomes an endpoint.

After t such insertions joining distinct clique-ended components, at most $2t - 1$ clique vertices are internal. Each subsequent join makes at most two former endpoints internal. Every other clique vertex is an endpoint of one component, with a singleton counted as one endpoint vertex. For $1 \leq t \leq m - 2$, the next vertex a_{t+1} has at least

$$k - (2t - 1) \geq 2m - 2 - 2t + 1 \geq 3$$

clique neighbors that are endpoints. A component has at most two endpoint vertices, so two such neighbors belong to distinct components. Join those components through a_{t+1}, maintaining the invariant.

After $m - 1$ core vertices have been inserted, the final vertex a_m has at least

$$k - (2m - 3) \geq 1$$

endpoint neighbor. Append a_m to that component. The resulting path has one independent endpoint and one clique endpoint. Concatenate every remaining clique-ended component at its clique endpoint, using clique edges. This is a spanning path. When $m = 2$, the intermediate insertion loop is empty and the same final step applies. □

17.4 Connectivity one

Assume $k = 1$. By (123), the nonempty columns have size at most two and pairwise symmetric difference at most one. If a two-element column occurs, every two-element column is identical to it and every singleton belongs to it. If none occurs, all singleton columns are identical. Every independent vertex has a neighbor, so $|I| \le 2 = k + 1$. Equation (124) proves traceability.

17.5 Connectivity two

Assume $k = 2$. The columns satisfy

$$1 \le |A_c| \le 3, \qquad |A_c \triangle A_d| \le 2, \qquad d_C(i) \ge 2. \tag{125}$$

If $|I| \le 3$, apply (124). Henceforth $|I| \ge 4$. The case $m = 1$ was handled in Theorem 17.5.

17.5.1 Two-element maximum columns

Two two-element columns must intersect. A pairwise-intersecting family of two-element sets either has a common element or is contained in a three-element ground set: from distinct sets $\{a, b\}, \{a, c\}$, a member not containing a must be $\{b, c\}$; its presence prevents any new ground element. Every singleton column belongs to every two-element column, by (125). A family with one distinct two-element member, or a family confined to three ground elements, would have $|I| \le 3$. Thus all two-element columns are $\{a, x\}$ for a fixed a, and their common intersection is exactly $\{a\}$. Singleton columns can only be $\{a\}$.

Write $I \setminus \{a\} = \{x_1, \dots, x_t\}$. Each x_j has two distinct clique neighbors p_j, q_j with column $\{a, x_j\}$, and these groups are disjoint for different j. The path

$$a, p_1, x_1, q_1, p_2, x_2, q_2, \dots, p_t, x_t, q_t$$

contains all independent vertices and ends in C. Append all unused clique vertices to obtain a spanning path.

17.5.2 Three-element maximum columns

Distinct three-element columns intersect in at least two elements. Such a family either has a common two-element subset or its union lies in a four-element set. Indeed, take distinct members $\{a, b, c\}, \{a, b, d\}$. A member not containing both a, b must be $\{a, c, d\}$ or $\{b, c, d\}$. Once such a member occurs, the intersection condition prevents any fifth element in any member. Every two-element column is a subset of every three-element column; every singleton belongs to every three-element column. A single distinct three-element type would force $|I| \le 3$.

Suppose first that all three-element columns have a common pair $\{a, b\}$. Their intersection is exactly this pair, since there are at least two distinct types. Every smaller column is a nonempty subset of $\{a, b\}$. For $I \setminus \{a, b\} = \{x_1, \dots, x_t\}$, where $t \ge 2$, choose distinct p_j, q_j with column $\{a, b, x_j\}$. The groups are disjoint, and

$$a, p_1, x_1, q_1, b, p_2, x_2, q_2, p_3, x_3, q_3, \dots, p_t, x_t, q_t$$

is a path containing I. Append all remaining clique vertices.

It remains to handle the four-element alternative, where $|I| = 4$. There are at least three clique vertices with three-element columns. Otherwise the two distinct column types would each have a private independent vertex occurring in no smaller column and of degree one. Choose distinct

t_1, t_2, t_3 with three-element columns not all identical. Their union is I and their intersection is nonempty.

For any specified x in that intersection there is an alternating seven-vertex path with endpoint x. To verify this, delete x from the three columns. On the remaining three points, up to relabelling and reordering, the multiset is either

$$\{ab, ab, bc\} \qquad \text{or} \qquad \{ab, bc, ca\}.$$

Order the columns respectively as ab, ab, bc or ca, ab, bc. Then

$$x, t_1, a, t_2, b, t_3, c \tag{126}$$

is the required alternating path.

If an unused clique vertex d has a three-element column, it sees at least one of the two independent endpoints of (126). Orient the path to end there, append d, and then append all other unused clique vertices. If all unused clique vertices have smaller columns, choose one of them, d. Its nonempty column is contained in the intersection of all three-element columns. Choose $x \in A_d$, construct (126) with endpoint x, orient it to end at x, and append d and the remaining clique vertices. If there are no unused vertices, (126) itself is spanning. This completes the connectivity-two case.

17.6 Connectivity three

Assume $k = 3$. We have

$$1 \le |A_c| \le 4, \qquad |A_c \triangle A_d| \le 3, \qquad d_C(i) \ge 3. \tag{127}$$

If $|I| \le 4$, use (124). Assume henceforth $|I| \ge 5$. Choose a maximum column $A = A_{c_0}$ of size m, and put $X = I \setminus A$. The identity

$$2|A_c \setminus A| = |A_c \triangle A| + |A_c| - |A| \le 3$$

shows that each clique vertex has at most one neighbor in X. Consequently the groups $C_x = N_C(x)$, $x \in X$, are pairwise disjoint, and each has at least three vertices. Put $C_0 = C \setminus \bigcup_{x \in X} C_x$; in particular $c_0 \in C_0$. For $c \in C_x$, let $S_c = A_c \cap A$. Comparison with A gives

$$m + 1 - |S_c| = |A_c \triangle A| \le 3, \qquad |S_c| \ge m - 2. \tag{128}$$

If $m \ge 3$, it is enough to find a path through all of A using at most one clique vertex from each C_x. At least two unused vertices remain in every group. Choose distinct unused $p, q \in C_x$; by (128), p has a core neighbor already on the path. The second operation of Lemma 17.3 inserts x. Different groups are disjoint, so this works successively for every $x \in X$. The first operation then inserts all unused clique vertices.

17.6.1 Maximum column size at most two

The case $m = 1$ was already handled. For $m = 2$, write $A = \{a, b\}$ and choose $r \in N_C(a) \setminus \{c_0\}$. The path a, r, c_0, b has a clique edge and reserves at most one vertex in each group. Use the third operation of Lemma 17.3 with two remaining neighbors of each $x \in X$. A clique edge remains after each insertion. Finally insert all unused clique vertices by its first operation.

17.6.2 Maximum column size three

Here $|X| \geq 2$. If some $d \neq c_0$ has two distinct core neighbors a, b, write $A \setminus \{a, b\} = \{c\}$. The path

$$c, c_0, a, d, b$$

contains A and reserves at most one vertex in a private group, so the absorption procedure applies.

Otherwise every clique vertex except c_0 has at most one core neighbor. Equation (128) says every private-group column is $\{a, x\}$ for some $a \in A$. Columns $\{a, x\}, \{b, y\}$ from different groups would have symmetric difference four if $a \neq b$. Since there are at least two groups, comparison with a fixed column in another group shows that all private-group columns use one common core vertex a. Write $A = \{a, b, c\}$. Both b, c have all their neighbors in C_0. Choose $p \in N_C(b) \setminus \{c_0\}$ and $q \in N_C(c) \setminus \{c_0\}$. They are distinct, because no column other than c_0 contains two core vertices. The path

$$a, c_0, b, p, q, c$$

contains A and reserves no private-group vertex. Absorb the groups at a and then insert the remaining clique vertices.

17.6.3 Four core vertices and at least two outside vertices

Assume $m = 4$ and $|X| \geq 2$. Each private column has at least two core neighbors. Suppose there are $p \in C_x$ and $q \in C_y$, $x \neq y$, such that $|S_p \cup S_q| \geq 3$. If S_p, S_q are disjoint, they are two-element sets, say $S_p = \{a, b\}$ and $S_q = \{c, d\}$; use

$$a, p, b, c_0, c, q, d.$$

Otherwise choose distinct a, b, c with $a, b \in S_p$ and $b, c \in S_q$; such a choice is possible because both sets have size at least two and their union has size at least three. For the remaining core vertex d, use

$$d, c_0, a, p, b, q, c.$$

Both paths reserve one vertex in each of two different groups. Absorption therefore applies.

If no such pair exists, all private core neighborhoods equal one and the same pair $B = \{a, b\}$. Indeed, comparison of columns from different groups forces equality of their core sets, each of size at least two; comparison with a fixed column in another group covers every column. Write $A \setminus B = \{c, d\}$. All neighbors of c, d lie in C_0. Choose $r \in N_C(c) \setminus \{c_0\}$ and $s \in N_C(d) \setminus \{c_0\}$, and choose any private-group vertex p. If $r = s$, use

$$a, p, b, c_0, c, r, d;$$

if $r \neq s$, use

$$a, p, b, c_0, c, r, s, d.$$

These paths contain A and reserve only the private-group vertex p. Again the absorption procedure completes the proof.

17.6.4 Four core vertices and one outside vertex

The final case is $m = 4$, $X = \{x\}$, and $|I| = 5$. A singleton column $\{x\}$ would differ from A in five elements, contrary to (127). Thus all of the at least three clique neighbors of x have columns of size at least two. Together with c_0, this gives at least four such columns.

Put $q = |C|$ and $n = q + 5$. For a missing incidence ic with $|A_c| \geq 2$,

$$d(i) + d(c) \geq 3 + (q - 1 + 2) = n - 1.$$

Apply Lemma 17.4 successively to make all these columns adjacent to all independent vertices. Every independent vertex then has degree at least four. Each remaining column is a nonempty singleton, so all of its missing incidences also satisfy

$$d(i) + d(c) \geq 4 + (q - 1 + 1) = n - 1.$$

Fill them using the same closure rule. The final complete split graph has five independent vertices and at least four clique vertices, and hence has a spanning path. Reverse the closure steps to obtain one in G. This completes the proof of Theorem 17.1.

18 Balanced incidence and the clique-size bound

Continue with $V(G) = C \dot\cup I$, $A_c = N(c) \cap I$, $q = |C|$, $h = |I|$, row degrees r_i, and column sizes $s_c = |A_c|$. The conditions in (123) may also be imposed with an integer lower bound k that is not necessarily the minimum row degree. We use that slightly more general incidence formulation below.

Theorem 18.1. *Let G be a connected noncomplete split graph satisfying (122). Choose a split partition with I maximum, and put $k = \kappa(G)$, $q = |C|$, and $h = |I|$. If $q \leq 2k$, then G is traceable. Moreover, $h = q$ always implies $q \leq 2k$, so every equal-size instance is traceable.*

Complete graphs are traceable without these restrictions.

For a missing incidence ic, the full degree of the clique vertex is $q - 1 + s_c$. The path-closure rule of Lemma 17.4 therefore permits its addition whenever

$$r_i + s_c \geq h. \tag{129}$$

Only incidences are added, and degrees never decrease.

18.1 The main incidence theorem

Theorem 18.2. *Let a split graph have nonempty clique columns and side sizes $1 \leq h \leq q \leq 2k$, where $k \geq 1$ is an integer. Suppose every independent vertex has degree at least k and every clique column has size at most $k + 1$. If $q = 2k$, suppose additionally that pairwise column symmetric differences have size at most k. Then the graph is traceable. For $q < 2k$ the symmetric-difference hypothesis is unnecessary.*

18.1.1 A sufficient descending order

Sort the original column sizes as

$$1 \leq s_1 \leq s_2 \leq \cdots \leq s_q, \qquad a = q - h \geq 0.$$

If $h \leq k+1$, every missing incidence satisfies $r_i + s_c \geq k+1 \geq h$, so closure immediately completes the graph. Assume henceforth $h \geq k + 2$. In particular $q - k \geq a + 2$. A sufficient condition for traceability is

$$s_j \geq j - a \qquad (1 \leq j \leq q - k). \tag{130}$$

For $j \le a+1$ this follows from nonempty columns.

To verify sufficiency, complete the columns in descending index order. For $j \ge q-k$, their sizes are at least $s_{q-k} \ge q-k-a = h-k$, so (129) follows from $r_i \ge k$. For each later index $j < q-k$, the $q-j$ larger-index columns have already been completed. Every independent vertex has degree at least $q-j$, and (130) gives

$$r_i + s_j \ge (q-j) + (j-a) = h.$$

All closure additions are valid. A complete split graph with $h \le q$ is traceable, and reversing the additions proves the criterion.

18.1.2 The deficit budget

Define

$$E = \sum_{i \in I} (r_i - k) \ge 0, \qquad W = \sum_{c \in C} (k+1-s_c) \ge 0.$$

Counting incidences yields

$$W = q(k+1) - kh - E = q + ka - E \le B := q + ka. \tag{131}$$

If (130) fails at j, then $a+2 \le j \le q-k$ and $s_j \le j-a-1$. The first j columns each have deficit at least $k+a+2-j$, so

$$W \ge f(j) := j(k+a+2-j). \tag{132}$$

The quadratic f is concave, and its minimum on $[a+2, q-k]$ occurs at an endpoint. Direct expansion gives

$$f(a+2) - B = 2k - q, \tag{133}$$
$$f(q-k) - B = (h-k-1)(2k-q). \tag{134}$$

If $q < 2k$, both differences are positive because $h \ge k+2$. Equations (131) and (132) then contradict each other. This proves Theorem 18.2 in the strict case.

18.1.3 The boundary and its two extremal patterns

Let $q = 2k$. Both endpoint differences are zero. If $E > 0$, then $W < B$ while $f(j) \ge B$, again ruling out any failure of (130). Suppose that condition fails and $E = 0$. Every independent vertex has original degree exactly k, and equality holds throughout (132). Strict concavity forces a failing index to be an endpoint of $[a+2, k]$, unless the endpoints coincide. Exactly j columns then have size $j-a-1$, and every other column has size $k+1$; any further deficit would make $W > B$.

Pattern A. For $j = a+2$, there are $a+2$ singleton columns and $2k-a-2 = h-2$ columns of size $k+1$. Since $h \ge k+2$, there are at least k of the latter. Choose a singleton $\{v\}$. Its symmetric difference with a size-$(k+1)$ column would have size $k+2$ unless that column contained v. Thus v sees every high column and its own singleton column, giving original degree at least $k+1$. This contradicts $E = 0$. Pattern A is impossible.

Pattern B. For $j = k$, put $\ell = k - a - 1 = h - k - 1 \geq 1$. There are k low columns of size ℓ and k high columns of size $k+1$. First complete every high column; its missing incidences satisfy

$$r_i + (k+1) \geq 2k + 1 \geq h.$$

Let U be the union of the original low columns. Every $i \in U$ retains at least one low-column neighbor, in addition to its k now-completed high-column neighbors. Therefore every missing incidence from U to a low column satisfies

$$r_i + \ell \geq k + 1 + \ell = h.$$

Complete all low columns to all vertices of U.

We claim $|U| \geq \ell + 1$. Otherwise all low columns equal the same set U of size ℓ. Each vertex of U originally had its k neighbors in those low columns, and had no high-column neighbor because its degree was exactly k. Every high column was therefore disjoint from U. Its symmetric difference with a low column had size $(k+1) + \ell = h > k$, a contradiction.

Now every low column has size at least $\ell + 1$, and every vertex outside U has degree at least k from the completed high columns. The remaining missing incidences satisfy

$$r_i + s_c \geq k + (\ell + 1) = h.$$

Complete them and reverse closure. If the two endpoint patterns coincide, Pattern A already rules out that configuration. This proves Theorem 18.2.

18.2 Consequences for the original split hypothesis

Lemma 18.3. *Under* (123), $h \leq q + 1$. *If* $h = q$, *then* $q \leq 2k$.

Proof. The incidence bounds give $kh \leq q(k+1)$. Counting unordered column differences by rows gives

$$\begin{aligned} \sum_{\{c,b\} \subseteq C} |A_c \triangle A_b| &= \sum_{i \in I} r_i(q - r_i) \\ &\geq k\left(qh - \sum_i r_i\right) \geq kq(h - k - 1). \end{aligned} \tag{135}$$

The same sum is at most $kq(q-1)/2$. If $h \geq q + 2$, incidence counting forces $q \geq 2k$, while (135) forces $q \leq 2k - 3$, a contradiction. Thus $h \leq q + 1$.

When $h = q$, equation (135) gives $q \leq 2k + 1$. Suppose equality holds. Both lower-bound steps in (135) must then be equalities. Hence every column has size $k + 1$, and every row degree is either k or q, the two equality possibilities in $r_i(q - r_i) \geq k(q - r_i)$. The total row excess is

$$E = q(k+1) - kq = q = 2k + 1.$$

If u rows have degree q, the same excess is $u(q-k) = u(k+1)$. This would require $(k+1) \mid (2k+1)$, which is impossible for $k \geq 1$. Therefore $q \leq 2k$. □

The remaining side-size possibility is supplied by the following explicit dependency. It is the balanced-deficit result; its complete proof is included in Subsection 18.3, so no unstated reduction to that case is assumed.

Lemma 18.4 (Balanced deficit)**.** *A split incidence graph satisfying* (123) *with* $h = q + 1$ *is traceable.*

Proof of Theorem 18.1. Lemma 17.2 supplies (123), and Lemma 18.3 gives $h \leq q+1$. If $h = q+1$, apply Lemma 18.4. Otherwise $h \leq q$, and $q \leq 2k$ permits Theorem 18.2. If $h = q$ was assumed instead, the second part of Lemma 18.3 supplies $q \leq 2k$ first. □

18.3 Proof of the balanced deficit dependency

Assume (123) and $h = q + 1$. Counting as in (135), but using this value of h, gives

$$kq(q-k) \le \sum_i r_i(q-r_i) \le \frac{kq(q-1)}{2}.$$

Thus $k \le q \le 2k-1$. Put

$$t = q-k \ge 0, \qquad d = 2k-q \ge 1, \qquad E = \sum_i (r_i - k) \ge 0.$$

The nonnegative total column deficit is

$$W = \sum_c (k+1-s_c) = q(k+1) - k(q+1) - E = t - E \le t. \tag{136}$$

Fix a column A of size s and let $E_A = \sum_{i\in A}(r_i - k)$. Including its zero self-difference, the sum of its differences with all columns is

$$\sum_b |A \triangle A_b| = \sum_{i\in A}(q-r_i) + \sum_{i\notin A} r_i = k(q+1) + s(q-2k) + E - 2E_A.$$

It is at most $k(q-1)$, and $0 \le E_A \le E$, so

$$sd \ge 2k + E - 2E_A \ge 2k - E.$$

Its own deficit $k+1-s$ is at most W, giving $E \le t - (k+1-s) = s-d-1$. Consequently every column satisfies

$$(s_c - 1)(d+1) \ge 2k. \tag{137}$$

Call a column high if $s_c \ge t+1 = q+1-k$. Its missing incidences satisfy $r_i + s_c \ge k+t+1 = h$, so complete every high column by Lemma 17.4. If ℓ columns were not high, each contributes at least $k+1-t = d+1$ to W. Hence

$$\ell(d+1) \le W \le t.$$

By (137) and $t \le k-1 < 2k$,

$$(s_c - 1)(d+1) \ge 2k > t \ge \ell(d+1),$$

so in particular every remaining column has size at least $\ell + 1$. Each independent vertex now sees the $q - \ell$ completed high columns. Thus every remaining missing incidence satisfies

$$r_i + s_c \ge (q-\ell) + (\ell+1) = q+1 = h.$$

Complete all remaining columns. The complete split graph with q clique and $q+1$ independent vertices has an alternating spanning path. Reverse closure to prove Lemma 18.4. For $k = 1$, the bound $q \le 2k-1$ gives $q = 1$, so the graph is the three-vertex path.

18.4 Padding the independent side

Corollary 18.5. *Suppose the incidence conditions (123) hold, and let $m = \max_c |A_c|$. If $0 \le b = q+1-h \le k+1-m$, the associated split graph is traceable.*

Proof. Add b independent vertices, each adjacent to every clique vertex. Every column size becomes at most $m + b \le k+1$, while all pairwise symmetric differences are unchanged. Old rows retain degree at least k, and new rows have degree $q \ge k$. The new graph has $q+1$ rows, so Lemma 18.4 gives a spanning path. Delete the added independent vertices from that path. Each internal deletion is spliced through the edge between its two clique neighbors; an endpoint deletion shortens the path. Added rows are pairwise nonadjacent, so these operations leave a simple path through every original vertex exactly once. □

19 Hall equality, exact factors, and alternating reach

We now identify the residual incidence obstruction after all the traceability cases above. An induced tree of a split graph contains at most two clique vertices: with one it is a star; with two it may use only their exclusive independent neighbors, and a maximum such tree uses all of them. Consequently

$$\tau(G) = \max(\{1\} \cup \{1 + |A_c| : c \in C\} \cup \{2 + |A_c \triangle A_d| : c \neq d\}),$$

omitting an empty collection of pairs. Thus the original premise is captured exactly by the following incidence bounds when $k = \kappa(G) = \min_i r_i$; the matrix results permit any lower bound k:

$$\begin{aligned} k \geq 1, \quad \varnothing \neq A_c \subseteq I, \quad r_i := |\{c : i \in A_c\}| \geq k, \\ |A_c| \leq k + 1, \qquad |A_c \triangle A_d| \leq k \quad (c \neq d). \end{aligned} \tag{138}$$

For $S \subseteq I$, write $N(S) = \{c : A_c \cap S \neq \varnothing\}$.

19.1 Hall inequalities and equality

The first counting reduction is included for completeness; it is not presented as a new matching theorem.

Lemma 19.1. *Under (138), every $S \subseteq I$ satisfies $|N(S)| \geq |S| - 1$. If $N(S) \neq C$, then $|N(S)| \geq |S|$. Consequently every proper subset of I satisfies ordinary Hall, and a deficient set can only be $S = I$ with $h = q + 1$.*

Proof. Fix nonempty S and put $p = |S|$, $t = |N(S)|$. Restrict to rows S and columns $N(S)$, writing $B_c = A_c \cap S$. Every row retains all its neighbors; the nonempty restricted columns still have size at most $k + 1$ and pairwise distance at most k. If $M = \sum_{i \in S} r_i$, then

$$kp \leq M \leq (k + 1)t.$$

Double counting pairs of columns separated by a row gives

$$\begin{aligned} \sum_{c<d} |B_c \triangle B_d| &= \sum_{i \in S} r_i(t - r_i) \\ &\geq k \sum_{i \in S} (t - r_i) = k(tp - M) \geq kt(p - k - 1). \end{aligned}$$

The left side is at most $kt(t - 1)/2$. If $p \geq t + 2$, the incidence bound gives $t \geq 2k$, whereas the pair count gives $t \leq 2k - 3$. This contradiction proves $p \leq t + 1$.

If $N(S) \neq C$, choose $d \notin N(S)$. Then $A_d \cap S = \varnothing$, and for every $c \in N(S)$,

$$|A_c \cap S| \leq |A_c \triangle A_d| \leq k.$$

Hence $k|S| \leq \sum_{c \in N(S)} |A_c \cap S| \leq k|N(S)|$. If instead $N(S) = C$ and $|S| > |N(S)|$, applying the first conclusion to both S and I forces $|S| = q + 1 = h$, so $S = I$. The empty-set cases are immediate. □

By Hall's marriage theorem [47], when $h \leq q$ there is a matching covering all of I. When $h = q + 1$, adjoining a new clique-side vertex adjacent to every row and applying Hall gives a matching missing at most one original independent vertex. Neither conclusion alone is a path theorem.

Theorem 19.2 (Tight proper-neighborhood sets)**.** *Suppose* (138) *holds and a nonempty* $S \subseteq I$ *satisfies*

$$|N(S)| = |S| = p, \qquad Q := N(S) \neq C.$$

Then there is a single vertex $x \in I \setminus S$ *such that*

$$\begin{gathered} I = S \,\dot\cup\, \{x\}, \\ A_d = \{x\} \quad (d \notin Q), \qquad A_c = \{x\} \cup X_c \quad (c \in Q), \qquad |X_c| = k, \quad X_c \subseteq S, \end{gathered} \tag{139}$$

and every row of S *belongs to exactly* k *of the sets* X_c*. Moreover* $p \leq 2k - 1$*.*

Proof. Equality must hold at every step of the final incidence count in Lemma 19.1. Therefore $r_i = k$ for $i \in S$ and $|A_c \cap S| = k$ for $c \in Q$. For any $d \notin Q$ and $c \in Q$,

$$|A_c \triangle A_d| = k + |(A_c \setminus S) \triangle A_d| \leq k.$$

Thus $A_c \setminus S = A_d$ for every such pair. All outside columns equal one common nonempty set B, and every inside column is B together with k rows of S. The rank bound gives $|B| \leq 1$, so $B = \{x\}$. Every row outside $S \cup \{x\}$ would have degree zero, proving (139). The remaining row and column counts are both k.

In this square p-by-p block, counting differences gives

$$pk(p-k) = \sum_{c<d\in Q} |X_c \triangle X_d| \leq kp(p-1)/2.$$

Since $p \geq k \geq 1$, division yields $p \leq 2k - 1$. □

Conversely, a system of the form (139) whose square block is k-regular and has pairwise column distances at most k satisfies (138). A block column and an outside singleton differ in exactly k rows. The theorem therefore describes the equality structure exactly.

19.2 Eliminating every nonempty Hall-equality set

Proposition 19.3. *Under* (138)*, the split graph is traceable if* $h \leq q \leq 2k$ *or* $h = q + 1$*. When* $h = q$*, the hypotheses imply* $q \leq 2k$*.*

Proof. These are Theorem 18.2 and Lemmas 18.4 and 18.3, whose complete proofs appear above. □

Corollary 19.4. *Under* (138)*, if some nonempty* $S \subseteq I$ *satisfies* $|N(S)| = |S|$*, then the split graph is traceable.*

Proof. First suppose $Q = N(S) \neq C$. Use Theorem 19.2 and retain the p columns of Q together with one outside column d. The retained graph has

$$|C'| = p + 1 = |I| \leq 2k.$$

Every row in S retains its k neighbors, and x has $p + 1 \geq k$ neighbors. The other bounds are inherited, so Proposition 19.3 gives a Hamiltonian path in the retained graph.

The column d is adjacent to x. Choose an edge incident with d on that path. Its other endpoint is either x or a clique vertex. Every omitted column is adjacent to both endpoints, and all omitted columns are mutually adjacent. Replace that edge by a path through all omitted columns.

If $N(S) = C$, then $|S| = q$ and Lemma 19.1 gives $h \leq q + 1$. The cases $h = q$ and $h = q + 1$ follow from the corresponding explicit inputs in Proposition 19.3. □

Corollary 19.5. *Any nontraceable split graph satisfying the original WOWII 199 premise, with the maximum-independent-set partition fixed above, would satisfy*

$$|N(S)| \geq |S| + 1 \qquad (\varnothing \neq S \subseteq I).$$

Proof. The balanced case $h = q + 1$ is traceable by Proposition 19.3. Otherwise Lemma 19.1 gives ordinary Hall on every set, and Corollary 19.4 excludes equality on any nonempty set. □

19.3 Strict Hall and an exact degree-two factor

Theorem 19.6. *Assume* (138)*, $k \geq 3$, and strict Hall expansion on every nonempty row subset. Then the bipartite incidence graph has a spanning subgraph F such that*

$$d_F(i) = 2 \quad (i \in I), \qquad 0 \leq d_F(c) \leq 2 \quad (c \in C).$$

Every nontrivial component of F is an alternating cycle or a path with both endpoints in C; isolated clique vertices may also occur.

Proof. We first prove the exact capacity inequalities

$$\mathrm{cap}_2(S) := \sum_{c \in C} \min(2, |A_c \cap S|) \geq 2|S| \qquad (S \subseteq I). \tag{140}$$

The empty-set case is immediate. For a singleton the left side is its degree, at least $k \geq 3$. Suppose $|S| \geq 2$. If no restricted column has size one, the left side is $2|N(S)| \geq 2|S| + 2$.

Otherwise choose d with $A_d \cap S = \{v\}$, put $S' = S \setminus \{v\}$, and write $r'_c = |A_c \cap S'|$. Restriction does not increase symmetric difference. Comparing a restricted column with $\{v\}$ gives $0 \leq r'_c \leq k$ for every c. Hence

$$\mathrm{cap}_2(S') = \sum_c \min(2, r'_c) \geq \frac{2}{k} \sum_c r'_c \geq 2|S'| = 2|S| - 2. \tag{141}$$

Adding v increases the contribution of column d from zero to one, so $\mathrm{cap}_2(S) \geq \mathrm{cap}_2(S') + 1$. If (140) failed, integrality would force equality throughout (141). For $k \geq 3$, the inequality $\min(2, r) \geq 2r/k$ is strict at every integer $1 \leq r \leq k - 1$. Consequently every positive r'_c equals k, and every row of S' has degree exactly k. Thus

$$k|N(S')| = \sum_c r'_c = k|S'|,$$

contradicting strict Hall on the nonempty set S'.

To construct the factor, use a network with source-to-row arcs of capacity two, incidence arcs from rows to columns of capacity one, and column-to-sink arcs of capacity two. Fixing the source-side row set S, the minimum cut capacity after choosing the source-side columns is

$$2(|I| - |S|) + \sum_c \min(2, |A_c \cap S|) \geq 2|I|.$$

The integral max-flow min-cut theorem [48] gives a flow of value $2|I|$. Its selected incidence edges have exactly the stated degrees. Since the selected graph has maximum degree two, its nontrivial components are paths or cycles. No independent vertex can be a path endpoint because all its selected degrees equal two. □

The lower bound $k \geq 3$ cannot be dropped from this factor statement. For $k = 2$, take rows $I = \{v, a, b\}$ and columns

$$\{v\}, \quad \{v, a, b\}, \quad \{a\}, \quad \{b\}.$$

The row minimum is two, the maximum column size is three, the maximum pairwise column distance is two, and strict Hall holds. But $\mathrm{cap}_2(I) = 1 + 2 + 1 + 1 = 5 < 6$, so an exact factor of the stated kind does not exist. This split graph is nevertheless traceable: labeling its columns c_0, c_1, c_2, c_3 in the displayed order, one spanning path is

$$a, c_2, c_0, v, c_1, b, c_3.$$

It is not a counterexample to the original conjecture.

19.4 Hypotheses and explicit inputs

Let I and C be finite disjoint nonempty sets, with $h = |I|$ and $q = |C|$. For each $c \in C$, let $\varnothing \neq A_c \subseteq I$. The associated split graph makes C a clique and I independent, with incidence neighborhoods A_c. Assume an integer $k \geq 3$ satisfies

$$r_i := |\{c : i \in A_c\}| \geq k, \qquad |A_c| \leq k + 1, \qquad |A_c \triangle A_d| \leq k \quad (c \neq d). \tag{142}$$

Write $N(S) = \{c : A_c \cap S \neq \varnothing\}$, and assume strict Hall:

$$|N(S)| \geq |S| + 1 \qquad (\varnothing \neq S \subseteq I). \tag{143}$$

In particular $q \geq h + 1$. Define

$$\mathrm{cap}_2(S) = \sum_{c \in C} \min(2, |A_c \cap S|).$$

The degree-two factor theorem (Theorem 19.6 and Corollary 19.5) establishes from (142)–(143) that $\mathrm{cap}_2(S) \geq 2|S|$ for every $S \subseteq I$, and that the incidence graph has a spanning subgraph F satisfying

$$d_F(i) = 2 \quad (i \in I), \qquad d_F(c) \leq 2 \quad (c \in C). \tag{144}$$

We use this exact factor theorem as an explicit input. A second input is that (142) implies traceability when $h \leq q \leq 2k$ (Theorem 18.2). No factor is silently regarded as a Hamiltonian path.

For connection with the original problem, let $\tau(G)$ be the maximum *vertex order of a connected induced tree* and $\kappa(G)$ the vertex connectivity. WOWII 199 asks whether a connected graph of order greater than one with $\tau(G) \leq \kappa(G) + 2$ must be traceable [2]. In a noncomplete split graph, choosing I maximum makes all columns nonempty and gives $k = \kappa(G) = \min_{i \in I} d_C(i)$. Induced stars and double stars then give (142). Corollary 19.5 shows that an unresolved nontraceable split instance must satisfy (143). The results below address this residual incidence formulation only.

19.5 Capacity equality

Lemma 19.7. *Under (142)–(143), if a nonempty $S \subseteq I$ satisfies $\mathrm{cap}_2(S) = 2|S|$, then $S = I$. Moreover, some whole column of size $k + 1$ is contained in S.*

Proof. There must be a restricted singleton column. Otherwise every nonzero $|A_c \cap S|$ is at least two and $\mathrm{cap}_2(S) = 2|N(S)| \geq 2|S| + 2$. If all restricted columns had size at most k, then $\min(2, r) \geq 2r/k$, with strict inequality at $r = 1$ since $k \geq 3$, would give

$$\mathrm{cap}_2(S) > \frac{2}{k} \sum_c |A_c \cap S| \geq 2|S|,$$

a contradiction. Thus some restricted column has size $k + 1$. The global rank bound makes it a whole column A contained in S.

Every column B meets S, because otherwise $|A \triangle B| > k$. If $B \cap S = \{v\}$, comparison with A first forces $v \in A$ and then forces $B = \{v\}$: an extra vertex outside S would increase the difference beyond k.

The exact factor (144) uses $2|S|$ incidences at S, equal to the sum of their column capacities. Hence every column meeting S at least twice uses both selected slots on S. A column meeting S once is a global singleton and has no outside neighbor. All columns meet S, so no row outside S could have selected degree two. Therefore $S = I$. □

Theorem 19.8. *Under* (142)–(143)*, global capacity equality* $\mathrm{cap}_2(I) = 2h$ *implies* $q \leq 2k$.

Proof. By Lemma 19.7, there is a column A of size $k + 1$. Let L be the number of singleton columns and H the number of other columns. Put $d = q - h \geq 1$. From $L + 2H = 2h$ and $L + H = q$ we obtain

$$L = 2d, \qquad H = h - d. \tag{145}$$

Every singleton label lies in A, by comparison with its $k + 1$ vertices. Let $t = h - k - 1$ count the rows outside A. For any column B,

$$2|B \setminus A| = |B \triangle A| + |B| - |A| \leq k.$$

The outside rows have no singleton neighbors, and one of the H nonsingleton columns is A itself. Thus

$$kt \leq (H - 1)\lfloor k/2 \rfloor \leq (H - 1)k/2. \tag{146}$$

Substituting $H = k + 1 + t - d$ gives $t \leq k - d$, hence

$$q = k + 1 + t + d \leq 2k + 1. \tag{147}$$

Suppose $q = 2k + 1$. Then $t = k - d$ and $H - 1 = 2t$. If $t = 0$, there is only one nonsingleton column and $L = 2k$. The total incidence count is then $3k + 1 < k(k + 1)$ for $k \geq 3$, impossible. Thus $t > 0$. Equality throughout (146) forces k even and every nonsingleton column other than A to have $k/2$ rows outside A. For each such column B, the distance condition gives

$$|B \cap A| \geq |B \setminus A| + 1 = k/2 + 1.$$

The rank bound forces $|B| = k + 1$. Therefore all H nonsingleton columns have size $k + 1$. The total row excess over the degree lower bound is

$$E = (k + 1)H + L - kh = 2k + 1 - kd \geq 0. \tag{148}$$

Since k is even and at least four, $d \leq 2$.

Let U be the set of singleton labels, with $u = |U|$. Comparison with each singleton shows that every nonsingleton column contains all of U. Also $U \subseteq A$, so $t > 0$ implies $I \setminus U \neq \varnothing$. The latter set has neighbors only among the H nonsingleton columns. Strict Hall gives

$$h - u + 1 \leq H = h - d, \qquad \text{hence } u \geq d + 1.$$

The rows of U alone contribute at least $u(H-k)+L$ to the total excess. Here $H-k = k+1-2d > 0$. Consequently

$$E \geq (d+1)(k+1-2d) + 2d.$$

The right side exceeds (148) by $(2d-1)(k-d) > 0$, a contradiction. Equality in (147) is impossible, so $q \leq 2k$. □

Corollary 19.9. *Under* (142)–(143)*, if* $q > 2k$*, then*

$$\mathrm{cap}_2(S) \geq 2|S| + 1 \qquad (\varnothing \neq S \subseteq I).$$

Proof. The earlier exact factor theorem gives the non-strict bound. A proper nonempty set cannot attain equality by Lemma 19.7, and the whole set cannot attain equality by Theorem 19.8. Use integrality. □

19.6 Directed alternating reach

Theorem 19.10. *Assume* (142)–(143) *and* $q > 2k$*, and fix any factor satisfying* (144)*. Orient nonfactor incidence edges from* I *to* C*, and factor edges from* C *to* I*. From the vertices of any factor cycle, a directed path reaches a clique vertex of selected degree less than two.*

Proof. Call a clique vertex *free* if its selected degree is less than two. Start reachability from all vertices of a factor cycle. Let $X \subseteq I$ and $Y \subseteq C$ be the reached sets, and suppose no free clique vertex is reached. Every $c \in Y$ then has selected degree two, with both selected neighbors in X. Every graph edge from X to a clique vertex outside Y must be a factor edge, or that vertex would be reached. Therefore, for every $c \in C$,

$$\min(2, |A_c \cap X|) = d_F(c, X).$$

For $c \in Y$, both sides are two; outside Y, all incidences to X are selected and there are at most two. Summing yields $\mathrm{cap}_2(X) = \sum_c d_F(c, X) = 2|X|$, contradicting Corollary 19.9, since $X \neq \varnothing$. □

The argument uses strict capacity, not ordinary strict Hall. For example, with rows $\{a, b, c, d\}$ and columns $\{a\}, \{b\}, \{c, d\}, \{a, b, c, d\}, \{a, b, c, d\}$, the hypotheses (142)–(143) hold with $k = 3$ and $q = 5$. A factor can consist of the cycle on rows c, d and the third and fourth columns, and the path through the first column, a, the fifth column, b, and the second column. The directed reach from this cycle stops short of both free path endpoints. Here $\mathrm{cap}_2(I) = 2h$ and $q \leq 2k$, as permitted by Theorem 19.8.

19.7 A shortest switch need not reduce the cycle count

Theorem 19.10 is not an absorption theorem. The following example satisfies even the original induced-tree premise, strict capacity, and $q > 2k$, but an arbitrary shortest switch can create a replacement cycle.

Let $I = \{x, y, a, b, c\}$ and let $C_0, \ldots, C_6$ form a clique, with

$$\begin{aligned} A_{C_0} = A_{C_1} &= \{x, y, a\}, & A_{C_2} &= \{a, b\}, \\ A_{C_3} &= \{y, a, b, c\}, & A_{C_4} &= \{x, a, b, c\}, \\ A_{C_5} &= \{a, c\}, & A_{C_6} &= \{a\}. \end{aligned}$$

Here $k = 3$, $q = 7$, and $h = 5$. The maximum column size is four and the maximum column distance is three. The split formulas give $\kappa(G) = 3$ and $\tau(G) = 5$. Directly checking the 31 nonempty row subsets gives minimum Hall slack two and minimum capacity slack one.

Choose a factor consisting of

$$x - C_0 - y - C_1 - x, \qquad C_2 - a - C_3 - b - C_4 - c - C_5, \qquad \text{isolated } C_6.$$

The free clique vertices are C_2, C_5, C_6. Neither cycle row sees one, so the directed route $x \to C_4 \to b \to C_2$ of length three is shortest. Toggle its edges and open the original cycle by removing xC_0. The resulting factor consists of

$$C_2 - a - C_3 - b - C_2, \qquad C_0 - y - C_1 - x - C_4 - c - C_5, \qquad \text{isolated } C_6.$$

All independent degrees remain two and all clique degrees remain at most two, but there is still one cycle.

This graph is traceable; one spanning path is

$$C_0 - x - C_1 - y - C_3 - b - C_4 - c - C_5 - a - C_6 - C_2.$$

Thus it is not a counterexample to WOWII 199, and the displayed initial factor is not a minimum-cycle factor. The example refutes only automatic cycle decrease for an arbitrary shortest alternating switch. It does not refute a carefully chosen minimum-cycle or multi-switch argument.

19.8 The precise unresolved step

Combining the preceding results, a nontraceable split graph satisfying the original premise would have $k \geq 4$, $q \geq 2k + 1$, $h \leq q - 1$, strict Hall on every nonempty subset of I, and

$$\mathrm{cap}_2(S) \geq 2|S| + 1 \qquad (\varnothing \neq S \subseteq I).$$

It has a factor with degree exactly two on I and at most two on C, and each cycle in such a factor has a directed alternating route to unused clique-side capacity. The displayed twelve-vertex example shows why an arbitrary shortest switch cannot be assumed to lower the number of cycles. A proof that suitable switches eliminate all cycles has not been supplied. Hence these results resolve explicit split subclasses and strengthen the residual obstruction; they do not resolve all split graphs or unrestricted Conjecture 199.

20 Even-distance traceability: reductions and obstructions for Conjecture 189

Let G be connected of order $n > 1$, and let $s = d_2$ be its second-smallest degree, counting multiplicity. For each vertex v, put

$$\mathrm{Even}(v) = \{x \in V(G) : \mathrm{dist}(v, x) \text{ is even}\}.$$

Distance zero is included, so $v \in \mathrm{Even}(v)$. The archived Conjecture 189 asks whether the hypothesis

$$|\,\mathrm{Even}(v)| \leq s + 1 \qquad (v \in V(G)) \tag{149}$$

implies traceability [2]. We first exclude every independent-set neighborhood obstruction to traceability, except for an explicitly traceable family. We then give exact examples explaining why several stronger automatic reductions are unavailable.

20.1 Saturation and deficient independent sets

Lemma 20.1 (Saturation)**.** *Assume* (149)*. If an independent set A of size $s+1$ is contained in $N(c)$ for some vertex c, then A dominates G. Moreover, each $a \in A$ is either complete or anticomplete to each component of $G - A$.*

Proof. For any $a \in A$, all of A is at distance zero or two from a. Thus $\mathrm{Even}(a) = A$ by (149). Suppose xy is an edge of $G - A$, with ax an edge and ay a nonedge. Then y has distance two from a, contrary to $\mathrm{Even}(a) = A$. Adjacency to a therefore propagates through every component of $G - A$. Connectedness ensures that each such component has a neighbor in A, so every vertex outside A is dominated by A. □

For an independent set, $N(I)$ denotes the union of its open neighborhoods; it is disjoint from I.

Theorem 20.2 (The only neighborhood deficiency)**.** *Assume* (149)*, and let I be independent with $|N(I)| < |I|$. Then*

$$|I| = s+1, \qquad |N(I)| = s, \qquad n = 2s+1.$$

All but at most one vertex of I are complete to $N(I)$, and G has an explicit Hamiltonian path.

Proof. Connectedness gives $s \geq 1$, and at most one vertex has degree below s. Write $t = |I|$, $q = |N(I)|$. The empty set is not deficient, and $t = 1$ gives $q \geq 1$. If $2 \leq t \leq s$, some vertex of I has degree at least s, so $q \geq s \geq t$.

Suppose $t = s+1$. At least s vertices of I have degree at least s. Since $q < t$, this forces $q = s$ and makes all those vertices complete to $N(I)$. Designate the possible exceptional vertex by x; if there is none, choose any $x \in I$. It has a neighbor $c \in N(I)$. The vertex c is adjacent to all of I, so Lemma 20.1 shows that I dominates G. Therefore $V(G) = I \dot\cup N(I)$ and $n = 2s+1$. Enumerate $N(I) = \{c_1, \ldots, c_s\}$ with $xc_1 \in E(G)$, and $I - \{x\} = \{a_1, \ldots, a_s\}$. The sequence

$$x, c_1, a_1, c_2, a_2, \ldots, c_s, a_s$$

is a Hamiltonian path, regardless of the edges within $N(I)$.

Finally suppose $t \geq s+2$. No vertex can have $s+1$ neighbors in I: such a subset A would dominate by Lemma 20.1, whereas the independent vertex in $I - A$ has no neighbor in A. Every vertex of $N(I)$ therefore has at most s neighbors in I. Counting incidences and using the positive degree of the possible exceptional vertex yields

$$sq \geq \sum_{a \in I} d(a) \geq s(t-1) + 1 > s(t-1).$$

Integrality gives $q \geq t$, a contradiction. All cases are covered. □

Corollary 20.3. *Every graph satisfying* (149) *has $\alpha(G) \leq \lceil n/2 \rceil$. If $n \neq 2s+1$, or if G is a nontraceable graph satisfying the hypothesis, then G has a perfect 2-matching: a spanning subgraph with components that are single edges or odd cycles. In particular it has a perfect fractional matching.*

Proof. An independent set larger than $\lceil n/2 \rceil$ is deficient, since its neighborhood is contained in its complement. Theorem 20.2 would instead give its size $(n+1)/2$, a contradiction. Under either extra hypothesis, every independent set J satisfies $|N(J)| \geq |J|$. For an arbitrary $S \subseteq V(G)$, here let $N(S)$ mean the union of open neighborhoods, possibly intersecting S. Put $J = S \setminus N(S)$. Then J is independent and $N(J)$ is disjoint from S, whence

$$|N(S)| \geq |S \cap N(S)| + |N(J)| \geq |S \cap N(S)| + |J| = |S|.$$

The bipartite double cover of G has left and right copies of $V(G)$, with $u_L v_R$ an edge exactly when $uv \in E(G)$. The displayed inequality is Hall's condition, so Hall's theorem [47] gives a perfect matching. It defines a permutation π with $v\pi(v) \in E(G)$ and no fixed points. A two-cycle gives a single edge, an odd cycle is retained, and an even cycle of length at least four is replaced by alternating edges. These components form the claimed perfect 2-matching. Give weight one to each single edge and weight one-half to each odd-cycle edge to obtain the fractional matching. This standard factor implication is included with its proof; it is not a new matching characterization. □

20.2 A tight family beyond a connectivity–independence shortcut

Proposition 20.4. *For every integer $s \geq 2$, there is a traceable graph J_s satisfying (149) with equality at every vertex and*

$$n = 3s + 1, \quad d_2 = s, \quad \kappa = s - 1, \quad \alpha = s + 1.$$

Thus the premise of Conjecture 189 does not imply the Chvátal–Erdős sufficient condition $\alpha \leq \kappa + 1$.

Proof. Take disjoint cliques $A = \{a_0, \ldots, a_s\}$ and $B = \{b_0, \ldots, b_s\}$, and an independent set $Z = \{z_1, \ldots, z_{s-1}\}$. Join every vertex of Z to $A - \{a_0\}$ and $B - \{b_0\}$, with no edges between A and B and no other added edges. The two special vertices have degree s, each ordinary clique vertex has degree $2s - 1$, and each vertex of Z has degree $2s$. Thus $d_2 = s$. For a_0, the even-distance set is $\{a_0, b_0\} \cup Z$, with b_0 at distance four and Z at distance two. For an ordinary A vertex, it is that vertex together with the s ordinary B vertices. For a Z vertex, it is itself, the other $s - 2$ vertices of Z, and a_0, b_0. The symmetric descriptions hold on B. Every count is $s + 1$.

Deleting Z disconnects the two cliques. Deleting fewer than $s - 1$ vertices leaves a member of Z and an ordinary vertex in each clique; all surviving vertices remain connected through these. Hence $\kappa = s - 1$. The independent set $Z \cup \{a_0, b_0\}$ has size $s + 1$. Any independent set meeting Z uses at most the two special clique vertices, while one disjoint from Z uses at most one vertex of each clique. Thus $\alpha = s + 1$. For a Hamiltonian path, start at a_0, append a_i, z_i for $i = 1, \ldots, s - 2$, then append

$$a_{s-1}, a_s, z_{s-1}, b_1, b_0, b_2, \ldots, b_s.$$

The initial loop is empty when $s = 2$. All displayed consecutive pairs are edges and every vertex is used once. □

The family was obtained by generalizing a graph in a previously public certificate list for $d_2 = 4$, credited in [49]. Its members are examples satisfying Conjecture 189, not counterexamples to it.

20.3 A Hamiltonian family excluding a linear-order extrapolation

Proposition 20.5. *For every integer $t \geq 2$, a Hamiltonian graph G_t satisfies (149) with $s = 3t-1$ and $n = 10t - 3 > 3s + 1$.*

Proof. Replace the vertices of a seven-cycle, in cyclic order, by cliques of sizes

$$(w_0, \ldots, w_6) = (t, t, t, 2t - 1, 2t - 1, 2t - 1, t),$$

joining bags completely exactly along edges of the seven-cycle. The order is $10t - 3$. The degrees in the successive bags are

$$(3t - 1, 3t - 1, 4t - 2, 5t - 3, 6t - 4, 5t - 3, 4t - 2).$$

The first two bags each have at least two vertices, so $d_2 = 3t - 1$. Distances between different bags equal the quotient-cycle distances; within a bag, distinct vertices are adjacent. A vertex in bag i therefore has $1 + w_{i-2} + w_{i+2}$ even-distance vertices. The seven counts are

$$(3t, 3t, 3t, 3t, 2t + 1, 3t, 3t),$$

all at most $s + 1 = 3t$. Nevertheless $n - (3s + 1) = t - 1 > 0$. Listing each bag consecutively in cyclic bag order gives a Hamiltonian cycle. □

This disproves an auxiliary extrapolation from small-s data, not the original conjecture or the quadratic order bound in the credited prior archive [49]. No computational solver is needed for the all-parameter statement above.

20.4 An explicit obstruction to two automatic sufficient-condition routes

The following twelve-vertex graph is taken from the same credited certificate list [49]; its graph6 string is `K?BELqwptrvN`. To specify it without external data, the table lists for each vertex only its neighbors with larger labels.

Vertex	Larger neighbors	Vertex	Larger neighbors
0	5,6,7,8,9,10,11	6	9
1	5,6,9	7	9,10,11
2	7,8,10,11	8	10,11
3	7,8,10,11	9	10,11
4	7,8,10,11	10	11
5	6,9	11	none

Its second-smallest degree is four. Breadth-first distance counts from vertices $0, \ldots, 11$ give even-distance cardinalities $(5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 4, 4)$, so the premise holds. The set $\{0, 1, 2, 3, 4\}$ is independent and $\{0, 9\}$ is a vertex cut. Thus $\alpha \geq 5 > \kappa + 1$, since $\kappa \leq 2$. For the independent triple $\{2, 3, 4\}$, all three neighborhoods equal $\{7, 8, 10, 11\}$, and hence

$$d(2) + d(3) + d(4) - |N(2) \cap N(3) \cap N(4)| = 8 < 11 = n - 1.$$

In particular neither the connectivity–independence threshold nor this modified triple-degree threshold follows automatically from the original premise. The path

$$1, 5, 6, 0, 8, 2, 7, 3, 10, 4, 11, 9$$

visits all twelve vertices along edges. The graph is therefore explicitly traceable. The accompanying standard-library checker reconstructs this table and verifies the distances, cuts, independent sets, and spanning path; no completeness claim about a broader census is needed.

20.5 Scope and prior work

Easwar and Johari prove traceability under several additional connectivity, order, and diameter conditions [50]. Those attributed results are not added to this paper's resolution count. The reduction proved here says that any counterexample to the original conjecture must have a perfect 2-matching and cannot arise from an independent set with a deficient neighborhood. Joining the factor components into a spanning path remains unproved. The example families identify limitations of proposed routes and do not refute Conjecture 189.

21 Further partial results and a formulation-dependent counterexample

This section records the further results obtained while revising the consolidation. The original-item ledger remains at fourteen conclusions. The new theorems enlarge proved subclasses of 40, 61, 165, and 199, restrict the residual profile of 384b, and refine a connected-domination equality case. For 389a a complete counterexample is obtained only after specifying the lower-rank quartile convention; it is kept separate from an unqualified resolution of the archived statement. None of the restricted-class results is asserted to settle its unrestricted entry.

21.1 A graph of supports with subdivided parallel links

The following extension transfers the simultaneous degree-switch argument of Section 10.5 to an arbitrary graph of supports. It gives a further subclass of statement 61 without a triangle-free hypothesis; it does not settle the unrestricted conjecture.

Let H be any connected nonempty simple graph on k vertices, called roots. For each edge uv of H, choose exactly one replacement: retain the edge uv, or replace it by $c_{uv} \geq 1$ internally vertex-disjoint paths of length two. Internal vertices are distinct between replacements. Attach at least one private pendant vertex to every root, and add no other vertices or edges. Write G for the resulting graph, and let Q be the spanning subgraph of H consisting of the edges replaced by at least two length-two paths.

Theorem 21.1. *For every graph constructed above,*

$$\nu(G) = k, \qquad f(G) \geq \operatorname{res}(G) + f(H) \geq \operatorname{res}(G) + \left\lceil \frac{\operatorname{diam}(G)}{2} \right\rceil .$$

Thus the original inequality in WOWII 61 holds on this class, with the stronger half-diameter term.

Proof. The roots form a vertex cover of size k. Matching each root u to one private pendant neighbor a_u attains this bound, so $\nu(G) = k$. Choose a maximum induced forest $H[W]$, put $S = V(H) \setminus W$, and let $t = \nu(Q[W])$. Since $Q[W]$ is a forest, its minimum vertex-cover size equals t.

Delete S and a vertex cover of size t from $Q[W]$. The surviving root skeleton is a subforest of $H[W]$. Every surviving link with at least two internal vertices loses one of its roots. Thus it reduces to a star or isolated vertices. A retained edge or a single subdivided edge cannot introduce a cycle over a forest skeleton. Pendant vertices also introduce none. Consequently

$$f(G) \geq n - |S| - t, \qquad n = |V(G)|. \tag{150}$$

Choose a matching of t edges in $Q[W]$. For each chosen uv, select one internal vertex x_{uv} of its link; delete ua_u and va_v, and insert uv and $a_u a_v$. Both inserted edges were absent, because uv was replaced rather than retained and the partners are private leaves. The five-vertex sets $\{u, v, x_{uv}, a_u, a_v\}$ are pairwise disjoint. All switches may therefore be performed simultaneously, yielding a simple graph G' with the same degree at every vertex.

In G' there are t triangles $ux_{uv}v$, t partner edges $a_u a_v$, and $k - 2t$ unchanged root-partner edges. These subgraphs are mutually vertex-disjoint, and additional edges between them do not invalidate the ensuing vertex-cover lower bound. Every vertex cover of G' has size at least

$$2t + t + (k - 2t) = k + t.$$

The classical residue inequality $\operatorname{res}(F) \le \alpha(F)$ and dependence of residue only on the degree sequence give

$$\operatorname{res}(G) = \operatorname{res}(G') \le \alpha(G') \le n - k - t. \tag{151}$$

Combining (150) and (151),

$$\operatorname{res}(G) + f(H) = \operatorname{res}(G) + k - |S| \le n - |S| - t \le f(G).$$

Finally every vertex of G is a root or has a root neighbor. Each edge of H is represented by a path of length at most two, so $\operatorname{diam}(G) \le 2\operatorname{diam}(H) + 2$. A diametral geodesic in H is an induced forest on $\operatorname{diam}(H) + 1$ vertices. Therefore $f(H) \ge \operatorname{diam}(H) + 1 \ge \lceil \operatorname{diam}(G)/2 \rceil$. □

Corollary 21.2 (Exact feedback number for a support tree)**.** *If $H = T$ is a tree and $t = \nu(Q)$, then*

$$\operatorname{fvs}(G) = t, \qquad f(G) = n - t, \qquad \operatorname{res}(G) \le n - k - t, \qquad f(G) \ge \operatorname{res}(G) + \nu(G).$$

Proof. Take $W = V(T)$ in the preceding proof. This gives a feedback vertex set of size t. Conversely, each edge in a maximum matching of Q supplies a four-cycle by selecting two internal vertices in its link. These t four-cycles are vertex-disjoint, so every feedback vertex set has size at least t. The residue bound is (151), and $f(T) = k = \nu(G)$. □

Example 21.3 (Beyond the earlier diameter and matching boundaries). For $r \ge 2$, take the support tree on $z, u_1, v_1, \dots, u_r, v_r$ with edges zu_j, u_jv_j. Retain each zu_j, replace each u_jv_j by two length-two paths, and add exactly one private leaf at each root. Then

$$n = 6r + 2, \qquad \operatorname{fvs}(G) = r, \qquad \nu(G) = 2r + 1, \qquad \operatorname{diam}(G) = 8.$$

Leaves at distinct v_j roots attain the diameter. For every $r \ge 2$ the feedback-vertex number exceeds one, the diameter exceeds six, and $\nu(G) > \lceil \operatorname{diam}(G)/2 \rceil = 4$. Thus the theorem reaches beyond all three earlier hypotheses. The feedback-vertex number and the excess matching number are unbounded.

The theorem uses the established degree-equivalent-realization method and the earlier simultaneous-switch mechanism. Its new content here is the transfer through a maximum induced forest of the arbitrary support graph H. This is an additional restricted-class theorem, not another fully resolved catalogue number or a claim of historical priority. Exact finite checks of the construction and the switch, forest, matching, and residue certificates accompany this section.

Proposition 21.4 (A pendant-neighbor subclass of statement 40)**.** *Let G be connected and have a nonempty vertex cover C of size k such that every vertex of C has at least two pendant neighbors. Then*

$$p(G) = n(G) - 2k, \qquad f(G) \ge \left\lceil \frac{p(G) + b(G) + 1}{2} \right\rceil.$$

Here p is the minimum number of vertex-disjoint paths covering all vertices, and b is the largest order of an induced bipartite subgraph. In particular the conclusion holds for the preceding support-graph construction whenever each root receives at least two private leaves.

Proof. A covering by p disjoint paths is a spanning linear forest with $n - p$ edges. Every edge meets C, and each member of C has degree at most two in a linear forest. Hence $n - p \le 2k$. Conversely, use two pendant neighbors at each member of C to form k disjoint three-vertex paths, and make all remaining vertices singleton paths. This gives $2k$ edges and proves $p = n - 2k$.

The set $V(G) \setminus C$ is independent. Adding any one vertex of C induces a star together with isolated vertices, so $f(G) \geq n - k + 1$. Since $b(G) \leq n$,

$$\left\lceil \frac{p(G) + b(G) + 1}{2} \right\rceil \leq \left\lceil \frac{2n - 2k + 1}{2} \right\rceil = n - k + 1 \leq f(G).$$

□

This elementary sufficient condition is not an unrestricted resolution of statement 40 and is not presented as a historically new path-cover principle.

21.2 A sharp partial result for the maximum-degree boundary in 165

For a connected graph G, let $L(G)$ be its maximum spanning-tree leaf number, let $M = \{v : d_G(v) = \Delta(G)\}$, and put $t = |N_G(M) \setminus M|$. The archived statement 165 asks whether

$$L(G) \geq \left\lceil \sqrt{2t} \right\rceil.$$

The following result establishes a class not explicitly treated in the supplied consolidated manuscript. It does not settle the unrestricted statement. Its forest-of-stars construction extends the full-star criterion used in the proof of Theorem 11.1.

Lemma 21.5. *Every finite loopless multigraph S of maximum degree at most three, with h vertices and m edges, has a feedback vertex set A with*

$$|A| \leq \min\{\lfloor h/2 \rfloor, \lfloor m/2 \rfloor\}.$$

Parallel edges are counted with multiplicity, and two parallel edges form a cycle.

Proof. Choose a cut maximizing the number of crossing edges. Moving one vertex cannot improve it, so each vertex has at most one neighbor of its own color, with multiplicity. Each color therefore induces a matching together with isolated vertices. Deleting the smaller color class leaves a forest and costs at most $\lfloor h/2 \rfloor$. For a second construction, repeatedly delete a vertex on a cycle until the remaining multigraph is a forest. Each deletion removes at least two edges that have not previously been removed. Thus this construction costs at most $\lfloor m/2 \rfloor$. Take the smaller of the two resulting deletion sets. □

Theorem 21.6. *Let G be connected with $\Delta(G) = 3$, and suppose its set M of degree-three vertices is independent. Then*

$$L(G) \geq 2 + \frac{|N_G(M) \setminus M|}{4}.$$

Consequently, the original inequality in statement 165 holds for G.

Proof. Write $h = |M|$. Form a loopless multigraph S on M by introducing one edge uv for each vertex of G adjacent to both $u, v \in M$. Such a vertex has degree two, since M is the entire degree-three set. Distinct such vertices may yield parallel edges. Every vertex of S has degree at most three. If $m = |E(S)|$, counting incidences from M to its external neighborhood gives exactly

$$t = 3h - m :$$

each external neighbor is counted once, except that the m vertices represented by edges of S are counted twice.

Take the feedback vertex set A from Lemma 21.5. The union of all stars centered at $M \setminus A$, including every edge incident with these centers, is a forest. Indeed, M is independent and every

vertex outside M has degree at most two; a cycle in this union would therefore project precisely to a cycle of the multigraph $S[M \setminus A]$. Add uncovered vertices as isolated vertices and extend this forest to a spanning tree T of G. Every center in $M \setminus A$ retains all three incident edges and has degree three in T.

Every nontrivial subcubic tree has exactly two more leaves than vertices of degree three. Hence

$$L(G) \geq L(T) \geq h - |A| + 2 \geq 2 + \max\{h/2, h - m/2\} \geq 2 + \frac{3h - m}{4} = 2 + \frac{t}{4}.$$

Finally $t \leq 4(L(G) - 2)$ yields

$$L(G)^2 - 2t \geq L(G)^2 - 8L(G) + 16 = (L(G) - 4)^2 \geq 0.$$

Since $L(G)$ is an integer, this is the required ceiling inequality. □

Connected graphs of maximum degree at most two also satisfy 165: paths have two spanning-tree leaves and at most two vertices in $N(M) \setminus M$, while cycles and K_2 have $M = V(G)$ and hence empty external boundary. The singleton case has both sides zero under the degree-one leaf convention.

Proposition 21.7. *The same conclusion $L(G) \geq 2 + t/4$, and hence statement 165, holds when $\Delta(G) \in \{4, 5\}$, the maximum-degree set M is independent, and every vertex outside M has degree at most two.*

Proof. Write $D = \Delta(G)$ and construct the auxiliary multigraph S as above. Then $t = Dh - m$, and $\Delta(S) \leq D \leq 5$. Color S with three colors so as to minimize the number of monochromatic edges, counted with multiplicity. If a vertex had at least two same-color neighbors, local optimality would force at least two neighbors of each other color, contradicting its degree being at most five. Each color class therefore induces a matching and isolated vertices. One of them is an induced forest of order at least $h/3$. The greedy cycle-deletion argument also leaves an induced forest of order at least $h - m/2$. Choose the larger such vertex set U. The union of its full stars is a forest and extends to a spanning tree retaining degree D at every vertex of U. The tree identity $L(T) = 2 + \sum_{v:d_T(v)\geq 3}(d_T(v) - 2)$ gives

$$L(G) \geq 2 + (D - 2)\max\{h/3, h - m/2\}.$$

For $D = 4$, a convex combination with weights $3/4, 1/4$ shows

$$2\max\{h/3, h - m/2\} \geq h - m/4 = (4h - m)/4.$$

For $D = 5$, the weights $5/6, 1/6$ give

$$3\max\{h/3, h - m/2\} \geq 4h/3 - m/4 \geq (5h - m)/4.$$

Thus $L(G) \geq 2 + t/4$ in both cases, and the preceding square identity establishes 165. □

Remark 21.8 (Sharpness and scope). The archived annotation of 165 already credits C. Foster with examples having $L = k + 2$ and $t = 4k$. The linear coefficient above is sharp even within the independent-maximum-degree class. An explicit realization uses vertices a_i, b_i for $1 \leq i \leq k$, two internally disjoint length-two a_i–b_i paths for each i, and length-three paths b_i–a_{i+1} with cyclic indices. For $k = 1$ this is the theta graph with path lengths $2, 2, 3$. All internal path vertices are distinct. The degree-three set is independent, $h = 2k$, and $t = 4k$. Every spanning tree must break each of the k disjoint four-cycles, so at least one of a_i, b_i loses degree three for each i. The subcubic tree identity gives $L \leq k + 2$, while the theorem gives the reverse inequality. Thus $L = k + 2$. The case $k = 2$ also gives equality $L^2 = 2t$ in 165. This sharpness pattern is consistent with the previously credited Foster parameters; it is not counted as a new discovery of that extremal parameter family. The unrestricted cases of 165, including adjacent maximum-degree vertices in general, remain unresolved here.

21.3 One further independence–connectivity boundary

The closure operation also removes one more residual boundary. This is an additional restricted case of statement 199, not a resolution of its unrestricted form.

Lemma 21.9 (A near-balanced row-degree criterion)**.** *Let G be a split graph with partition $C \dot\cup I$, where $h = |I| \geq 4$, every vertex of C has a neighbor in I, and*

$$|C| \geq h - 1, \qquad d_C(i) \geq h - 2 \quad (i \in I).$$

Suppose also that $|N_C(S)| \geq |S|$ for every nonempty proper subset $S \subsetneq I$. Then G is traceable.

Proof. Repeatedly add a missing incidence edge ic whenever $d_C(i) + |N(c) \cap I| \geq h$. The total degree sum of its endpoints is then at least $h + |C| - 1 = |V(G)| - 1$. Thus each addition preserves traceability in both directions by the path-closure lemma (Lemma 17.4). The split partition is unchanged; all the stated degree and neighborhood inequalities persist.

In the resulting closed graph, any column with at least two independent neighbors is adjacent to all of I: otherwise its missing incidence satisfies $d_C(i) + |N(c) \cap I| \geq (h-2) + 2 = h$. Consequently every column is either universal to I or has exactly one independent neighbor. Let U be the set of universal columns, put $u = |U|$, and let L_i be the set of singleton columns adjacent to i. These sets partition C, and $u + |L_i| \geq h - 2$ for all $i \in I$.

If $u \leq h - 4$, every L_i has at least two vertices. Choose distinct $a_i, b_i \in L_i$ for each i, and concatenate the disjoint paths a_i, i, b_i by clique edges. This gives a path through every vertex of I.

If $u = h - 3$, each L_i is nonempty. Label four distinct rows i_1, i_2, i_3, i_4 and choose $c_j \in L_{i_j}$ for $1 \leq j \leq 4$. In an ordering $i_1, \ldots, i_h$ of the rows, realize the first and third gaps by

$$i_1, c_1, c_2, i_2 \qquad \text{and} \qquad i_3, c_3, c_4, i_4.$$

Realize each of the remaining $h - 3$ gaps with one distinct vertex of U. This is again a simple path through all rows.

If $u = h - 2$, at least two rows have nonempty singleton sets. Indeed $|C| \geq h - 1$ ensures at least one such row, and if it were unique, say x, then $S = I \setminus \{x\}$ would have $|N_C(S)| = u = h - 2 < |S|$, a contradiction. Choose distinct such rows x, y and columns $c_x \in L_x, c_y \in L_y$. The $h - 2$ universal columns give an alternating path starting at y through all $h - 1$ rows of $I \setminus \{x\}$. Prepend x, c_x, c_y to this path.

Finally, if $u \geq h - 1$, alternate all h rows with $h - 1$ distinct universal columns. In every case the resulting path contains I. Each unused clique vertex has a neighbor i on this path and can be inserted into a path edge ic by replacing it with i, d, c. Repeating this operation includes every unused clique vertex. The closed graph is traceable, so the original graph is traceable. □

Corollary 21.10 (The independence gap two case of 199)**.** *Let G be a connected split graph of order greater than one, satisfying the original premise $\tau(G) \leq \kappa(G) + 2$. If $\alpha(G) \leq \kappa(G) + 2$, then G is traceable. Consequently any remaining nontraceable split instance must have*

$$\alpha(G) \geq \kappa(G) + 3.$$

Proof. Complete graphs and the cases $\kappa(G) \leq 3$ are already covered above. Put $k = \kappa(G) \geq 4$, choose I maximum, and write $h = |I| = \alpha(G)$. The Chvátal–Erdős path theorem covers $h \leq k + 1$, so suppose $h = k + 2$. The incidence conditions (138) hold, and Lemma 19.1 gives $|C| \geq h - 1$ and ordinary Hall on every proper row subset. The row-degree lower bound is $k = h - 2$. Apply Lemma 21.9. □

This boundary is not subsumed by the previously listed small-connectivity, small-column, clique-size, balanced, padding, or Hall-equality cases. For example, for any $k \geq 4$, take $h = k + 2$ rows and $q = 2k + 1$ columns, each consisting of all rows except one. Let $k + 1$ columns omit one fixed row x, and let the other k columns omit distinct rows different from x. Then the minimum row degree is k, the maximum column size is $k + 1$, and pairwise column distances are at most two. Strict Hall holds, $q > 2k$, and the padding allowance $k + 1 - \max_c |A_c|$ is zero. The corollary applies to this infinite family. It supplies an additional sufficient condition, not a claim that this easy family was previously of unknown traceability.

21.4 Extending the boundary by the Tan–Hung classification

The following improvement uses a published Hamiltonicity theorem. Its contribution here is an application to the residual incidence system of statement 199; we do not claim a new general classification theorem.

Lemma 21.11 (A consequence of Tan–Hung)**.** *Let G be a split graph with independent part I, where $h = |I| \geq 6$. If every $i \in I$ has degree at least $h - 3$ and*

$$|N_C(S)| \geq |S| + 1 \qquad (\varnothing \neq S \subseteq I),$$

then G is traceable.

Proof. Add a vertex w adjacent to every vertex and put $F = G + w$, with split partition $I \dot\cup (C \cup \{w\})$. Strict Hall on I gives $|C| \geq h + 1$. Hence every clique vertex of F has degree at least $h + 1$, every independent vertex has degree at least $h - 2$, and

$$|N_F(S)| \geq |S| + 2 \qquad (\varnothing \neq S \subseteq I).$$

We apply Tan and Hung's Theorem 1[1]. For an independent part of size $h \geq 6$, their theorem says that a split graph of minimum degree at least $h - 2$ satisfying the stated strict neighborhood inequalities is Hamiltonian unless it is a spanning subgraph of their exceptional graph G_t^h, where t is the clique size. In that graph, the $h - 3$ rows $u_4, \ldots, u_h$ have all their neighbors in the same set of $h - 2$ clique vertices (Table 1 of that paper). Such a spanning subgraph has those $h - 3$ rows of degree at most $h - 2$. Under any exceptional embedding they must therefore lie in our independent part, since our clique vertices have degree at least $h + 1$. It cannot satisfy the displayed inequality on $S = \{u_4, \ldots, u_h\}$: it would require at least $h - 1$ neighbors. Thus F is Hamiltonian. Deleting w from a Hamilton cycle of F gives a Hamiltonian path of G. □

Corollary 21.12 (The independence gap three case of 199)**.** *Let G be a connected split graph of order greater than one satisfying the original WOWII 199 premise $\tau(G) \leq \kappa(G) + 2$. If $\alpha(G) \leq \kappa(G) + 3$, then G is traceable. Consequently every remaining nontraceable split instance must satisfy*

$$\alpha(G) \geq \kappa(G) + 4.$$

Proof. Complete graphs and the cases $\kappa(G) \leq 3$ are already covered. Write $k = \kappa(G) \geq 4$ and fix a maximum independent part I of size $h = \alpha(G)$. Corollary 21.10 handles $h \leq k + 2$, so assume $h = k + 3 \geq 7$. If G were nontraceable, Corollary 19.5 would give strict Hall on every nonempty row subset. The minimum row degree is $k = h - 3$. Lemma 21.11 is then a contradiction. □

This extends a restricted sufficient condition from independence gap two to gap three. It does not close statement 199 for arbitrary split graphs, and it does not address the nonsplit residual case.

To see that the new boundary is not contained in the earlier listed regimes, fix $k \geq 4$ and let $I = \{0, 1, \ldots, k+2\}$. Take $2k+1$ columns whose omitted row pairs are

$$\{0, j\}\ (1 \leq j \leq k+1), \qquad \{1, j\}\ (2 \leq j \leq k+1).$$

Every column has size $k+1$, the minimum row degree is k, and two columns have symmetric difference at most four. Thus the original premise holds, with $\alpha = k+3$ and $\tau = k+2$. Strict Hall holds: a singleton has at least k neighbors; a pair misses at most one column; and a set of at least three rows meets every column. Here $q = 2k+1$, while the padding allowance is zero. In particular this family lies beyond the gap-two, small-clique, small-column, balanced, and Hall-equality cases. Its explicit traceability is not itself claimed to be previously unknown.

21.5 A further restriction at high-degree order thirty-three

The fixed-active method yields an additional structural restriction without resolving statement 384b. Degrees in the following proposition are full degrees in G, not degrees in the induced high-vertex graph H.

Proposition 21.13. *Every counterexample to 384b with exactly $h = 33$ vertices of degree at least three has exactly two vertices of degree three. Their triangle counts are two and three. In particular, the latter vertex and its three neighbors induce a K_4. Moreover, no pendant triangle occurs, and the high-root edge of every ordinary low triangle is avoidable by a maximum matching of H.*

Proof. Use the critical profile and matching lemmas from Section 16. The high graph has a near-perfect matching of sixteen edges, the forbidden high contribution to a 2-dominating set is sixteen, and all thirty-three high triangle counts are distinct and at least two. The rank-profile multiplicities for types 3 through 10 are

$$(2, 3, 4, 5, 6, 6, 5, 2).$$

Choose exactly twelve matching edges uniformly, select one endpoint uniformly on each active edge, and independently select the unmatched vertex with probability $2/5$. Thus a matched vertex has marginal $3/8$, and the expected initial high selection count is $62/5$.

Use the artificial neighborhoods and pair repair of Lemma 16.16. Conditional on focal inactivity, the external matching has twelve active edges out of fifteen; conditional on focal activity, it has eleven active edges out of fifteen. If X, Y are capped outside support counts, the expected focal repair is

$$W = \frac{1}{4}\mathbb{E}_{12} g(X, Y) + \frac{3}{8}(\Pr_{11}(X = 0) + \Pr_{11}(Y = 0)),$$

with the same g as in (109) and with the singleton mixed at probabilities $3/5, 2/5$. The finite-capacity coefficient computation in (117)–(118), now with fifteen external edges, proves $W \leq \lambda_a + \lambda_b$ for

a	3	4	5	6	7	8	9	10
$10^9\lambda_a$	346108345	246668957	171497253	116878435	77326580	50571772	34135474	24984333

and the unmatched repair is at most $\lambda_a + \zeta$, where $10^9\zeta = 88891656$. The exact verifier checks all $36,985$ feasible color-pair patterns, all thirty-six unordered type pairs, and all unmatched double-neighbor placements. Its enumeration uses precisely the six-count classification in (111); it fills the unused positions to a total of fifteen external edges, rather than adding arbitrarily

many zeros. The displayed rounded-up rational potentials are checked directly, so no numerical optimizer is a proof dependency. The resulting high-cost bound is

$$B = \frac{62}{5} + \sum_a n_a \lambda_a + \zeta = \frac{2022441569}{125000000} = 16.179532552.$$

A low triangle must occur, since the critical profile includes the low count one. If it is a pendant triangle, omit one private vertex when its root is initially selected. This gives the upper bound

$$B - \frac{3}{8} = 15.804532552 < 16,$$

so no pendant triangle can occur.

Consider an edge attachment on roots u, v. If uv is in every maximum matching of H, use the cubic repair in Lemma 16.21. Conditional on the focal pair being inactive, its third high neighbor has selection probability $2/5$, whether it is on another matching edge or is unmatched. The special net repair is therefore $(1/4)(3/5) = 3/20$. Replacing the generic pair potential gives

$$B - \lambda_3 - \min_a \lambda_a + \frac{3}{20} = 15.958439874 < 16.$$

Hence every such edge can be avoided by a maximum matching.

Finally suppose there is at most one full-degree-three high vertex. A cubic vertex has at most three incident triangles. Since all high counts are distinct and at least two, the only two rank positions that can be cubic are the first two. At least one of their profile types can therefore be increased from three to four. This decreases the expected bound by $\lambda_3 - \lambda_4 = 0.099439388$. Choose a maximum matching avoiding the root edge uv of any low triangle. If both roots are matched, their joint selection probability is

$$\frac{12 \cdot 11}{16 \cdot 15} \cdot \frac{1}{4} = \frac{11}{80}.$$

If one is unmatched, it is $(2/5)(3/8) = 3/20$, which is larger. Omit the private low vertex whenever both roots are initially selected. This is compatible with every pair repair and with the profile upgrade, since only already selected high roots lose a low neighbor. The bound is

$$B - \frac{11}{80} - (\lambda_3 - \lambda_4) = \frac{3985648291}{250000000} = 15.942593164 < 16,$$

a contradiction. There are consequently exactly two cubic high vertices, of distinct counts two and three. Three triangles through a cubic vertex mean its three neighbors are pairwise adjacent, yielding the stated K_4. □

This is a restricted-case exclusion and a necessary profile for any remaining $h = 33$ counterexample. It does not remove $h = 33$ entirely, change the order-37 representative bound, or add a resolved original conjecture. The accompanying standalone `verify_h33_restriction.py` checks the rational certificate and compares small fixed-active cases with direct enumeration.

21.6 The first-quartile-one case of 389a

Retain the notation and source distinction of Section 12. The following bounds treat the small quartile value analytically.

Proposition 21.14 (The first-quartile-one boundary)**.** *Let G be a connected graph of order $n \geq 4$, let ℓ be its number of leaves, let a count the vertices of degree at most two, and let μ be the largest degree frequency. If $\ell \geq 1$ and $n \leq 4\ell + 1$, then*

$$\gamma_2(G) \leq \mu(a+1) - 1.$$

Consequently, WOWII389a holds when its first-quartile degree is one and that quartile is taken at rank $\lceil n/4 \rceil$. In particular this proves the quartile-one case for every unambiguous source order $4 \mid n$.

Proof. Use the cut bound of Lemma 12.1 $\gamma_2 \leq \lfloor (n+a)/2 \rfloor$ and the inequalities $a \geq \ell$ and $\mu \geq \max\{2, \ell\}$. If $\ell = 1$, then $\lfloor (n+a)/2 \rfloor \leq \lfloor (5+a)/2 \rfloor \leq 2a+1$ for every $a \geq 1$. If $\ell = 2$, then $\lfloor (n+a)/2 \rfloor \leq \lfloor (9+a)/2 \rfloor \leq 2a+1$ for every $a \geq 2$. If $\ell \geq 3$, then

$$\mu(a+1) - 1 - \frac{4\ell + 1 + a}{2} \geq \left(\ell - \frac{1}{2}\right) a - \ell - \frac{3}{2} > 0,$$

since $a \geq \ell \geq 3$. Integrality completes all cases. At quartile rank $\lceil n/4 \rceil$, the value one implies $\ell \geq \lceil n/4 \rceil$, and hence $n \leq 4\ell$. □

Corollary 21.15 (Residual orders under the weaker floor-rank condition)**.** *Suppose instead that the quartile-one condition is interpreted only as $\ell \geq \lfloor n/4 \rfloor$. Any failure of 389a in this case must have one of the profiles*

$$(n, \ell, a, \mu) = (7, 1, 1, 2), \qquad (10, 2, 2, 2), \qquad (11, 2, 2, 2).$$

This is a finite parameter reduction, not an asserted resolution of the remaining three profiles.

Proof. Now $n \leq 4\ell + 3$. If $\ell \geq 3$, the same cut bound works, since

$$\mu(a+1) - 1 - \frac{4\ell + 3 + a}{2} \geq \left(\ell - \frac{1}{2}\right) a - \ell - \frac{5}{2} > 0$$

for $a \geq \ell \geq 3$. For $\ell = 1$, $n \leq 7$ and the bound $\lfloor (7+a)/2 \rfloor \leq 2a+1$ holds for $a \geq 2$. If $a = 1$ and $\mu \geq 3$, the right side is at least five and the cut bound is at most four. For $a = 1$, $\mu = 2$, and $n \leq 6$, the cut bound is at most three, exactly the required threshold. For $\ell = 2$, $n \leq 11$ and $\lfloor (11+a)/2 \rfloor \leq 2a+1$ for $a \geq 3$. If $a = 2$ and $\mu \geq 3$, the right side is at least eight and the cut bound is at most six. When $a = 2$, $\mu = 2$, and $n \leq 9$, the cut bound is at most five. The displayed profiles are all that remain. □

Proposition 21.16 (Exclusion of the seven-vertex residual profile)**.** *No counterexample occurs in the profile $(n, \ell, a, \mu) = (7, 1, 1, 2)$. Consequently the floor-rank quartile-one case can fail only in the profiles $(10, 2, 2, 2)$ and $(11, 2, 2, 2)$.*

Proof. Let p be the sole leaf and let v be its neighbor. Every other vertex has degree at least three, and each degree appears at most twice. Any degree-six vertex must be v, because it is adjacent to p. If there were no degree-six vertex, the other six vertices would have degrees $3, 3, 4, 4, 5, 5$; including p would give odd degree sum 25, which is impossible. Thus v has degree six. Among the remaining five vertices there must be a degree-five vertex w, since degrees three and four each appear at most twice. The vertex w is adjacent to all vertices except p. The set $\{p, v, w\}$ is therefore 2-dominating, and its size is the required value $2(1+1) - 1 = 3$. □

21.7 The lower-rank quartile interpretation fails

The quartile convention is not a harmless presentational choice. The following example refutes the interpretation $q = d_{\lfloor n/4 \rfloor}$, while not refuting the upper-rank interpretation $q = d_{\lceil n/4 \rceil}$. Since the archived source specifies rank $n/4$ without resolving nonintegral ranks, this is a *formulation-qualified result*, not an additional unqualified resolution of the original numbered entry.

Theorem 21.17 (An eleven-vertex lower-rank counterexample)**.** *There is a connected graph G on eleven vertices with*

$$\gamma_2(G) = 6, \qquad a = 2, \qquad \mu = 2, \qquad d_{\lfloor n/4 \rfloor} = 1.$$

Consequently the lower-rank reading of 389a would require $6 \leq 5$.

Proof. Use vertices $0, \ldots, 10$. The following table lists each vertex's neighbors with larger labels and specifies all edges.

Vertex	Larger neighbors	Vertex	Larger neighbors
0	2,3,5,6,7,8,9,10	6	7,8
1	4,6,8	7	8
2	5,7	8	none
3	4,6,8	9	none
4	5,7	10	none
5	7,8		

The graph is connected and has degree multiset

$$(1, 1, 3, 3, 4, 4, 5, 5, 6, 6, 8).$$

Hence $a = 2$, $\mu = 2$, and the lower quartile is $d_2 = 1$. The leaves $9, 10$ force both of themselves into every two-dominating set. The set $\{0, 1, 2, 3, 9, 10\}$ is two-dominating, so it remains to rule out a set of size five. It would consist of both leaves and three vertices of the core $\{0, \ldots, 8\}$.

First suppose vertex 0 is selected and let T be the other two selected core vertices. The vertices $1, 4$ are the only core vertices not adjacent to 0; their neighborhoods $N(1) = \{4, 6, 8\}$ and $N(4) = \{1, 3, 5, 7\}$ are disjoint. If neither belongs to T, both would require all of T in their disjoint neighborhoods, an impossibility. If both belong to T, vertex 2 has only one selected neighbor, namely 0. If $1 \in T$ but $4 \notin T$, the other member of T must be in $\{3, 5, 7\}$ to dominate 4 twice. Choosing 3 leaves vertex 2 with only the neighbor 0; choosing 5 or 7 leaves vertex 3 with only the neighbor 0. If $4 \in T$ but $1 \notin T$, the other member must be 6 or 8, and vertex 2 again has only the neighbor 0. All cases fail.

Now suppose 0 is omitted. The three selected core vertices would have to two-dominate $K = G[\{1, \ldots, 8\}]$, since the two leaves have no neighbors in K. In K,

$$N_K(2) = \{5, 7\}, \qquad N_K(1) = N_K(3) = \{4, 6, 8\}.$$

If 2 is omitted, 5 and 7 must both be selected, leaving at most one selected neighbor of whichever of $1, 3$ is omitted. Thus 2 is selected. Selecting exactly one of $1, 3$ leaves the other without two selected neighbors; selecting both leaves 5 with only the selected neighbor 2. If neither is selected, the remaining two selections must be a pair from $\{4, 6, 8\}$. The pairs $\{4, 6\}$, $\{4, 8\}$, and $\{6, 8\}$ fail respectively at vertices 8, 6, and 4. Thus no three-set two-dominates K. This proves $\gamma_2(G) = 6$, and $\mu(q + a) - 1 = 2(1 + 2) - 1 = 5$. □

For the same graph, the upper-rank quartile is $d_3 = 3$, giving the nonviolated proposed upper bound $2(3+2)-1 = 9$. Its 2,048 vertex subsets were independently checked by the standard-library program `verify_389a_rounding.py`; the construction and case proof above do not depend on the SAT solver used to discover the graph. This graph is not claimed to have minimum possible order.

21.8 Strict neighborhood containment forces positive slack

Here $b(G)$ is the maximum order of an induced bipartite subgraph and $\gamma_c(G)$ is the connected domination number.

Proposition 21.18 (A forbidden inclusion at equality)**.** *Let G be a finite connected graph. If $N_G[u] \subsetneq N_G[v]$ for two vertices u, v, then*

$$b(G) \geq \gamma_c(G) + 2.$$

Consequently, in an equality graph $b(G) = \gamma_c(G) + 1$, any two closed neighborhoods that are comparable by inclusion are equal.

Proof. The vertices u, v are adjacent. Choose $x \in N_G[v] \setminus N_G[u]$; then u, v, x induce a path. Use this path to initiate the rooted bipartite-growth construction of Subsection 9.2.1, and denote its first maximal connected induced bipartite piece by B_0. Every neighbor of u other than v is also adjacent to v. Such a vertex cannot belong to B_0, since it would form a triangle with u, v. Thus u has degree one in $G[B_0]$.

Take a spanning tree of $G[B_0]$. Its vertex u is a leaf, while v is not a leaf: it joins u to the nonempty remainder of the tree. Choose another leaf w. Deleting u, w leaves a nonempty connected set that dominates B_0. It also dominates every vertex outside B_0 having a neighbor in B_0. Indeed, maximality implies that such a vertex sees both color classes of B_0. If all its B_0-neighbors belonged to $\{u, w\}$, it would have to be adjacent to u, and therefore also to the retained vertex v, a contradiction.

Continue the cited construction with pairwise anticomplete pieces $B_1, \ldots, B_k$ and at most k connector vertices. From each later nonsingleton piece delete one spanning-tree leaf different from its attachment root; delete a later singleton piece entirely. Keep all connectors. Each connector retains a neighbor in the first earlier piece it meets: for B_0 this was just proved, and for a later piece it follows from the two-color property after deletion of only one vertex. A singleton piece cannot be that first piece. Every retained piece is therefore connected to the initial retained piece. The same neighbor property dominates all uncolored vertices, deleted tree leaves have retained neighbors, and a deleted singleton root is dominated by its connector. We obtain a connected dominating set of size at most

$$\sum_{j=0}^{k} |B_j| + k - (2 + k) = |B| - 2 \leq b(G) - 2,$$

since the union B of the pairwise anticomplete bipartite pieces is itself induced bipartite. This proves the inequality. □

This is a refinement of the credited DeLaViña–Waller construction. It supplies an additional structural restriction at equality; it is not a complete resolution of any further numbered statement.

A Archive provenance and coverage

The five supplied outer archives were opened recursively through their twenty-two round archives. Their extracted files were compared with the original member bytes by SHA-256. All 1,333 compared files matched; all 51 main LaTeX manuscripts were among the matched files. No original source archive was edited. The main source corpus contains 15,798 lines of LaTeX, before consolidation. Supporting proof notes and supplemental manuscripts are included separately in the source map.

A.1 Outer-archive identity

The following full SHA-256 values identify the exact supplied bundles. The final "(1)" in each local upload name is only the supplied filename suffix. Each digest is printed on two lines, with no character omitted.

Rounds	Bytes	SHA-256
1–5	2,196,897	`9277d9ea942b9dd21d43c947c87de211` `00363890548a68f3076071611eb57878`
6–10	3,273,667	`2779810d2cce8e8e0a5a488d347c825f` `a2bf81973a10ae30125c4c562e574dff`
11–15	11,544,658	`31c39ef6c8ea7b0992ea09e865b26bad` `1dae5603441f0249ef074d9300bbadc7`
16–20	10,857,175	`cbb57fdfbc2852d30428b45cc2611f86` `b3ead07431f605cf6e771e47f6a444a4`
21–22	1,367,329	`8f86fb2135deadecd620fc047ca2813d` `14963b8cfe795a128ff1a81bf1418a0b`

A.2 Round-by-round mathematical coverage

The table maps all rounds to retained content. Repeated improvements of the same item are integrated into their strongest terminal statement; their reusable lemmas and exact certificate dependencies are retained.

Round	Main papers	Retained subject matter
1	2	434d proof; 394 counterexamples
2	1	392e/f counterexamples; auxiliary 133 and 269 reductions
3	1	395a counterexamples
4	2	434e proof; 382c counterexamples
5	1	269 counterexamples; auxiliary 189 and 384b reductions
6	1	302 counterexamples; auxiliary 389a sufficient conditions
7	1	298 and the prescribed $4k$-cycle theorem
8	1	384b critical profiles, kernels, and alteration bounds
9	2	133 density and equality bounds; 384b refinement; 189/199 notes
10	2	177/180 claw-free cases; 384b refinement; matching padding
11	3	133 path restrictions; 171 counterexamples; 384b small kernels
12	3	133 radius four; 382e counterexamples; forest and residue results

Round	Main papers	Retained subject matter
13	2	133 dominating-five-cycle branch; 154 counterexamples
14	3	133 nondominating branch; historical 354 refutation; 384b core connectivity
15	2	133 global P_7-free density; 384b rank refinement
16	4	133 even-cycle classification; 384b refinements; unigraph inheritance
17	4	155 low-leaf case; 199 split cases; 384b refinement; 61 triangle-free equality
18	4	133 P_8 cores and radius five; 199 balanced case; 384b refinement
19	5	133 P_9 cores; 199 clique bound; 384b sampling/order; 61 apex forests
20	3	384b sampling/order refinements; 61 matching–geodesic case
21	2	199 Hall equality and exact factor; 384b order-37 representative
22	2	199 strict capacity and reach; 384b exclusion of $h = 35$

The early low-average result for 133 is contained in the later unrounded-average theorem. The Round 11–14 five-, six-, and seven-cycle arguments are included in the full global P_7-free proof. The separate radius-four attachment proof is replaced by the already proved general even-cycle theorem at $k = 3$, after the complete radius-four reduction; this substitution introduces no circular dependency. The earlier 384b representative bounds are superseded by 37, while their structural and sampling lemmas are proved in the combined chain. The balanced 199 proof is supplied once and then reused for padding and Hall equality.

A.3 Main-manuscript index

Each filename below is relative to the indicated round's original `paper/` or `papers/` directory. A twelve-hex-character digest prefix is printed for convenience; the full hashes and exact paths are in `provenance/source_manifest.json` in the package. The original source files are preserved there for comparison.

Round	Main manuscript	SHA-256 prefix
1	`KEIO_WOWII_R01_394.tex`	`31685100757c`
1	`KEIO_WOWII_R01_434d.tex`	`e3efdd2d0a62`
2	`KEIO_WOWII_R02_392ef.tex`	`48be15dff345`
3	`KEIO_WOWII_R03_395a.tex`	`2eb5c2ccc475`
4	`KEIO_WOWII_R04_382c.tex`	`4bd120d314ed`
4	`KEIO_WOWII_R04_434e.tex`	`b69324b11f4a`
5	`KEIO_WOWII_R05_269.tex`	`047f4f179ca7`
6	`KEIO_WOWII_R06_302.tex`	`00dd7a59883d`
7	`KEIO_WOWII_R07_298.tex`	`a61253f072c7`
8	`KEIO_WOWII_R08_384b.tex`	`369efb33c34a`
9	`KEIO_WOWII_R09_133.tex`	`3214b6b7ddd1`
9	`KEIO_WOWII_R09_384b.tex`	`49bab377de00`
10	`KEIO_WOWII_R10_177_180.tex`	`9f0b3c608b36`
10	`KEIO_WOWII_R10_384b.tex`	`fd9b6aa4811a`
11	`KEIO_WOWII_R11_133.tex`	`bb181f8b0267`
11	`KEIO_WOWII_R11_171.tex`	`4034f71ff698`
11	`KEIO_WOWII_R11_384b.tex`	`fb46660351c3`

Round	Main manuscript	SHA-256 prefix
12	KEIO_WOWII_R12_133.tex	1e2154125f3d
12	KEIO_WOWII_R12_382e.tex	1716684c54d7
12	KEIO_WOWII_R12_Forest.tex	7329132992b0
13	KEIO_WOWII_R13_133.tex	1a0d09cd05e4
13	KEIO_WOWII_R13_154.tex	8e47357e3726
14	KEIO_WOWII_R14_133.tex	8f7f56bd93c3
14	KEIO_WOWII_R14_354.tex	2943dcaf2beb
14	KEIO_WOWII_R14_384b.tex	22355a2d2fee
15	KEIO_WOWII_R15_133.tex	7471dcc46945
15	KEIO_WOWII_R15_384b.tex	61e6911853ce
16	KEIO_WOWII_R16_133_cycles.tex	be2fa44cd46d
16	KEIO_WOWII_R16_384b_h41.tex	0ca444303059
16	KEIO_WOWII_R16_384b_order42.tex	a78dcbeccd5a
16	KEIO_WOWII_R16_unigraph_auxiliary.tex	6677f073c5da
17	KEIO_WOWII_R17_155_low_leaves.tex	a62b9f6d70a7
17	KEIO_WOWII_R17_199_split.tex	4b759b011451
17	KEIO_WOWII_R17_384b_h40.tex	3322baf6c089
17	KEIO_WOWII_R17_61_trianglefree.tex	de1deae838ad
18	133_p8.tex	7ba353f8879f
18	133_radius5.tex	0fc11b81a579
18	199_balanced.tex	a015fec418cd
18	384b_h39.tex	3c8e13f187ee
19	133_p9.tex	bab20ad65bf5
19	199_clique_bound.tex	2ac3670001a1
19	384b_order40.tex	25b8fe84a922
19	384b_paired.tex	22b519f94531
19	61_apex_forests.tex	4df8b15b1e8f
20	384b_h37.tex	790789191ef2
20	384b_order39.tex	3e7b6113c1c0
20	61_matching_geodesic.tex	0fa44736f3dd
21	199_hall_equality.tex	650065c59273
21	384b_order37.tex	fe7dea9c3166
22	199_strict_capacity.tex	1054d26069b7
22	384b_h35.tex	3ff9cc3e34b0

B Verification and reproducibility

The accompanying package contains the standalone manuscript, the unmodified source materials retained for the proof record, the exact certificate data, and rerun outputs from this consolidation. The machine-readable manifests distinguish original source hashes from outputs newly generated by a verifier. Exploratory search histories do not substitute for the proved extension, reduction, or coverage arguments in the paper.

B.1 What was rerun

The verification work is organized into the following groups. A program count is an execution count within the specified group, not a count of solved conjectures or independent human reviews.

Group	Programs	Scope
Early conclusions	10	434d/e checks; exact 394, 392e/f, 395a, 382c, 269, and 302 constructions; 298 coloring and the credited antecedent chain's finite checks
Later structural work	6	171, 382e, 154, and 354 witnesses; exact 389a parameter inequalities; matching-padding and weak-majorization checks
384b	21	Small kernels and their independent input coverage; exact rank, sampling, matching-repair, and representative-order certificates
133, 189, and 199	16	The two closed core certificates; the general even-cycle path checks; four 189 reduction/example guards; nine 199 incidence, closure, factor, capacity, and warning-example checks

All listed final runs passed. The original mathematical arguments, rather than the breadth of these finite tests, justify the analytic theorems. The finite proof dependencies are identified in the body.

The P_8 certificate has 192 graph states, maximum state order 25, and 93,895 required one-vertex extensions. The P_9 certificate has 13,930 graph states, maximum state order 45, and 442,204 contact-tree nodes. Both checks account for every permitted branch, with zero unexplained frontier. The maximum state orders are observed properties of closed certificates, not assumptions on target graph order. Their standard-library verifiers check paths and permutations directly.

For 384b, the 21 reruns include coverage of the small graph kernels, not merely arithmetic on a list of unspecified completeness. Exact integer and rational checks verify the potentials used for the later matching repairs. The $h = 35$ computation uses exactly thirteen active matching edges, with the correct conditioned probabilities; it does not replace them by independent-edge probabilities. The final representative theorem additionally uses the analytic low-vertex trimming and simultaneous-omission lemmas.

B.2 Restored dependencies and explicitly limited reruns

Two packaging issues required particular care. The Round 17 $\kappa = 3$ checker expected a companion family JSON file that was absent from its main verification directory but present in that round's independent-audit snapshot. That original file was restored to the rerun directory, with its hash recorded; the checker was unchanged and then passed.

The Round 7 independent checker completed its mathematical tests but then attempted to hash two prior manuscript snapshots absent from the supplied archives. In a separately recorded copy, only those final metadata hash operations were omitted. The mathematical tests were unchanged. The original and adapted checker hashes and this limited adaptation are recorded in the audit. That limited adaptation belongs to the original consolidation record. For the revised edition, the primary manuscript archive was independently retrieved and the written proof checked. The archive digest is

`d067e88bdbf41f4fd4405b61e7cc71acfc9e4a6a6e64d7ca8d55f35b289d2afc`

The source URL, retrieval record, and dependency audit are retained in `provenance/revision/298_primary_audit.md`. The classical transversal bound and all author attributions remain unchanged.

No external Lean certificate was rebuilt. No claim is made that every historical exploratory enumeration or discovery program was rerun. In particular, a historical output's older residual range or "pending" label is not used as the terminal theorem's status when the subsequent audited proof gives a stronger conclusion.

B.3 Checks added for the revised edition

The original 53-program verification record remains a historical record of the supplied consolidation. In this edition the additional 394 checker was rerun on the triangle-free 15- and 19-vertex witnesses, enumerating all 2^{15} and 2^{19} vertex subsets respectively and independently checking maximum matching, local independence, and odd-cycle-transversal numbers. The new results have direct written proofs for arbitrary family size. Additional independent checks and research-specific certificates are identified under the `extensions` verification group. Original proof-certificate data and source texts are preserved.

B.4 Using the package

The root-level `README.txt` identifies the paper, source archive, verification groups, and exact commands. The companion `verify.py` lists or runs the recorded checks and compiles the small C++ verifiers when needed. Python code uses either the standard library or the explicitly listed graph-library dependency; C++ checks use the documented compiler options. The paper itself is a single LaTeX source file with its bibliography inline and no external figure or input-file dependency. It can be rebuilt by running `pdflatex` until the cross-references stabilize.

C Attribution and consolidation corrections

The following distinctions are maintained throughout the paper. The total-domination–annihilation antecedents are credited to the research record attributed to Gebendorfer; the classical transversal bound is credited to Chvátal and McDiarmid. The half-coefficient 354 counterexample modifies a construction credited to Feit. The precursor for the Maxine mechanism is credited to Cao and Eppstein. The common-neighbor observation used in the induced-forest discussion is credited to Gunderson and collaborators, as corrected in the later source audit. The triangle-free P_8 classification is credited to Chudnovsky and Stacho, and the Y–Δ–Heawood graph is a known graph. Hall's theorem, Hamiltonian closure, elimination-sequence majorization, and the published matching bounds retain their original attributions.

The consolidated 133 statements explicitly restore the connectedness assumption of the original deficit and equality theorems. The phrase "distance-two dominating" is used for distance at most two from a cycle, to distinguish it from two-neighbor domination. The local triangle count is denoted consistently by $t(v)$, while the total local independence is Λ. Graph6 data strings were compared with the source strings and preserved. The 199 incidence formulation states that an induced tree with two clique vertices may use only exclusive neighbors, and a maximum such tree uses all of them. These are clarifications of scope and notation during integration, not new conjecture conclusions.

The revised manuscript preserves every complete original-item conclusion and the proved partial programs through Round 22. The subsequent structural extensions are identified separately. No unproved cycle-elimination claim, finite failure to find a counterexample, or exploratory search is promoted to a full resolution. The 392e/f family statement now explicitly restricts "fails" to positive-deficit parameters: $r \geq 3$ and $r \geq 4$ respectively for J_r; the original principal witnesses are unchanged.

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